

The Core Throttling

— BOOK TWO OF THE ENERGY NEXUS TRILOGY —



BY KALAMAZOO

Kalamazoo

The Core Throttling

The Core Throttling

Book Two of the Energy Nexus Trilogy

A Novel

By *Kalamazoo*

July 2026

Deep beneath the Antarctic ice, something ancient is still running.

After the events of *Iceland*, intelligence analyst Jason Cavanaugh and physicist Dr. Joanne Sanderson are transferred under military escort to Operation Cold Horizon, a classified drilling site built above a vast subterranean structure. The object is not inert. It is processing, drawing energy from the planet, and producing the same impossible signal first detected in the sub-zero event.

As Sanderson studies the structure's thermal behavior and Cavanaugh reconstructs the operation's concealed data trail, they discover that the site's commanders have been filtering evidence, falsifying timelines, and preparing to force a direct interface with technology they do not understand.

The deeper they descend, the clearer the danger becomes. The structure is not merely cooling the Earth's interior. It is regulating a system that extends far beyond the planet—and humanity may already be inside it.

The Core Throttling is the second book in the *Energy Nexus* trilogy, a hard-science techno-thriller about hidden infrastructure, institutional control, and the consequences of mistaking access for understanding.

The Core Throttling

Kalamazoo

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The Core Throttling

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Note on AI Collaboration

This novel was created through collaboration between the author and artificial intelligence tools. Artificial intelligence was used during portions of the planning, drafting, revision, and editorial process. The author directed the conception and development of the story, selected and revised the material, resolved narrative and factual decisions, and approved the final manuscript.

Responsibility for the characters, plot, themes, language, and published work rests solely with the author.

Contents

Chapter 1: The Borehole Camera..... 10

Chapter 2: Military Escort South..... 22

Chapter 3: Arrival at Cold Horizon 41

Chapter 4: Intake and Compartmentalization 56

Chapter 5: Baseline 74

Chapter 6: First Contact..... 96

Chapter 7: First Data Bay Pass 112

Chapter 8: Contact Angle 132

Chapter 9: Diagnostic Spike..... 148

Chapter 10: Ancillary..... 164

Chapter 11: Curated Logs..... 186

Chapter 12: Heat Sink Model..... 209

Chapter 13: Maintenance Outage..... 233

Chapter 14: Reality Blink Files..... 255

Chapter 15: Sanderson’s Independent Storage Core..... 278

Chapter 16: Underwood’s Utility Framing 304

Chapter 17: Sanderson’s Warning 327

Chapter 18: Proof Streams 351

Chapter 19: House Arrest..... 372

Chapter 20: Lock Bypass 391

Chapter 21: Descent Under Countdown 414

Chapter 22: Cavern Crossing 433

Chapter 23: Direct Interface 457

The Core Throttling

Chapter 24: Unredacted Terminal482

Chapter 25: The Grid in the Sky.....505

Chapter 1: The Borehole Camera

At 03:12 station time, Halden noticed the counter before he noticed anything was wrong.

Monitor Four held on 6,481 meters. The last digit didn't roll over. It sat between two numbers, flickering, the way a scale needles between ounces when no one has decided to trust it yet.

He set his coffee down without drinking it.

"Winch status."

The operator tapped the gauge casing with two fingers, an old habit that had never once fixed anything. "Motor idle. Brake engaged. Cable's not hung."

"It's hung somewhere."

"No, sir." The man's voice stayed level. His shoulders had climbed half an inch toward his ears. "If it were hung, we'd be pulling load. We're not."

Halden looked at the tension readout. Three percent up. Five down. Back to nominal, no motor correction logged anywhere in between. He'd run winches on two continents and never seen a line move like that on its own.

The Core Throttling

The overnight tech stopped chewing the inside of his cheek. The seismic operator leaned in toward her screen without seeming to decide to. Above the east bulkhead, the heater clicked off and did not click back on, and the room settled into the particular cold that always found the seams in insulated walls down here, a cold with texture to it, dry and specific.

Nobody had asked for quiet. It arrived anyway, one dropped sound at a time.

On the main display, the borehole feed showed gray meltwater turbulence and the oval ghost of the camera's own light thrown back off ice-slicked stone. The image rolled, corrected, settled a few degrees off level. Along the bottom edge, a data string updated in its usual scroll: temperature, pressure, orientation, cable load, casing integrity, lens heater draw.

The lens heater draw went red.

"Why is the heater cycling?"

"It isn't cycling." The video tech had both hands flat on the console edge. "It's being pulled down."

"By what?"

Nobody had an answer for that one either.

The control module wasn't much to look at — steel decking, composite wall panels somebody had painted a

corporate gray that had never once made the place feel less temporary, cable trays overhead furred with frost where the ceiling seals had shrunk in the night. Through the narrow window, the gantry stood black under floodlight, its lattice softened by blowing ice.

Eleven months to drill. Six line items buried across three appropriations cycles that Halden had stopped trying to reconcile in his head. Four miles of ice, refrozen melt, sediment, fractured basalt, and — according to the geologists who'd signed off on the survey — more thermal noise at depth than any public model accounted for. The file called it a geothermal anomaly assessment. Nobody in the room had used that phrase without irony since the camera passed the lower shaft marker.

Halden pulled his headset mic an inch closer to his mouth. "Camera team, confirm local mechanical."

Wind answered before the voice did. "Guide wheel iced again. We're clearing it."

On the auxiliary feed two suited figures worked the exterior winch housing, one of them swinging a rubber mallet in short economical arcs. The first strike vanished into the gale without a sound reaching the mic.

The Core Throttling

The second sent a fan of ice crystals scattering white across the floodlight cone.

Inside, the depth counter moved.

6,482.

6,487.

6,501.

"Brake just slipped," the winch operator said.

Halden turned to him.

The man's hands were already on the controls, moving through a sequence Halden had watched him run a hundred times without a hitch. "I didn't release it."

On Monitor Four the camera dropped through gray and then through black, the image smearing into vertical bands as the borehole walls fell out of frame. The pressure value dropped out of range and stayed there. The orientation compass spun through headings that didn't correspond to any direction the interface had ever been asked to render, then settled on a string of characters that weren't characters at all, just placeholder glyphs standing in for something the software had no slot for.

The seismic operator pushed her chair back six inches. "We lost wall contact."

"Void?"

"Could be. Could be collapse."

"A cavity doesn't stay open at this depth."

"It's open."

The camera light swept a slow arc across nothing. For a stretch of two, three seconds there was no scale to hang a guess on — ice crystals hanging in suspension, black vapor drifting through the beam, no far wall giving the light anywhere to land. The rig turned slowly on its own cable, unassisted, the beam finding nothing, nothing, nothing.

Then it found a surface.

Not the basalt the survey had promised. The nearest wall rose past the top of frame, faceted, dark, ridged in repeating planes that met each other at angles too clean to be a fracture pattern. The beam struck it and didn't come back. No glare. No scatter. No wet mineral sheen, no ice shine. The light went into the surface and stopped existing there.

"That's not basalt," the video tech said, to no one, and no one told him he was wrong.

The camera kept rotating. More of the cavern resolved out of the dark as it turned — broad metallic growths rising from floor and wall in nested, layered geometry, the kind of shape you'd get if you asked a

The Core Throttling

machine to build a crystal and told it symmetry was optional. Some of the formations climbed past the reach of the light. Others bridged the open space in ribs thick enough to carry four lanes of traffic.

A pulse of blue-white light moved through one of them.

It didn't travel the way current traveled. It went in rectangular segments, each one lighting a fraction of a second after the one before, the way a marquee sign refreshes panel by panel rather than all at once.

The room's monitors dimmed, all of them, together.

"Power."

"Station feed's stable." The electrical tech already had his hands on the board. "Local draw's stable too."

"Then why did my screens just dim."

The video tech swallowed audibly. "Feed gain dropped."

"Restore it."

"I didn't reduce it."

The camera dropped another meter with no command behind it. Cable tension rose one percent and held there.

The exterior team's voice cut back in over the wind.

"Control, guide wheel's clear. You should have normal movement now."

The winch operator was still staring at his panel.
"We don't have normal anything."

On Monitor Four the light stuttered. The lens heater failed, rebooted, failed again. Frost crept in from the frame edge — not on the housing where frost belonged, but inside the image itself, white branching lines spreading across the pixels in patterns that looked like ice for exactly as long as it took them to start meeting at right angles.

The seismic operator went still in a different way than before. "I'm getting vibration."

"From the winch?"

"No. From under us."

The floor gave a low shiver, not enough to spill anything, enough that every cable hanging in the overhead trays tapped once against its neighbor and went quiet again.

The feed broke into static, cleared, and came back on a new angle. Something below the camera's position gave off a faint internal glow of its own, low and steady, and the nearest structures answered it in sequence, blue-white lines running the length of their seams and disappearing into a darkness where the cavern

The Core Throttling

roof should have closed the space off and didn't. The camera's exposure adjusted, failed, adjusted again, chasing a light level that kept moving out from under it.

For three seconds the room saw the shape of the thing.

It wasn't a machine, not in any sense the word usually did work for. It wasn't a wall or a vein or a chamber built to hold something. It was a fragment — visible, lit, and clearly only a fragment — of a matrix of metal and crystal grown or built or extruded through stone and ice in a way that made the crust look like it had closed around the object afterward rather than the other way around.

The low vibration came again, and this time it wasn't only in the floor. It came up through the audio feed too — a hum under the register of speech, under the register of any machine Halden had ever stood next to, low enough that it found the desk panels before it found the ear and made breathing feel shallower than it had a minute ago for no reason anyone could name.

The acoustics station's waveform display opened on its own. A technician who hadn't said a word all shift pulled a glove off with his teeth when the fabric wouldn't register on the touchscreen, and worked the display bare-handed.

"Frequency," Halden said.

The tech didn't answer fast enough.

"Frequency."

"Seventeen point eight hertz, primary.

Harmonics above forty." He looked up. "There's EM coupling. It's not just acoustic."

"Source."

He looked at the waveform, then at the borehole feed, then back. "The cavern."

The camera light bloomed white — not from brightness, from something closer to the opposite, the image overexposing until the structure became a black shape cut out of glare rather than a lit one. The tension alarm sounded once and cut off in the middle of its own tone. The telemetry strip collapsed into blocks of characters nobody's software had a use for. Lens heater draw dropped to zero. Internal temperature dropped to zero. Then the temperature field went blank, no reading at all, not even an error code.

"Bring it up."

The winch operator engaged retrieval.

The motor hummed. The cable didn't move.

"Bring it up."

"I'm trying."

The Core Throttling

Outside, on the auxiliary feed, the winch housing shook hard enough to blur in the floodlight. One of the suited figures stepped back, slipped on the ice sheeting the deck, caught the railing with both hands, and looked down toward the borehole the way a person looks at something that has just pulled at them and let go.

Inside, the east bulkhead heater clicked back on. Warm air coughed out of the vent, carrying the smell of dust cooking off a coil that had sat cold too long.

The borehole feed died.

Monitor Four went black, and for one full breath the only sound in the room was the ventilation system and the tick of metal cooling somewhere in the housing behind the wall.

Then a line of text appeared in the center of the dead screen.

Not the camera interface. Not the drilling software's font, its spacing, its palette. White letters, fixed-width, all capitals, set against black with nothing else on the screen to compete with them.

SIGNAL ACQUIRED

The video tech pushed his chair back from the console.
"That's not our overlay."

"Record it," Halden said.

"We are."

The line disappeared. A second one replaced it.

PATTERN COMPARISON: ARCHIVAL
MATCH

The seismic operator's mouth moved once with no sound behind it. She pulled up a side window, her hands moving faster than the system wanted to follow — directories opening, closing, reopening, permission flags flashing red and then gray on a half-second delay that had nothing to do with her typing speed. A classified archive index populated under a routing label Halden had been told, in writing, never to say aloud outside a cleared compartment.

He felt the room notice him noticing it.

"Kill external feeds."

Nobody moved.

"Now."

The comms tech cut the outgoing relay. The small red transmission light on the panel went dark, flickered once on its own a half second later, as if something on the other end of the line had wanted the last word on whether it was really off.

The Core Throttling

On Monitor Four, one more line resolved out of the black.

SIGNAL MATCH: ICELAND SUB-ZERO EVENT

The borehole camera did not come back online. Nobody in the room suggested trying again.

Below the station, under the shaft and the ice and every instrument built to put a number on the world, the hum went on.

Chapter 2: Military Escort South

The aircraft dropped hard enough to lift Cavanaugh's shoulder blades off the webbing, and the restraint caught him across the chest before he understood he was awake.

For a half second the cabin was red light, turbine noise, and the metal taste that came from sleeping with his mouth open in dry recycled air. Someone forward swore into an oxygen mask. Cargo straps snapped against the crates behind the bulkhead, once, twice, then went quiet as the aircraft leveled with a long groan that traveled the length of the fuselage.

He did not reach for his laptop.

Thirty-six hours ago that would have been automatic. It wasn't anymore. The laptop sat twelve feet away, sealed in a transparent evidence bag with a red tamper strip, clipped to a cargo net beside his phone, his field notebook, his encrypted token, his watch, and the pen he'd taken from the NRO conference room because the government pens skipped on cold paper and this one didn't. Each item had its own tag. Each tag carried a

The Core Throttling

classification line that didn't match anything in his training.

He'd checked twice at Andrews. Once more at Thule. The markings hadn't changed.

Not a clerical error repeated three times running. A category built for this specifically, applied consistently, by people who knew what they were doing.

He'd spent most of the last thirty-six hours cataloguing what he was no longer allowed to carry. His badge, surrendered at Andrews and replaced with a visitor tag that had no expiration date printed on it. His secure phone, swapped for a loaner that dialed exactly one number. His access to systems he'd used every working day for six years, revoked without a memo shown to him, which was itself the kind of gap he used to flag in other people's files before anyone thought to ask about his own. He had built a career on noticing when an operation was being run around someone instead of through them. He had never before been the subject line.

The aircraft hit another pocket of air and the seatbelt latch dug into his hip. He thumbed the release, meaning to loosen it half an inch, no more. The mechanism didn't move. He tried again with less force, in case the first attempt had jammed something. Still nothing. Ice in the hinge, or a bent pawl left by whoever sat here last, or the ordinary talent military hardware

had for failing exactly when a man needed not to look like he was fighting his own chair.

He stopped pulling at it and made himself sit still instead, which took more effort than the latch had.

Two rows forward, across the aisle, Dr. Joanne Sanderson was already awake.

She sat with one shoulder braced against the vibrating wall, hair pulled back badly after too many handoffs and no mirror, chin down so the red cabin light caught her jaw and not her eyes. Her sensor cases rode in the cargo section behind her, and when the loadmaster shifted his weight Cavanaugh could see one stenciled corner: SANDERSON / DTRA / THERMAL FIELD PACKAGE A. A second label had gone over it since Thule, black block text, fresh enough that the adhesive edge still showed a seam.

CONTROLLED ASSET.

She'd seen it. He could tell from how carefully she wasn't looking at it.

Cavanaugh had watched her argue with a DTRA logistics officer at Thule for eleven minutes over the calibration schedule on that same equipment, insisting it be run within four hours of arrival or the baseline

readings would be worthless. The officer had said yes to everything and changed nothing, the particular kind of yes that meant a decision had already been made above him and he'd been told not to complicate it with details. Sanderson had stopped arguing exactly when Cavanaugh would have stopped, at the point where the argument had become information rather than a real negotiation.

A soldier sat between them, in the seat that would have let Cavanaugh and Sanderson speak without shouting. No unit patch. No name tape visible above the cold-weather harness. He had the flat, watchful boredom of a man given one order and nothing to explain it, and he'd spent most of the flight pretending not to watch either of them, badly.

Cavanaugh had tried to place him somewhere around the second hour out of Thule and given up. Not Underwood's people — he was guessing at that, since he'd never met Underwood, only heard her name attached to two different transfer orders and one clipped radio transmission that had used his name like a line item. This man's gear was too generic, his boots too clean for someone who lived on the ice, his patience too total for someone doing a job he'd trained years for. He had the look of a man borrowed for the flight and told to say as little as possible, which put him somewhere below the level that

mattered and made him, for Cavanaugh's purposes, mostly useless except as a barometer.

Cavanaugh leaned as far as the stuck latch allowed. "Where are we in the chain?"

The soldier looked at him and said nothing.

"Custody chain," Cavanaugh said. "My property's been inventoried under a security authority. I'm asking which one."

"You'll be briefed on arrival."

"I've been told that at Andrews. At Thule. Somewhere over the South Atlantic, by a colonel who kept calling me Doctor."

Nothing.

"Either the paperwork's bad," Cavanaugh said, "or whoever's moving us doesn't want a clean record of who had custody when."

The soldier's eyes flicked once toward the evidence bags on the cargo net. Too fast to write down. Long enough to count.

"Sir. You'll be briefed on arrival."

"That answers a different question." Cavanaugh let his head settle back against the seat frame and said nothing else for a while, because the soldier had told him

something true without meaning to, and there was no reason to spend it on a second answer he already expected.

Overhead, the red lights dimmed, came back, dimmed again in a pulse that didn't track with the engine note. A low alarm chirped somewhere forward and was silenced within half a beat, the way a sound gets killed by someone who already knows what it means and doesn't want it explained to anyone else on the aircraft.

Cavanaugh counted the pulse anyway. Three dims, uneven spacing, then nothing. He had no baseline for aircraft electrical behavior and no way to log it, so he did what he could still do without a recording device: he held it in his head, the way he'd once held a filename until he could reach a terminal to write it down properly.

Sanderson turned toward the loadmaster, who was checking a manifest on a tablet bolted to the bulkhead. "Are those capacitor banks?"

He looked up as if he'd forgotten she was allowed to talk. "Station support."

"That's not a generator crate."

"Ma'am, I'm not cleared to discuss cargo."

"You don't have to discuss it. I can read the stencil."

Cavanaugh angled his head past the soldier's shoulder, past the bundled straps and the frost scraped thin along the door seal. Three long crates sat chained double to the floor of the bay. Not drilling spares. Not instrument cases. Hazard diamonds stenciled at the corners, a coil-warning symbol, a discharge warning printed twice on the same lid as if once hadn't been enough for whoever loaded it. The shape of something built to move a great deal of stored electrical force in very little time.

Sanderson looked back at him. There was no room for a sentence, no privacy in it at all with the soldier sitting between them, so she lifted two fingers off the armrest and lowered one.

One controlled pulse.

He gave her the smallest nod he could manage without moving his shoulders.

The soldier saw that too.

"Mr. Cavanaugh."

"That's the first time anyone on this aircraft has used my name."

"Please remain seated."

The Core Throttling

"I'm trying." He pressed the latch once more, felt it hold, and let his hand drop away as if the problem had simply bored him. The last thing he wanted was a man trained to restrain people in moving aircraft deciding to be helpful.

The intercom cracked and resolved into the pilot's voice. "Ten minutes to descent corridor. Remain strapped in. No unsecured equipment. Cold-weather masks on before door cycle."

A second voice came in under it, thinner, routed through a different channel. "Horizon Control, this is Victor flight, two controlled civilians aboard with cargo package—"

The line cut.

Cavanaugh looked toward the cockpit bulkhead, not because he expected to see anything through it but because the sentence had left a shape he wanted to keep looking at.

"Victor flight," he said. "That's not a cargo designator."

The soldier's jaw tightened a fraction, which Cavanaugh took as its own kind of confirmation, though he didn't say so.

"It's an escort designator."

The soldier's gloved hand shifted an inch toward the retention strap over his sidearm. Not a threat, or not only that. A reminder, delivered without heat, which made it more useful

to Cavanaugh than any answer would have been. Men didn't reach for reminders like that over station support crates.

Cavanaugh let the silence sit rather than fill it. In his experience a man who'd just given something away without meaning to would rather talk past it than sit inside it, and the ones who could sit inside it were the ones worth watching later. This one shifted his weight, checked the aisle in both directions, and said nothing further, which put him in neither category cleanly. Cavanaugh kept it anyway, without much confidence he'd ever get to use it.

"Jason."

He turned. Sanderson hadn't raised her voice, and the engine should have swallowed it at that volume, but the cabin had dropped into one of those brief pockets where the vibration canceled itself and sound traveled farther than it should have any right to.

Her eyes went to the evidence bags. Then to the soldier. Then down to her own wrist, where her watch should have been and wasn't.

Time taken. Tools taken. No channel left between them but this one, and it wasn't much of one.

The Core Throttling

"Understood," he said, because it was the only word that fit all three at once.

"No cross-cabin conversation," the soldier said.

"That wasn't one," Sanderson said.

"It is now."

She looked at him the way she might look at a technician who'd mishandled a sample. "Then write it down."

The loadmaster turned his face toward the manifest and failed, not entirely, to hide a smile.

Cavanaugh found the latch again with his thumb, worked one glove seam under the release lip, and eased it sideways, a fraction of a turn at a time, until the iced hinge gave with a small click he felt more than heard. The belt loosened enough to let him breathe without the strap cutting into his ribs. He left it fastened.

The soldier watched him do it and said nothing, which Cavanaugh took as the closest thing to a concession he was going to get from this particular man. Somewhere behind them the loadmaster returned to his tablet, thumb moving down a column Cavanaugh couldn't see the header of, and the cabin settled into the particular quiet of people who had run out of things they were allowed to say to each other.

The descent began in the floor first, a change in the engine pitch that reached his stomach before it reached his ears.

The fuselage wall cooled through his sleeve, faster than the cabin air, which stayed dry and overheated and smelled of dust cycling through the same ducts it had cycled through since Thule. Frost feathered out from a rivet head near the window. Below them the cloud layer broke into pieces and let the ground show through.

White shelf. Black open water somewhere at the edge of it. A long blue wound of crevasse field under a sun that gave almost no heat at this angle. Then nothing but wind-scoured surface ice, flattened by altitude into something with no texture at all, no roads, no vehicle tracks, nothing to tell him a human being had ever crossed it under their own power.

He'd read the country before the flight, in the thin briefing packet they'd let him keep for four hours before collecting it again. Coordinates redacted. Elevation redacted. Nearest station redacted, though the packet had at least admitted there was one, which was

The Core Throttling

more candor than he'd gotten from anyone in a uniform since Andrews.

Sanderson leaned to see past the row opposite hers.

"That isn't McMurdo."

The loadmaster didn't look up from his tablet.

"Then it's not a standard support station," Cavanaugh said.

"Final warning," the soldier said.

"For what."

"For continuing to ask questions."

"That wasn't a warning," Cavanaugh said. "That was a policy preference." He let it stop there. The point of pressing an escort wasn't to win an exchange with him, and it certainly wasn't to be liked by him. It was to find out where the wall started.

The wall had started earlier than location. Earlier than custody authority. Earlier, even, than the moment someone decided a physicist and an analyst needed a joint escort instead of separate transport, which was its own kind of decision, made by someone weighing the risk of letting them travel apart against the risk of letting them travel together. Whoever had made that call had judged the second risk smaller. That meant they expected something from the two of them working in tandem,

or they expected to control that tandem more easily than two separate lines of movement, and either reading put their expertise somewhere below their containment on whatever priority list governed this operation.

He watched the window and said none of it out loud.

The aircraft banked. Lights resolved below, first as a smear the eye tried to flatten into something ordinary, then, as the aircraft dropped, into a hard, even line across the ice — too regular for a research camp that had grown the way research camps grow, expansion by need, one module bolted onto the last. Darker rectangles took shape around the line: modules, gantries, a vehicle shelter, fuel berms, antenna shrouds. An inner perimeter marked by towers spaced with the kind of geometry that belonged to a prison yard and not a science station — even intervals, overlapping fields of view, no gaps a planner would tolerate if the only concern were weather and wildlife.

He counted the modules out of habit, the way he'd count server racks or filing cabinets, anything that might later resolve into a floor plan. Eleven structures visible from this angle, not counting the vehicle shelters,

which put the station well past the size any unclassified manifest had ever mentioned for a seasonal drilling camp. Antenna shrouds clustered near the largest module rather than spread for coverage, which read less like communications infrastructure and more like something built to listen in one direction only.

At the center of the light line stood something like a derrick that had been swallowed by armor plating.

A drilling tower. Enclosed. Floodlit from four directions so it threw almost no shadow. Guarded.

Sanderson's hand tightened on the armrest. The glove seam near her knuckle had already torn somewhere over the last two days, and the skin under it looked raw in the cabin light, though she didn't look down at it and neither did he, out loud.

Cavanaugh turned back toward the cargo bay without meaning to make it a decision. Capacitor banks. Two controlled civilian experts under a security authority nobody would name. No outside channel. A black-site drill camp lit in straight lines by people who did not design light discipline for geologists.

Somewhere behind him a crew chief moved down the aisle checking mask seals, working from the tail forward. He stopped at the soldier first, then at Cavanaugh, tugging the strap at his temple hard enough to leave a mark, saying nothing about it either way. The mask smelled of the last person who'd worn it

and something chemical underneath that, rubber and cold plastic, and Cavanaugh breathed through it experimentally and got a thin, over-filtered version of the recycled cabin air, no better and no worse.

The crew chief reached Sanderson last. She'd already seated her own mask before he got there and sat with her hands folded over the strap adjustment, waiting rather than asking, the way she'd apparently decided somewhere over the South Atlantic that fewer requests meant fewer refusals. The crew chief checked her seal with two fingers at the temple, nodded once, and moved on without a word, and Cavanaugh understood from the economy of it that this was a man who'd fitted a hundred masks in his career and had opinions about none of the faces under them.

Outside the window the light had changed again, gone flatter, the way it did in the last minutes before an aircraft committed to a descent it couldn't abort. He watched his own reflection ghost over the glass, mask strap crossing his jaw at an angle that made him look like someone else's photograph, and thought, without much conviction, that whatever paperwork existed to explain this flight would say none of this. It would say two

The Core Throttling

subject-matter experts had been transported to a research annex for consultation. It would not say anything about capacitor banks, or a soldier's hand near his sidearm, or a designator that didn't belong on a cargo run.

The landing gear dropped with a heavy mechanical clunk that ran up through the floor.

His phone shifted inside its evidence bag and tapped once against the laptop case beside it, glass against aluminum — a small ordinary sound, the kind two devices make bumping in a drawer at home, out of place here in the belly of an aircraft carrying them farther from any authorized channel either of them had left.

The pilot brought them down hard enough that Cavanaugh's teeth clicked together. Snow blew sideways across the window in a flat white sheet, then thinned, and the aircraft rolled and slowed and turned toward a floodlit apron where two files of armed personnel stood waiting, faces hidden behind masks and amber-tinted goggles, breath fogging out from under the mask seals in short bursts that didn't quite sync between the two lines.

Cavanaugh counted twelve in the near file, ten in the far one, and stopped counting when the aircraft's angle put half of them behind the wing. Fuel trucks idled beyond them with their

running lights on and their hoses still coiled, which told him refueling wasn't the aircraft's next order of business. A low tractor unit sat between the trucks with a flatbed trailer behind it, empty, backed into position as if someone had already decided what it would carry before the aircraft touched down.

The intercom clicked once.

"Welcome to Cold Horizon," the pilot said, in the flat voice of a man reading a line he'd been handed rather than one he'd chosen.

No one in the cabin answered him.

Beyond the personnel and the fuel trucks and the rigid station lights, the dark seemed to carry something under it — not a sound Cavanaugh could have named on a form, more a pressure behind the ears that came and went with the wind and didn't quite belong to the wind. He didn't try to describe it to himself further than that. There wasn't a category for it yet, and he'd learned enough in the last two days not to manufacture one just to have something to hold.

The soldier unbuckled first, stood, and reached for the release on Cavanaugh's harness before Cavanaugh could do it himself. Cavanaugh let him.

The Core Throttling

There was no version of this where refusing a hand on a belt buckle bought him anything.

"Personal effects stay with the aircraft," the soldier said.

"You'll be issued cold-weather gear at the ramp."

"And the evidence bags?"

"Inventoried separately."

"By whom."

The soldier didn't answer that one either, which told Cavanaugh most of what he needed for now.

Down the aisle, the loadmaster had already begun unclipping the cargo net that held the sealed bags, working with the brisk indifference of a man handling freight rather than someone's confiscated life. Cavanaugh watched him lift the laptop bag by its corner, glance at the tamper strip, and set it into a hard case with a foam cutout shaped for it exactly — a cutout that had been cut before this flight, not after, which meant someone had known the dimensions of his laptop before he'd been told he was coming here at all.

He stood, joints stiff from thirty-some hours of transport seating. Ahead of him the ramp door began its cycle, a heavy hydraulic sigh followed by a blast of air so cold it registered as a physical weight before it registered as temperature, and the soldiers on the apron shifted their footing

Kalamazoo

in near unison against it, a small collective flinch that told him this wind was ordinary to the station and still worth flinching at.

A figure near the front of the near file lifted a hand, not a wave, more a signal to whoever was watching from inside one of the modules that the aircraft had emptied its manifest. Cavanaugh watched the gesture and let it go, because it wasn't anything yet, only a hand raised in cold air by a man whose face he couldn't see.

He looked once more across the aisle at Sanderson.

She wasn't looking at the soldiers filing toward the ramp door, or at the drill tower lit up white beyond them, or at the vibration in the ice that neither of them had a word for yet.

She was looking at the capacitor crates.

Chapter 3: Arrival at Cold Horizon

The stairs were still shuddering under the last passenger when the wind found the gap at Sanderson's collar.

She had budgeted for cold. Utah before sunrise, the overcooled clean room in Nevada where her fingertips had gone numb through two pairs of gloves, a February in Fairbanks where every BNC connector became something to be handled like a loaded weapon. She had a working theory of cold, built over fourteen years, and it did not survive contact with the stairs of a C-17 parked on the Ross Ice Shelf. The wind did not blow so much as arrive, continuously, carrying ice in grains fine enough to work into the seam of her hood, under the flap of the zipper, into the half-centimeter where her left glove had not sealed against her wrist.

A sergeant at the base of the stairs was saying something. She caught none of it. She gripped the rail and went down anyway, because staying on the stairs was not an option the wind was offering.

The station was half a kilometer past the strip, low, armored, wrong. From the air it might have photographed as a research camp — modules on skids, corridors elevated on cable trays above the ice to keep the permafrost from swallowing

them. On the ground it read as something else entirely. The perimeter fence ran in two lines, ten meters apart, hung with thermal-imaging housings at intervals too tight for a weather perimeter. Two tracked vehicles sat idling by a snow berm, their exhaust shredded flat and sideways by the wind before it cleared the hull. Personnel stood post at spacing that had nothing to do with visibility and everything to do with interlocking fields of fire.

At the center of it, above what should have been a borehole, a gantry rose forty meters into the floodlit dark.

It was not proportioned like anything she had drilled before. The steel legs braced wide against the ice, oversized for the load a scientific core sample would ever put on them. Heated cable ran in black arcs between anchor points, thick as her forearm, kept from freezing by systems that had their own power draw and their own maintenance schedule. A lift housing crouched under an insulated shell at the center, doors striped in hazard yellow, three separate access panels bolted rather than latched. The floodlights turned the blowing snow

into static, a visual noise that made the whole compound look like something transmitting badly.

Sanderson stopped at the bottom of the stairs.

One of her crates was already forty meters away, moving on a sled behind a snowmobile, its runners kicking a rooster tail of ice into the wind.

"Hey." She went after it before she had decided to.

The sergeant stepped into her path, not touching her, close enough that the choice became whether to go through him. "Dr. Sanderson. This way."

"That's my equipment."

"It's been received."

"It's not luggage."

"It's been received." He said it the same way both times, without inflection, the way people repeat a phrase they've been told to repeat.

The sled didn't slow. It hit a pressure ridge hard enough that she felt the jolt in her own jaw, thirty meters off, an old reflex from years of driving instruments over bad roads. Yellow tags slapped against the crate's flank in the wind. She couldn't read them from here, but she knew her own labeling, and these weren't hers.

She started forward again. The sergeant's arm came up — not a block, a suggestion of one, close enough to require a decision. Behind the fogged eye protection his face gave her nothing. His rifle stayed angled at the ice, unthreatening, professional, which was its own kind of message she chose not to finish reading.

"Ma'am. Colonel Underwood will brief you inside."

"If the accelerometer package in that crate goes out of calibration in transit, Colonel Underwood can brief the wreckage."

No change in his face. "Inside."

Cavanaugh came off the stairs behind her with his coat half-zipped, more interested in the compound than the weather. His eyes went over the fence line, the escort spacing, the gantry, the separate route her crates were taking — a fast pass, the kind he did without seeming to try. He looked pale from the flight and irritated in the specific, contained way she'd learned meant he was counting something.

"Problem?"

"They're impounding my instruments."

"They impounded us first."

"That wasn't a joke."

"Wasn't one when I said it, either."

A gust took the runway lights out for two full seconds. When the snow thinned enough to see again, the sled and her crates were gone behind a rolling door in the station's flank.

She followed the sergeant because there was nothing useful in the other direction.

Inside, the air was warm enough to hurt. Diesel heat, wet wool, cold electronics, old coffee — the particular composite smell of a facility that ran its own atmosphere because the one outside would kill you. Melt ran off her boots in thin lines toward floor grates. The corridor walls were modular composite, ribbed with conduit and fiber trunking, labeled and relabeled: printed placards over stenciled ones, adhesive ghosts where older labels had been peeled and not quite finished removing. Not elegant. Not temporary, either. A structure that had been lived in past its intended lifespan.

A folding table blocked the corridor thirty feet ahead. Two security personnel stood beside it with evidence bags, a handheld scanner, a mounted clipboard in a weatherproof case.

"Outer gloves, bags, personal electronics, loose media," one said, without looking up.

"My electronics were taken on the aircraft."

"Then declare none."

"My instruments?"

"Controlled assets."

"You keep using that phrase."

He turned the clipboard toward her. "Sign acknowledging transfer of field equipment to site custody."

She looked at the page. Each crate carried a number now instead of a name. Her calibration references — three reference masses and a laser interferometer she'd spent eleven months getting funded — were listed under COMPONENTS. Her independent battery packs, chosen specifically because they weren't tied to station power and couldn't be cut from a breaker panel somewhere she'd never see, were listed as BATTERIES. The diagnostic rig, four years of her working life condensed into a case with a pressure-sealed lid, had been reduced by administrative language to PORTABLE TEST PACKAGE, NON-PRIMARY.

She did not sign.

The guard waited. He had the patience of someone who got paid the same whether this took ten seconds or ten minutes.

Cavanaugh moved in beside her, close enough that his shoulder came between her hand and the corridor camera's angle. "They'll log the refusal as noncompliance."

"They can log physics as weather if they want. It won't make it true."

"Doctor." A woman's voice, unhurried, from the far end of the corridor. "Your equipment is being secured. Not confiscated."

Colonel Jacqueline Underwood came toward them without any particular speed, cold-weather layers worn like they weighed nothing. Hair pulled back tight enough to leave no loose strands for the wind to have found. Her face was wind-reddened at the cheekbones and nowhere else, and her eyes moved over Sanderson with the specific attention of someone who had already decided, in advance, how much of this conversation would matter.

"Colonel."

"Dr. Sanderson. Mr. Cavanaugh." No hand offered. "Welcome to Operation Cold Horizon."

"Is that the research designation or the custody designation?" Cavanaugh asked.

Underwood looked at him half a second longer than the question required. "It's the site designation."

Sanderson tapped the clipboard with one gloved finger. "My instruments require documented custody continuity. Break the calibration chain and everything I tell you from this point forward carries an asterisk."

"Chain of custody is being maintained by my technical control officer."

"Your technical control officer didn't build my rig."

"No. But he controls the environment it operates in."

That was the first sentence since the tarmac that Sanderson believed without needing to check it.

She let the clipboard hang. "Then I need immediate access to the crates, the borehole telemetry, and the raw environmental baseline readings from before breach."

"You'll have access, under escort."

"Raw baselines."

"To relevant data."

"No. Raw."

"Relevance will be determined by site command."

Cavanaugh made a short, dry sound that wasn't a laugh.

Sanderson kept her eyes on Underwood. "You brought me here because something under this station is producing readings your people can't defend."

"We brought you here because your prior work intersects a classified technical condition."

"That's one way to phrase it."

"It's the only way I'll phrase it in this corridor."

The overhead heater cycled down between blowers, and for a few seconds the corridor seemed to hold still and listen to itself — water ticking into the grates, machinery working somewhere under the deck plating, the gantry outside grinding faintly against the wind.

Then she felt it.

A vibration too low to register as sound came up through the soles of her boots, through her calves, into the base of her teeth. It didn't have the signature of station machinery — motor imbalance had a rhythm, bearing wear had an edge, pump cavitation had a stutter she could have named on an oscilloscope without looking at the screen. This didn't. It was smooth, continuous, and wrong in her body a full second before it registered anywhere else.

She turned toward the wall.

"What?" Cavanaugh said.

"Quiet."

The guard at the table frowned. "Ma'am?"

She pulled her right glove off with her teeth and put bare fingers against the metal handrail beside the table.

The vibration was there in the rail, too. Low. Stable. It didn't change when the outer door cycled and sealed behind two passing technicians, and it didn't change when a gust slammed the module skin hard enough to rattle the light fixtures. It moved through the structure the way sound moves through water — like the building had been raised on top of the diaphragm of something enormous, and every wall in it was just skin stretched over the same drum.

"How long has this been present?" she asked.

Underwood didn't ask what she meant. That told her more than any answer would have.

"Intermittently," Underwood said.

"Since breach?"

"You'll be briefed, Doctor."

"Since breach?"

Underwood's eyes moved to the guard. "Release Dr. Sanderson's primary crate to escorted technical use. No independent recording. No external transmission. No unsupervised equipment access."

The guard hesitated. "Colonel, the intake order specifies full—"

"I'm aware of the order."

He stepped back from the table.

Underwood turned toward the sealed door at the far end of the corridor. "You wanted the anomaly, Doctor. You'll have your first descent in twenty minutes."

"Compromised equipment. Escorted access. Filtered data."

"You'll have a lift, an escort, and a living team, if you respect the limits I set."

"And if the limits are the reason your own readings don't make sense?"

Underwood stopped with her hand near the door control. For one moment — controlled, brief, gone before Sanderson could be sure she hadn't invented it — something crossed her face that looked like fatigue.

"Then I expect you to prove it carefully."

The door opened on a wash of colder air rolling up out of the lift shaft beyond it, air that had come from somewhere the station's heaters hadn't reached.

They gave her back the crate at a staging bay two corridors on — not the reference masses, not the interferometer, but the diagnostic rig itself, its case scuffed along one edge in a way it hadn't been that morning. A technician in a site coverall unlatched it under the guard's supervision and ran a wand over the interior before letting her touch anything, as if her own equipment might be smuggling something in.

"Seals are intact," the technician said, mostly to the guard, not to her.

"I'll be the judge of the seals," Sanderson said, and crouched to check them herself. They were intact. The foam cutouts had shifted half a centimeter from vibration in transit, not damage, but she noted it anyway, because a rig that had been jostled read differently for the first hour after handling and she needed to know when that clock had started.

Cavanaugh stood near the bay's inner door with a folder of intake paperwork he hadn't been allowed to

keep and a face that suggested he was memorizing it instead. He didn't ask what she'd felt in the corridor. She hadn't told him yet, in words. She thought he already had some version of it — he'd gone quiet exactly the way he did when a room told him something the people in it hadn't said out loud.

"Twenty minutes," he said. "Is that the actual number, or the number they tell people who ask twice?"

"I don't know yet."

"That's not nothing."

"It isn't."

A staff sergeant appeared at the bay door with two sets of cold-weather descent gear draped over one arm — insulated coveralls, harnesses, headlamps rated for the shaft environment — and told them fitting would take ten minutes and the lift would not wait past its window. Sanderson took the coveralls he handed her and checked the harness stitching out of habit before she put a hand through it. It was sound. She noted that, too, the way she noted everything now: not as reassurance, just as data, filed against whatever came later.

She flexed her bare fingers once more before she put the glove back on. The handrail two corridors back was likely still humming under whatever hand touched it next. Her own fingertips held the memory of it, a faint residual ache low in the

bone, the kind of thing she'd have dismissed as cold if she hadn't felt the frequency of it separately, distinctly, underneath.

The lift housing waited at the center of the compound, floodlit, hazard-striped, larger than any borehole apparatus had a structural reason to be. Two escorts walked them out to it across packed snow scoured smooth by generator exhaust and boot traffic. The wind had not slackened. It came off the gantry legs in gusts that carried a low overtone she hadn't noticed on the walk in, buried under engine noise and voices and the general chaos of arrival — a tone too steady to be wind alone, threading underneath it, present in the ice through her boots the same way it had been present in the handrail.

At the lift doors, a technical control officer she hadn't been introduced to ran a manifest check against her name and Cavanaugh's, checked their harnesses, and logged the descent time on a tablet without offering to let her see the screen.

"First descent," the officer said. "You'll have telemetry feed to your headset. No independent recording."

"Understood," Sanderson said, in the tone she used when she meant the opposite and intended to act on it the first chance she got.

The lift doors opened onto a cage lit in flat white, cables running up into the shaft housing above and down into dark she couldn't measure from here. Cold air moved out of it in a slow, steady exhale, colder than the surface air by a margin her body registered before her mind assigned it a number.

She stepped in. The vibration met her again through the deck plate the instant her boots touched it — not building, not fading, simply present, the way it had been present in the corridor and the handrail and the ice outside, as if the entire station sat balanced on top of a single sustained note that had been playing long before any of them arrived and had no evident intention of stopping now that they had.

The doors closed. The cage began to move.

Chapter 4: Intake and Compartmentalization

The corridor had been built to make people forget they had arrived anywhere. White composite panels. Gray flooring. Red emergency strips underfoot like a suggestion nobody had followed through on. No windows, no unit patches, no names on doors. Camera domes sat black in the ceiling every fifteen paces, and somewhere behind the wall a compressor cycled with a tired metallic cough.

Cavanaugh counted the domes because the corridor gave him nothing else honest.

Seven visible. Two blind angles that weren't blind if the ceiling tiles had pinholes, which they usually did. One pressure door ahead, access pad mounted too high to be convenient and exactly high enough to make a person stop and tilt his face into the lens.

Sanderson walked three meters ahead, flanked by two security officers and a civilian tech who kept glancing at the tagged sensor crates on the rolling cart like they might incriminate him by association. She

hadn't looked back since Underwood split them off the tarmac. He read that as deliberate. She didn't waste gestures under glass.

The officer behind him carried Cavanaugh's phone and laptop in a transparent evidence case. Not a bag. A case — foam cutouts, tamper tabs, an RF shield lining sewn into the seams. Somebody had upgraded the custody procedure between the aircraft and the station.

Which meant the flight hadn't been transport.

It had been intake.

"Where are you taking Dr. Sanderson?"

"Technical processing," the officer beside him said.

"She's already processed."

"She's being reclassified for subsurface access."

"By whom?"

Nothing.

He looked at the man's shoulder. The patch was subdued, rank visible, unit blank. Cold-weather gear covered too much and not enough — the radio clipped under the left collar carried a channel label on adhesive tape instead of stamped issue. Temporary integration. Mixed command.

He set that next to the badge colors. Sanderson's had a blue stripe. His, gray. Underwood's, black, no printed name at any distance he'd gotten. The engineer who'd tried to take

Kalamazoo

Sanderson's crates wore yellow. The soldiers wore red pips, no stripe at all.

A local permission system wearing an ID badge's clothes.

Good systems told you who a person was. Bad systems told you what a person was allowed to touch. Compartmented systems told you both, when it served them, and neither, when it didn't.

At the junction Sanderson stopped. The tech with her crates peeled left toward an equipment vestibule. She didn't.

"My instruments go where I go."

The lead officer didn't slow. "Your instruments go through environmental inspection."

"They've already been inspected."

"Station inspection."

"Then I go with them."

He looked at the camera above the junction, not at her. "Negative."

Her right hand closed once against the torn seam in her glove. Cavanaugh caught the motion before she stilled it.

"Dr. Sanderson."

She turned. Not long. Not warm. Half a second, and in it he read three things — *they're splitting us, don't escalate here, remember the crates* — and then the pressure door opened between them and took the half-second with it.

Colder air came through from the other side. Sanderson and the cart went left. His escort turned him right. The door sealed with a thump that traveled through the floor and up through the soles of his boots.

"Efficient," he said.

The officer didn't answer.

They walked him through a second corridor, shorter, warmer, into a briefing room built for a table too large for it. One wall held a screen dark enough to throw back the room in reflection — Cavanaugh, two officers, an empty chair across from him, no Underwood. The ceiling vent rattled every time the compressor kicked on. A coffee pot sat on the side counter, carafe full, no cups anywhere near it.

He almost laughed at that.

They put him in the chair facing the screen. The officer with the evidence case set it on a shelf behind him, out of reach, in the reflection. His phone lay faceup, dead black. The laptop's tamper strip had been replaced — the original was white. This one was pale green.

Someone had opened it.

He sat down.

"Am I being detained?"

"No."

"Then I'd like my property back."

"Your devices are under temporary
communications quarantine."

"Quarantine has a medical definition."

"Here it has a security one."

"Security definitions usually come with an
authority attached."

"Colonel Underwood will brief you."

"She's not here."

"She will be."

The man moved to the door. The second officer
stayed put, a step behind Cavanaugh, close enough to be
present in the glass.

The screen woke. Not to Underwood. A
grayscale document, already redacted — black blocks
over the header, the originating office, the routing line,
half the subject field, every signature. What remained
legible:

CONTROLLED SPECIALIST INTAKE
TEMPORARY ACCESS BRIEFING SUBSURFACE
ANOMALOUS MATERIAL PROGRAM

"That's a lot of black ink for temporary access."

No response.

The page turned itself. A list of acknowledgments: provisional analytic access to site telemetry, historical anomaly comparisons, command-filtered sensor outputs, approved archival fragments. Prohibited from independent communications, unsupervised data transfer, external reporting, removable media, handwritten duplication, photography, signal mapping, source-chain reconstruction, and *unauthorized correlation across compartment boundaries*.

He stopped there and read it again.

Not disclosure. Not exfiltration. Not copying.

Correlation.

They weren't worried he'd steal a file. They were worried he'd notice that two files belonged together.

The officer behind him shifted his weight. A boot sole squeaked against the floor tile.

"Do you have a pen?"

"No."

"Then how am I acknowledging this?"

Kalamazoo

"Verbal confirmation."

"Convenient."

The document scrolled without his hand near the desk. It stopped on the last line. A microphone icon lit in the corner.

I acknowledge that access to Operation Cold Horizon materials is conditional, revocable, and subject to command interpretation.

"That says command interpretation."

"Yes."

"Not legal interpretation."

Nothing.

"Not agency interpretation."

Nothing.

"Not scientific review."

The screen waited.

He thought about refusing. It would accomplish nothing beyond confirming, for whoever watched this feed, that he hadn't adapted yet. Underwood wanted him constrained, irritated, and formally complicit — refusal handed her one more excuse to wall him off from the data layer entirely.

He looked at the reflection of the laptop. Green strip. Opened, resealed, cataloged. They hadn't taken it for what he might send out. They'd taken it for what he'd carried in from Iceland — notes, fragments, timestamps, recovery hashes, all legal-gray, all deniable, dangerous only if matched against something under this ice.

"I acknowledge receipt."

The icon blinked once and went dark.

The screen changed to a station map. Not a full one — a permission map. Corridors without destination labels. Elevators color-coded by clearance. Subsurface levels drawn as rectangles with their centers cut out. The borehole shaft ran through the diagram as a single vertical blue line and disappeared into a blank space labeled GEOLOGICAL INTERFACE.

Three authorized locations lit up: quarters, escorted analysis bay, medical. Everything else went black.

He pointed at the blank around the shaft. "That's not geology."

Nothing from the man behind him.

"It's fine. Most maps lie. This one's just embarrassed about it."

A cursor settled on the analysis bay.

Briefing begins in eleven minutes.

The screen died.

The officer at the door opened it. "This way."

Cavanaugh didn't stand right away. He let the silence run long enough to make the escort decide whether to order him up. Men with real authority waited, because they didn't need to prove they had it. Men with borrowed authority filled silence.

"Mr. Cavanaugh."

Borrowed, then.

He stood.

They took him deeper. The second corridor sloped, barely — nothing a visitor would clock unless he was already counting his own steps, but enough to mean the surface modules connected to something set *into* the ice rather than sitting on top of it. The wall panels changed at a seam: newer composite giving way to older metal, paint layered over paint. A placard had been removed from one stretch of wall, leaving four screw holes and a rectangle of cleaner surface where dust hadn't settled yet.

He slowed there.

The officer touched his elbow. He let himself be moved.

The Core Throttling

At the next junction a maintenance tech came out of a side door with a coil of fiber cable over one shoulder. Yellow badge, no escort. The man clocked Cavanaugh's gray stripe, looked away a beat too fast, and turned the coil so the casing label faced his own chest instead of the corridor.

Not fast enough.

FIELD-DAMPENED LINE / CAVERN USE ONLY
Cavern.

So there was a void. Not a theoretical borehole return, not an acoustic echo somebody had dressed up in a report. A space large enough to need field-shielded fiber run into it.

They passed a half-open door on the right. Two engineers stood over a workstation watching a looped video, contrast crushed almost to black. He caught a blue-white angular pulse before one of them reached over and killed the display.

Late.

The rhythm of the light stayed with him. Not the image — the timing.

Iceland had produced timing like that, in transaction logs that had no business sharing a clock: trades settling before the orders that should have caused them, reconciliations arriving from nowhere, a network behaving as if latency were something local rather than a distance to be crossed.

Kalamazoo

The escort stepped in front of him. "Keep moving."

He looked at the badge instead. Red pips, no stripe. Enforcement, not access.

"You've seen it."

The man's jaw set.

"Not enough to understand it. Enough to know where not to stand."

"Keep moving."

There it was — not fear, exactly. A trained man's resentment at being reminded his orders hadn't made him safe.

The analysis bay door opened on a room colder than the corridor, workstations and server cabinets behind glass, a fine interior fog on one cabinet window that shouldn't have survived in a controlled environment. Every station had a privacy hood. Every chair faced inward, away from the walls. Two cameras on the entrance, one on the emergency exit, one on the keyboards.

A junior analyst waited at the nearest station. Civilian, pale from station light rather than weather,

yellow badge with a temporary white tab stapled below it.

"Mr. Cavanaugh. You'll work here."

"Will I?"

Her eyes flicked to the escort and back. "Your access is live for command-filtered telemetry and prior-event comparison sets."

"Prior events."

"That's the terminology."

"Whose terminology?"

She checked her tablet. "The program's."

"What's your name?"

She hesitated.

"Names aren't required for task support," the escort said behind him.

He watched her hate that. Not visibly — not enough for the camera — but her thumb went white against the tablet's edge.

"Fine. What's my login."

She set a card on the desk. "Session credential. Don't remove it from the reader. Don't attempt transfer. Don't query outside the assigned structure. Don't correlate across—"

"Compartment boundaries."

She looked at him then. Really looked.

He sat. The keyboard was standard military procurement. The card reader wasn't — custom housing, thermal tape along one edge, a status light blinking a half-beat out of sync with the rest of the station. The credential took seven seconds to authenticate. Three folders populated the desktop.

LIVE SITE TELEMETRY ARCHIVAL
THERMAL EVENTS ICELAND COMPARISON
SET

He put both hands flat on the desk and didn't open any of them yet.

"Is there a problem?"

"Yes."

"What problem?"

"You gave me the Iceland comparison set."

"That's approved."

"You're not asking whether this place is connected to Iceland."

The cameras kept their black, patient watch.

He opened the Iceland folder. Sanitized waveform overlays, power-consumption abstracts, surface facility maps, a summary scrubbed of names, agencies, causal language. The waveform file showed

The Core Throttling

blue-white peaks marching across the screen in narrow, repeating intervals.

His own notes had been messier. Incomplete, but alive. This file had been cleaned until it stopped accusing anyone.

He opened live site telemetry. Dozens of columns populated: borehole temperature, ambient pressure, shaft integrity, cable vibration, deep acoustic resonance, electromagnetic coupling, maintenance lockouts, power draw, seismic variance.

One column header sat above nothing.

LOCAL_CONSTANT_DEVIATION

He stared at the empty field long enough that the analyst noticed.

"That field's deprecated."

"Deprecated means replaced. What replaced it?"

"I don't know."

"Yes, you do. You've been told not to say."

The escort shifted his weight against the floor.

Cavanaugh leaned back, no warmth in it. "That wasn't a question."

The workstation chimed. A notification dropped in at the bottom edge of the screen.

SUBSURFACE ACCESS TEAM MUSTER:
DR. J. SANDERSON — CALIBRATION HOLD

Gone before he could click it.

Calibration hold. They were sitting on her instruments.

He opened the maintenance log, because nobody ever secured maintenance logs the way they secured everything else. Scrubbed, but scrubbed by someone protecting nouns and not verbs. He searched *hold*. No results. *Calibration*. Three hits. *Sanderson*. Denied. *Sensor crate*. Denied. *Cavern*.

The system froze.

For half a second the query field stayed on screen with the word sitting there in plain text, unhidden, before the desktop refreshed and returned him to the three folders.

The analyst's voice dropped. "Don't do that."

"Stand up," the escort said.

He didn't move yet. "That wasn't an external query."

"Stand up."

"It was a local term search on an assigned workstation."

The escort's hand drifted near his sidearm. Not on it. Near it — enough for the camera, enough for the report.

The analyst wasn't looking at the weapon. She was looking at the screen, and her fear had changed shape.

He followed her eyes. A fourth icon had surfaced beneath the three folders, held there for no more than two seconds before the desktop corrected itself.

UNREDACTED INTAKE FEED

Then it was gone.

A mistake. The kind a compartmented system made when two permissions collided somewhere behind the glass.

He stood before the man had to say it twice, face flat. The trick was never to look like you'd won something. Winning made people close doors.

The escort lifted the badge off the desk and pulled the credential from the reader. "Temporary access suspended pending supervisor review."

Cavanaugh looked once at the analyst. She didn't shake her head. She didn't nod. She dropped her eyes to the tablet and turned it toward him a few degrees with one finger.

Not enough for the camera. Maybe.

Enough for him.

Under station scheduling, between power checks and a security shift change, one line sat by itself.

FIRST DESCENT WINDOW — 24
MINUTES SANDERSON / CONTROLLED RIG /
ESCORT REQUIRED

He let his gaze slide past it like it meant nothing, and the officer pushed him toward the door.

In the corridor, the hum found him for the first time.

Low, almost below hearing — not machinery, not wind. A pressure behind the teeth that rose up through the floor and the walls and the soles of his boots on a rhythm that made the fluorescent tubes overhead seem late to their own flicker.

He looked down the corridor toward the sealed lift bay he had no clearance to see.

Somewhere below them, Sanderson was being allowed to bring compromised instruments to something the map called geology.

Cavanaugh touched the gray stripe on his badge and felt the cheap plastic give a little under his thumb.

The Core Throttling

They'd built the station to keep people separated from evidence, evidence from context, context from consequence.

That made it a system.

Systems had seams.

Chapter 5: Baseline

The equipment bay had been built for compliance, not science. Sanderson knew it before she crossed the threshold.

Steel tables, bolted in two rows. Yellow lines painted across the floor in a color that had clearly been chosen for visibility on camera rather than for anything a technician would need. The overhead cameras sat high in the corners, angled down at the tables and not at the doorway, which told her what the room considered worth watching. Hands. Not faces.

A working lab left mess. Cases half-open, cable spools kicked into corners, someone's coffee going cold on a shelf that had nothing to do with the experiment. This room had none of that. It had inventory.

The air smelled of cold metal and floor sealant, and underneath both, faintly, of diesel exhaust carried in through a duct someone hadn't bothered to baffle. Fluorescent tubes buzzed at a pitch just above hearing, the kind of sound she usually stopped noticing after an hour in a new room. She noticed it now. She noted

everything now, on principle, because a site that hadn't earned her trust didn't get the benefit of her habituation.

Her crates sat past the yellow line, tamper seals intact across the latches, in a configuration too neat to be anything but recent. Two technicians in gray thermal liners stood over a tablet, reading serial numbers aloud to each other in a flat call-and-response rhythm, a cadence built for an audit trail rather than for understanding what the numbers belonged to. A third had already opened her primary diagnostic case. Its contents were laid out on foam under inspection lights, spaced with the kind of care usually reserved for firearms — each component given its own square of space, none of them touching, as if contact itself were the hazard rather than the handling.

She stopped walking. The escort behind her stopped a half-step later, close enough that she felt the shift of his weight rather than heard it.

A fourth technician, younger than the others, stood near the wall with a clipboard, logging each removed item against a manifest before it was allowed to touch the inspection table. He hadn't looked up once since she'd entered. That, too, went into the tally she was keeping without meaning to keep it — who watched the equipment, who watched her, who watched nothing at all.

"That probe head is not a firearm," she said.

The technician nearest the case looked up.

"Ma'am?"

"The ceramic thermal head. The one you're holding with ungrounded gloves."

He looked down at his own hand as if it belonged to someone else, then at the probe resting across his palm.

She took one step toward the line. The escort moved with her, a small correction, nothing dramatic.

"Doctor." Underwood's voice came from the doorway behind them. "Your equipment is being processed."

"It's being contaminated."

Underwood walked in without hurrying, already in a white outer parka with no rank showing, none needed. The room made space for her the way rooms do for people who have stopped expecting resistance.

"Define contaminated."

Sanderson pointed at the table. "That sensor was packed under dry nitrogen. Its contact surface is rated for microthermal gradients. Your technician handled it after touching a cargo latch."

The Core Throttling

The technician froze, the probe still balanced in his hand like something he'd forgotten how to set down.

"Put it down," Sanderson said.

He looked at Underwood first. Not at the instrument. Not at her.

She filed that without comment and kept her eyes on the probe.

"Set it down," Underwood said.

He lowered it onto the foam slowly enough to look careful now, which did nothing about the fact that he hadn't been careful thirty seconds earlier.

Sanderson crossed the line.

This time the escort's hand closed around her sleeve, above the elbow. Not hard. Enough.

She looked at the hand. Then at Underwood.

"If you want anything from me better than theater," she said, "he takes his hand off me and I rebuild my own rig."

The bay heater kicked on with a hollow metallic cough, and warm air rolled down from a ceiling vent, stirring a loose packing label against the side of the open case. The label caught on a connector and stayed there, fluttering once before it settled.

Underwood watched it for a moment longer than the label deserved. "You will rebuild under observation."

"I'll rebuild under a clean protocol."

"You will rebuild under observation."

No anger in it. That was the part that cost her something to hear — not because it surprised her, but because anger would have meant another will in the room competing with Underwood's own, and there wasn't one.

Sanderson held the eye contact two seconds past what the room's protocol would have called polite. Then she turned to the escort. "Hand."

He let go.

She stepped over the line.

—

The rig had been built for hostile sites. Not for hostile custodians.

The frame was compact, black composite over shock-mounted rails, sensor ports arranged in a crescent so she could swap channels without cracking the protected core. Thermal. Electromagnetic. Seismic. Acoustic. Radiation. Atmospheric pressure. More instrumentation than any field pack should reasonably carry, and still, she suspected, not enough for whatever sat under four miles of ice below this bay.

The Core Throttling

The inspection team had pulled the cable bundles and sorted them by category, which made sense if you were cataloguing a weapons locker and made no sense at all if you meant to measure anything. The labels were intact. Two of the fiber leads had been coiled too tight, the kind of tight that stressed the jacket at the bend radius and showed up later as intermittent signal loss when you least wanted it, usually forty minutes into a run when there was no time left to trace it. A pressure transducer lay next to a screwdriver that wasn't hers. The static mat had no ground wire at all.

She went through the case a shelf at a time. The seismic accelerometers were still in their foam, undisturbed, which told her the inspection team had known enough to recognize what didn't look interesting. The radiation badge had been removed from its slot and clipped back in upside down. The atmospheric sensor's intake filter had a thumbprint pressed into the mesh, a clean whorl of oil against gray fiber, the kind of print that would read as background noise in a pressure trace until it didn't.

She catalogued each item without saying anything about most of them, because saying something about all of it would have taken longer than she had, and because a few of these were recoverable in the field and some were not, and she needed to

Kalamazoo

know which were which before she decided what was worth a fight.

She exhaled once through her nose.

"Ground strap."

A technician blinked at her. "We have anti-static gloves."

"Ground strap."

He looked toward Underwood.

"Give her what she asks for," Underwood said, "if it doesn't compromise security."

"It compromises security if my readings are garbage," Sanderson said.

Underwood didn't answer that. Sanderson noted the shape of the silence and moved on, because there was nothing more to extract from it yet.

She pulled the rig toward her across the table. The case dragged with a rubber squeal against the steel. Main seal, intact. Corner shock indicators — two yellow, one green, one red.

She tapped the red one with a fingernail. "Which crate was dropped?"

"No crate was dropped," the inventory technician said.

The Core Throttling

She looked at the red window and said nothing.

He checked the tablet. "One crate experienced a hard set-down during unloading."

"That's a drop with paperwork."

"It was within tolerance."

"Whose tolerance?"

Nobody answered that one either.

She opened the side compartment and lifted out the calibration block — tungsten core, ceramic shell, a threaded insert at the crown, the size of a fist and heavier than it looked, always heavier than people expected the first time they held it. The familiar weight settled her hands more than she wanted to admit in a room full of people watching her hands.

It had crossed four continents with her over nine years. She had recalibrated instruments against it in a server farm outside Reykjavík with frost forming on the inside of her collar, and in a testing bay in New Mexico where the desert heat had done the opposite kind of damage. She knew the exact temperature at which the ceramic shell began to conduct unevenly, and the specific drag of its threaded insert when the seal was seated correctly. None of that was superstition. It was just data she carried in her hands instead of a logbook.

She set it on the mat. Then stopped.

The tamper strip on the underside had been lifted and pressed back down. Not removed. Not obvious, unless you knew the adhesive pattern the way she did, from years of checking it before every run.

She turned the block over in her palm. The edge of the strip caught under her thumbnail, faintly wrinkled where it shouldn't have been.

"What happened to this?"

The nearest technician leaned in. "Calibration weight?"

"I know what it is."

"It was cleared."

"That wasn't my question."

Underwood stepped closer, close enough that Sanderson could smell the cold still clinging to the parka. "Is it functional?"

Sanderson almost laughed. What came out instead was a short, sharp breath through her nose. "You don't know what functional means yet."

Underwood's eyes moved from the block to her face. "Then tell me what you need before descent."

"Unsupervised equipment access."

"No."

"Clean bench."

"You have one."

"Grounded bench."

Underwood glanced at the technician, who was already moving toward the wall panel before she finished the glance.

"Full raw recording privileges."

"No."

Sanderson set the block down, carefully, the way you set down something you don't yet trust not to have a hairline crack in it. "Then you don't need me. You need a badge with a doctorate on it."

The room went still in a different register than before. The escort by the door looked at Underwood. The technicians found reasons not to look at anyone. The cameras kept doing what cameras do.

"You will have local raw capture on approved channels," Underwood said.

"Local isn't raw if command filters it."

"It will not be filtered before local storage."

"Who holds the storage?"

"Site custody."

Sanderson shook her head. "No. I hold the storage."

"Doctor —"

"If you're asking me to stand next to an unknown structure under four miles of ice and tell you whether it's safe to touch, I hold the storage."

Underwood's face didn't change. Something behind it did — a narrowing, not of temper, of calculation.

"You are not being asked whether it is safe to touch," Underwood said.

There it was.

Sanderson felt the room's temperature shift, though she knew perfectly well it hadn't. Her hand stayed flat on the calibration block. "What am I being asked?"

"You are being asked to characterize observable properties during controlled access."

"Controlled by whom?"

"By this command."

"Then you already have the answer you want."

Underwood didn't answer that directly. Instead she looked past Sanderson at the open case, at the components spaced out on their foam like evidence at a scene. "How much of that equipment is operational as configured?"

"None of it. That's the point."

"Then what would you take down as-is?"

"Nothing worth the trip." Sanderson picked up the pressure transducer, turned it once, set it back down exactly where it had been. "You want a number, I can give you a number. Forty minutes gets me a partial thermal channel and an acoustic pickup, if the ground holds. Everything else is a guess with a serial number attached."

"That is still more than we had an hour ago."

"That's not the same as enough."

"I didn't say it was enough." Underwood's voice hadn't moved from its register the entire exchange, which was its own kind of pressure, steadier than shouting would have been. "I said it's more."

Underwood let the silence run long enough to become its own kind of procedure. "You will descend in forty minutes. You may inspect, reassemble, and calibrate what you can carry. Recording cores remain logged. Transmission stays disabled. No external storage. No private comms. No unsupervised sampling."

Sanderson looked down at the stripped rig on the table.

Forty minutes wasn't enough to recover a baseline. It was enough to find the worst of the damage and put off dealing

with the rest until it mattered, which in her experience was always sooner than the schedule implied.

She pulled the thermal probe back into its cradle and checked the contact surface under the bench light. A faint smear of skin oil cut across the ceramic in a half-moon, exactly where a bare thumb would land reaching for the housing. She reached for a solvent swab.

The wrapper resisted her gloves. She tugged harder than she meant to. The plastic tore too fast and the swab dropped, bounced once, and landed past the yellow line on the concrete.

Nobody moved to retrieve it.

She looked at the fallen packet for a second longer than it required.

Then she opened another, more carefully this time, peeling the seal instead of pulling it, and wiped the ceramic surface in three passes, each one shorter than the last, checking the cloth under the light between passes for anything that came away. Nothing did, on the second pass. A faint gray smear did, on the first. She logged the swab as used and set it aside on a clean square of foam rather than in the tray the technicians

The Core Throttling

had left out, which already held two other swabs from equipment that wasn't hers.

The ground strap arrived, carried by the technician like something he wasn't sure he was allowed to be holding. She clipped it to the bench, then to the rig. The alligator clamp slipped twice on the painted metal edge before it finally bit down. When the continuity light came on, it flickered rather than holding steady.

"Your ground floats," she said.

The wall technician frowned at his panel. "It shouldn't."

"Useful sentence. Not a measurement."

He didn't say anything else.

She ran the first calibration pass. Channel one returned a clean, flat room-temperature line — a reassuring lie, the kind every system told you before it told you the truth. Channel two drifted high, climbing half a degree over ninety seconds before she cut it off and made a note to check the lead insulation once she had a bench that wasn't shared with three other people's equipment. Channel three pulsed at regular intervals, tracking the bay heater instead of the reference block, which was ordinary enough on its own — heaters put noise into everything if you didn't shield for it, and she hadn't had the shielding kit out of the case for more than four minutes.

She reset the input. Swapped the lead for a spare from her own kit rather than the one the technicians had recoiled. Ran it again.

The pulse stayed.

She checked the connector seating, the reference resistor, the cable run itself for a pinch point that might be acting as an unintended antenna. All of it read clean. She was left with the instrument, the block, and the room, and she had learned a long time ago that when you ran out of instrument explanations, you were supposed to start suspecting the room. Most of the time the room was the answer. Loose grounding, a nearby transformer, a technician's radio left transmitting in a coat pocket somewhere out of sight.

She asked, without much hope, whether anyone had a radio on.

Nobody did. Or nobody admitted to it, which in this room might not be the same thing, and she filed that distinction away rather than deciding which one it was.

She leaned closer to the screen. The period was wrong for the heater. Heaters cycled on thermostatic hysteresis, not clean intervals; this was cleaner than that, almost metronomic.

The Core Throttling

"Cut the vent," she said.

The wall technician looked at Underwood.

Underwood nodded.

The heater shut down somewhere behind the wall panel, the sound dying mid-breath. Air stopped moving through the bay. The label on the table, still caught on its connector, stopped fluttering.

Channel three kept pulsing.

Her irritation thinned into something closer to attention, the kind that made the room recede a little around the edges of the screen.

She moved the calibration block six inches to the left.

The pulse amplitude dropped.

She moved it back.

It rose again.

The technician nearest her crossed his arms. "Could be interference from the wall conduit."

"Could be," she said, and left it there, because agreeing wasn't the same as believing him.

She rotated the block ninety degrees on the mat. The pulse phase shifted with it, a fraction of a second, but visible on the trace.

The room seemed to lean toward the screen, technicians included, though none of them moved their feet.

She didn't like that. Curiosity in a room like this wasn't the same thing as permission, and everyone here wanted permission for something — to watch, to report, to understand before they were supposed to.

She disconnected channel three, capped the port, and switched to acoustic input instead. The line drew itself flat across half a second of room noise, then lifted into a low, steady vibration.

She checked the gain setting. Correct. She checked the input impedance against the reference card taped inside the case lid, an old habit from a lab director who'd drilled it into her before she'd ever touched real fieldwork. Correct. She dropped a second acoustic channel in parallel, cross-wired through a different preamp, to rule out something failing quietly inside the first.

The second channel returned the same trace.

Seventeen point eight hertz.

Her thumb stopped on the dial.

Underwood had seen it too. "What is it?"

"Noise."

"Doctor."

Sanderson kept her eyes on the rig. "Low-frequency background."

"From the cavern?"

"I haven't been to the cavern."

"Is it consistent with the borehole recording?"

She looked up at that.

So Underwood knew there was a borehole recording. Knew enough, at least, to ask the question in exactly that shape.

"You have a borehole recording," Sanderson said.

"You will review relevant portions after descent."

"After?"

"The descent will establish independent confirmation."

"That's backward."

"That's controlled."

Her hands had gone cold inside the gloves, colder than the room accounted for. She flexed them once, not for warmth. To confirm she still could.

The acoustic line held steady on the screen. A hum, reaching up through steel and ice and concrete and whatever classification sat over all three, into a room built to inventory hands.

She started a local capture. The file name generated itself before she'd finished reaching for the keyboard, a custody tag stamped onto it automatically:

CH-OBSERVED_ASSET / SANDERSON_J /
DESCENT_PREP / COMMAND_REVIEW

She deleted the default note field. Typed three words.

BASELINE NOT CLEAN.

Saved it. The software took two seconds longer than it should have to confirm the write.

Across the table, the calibration block sat where she'd left it, heavy, unremarkable to anyone who hadn't spent a decade learning its edges. The lifted tamper strip had settled almost flat again. Almost.

She picked it up and turned it in her hands, angling her body so the motion read as checking the threaded insert rather than what it actually was. With her thumbnail she pressed the loose edge of the strip down along its seam until the adhesive caught and held. If whatever recorded inside that shell had survived the hard set-down, nobody in this room needed to know it yet. Including, for the moment, her.

The Core Throttling

A buzzer sounded somewhere in the corridor beyond the bay doors.

The technicians straightened at the sound the way people do when a routine has been trained into their spines rather than their minds. The escort moved to the door. Underwood checked her watch, an old habit that told Sanderson more about her than any file could — someone who didn't fully trust the site's synchronized clocks, or hadn't always had access to them.

"Descent team is ready," Underwood said.

Sanderson closed the rig case and latched it herself.

One latch caught. She pushed harder. It resisted a half-second longer than it should have, then snapped shut with a sound like a small bone setting wrong.

"I'm not," she said.

Underwood opened the door anyway. Cold corridor air moved into the bay and pushed the last of the dead warm air out ahead of it, the way outgoing tide clears a room of anything not fixed down.

"Bring what you can carry."

She lifted the diagnostic rig. The strap bit into her shoulder immediately, the weight distributed wrong without the padding she'd have added given the full forty minutes she hadn't

gotten. Too much rig. Missing isolation. A dirty baseline she hadn't had time to fully chase down. A ground that floated when it should have held steady. A hum already sitting in the room before she'd even laid eyes on whatever produced it.

The corridor beyond the bay doors ran narrower than the one they'd come in through, unlined pipe conduit fixed along one wall at shoulder height, frost creeping along the seams where the insulation had failed or never been installed at all. Her breath showed here in a way it hadn't in the bay. The escort fell in a half-step behind her again, close enough to matter, and she found she'd stopped registering him as anything other than a fixed distance, the way you stop registering the weight of a coat after the first hour outside.

Underwood walked ahead without looking back to see if she followed, which was its own kind of confidence, the sort that didn't need confirming.

At the far end, past two sealed doors and a pressure station that hissed faintly as she passed, the descent lift waited with its interior lights burning a flat blue-white, the kind of light that made everyone standing under it look slightly wrong, slightly bloodless,

before they'd even gone anywhere. A technician already stood inside, holding the door open with one boot, watching her approach with the particular blankness of someone who had been told not to make conversation.

Sanderson shifted the rig higher on her shoulder and stepped over the threshold.

Through her teeth, through the floor, through four miles of ice she hadn't measured yet, she felt the vibration again.

Chapter 6: First Contact

The lift did not feel like descent. It felt like the station letting go of them.

Sanderson watched the overhead panels shrink through the grated ceiling, bright rectangles folding smaller until they were the size of playing cards, then gone. The smell went next — diesel and disinfectant thinning into something colder, mineral, threaded with the ozone-and-hot-copper smell of cable under load. She kept one boot braced against the lift wall and one hand on the strap of the diagnostic case, and she watched the depth numbers count down on the panel beside the doors because it gave her eyes something to do besides find Cavanaugh's face.

Cavanaugh stood across from her, two security personnel between him and the control console. No laptop. No phone. No folder. Just a badge clipped crookedly to the front of his parka, because the magnetic backing wouldn't sit flat over the insulation.

He caught her looking and touched the badge once, two fingers, like checking a seam.

Not hers to read. Not his to trust.

The Core Throttling

The lift shuddered at 1,800 meters.

One of the soldiers adjusted the rifle against his chest, a motion so practiced it had stopped meaning anything. Sanderson looked at the rifle, then at the frost building along the door seam, then back to the case at her feet. It carried three new tags she hadn't put there: CONTROLLED ASSET. LOCAL RECORDING PROHIBITED. ESCORT RELEASE ONLY.

They had let her keep two thermal probes, one low-frequency electromagnetic sensor, one seismic puck, and the main logging unit — after stripping the wireless module and swapping two of her cables for Cold Horizon inventory.

Not ideal.

Not clean.

Enough to know when someone was lying, if she was careful.

The vibration reached her before the doors did. She'd felt it faintly in the surface lift, before the machinery even engaged. Down here it came through the floor and into her teeth, below the register of sound, a pressure with a rhythm inside it.

"Is that the lift?" Cavanaugh asked.

The lead escort, Sergeant Pell, answered before she could. "Drive resonance."

"No," Sanderson said.

Pell looked at her.

She kept her hand flat on the case so she wouldn't make the mistake of pointing at anything.

"Drive resonance changes with load, speed, shaft contact. This doesn't change."

The lift slowed. Below the floor grating, the dark opened up in a way that had nothing to do with the shaft.

The doors parted on a groan that sounded reluctant, mechanical, tired, and the cold that came through was not the cold from the surface. Surface cold cut. This pulled. It found the moisture behind her eyes and the lining of her nose and took it. It found the damp inside her glove, where the cut from packing the case that morning had bled through the liner, and it drew the wound tight into a hard crescent against her knuckle.

Beyond the doors, the cavern did not fit inside the lift's light.

A platform of bolted steel ran out into a space so large the station's lamps couldn't find a far wall. Someone had strung floodlights along gantries and cable frames and temporary scaffold towers, and all of it

The Core Throttling

looked recent, human, and afraid. The infrastructure ended in hard, deliberate lines. Past those lines, the cavern kept going.

Then the matrix started.

Her mind rejected the word *structure* on contact, because structure implied a boundary between the thing and the space around it, and this had none. The exposed faces rose out of the cavern floor and wall in flat, faceted planes, black-blue under the work lights, seamed in nested geometric ridges. Not ice. Not basalt. Not any metallic growth pattern she'd seen outside a simulation or bad concept art. It read as crystalline until one seam brightened and sent a low blue-white pulse across a face the size of an aircraft hangar.

The pulse didn't reflect.

It moved through the material from inside it.

Cavanaugh came up beside her despite Pell's hand lifting toward his shoulder. "That's what the camera saw."

She didn't answer.

She had braced for scale. She hadn't braced for this kind of wrong at this size. Not theatrical wrongness — no tower, no shape built to frighten. It held still. It let the work lights wash across it and gave almost nothing back.

A junior engineer waited near a portable terminal on the platform, face raw from recycled air and short sleep. His badge

Kalamazoo

read ORTIZ. He looked relieved to see her cases and annoyed to see her.

"Doctor Sanderson." He didn't offer a hand.

"We have a twelve-minute cavern window."

"You have a what?"

"Command authorization for baseline contact.

Twelve minutes before lift return."

"I can't baseline an unknown thermal system in twelve minutes."

"That's the access window."

"That's a political window. Not an instrument window."

Ortiz's eyes flicked to Pell and came back without help.

"Doctor," Pell said, "you're authorized to take readings from the marked contact zone."

"Marked by whom?"

"Site engineering."

"With what criteria?"

"Surface stability. Approach safety. Prior sensor placement."

"Prior sensor placement is contamination."

The Core Throttling

Ortiz opened his mouth. Whatever answer he'd been issued didn't arrive fast enough to fill it, and he closed it again.

Sanderson crouched by the case. Her knees complained against the frozen grating. The latch she'd loosened during prep had iced over the hinge and stuck again. She got her thumb under the release and pulled, and the motion reopened the cut in her glove against the edge of the case. The pain came sharp, then went somewhere distant and manageable.

Pell took a half step toward her. "You need assistance?"

"No."

The latch gave with a crack.

Inside, the foam had gone stiff with cold around the remaining instruments. She lifted the replacement cable and felt the insulation resist her hand — not frozen, not brittle, but tensed, the way it might resist a direction it didn't want to go.

She stopped.

"What?" Cavanaugh's voice, low.

"Cable memory shouldn't behave like this at this temperature."

"That sounds like a small problem."

"It's a small problem if the cable is the problem."

The low-frequency sensor came up green, then amber, then green again. She clipped it to the mounting bracket and set

the seismic puck on the platform decking. It skidded a quarter inch before it settled.

She stared at it.

The platform read level on the bubble gauge welded to the rail.

"Did you see that?" she asked.

Cavanaugh's gaze had already gone to the puck.
"Yes."

Ortiz looked between them. "See what?"

"Nothing yet." She hated the answer as soon as it left her mouth, but she didn't have a measurement. She had a motion. A motion wasn't proof. A motion was where proof started, if it started at all.

She carried the thermal probe toward the marked contact zone.

The gantry from the lift to the matrix ran narrow, bolted into anchors drilled straight into the cavern floor. Every third slat had been sanded for grip. Every fourth had refrozen since. Below the grating, cable bundles and pylon shadow dropped into dark she couldn't measure by eye. The matrix rose ahead, matte, angular, and close enough now that the work lights made less and less sense against it — the beams went in

and didn't come back with the brightness they'd gone in with.

Ortiz kept two paces behind her. Pell kept behind Cavanaugh. The other soldier stayed near the lift, rifle turned out toward the cavern, which reassured her less than no rifle at all.

"Contact point's there." Ortiz pointed to a small yellow bracket clamped near one seam.

She stopped short of it. "That bracket's touching the surface."

"Yes."

"For how long?"

"Since first access."

"How long, specifically."

He checked his tablet. "Thirty-one hours."

"Then it isn't a clean contact point."

"It's where command wants the reading."

She let that sit in the cold a moment.

"Doctor?" Cavanaugh said.

"I'm taking two readings. One from their contaminated point. One ten centimeters off, from untreated surface."

Ortiz stiffened. "That's not authorized."

"Then log the first reading as compromised and the second as necessary."

"Doctor—"

She looked at him. "If you want a number badly enough to ruin it before you get it, you don't need me down here."

The hum deepened — not louder, lower, a shift she felt in the floor of her sinuses before she heard it. The hand holding the probe trembled once. She tightened her grip until it stopped.

Cavanaugh had drifted nearer to the bracket, not to the matrix. His attention had gone to the clamp, the serial tag, the small sensor blister mounted just above it.

"What's that logging to?" he asked.

Ortiz hesitated. "Site telemetry."

"Which layer?"

"Excuse me?"

"Raw. Engineering. Command summary."

"Mr. Cavanaugh," Pell said, "your role here is observational."

"Then I'm observing a routing question."

Sanderson pressed the first probe to the bracketed zone.

The instrument chirped once, an unhappy sound she hadn't heard it make in four years of fieldwork. The

The Core Throttling

readout stabilized at minus eleven Celsius. Then minus nineteen. Then a calibration warning flashed across the screen and cleared itself without her touching anything.

She held the probe where it was, jaw set against the vibration coming up through the platform. Two meters back, near a fissure in the cavern wall, she'd felt actual geothermal heat radiate through the insulated knee of her suit when she'd crouched to check her footing. The cavern should have been a contest between that warmth and ice loss into open air. It read instead like a controlled absence, something subtracted rather than simply cold.

The number dropped again.

Minus forty-three.

Ortiz leaned in despite himself.

"Surface frost contamination," he said.

"Frost doesn't do that."

"Probe seating error?"

"Maybe." She didn't look away from the display. "That's why there's a second reading."

Minus seventy-one.

The probe casing popped, a soft contraction click she felt through her fingers more than heard. She pulled it back before the sensor head could crack against the cold. A thin white

scar of frost sat on the black-blue surface where the tip had rested. It vanished in under a second — not melting outward into the air the way frost should, but drawing inward, disappearing into the material itself.

Her mouth went dry.

"Bad?" Cavanaugh asked, watching her face rather than the readout.

"Interesting."

"That's worse."

She moved ten centimeters to the right, off the bracket, off thirty-one hours of prior contamination, off anything the site could later use to discredit the number. The replacement cable pulled taut behind her. Not caught on anything. Tightened, as if by intention rather than geometry.

"Hold the slack," she said.

Cavanaugh reached for the cable. Pell's forearm came up and stopped him before his hand closed on it.

Sanderson looked at Pell. "If that cable shifts while I'm in contact, your cleanest reading dies with it."

Pell didn't move.

"Sergeant." Ortiz's voice had gone smaller.

Pell released Cavanaugh's arm with a short nod, timed carefully enough to look like his decision and not hers.

Cavanaugh took the cable in both hands and fed it out slow. "How much slack?"

"Enough that it doesn't pull me off the contact point."

"Scientific precision, noted."

She almost smiled. Almost.

The second contact made no chirp at all.

The readout went blank — not zeroed, blank, the way a screen goes when the unit can't decide what field to populate. Then every value on the logging display copied itself into the adjacent fields, a smear of numbers filling space where labels should have been. She tapped the casing once, then again. "No. Don't do that."

The display recovered.

Thermal emission: 0.00.

She waited for the unit to reject the value, the way it usually did with anything outside a sane physical range.

It didn't.

She cycled modes by hand. Passive infrared. Contact differential. Surface flux approximation. The instrument complained about geometry, about emissivity assumptions,

Kalamazoo

about calibration range falling outside spec. It complained about everything except the number itself, which held.

The surrounding deep-crust rock ran warm enough, over time, to damage unshielded equipment left in prolonged contact. She'd felt that heat with her own knee twenty minutes ago.

The exposed matrix gave off nothing.

Not low heat.

Not masked heat.

Nothing the instrument could register at all.

"That can't be right," Ortiz said.

She hated that sentence more than any other sentence in the discipline. Most impossible things started life as bad readings. The work wasn't believing them. The work was cornering them until they either died on inspection or changed what you thought you knew.

"It may not be," she said. "Get me the control reading off the basalt seam."

Ortiz didn't move.

"Now."

He went.

The Core Throttling

Cavanaugh still had the cable in both hands, eyes on the logger. "Zero?"

"Zero thermal emission within instrument range."

"Zero like broken sensor, or zero like—"

"Don't finish that."

The hum passed through the platform again, and this time her low-frequency sensor spiked and, without her command, overlaid a stored comparison trace she'd loaded into the unit during prep — a trace she hadn't meant the rig to call up on its own. The device shouldn't have had authority to reach that archive without her pressing anything.

Two lines drew across the small screen. One live. One historical.

They matched.

She felt the cold biting through the torn glove, distant and specific at once. "Jason."

He heard the change before he looked up.

She turned the logger so the screen faced him. For once he didn't ask a question first.

The live cavern waveform sat almost exactly over the recorded trace from Iceland — the one she'd pulled two months and one continent ago, before either of them had heard the word *Cold Horizon*.

Not similar. Not the same family of machine noise.

Match.

Ortiz came back with the basalt control reading and stopped short when he saw their faces. "What?"

Sanderson lifted the probe off the matrix. The cable went slack in Cavanaugh's hands with a small, unpleasant twitch, as though something on the other end had let go reluctantly.

She looked past Ortiz, past the bracket, out at the faceted black-blue surface running off into a dark her lights couldn't close. The blue-white pulse moved again along a seam, stepping rectangular sections of itself off into a scale her mind refused to finish holding.

"This isn't dormant," she said.

Ortiz's tablet gave a priority tone. Pell's radio crackled at nearly the same instant. He turned, listened, and turned back with a different face than the one he'd worn all morning.

"Lift window's closed," he said. "Command wants you topside."

She kept the logger against her chest, screen inward, and didn't move yet.

Cavanaugh's eyes had gone from the routing tag on Ortiz's tablet to the sensor blister above the clamp to the platform camera watching them off the gantry rail.

"What did they see?" he asked.

Pell's jaw tightened. "Move."

The cavern lights flickered once, all of them, in the same half-second.

On the matrix surface, the frost scar where the first probe had touched reappeared — a clean white line, then a set of small right-angled branches spreading off it before the whole thing sank back out of sight beneath the surface.

Sanderson didn't move until the logger finished saving the file.

The timestamp took three tries to write.

When it finally appeared, it read eight seconds ahead of station time.

Chapter 7: First Data Bay Pass

The data bay ran ten degrees warmer than the corridor outside it, which told Cavanaugh more than the door did, and the door told him plenty.

Three locks, stacked in visible sequence. A biometric plate, newer than the frame it sat in. A badge reader wired in below that, cable run exposed along the jamb rather than through it. And under both, a mechanical keyway nobody had bothered to strip out of the original housing. Original construction, then a retrofit, then a hardening pass added by people in a hurry. Three administrations of the same door, none of them removing what the last one left behind.

That was where he always started. Not with what a place claimed to be. With what it had been built out of, in order, and what nobody had gone back to fix.

The soldier was already inside when the door opened for him, which was the second thing.

Not stationed at the threshold. Not covering the corridor. Standing four feet past it, at an angle that put him in reading distance of the primary workstation without quite facing it, rifle slung muzzle-down, weight

settled onto his heels the way people stand when they expect to be there for hours rather than minutes. Windburn sat under both eyes in a raw double crescent. A strip of white medical tape wrapped the first joint of his right index finger, snug enough to have been done by someone competent, recent enough that the adhesive hadn't yellowed.

Cold-weather fracture, most likely. A hatch bite, if he'd been unlucky with a hinge. Not a fight.

He looked bored. Cavanaugh filed that as the least comfortable read in the room. Alert people watched for threats. Bored people watched because there was nothing else to do, and they noticed things threats didn't, out of nothing but irritation with the wait.

"Analyst Cavanaugh."

The woman at the center console didn't rise to say it. Headset looped down around her neck, station fleece zipped under a gray command vest, and a color to her skin that came from months of overhead light and not enough of anything else. Her badge read P. ARDA — SYSTEMS ENGINEERING, three colored stripes across the bottom edge, two of them blacked out under a strip of tape gone slightly tacky at the corners.

Recent. Not manufactured that way. Applied.

He set his tablet on the desk and left the chair alone for a moment.

"Senior analyst," he said.

She looked at him without expression changing. "Your intake officer said analyst. Your screen says civilian technical. My transport manifest said temporary compartmented asset." He nodded at the monitor without looking at it. "I've been trying to work out which one carries the least liability, in case something in this room goes wrong today."

"The workstation's live," Arda said. "Your permissions are already loaded."

"That usually means somebody already decided what I'm allowed to find."

"It means I've been up since Thursday."

He sat with that a second. It was the first sentence anyone at Cold Horizon had handed him that didn't sound rehearsed before it left her mouth.

He took the chair.

It sat too low for the desk. He adjusted the lever once, felt the mechanism catch halfway through its travel, and left it where it stopped rather than force it. The surface in front of him smelled like reheated coffee

and the particular staleness of plastic that had been run too hot for too long. Someone had scraped a sticker off the monitor bezel with a thumbnail instead of solvent, leaving a rectangle of adhesive residue that caught the overhead light at an angle and held it. The keyboard was a standard commercial model, civilian labeling intact, except for five function keys relabeled in white tape marker: TLM, GRID, DIA, ARCH, SEC.

He touched the space bar. The screen woke without a prompt.

No login. No credential check. A session already open, already his, apparently, before he'd walked in.

That wasn't convenience. Convenience didn't need a soldier standing where he could read the glass.

"Who authenticated me?" he asked.

Arda pointed at the ceiling, still watching her own display. "Command."

"Command has my credentials."

"Command has enough."

"Enough to generate actions under my name without me touching a key."

That one landed. She turned her head a fraction. "Every keystroke in this bay mirrors out. You should assume that from the start."

"Mirrors to command?"

"To compliance."

"Different office."

"Different word," she said, and went back to her screen.

He kept both hands flat on the desk beside the keyboard until he registered, in the edge of his vision, that the soldier had stopped tracking his fingers. Then he read what was actually in front of him.

Cold Horizon didn't run a single dashboard. It ran several systems that had all been skinned to match, stitched together after the fact by someone who cared more about appearing unified than being unified. Seismic models sat in one frame. Thermal maps in another. Power load in a third, borehole integrity in a fourth, cavern atmospheric estimates crowding the margin. A narrow telemetry ribbon ran along the bottom, fed from the instrument package Sanderson had lowered into the breach two days prior. The column headers had that particular confidence engineers used when they were naming something they hadn't fully characterized and didn't want to admit it.

MATRIX_SURFACE_TEMP.

MATRIX_EM_COUPLING. DEEP_THERMAL_DELTA.

ACOUSTIC_PRIMARY_HZ. FIELD_RESPONSE_INDEX.

The last one carried no unit label at all.

He clicked it once.

A gray banner unrolled across the top of the frame.

INSUFFICIENT AUTHORIZATION: DERIVED METRIC.

"Derived by whom?" he asked.

"Site engineering."

"From what inputs?"

"Multiple sources."

"That's a description, not an answer."

"It's the answer you're cleared for."

The soldier's attention had drifted to the dark server-rack panel across the room, watching the screen's reflection in the glass rather than the screen itself. Cavanaugh let his own posture shift a few degrees, enough to catch the same reflection at a better angle without appearing to look for it.

A crescent of condensation had formed high on the inner pane of the rack door, thin, trembling faintly with the room's ventilation cycle. The reflected screen sharpened where the dark glass sat behind it. Headers, mostly. Not values, not at that distance, not through that much distortion. Enough.

He opened the raw feed from the first descent. Sanderson's numbers were there, which surprised him for the length of one breath, until he saw what had happened to them on the way in. The individual sensor channels had been folded into summary bands. Names stripped. A dozen calibration flags collapsed into a single confidence score with no visible method behind it.

He clicked export.

Nothing.

He clicked again. The mouse double-fired on the release and opened an empty help window instead, unrequested, hovering over the corner of the frame like a mistake apologizing for itself.

Arda exhaled through her nose without turning.
"Mouse sticks."

"It's not sticking. It's double-registering on release."

"Been doing it three weeks."

"This is a data bay."

"This is Antarctica," she said, and didn't elaborate, and he decided not to make her.

He moved to the keyboard instead, cycling through menus by shortcut rather than cursor. The

soldier's mouth tightened, almost nothing, the reaction of someone who understood that a person navigating by keys moved faster than a person following a pointer and could not always be read from across a room. Cavanaugh slowed his pace deliberately, letting each keystroke land with a beat of hesitation between it and the next, and watched the tension across the soldier's jaw ease back down by that same small margin.

He opened the telemetry schema instead of the data itself.

That loaded.

Not the readings. The shape of the place where the readings should have lived — field names, table dependencies, ingest timestamps, checksum intervals — the skeleton of a system, which was sometimes more honest than the system's face. Every controlled environment had an architecture, and every architecture carried a history, and every history left behind small accidents nobody had gone back to clean up. He scrolled.

One hundred and fourteen columns listed in the active telemetry table. Thirty-nine of them rendered anywhere on the visible dashboard.

That, on its own, meant nothing. No user needed every field a system tracked.

He sorted by status.

Kalamazoo

Seventeen columns marked inactive. Also unremarkable, if the disabled fields were legacy artifacts or reserved placeholders nobody had gotten around to retiring.

These were not that.

CONST_DRIFT_LOCAL.
GRID_INSTABILITY_CORR.
EVENT_PHASE_OFFSET.
REMOTE_LOAD_ECHO.
COHESION_ANOMALY_FLAG.

He held his breath for one count, felt the soldier's attention drift back toward him, and let it out on schedule before it became a tell.

He didn't copy anything yet. He opened the metadata for the first inactive field.

INSUFFICIENT AUTHORIZATION:
ARCHIVAL FIELD.

Second field. Same banner.

Third. Fourth. Same, every time, the system refusing him in exactly the same words with exactly the same delay, a machine incapable of getting bored with its own answer.

The stray help window from his earlier misclick still sat open behind the schema frame, its search bar live. Because it was active, the session had logged his last focus point there rather than inside the restricted metadata panel. A small mercy, built entirely out of sloppy software rather than anyone's intention. He typed the column name into the help search instead of the system search, on the chance that the boundary between the two hadn't been drawn as cleanly as the banners suggested.

No results.

He added an underscore where he half expected one, correcting a guess rather than making a deliberate move, and the system returned a document he hadn't asked for by name: an internal naming-convention guide, unclassified within the site, written years earlier by engineers who had apparently grown tired of inconsistent labels and said so in the changelog.

He opened it.

"You won't find interpretation in user documentation," Arda said, without looking over.

"I'm looking for grammar."

"That's an interpretation."

"Usually is," he said, and let the sentence sit there instead of defending it further.

The guide listed prefix families in a plain table. THERM for thermal. SEIS for seismic. EM for electromagnetic. GRID for facility and regional electrical systems. And CONST, defined in four words that he read twice to be sure he hadn't misread them the first time.

Localized physical reference values.

Not instrument constants. Not calibration constants, which would have meant the numbers described the sensors rather than the world. Physical reference values, unmodified, sitting in a maintenance document like something left on a cafeteria tray that nobody had thought to hide because nobody expected a visiting analyst to read past the header row.

He didn't highlight it. He didn't slow his scroll speed to linger. He kept moving through the page at the pace of a man skimming for something else entirely.

CONST_G — gravitational acceleration estimate, local reference frame. CONST_EPS — effective permittivity estimate, local reference frame. CONST_MU — effective permeability estimate, local reference frame. CONST_DRIFT_LOCAL —

aggregate variance from expected reference behavior during active matrix cycles.

He closed the document.

A half second too late.

A small notification unfolded in the lower right corner of the screen and folded itself away again before a careless reader would have caught the text.

QUERY CLASSIFICATION ELEVATED.

The soldier straightened by an inch, weight shifting forward off his heels.

Cavanaugh opened the weather log.

Outside temperature. Wind speed. Visibility. Ice accumulation rate. A rolling estimate of safe surface movement windows updated every six minutes by an automated feed with no restriction banner attached to any field in it.

"You've got katabatic gusts over eighty kilometers an hour topside and a lift shaft running four miles into a cavern," he said, mostly to the room. "I'm not convinced the thing at the bottom of the shaft is the most hostile feature of this site."

"Weather doesn't require classification," Arda said.

"Give it time."

The soldier's shoulders eased back down, not all the way, but enough that Cavanaugh registered the difference and filed it without comment.

He left the weather window open and used the keyboard shortcut for backward application focus, cycling through his own recent history rather than opening anything new. The telemetry schema reappeared. Then the naming guide. Then, without his having requested it, a system cache view that should not have been reachable through that keystroke at all.

He sat with that for a second before touching anything else.

The cache logged recent queries against the workstation profile, not just his own. Other names, other timestamps, all pulling the same seventeen disabled columns he'd just been refused. Repeated attempts, clustered — one dense block around the hour of the borehole breach, a second block during Sanderson's descent two days back. After each cluster, an export failure. After each export failure, a reclassification event, the same fields shifting status within minutes of being denied.

Nobody had hidden this data at the start.

The Core Throttling

They had gone back and buried it once it started meaning something.

He scanned the username column. Mostly service accounts, generic and interchangeable. A handful of engineering initials repeated throughout. One entry sat apart from the rest, tagged at command level.

UNDERWOOD_MA_AUTH.

He didn't react to it, or told himself he hadn't, and kept scrolling before the soldier could measure his stillness against anything.

Near the bottom of the log, one line held his eye longer than the rest.

CONST_DRIFT_LOCAL correlated against
GRID_EXT_MICROCOLLAPSE — result archived.

External. Not station power, not internal load shedding.
A grid failure somewhere outside the wire.

"Do you have regional electrical monitoring in this bay?"
he asked.

Arda's eyes stayed on her own console. "We monitor a great deal."

"Antarctic stations specifically?"

"Yes."

"Southern Hemisphere infrastructure more broadly?"

The answer came a half beat slower than the one before it. "Not from this terminal."

"Is there a terminal it does come from?"

"Stay on assigned task," the soldier said, the first full sentence he'd offered since the door closed.

Cavanaugh looked at him directly for the first time since sitting down. "My assigned task is determining whether the phenomenon Dr. Sanderson measured two days ago correlates with a prior signal detected in Iceland. If this facility holds external grid correlation data, that falls inside the task, not outside it."

The soldier's jaw worked once and stopped.

"He isn't the person who can authorize that release," Arda said.

"Who is?"

"Command."

"Colonel Underwood."

"Command," she said again, and the repetition told him more than a name would have. Underwood was the face people were handed when they asked. Whether she was the source of the answer was a separate and unresolved question.

The Core Throttling

He returned to the dashboard. The mouse fired twice on a single click again, opening a secondary graph window he hadn't requested and hadn't meant to keep open. Before he could close it, he registered a gray trace running beneath the thermal display, unlabeled, clearly never intended for a standard user's screen. A residual plot. Matrix activity charted against regional grid noise, two lines that should have run independent of each other and did not.

The peaks weren't simultaneous.

The grid line stuttered first, by a margin small enough that he had to hold his focus on it to be sure he wasn't inventing the gap. Then, after the interval, the matrix activity line rose.

Not command's preferred direction of cause and effect. A response, not an initiation, if the timing held up under anything sturdier than a glance at a screen he was about to lose.

He leaned in half an inch.

The plot vanished.

A red banner replaced it, centered and unambiguous.

SESSION PERMISSIONS UPDATED.

Arda said something under her breath that might have been a curse and might have been a name.

The soldier's hand moved to his radio.

Cavanaugh lifted both palms an inch off the desk, slow, visible. "I didn't touch anything."

"No one said you did," the soldier said.

"Then set the radio down before this gets more interesting than it needs to be."

The soldier's eyes narrowed, and his thumb stayed on the transmit key without pressing it.

Arda stood. "Sergeant. Outside."

"Ma'am—"

"Outside. If compliance wants him locked out, compliance can come down here and do it themselves instead of routing it through you."

He hesitated a beat longer than protocol required, long enough to confirm that whatever authority Arda carried on the ground outweighed the stripes taped over on her badge. Then he stepped into the corridor and left the door standing open behind him.

Not privacy, Cavanaugh understood. A witness kept close enough to hear voices and far enough not to see the screen. Deniability, arranged for whoever asked about it later.

Arda leaned over his shoulder and typed a string too fast for him to hold onto cleanly. The dashboard

reset itself to the reduced, sanctioned view he'd started with. She didn't look at him while her hands moved.

"You didn't see external grid correlation."

"I saw a disabled residual plot."

"You didn't see that either."

"Who disabled it?"

Her fingers paused over the keys for a fraction of a second. The overhead lights in the bay dimmed, held there, and came back up. From somewhere below the floor plate came a low vibration, not quite the frequency Sanderson's recorder had caught days earlier but close enough that he felt it settle into his back teeth before it faded.

"Fields don't get disabled unless the answer they were giving became inconvenient," Arda said, quiet enough that the words barely carried past the console.

"Before or after the borehole camera went dark?"

"Before Iceland."

The sentence sat in the room longer than its length accounted for.

"Cold Horizon saw this before Iceland," he said, keeping his voice level, keeping his face still.

She closed the system cache window without answering that directly. "Cold Horizon saw a great deal before Iceland."

"Do you have the unfiltered records?"

"No."

"Does Underwood?"

She glanced toward the open door before answering. "You're asking the wrong question."

"Then give me the right one."

"Why would a drilling site carry a column for local physical constant drift before anyone officially found something capable of changing a local physical constant?"

The vibration under the floor faded out. The screen refreshed on its own cycle, and for one full second before the permission layer caught up with itself, the dashboard rendered all one hundred and fourteen telemetry columns at once, unfiltered, before the forbidden fields stripped themselves back out of view. He caught most of it only as motion, names flickering past too fast to fix in memory, except for one that stayed lodged behind his eyes after the rest had gone.

RESOURCE_THROTTLE_STATE.

Arda straightened and stepped back from the console before the door filled again.

Cavanaugh opened the weather log, unprompted, and left it open.

When the sergeant came back through, the screen showed wind speed, surface visibility, and a standard cold-weather exposure advisory, nothing else.

"Productive?" the soldier asked.

Cavanaugh looked at the monitor rather than at him.

Outside, the katabatic gusts had climbed another eight kilometers an hour since he'd last checked the figure. Whiteout conditions were projected before local morning. The station would seal itself tighter against the weather, which meant fewer people moving between levels, which meant fewer excuses to be anywhere at all.

"Not yet," he said.

In the dark glass of the server rack, Arda's reflection watched him without turning her head to do it directly.

At the bottom edge of the weather window, a small notification surfaced and dissolved before either of them could have been expected to read it.

UNAUTHORIZED QUERY MIRRORED TO
COMMAND.

Chapter 8: Contact Angle

Sanderson made them wait seventeen minutes, and the escort lieutenant kept his displeasure folded into the set of his shoulders instead of his voice. The two site engineers were less disciplined about it. The lift crew, back at the shaft collar, kept turning to check the cavern behind them, as if patience itself might provoke something in the dark.

She did not look at any of them. She knelt on the ribbed mat laid across the cavern floor, felt her left knee go numb through the insulated fabric, and watched the thermal baseline crawl sideways across her tablet screen instead of settling.

It should have settled by now.

The probe anchored into the exposed basalt showed a slow, steady climb — deep-crust heat working up through fractured rock, the kind of number she could have predicted from a chart before she ever came down the shaft. The air column above it stayed cold enough to frost the edge of her breath shield. The surface probe clamped to the titanium-silicon matrix read lower than both of them.

That part she could explain six different ways before breakfast. Instruments lied constantly this close to a power system. Contact patches failed. Calibration drifted under cold-soak. A film of condensation could bridge two leads and hand her a number that meant nothing.

What she couldn't explain was that the rock around the matrix was bleeding heat into it — she could see the loss in the basalt reading, a fraction of a degree at a time — and the contact surface wasn't gaining any of it.

Not a tenth of a degree in six minutes.

"Doctor." The lieutenant's voice came flat through the suit comm. "We're inside the authorized window."

She pinched the probe lead between thumb and forefinger and felt the cable resist her, not frozen, exactly, but reluctant, the insulation gone stiff in a way that had nothing to do with the ambient temperature on her sensor log.

"We're inside a compromised measurement environment," she said. "That's a different window."

Okoro shifted on the gantry plate behind her, a big man, red around the eyes, both hands wrapped around a field tablet like it might be taken from him. His jacket still carried the half-peeled Cold Horizon engineering patch from whatever previous

rotation had assigned it to him. "The baseline's good enough for initial classification."

She turned her head just far enough to find him.
"Classification of what?"

His eyes went to the lieutenant and came back.
"Thermal absorption profile."

"That's a behavior. It isn't a classification."

"We need a working term."

"You need a clean reading. You don't have one."

The lieutenant exhaled through his nose, and the cavern did something unkind to the sound, flattening it, delaying its arrival by a fraction of a second that her ear kept trying to correct for and couldn't. Voices behaved strangely down here. Footsteps arrived late. The low hum coming off the exposed structure filled every silence between words until the silences started to feel deliberate, engineered, instead of merely empty.

Past the work lights the matrix rose in dark, faceted planes, black where the light struck it, threaded with blue-white seams that woke and faded on some interior schedule of their own. No dust sat on it. No frost. No pooled meltwater, no film of vapor, nothing. The rock immediately around it steamed faintly where

the cold air met residual warmth. The matrix itself stayed dry, and had stayed dry through three separate descents, and she still hadn't stopped noticing it every time.

Cold surfaces collected evidence. That was the whole quiet miracle of doing this kind of work — ice held a record, moisture held a record, even a dirty glove left something behind. The matrix collected nothing. It gave nothing up for free.

She checked the timestamp on her tablet, then the timestamp on the site recorder bolted to the gantry rail, then the third clock running inside the calibration-weight housing she'd hollowed out and repurposed without telling anyone. The official feed and her hidden module disagreed by twenty-one milliseconds.

That number was new. It hadn't been there on the last descent.

She tapped the delta marker with a fingernail, more to confirm it than to fix it. "When did you start routing my instrument clock through site timing?"

Okoro's mouth tightened. "All field devices sync through facility control."

"Mine didn't sync through anything on the last descent."

"The last descent was initial exposure."

"And this is what."

Nobody filled the gap fast enough to make it sound like an answer instead of a delay.

She let the silence run. Years of bad briefings had taught her that people confessed in the seconds before they spoke more reliably than in anything they actually said.

Okoro looked at his own tablet like it might rescue him. "Command wants everything normalized against site telemetry."

"Command can want gravity to file a status report. That doesn't make it a valid reference frame."

The lieutenant stepped closer, boots clicking once on the metal mat, then catching where frost had built between the ridges. "Doctor, you are not here to audit command procedure."

"I'm here because your procedure is contaminating the only data that matters."

She pulled the surface probe free of its clamp.

The rig alarmed at once — a thin double chirp through her comm, cut off midway, as though the system had reconsidered whether it had permission to warn her at all. On the tablet the matrix-surface reading vanished, then reappeared below the calibrated floor as a

negative value that meant nothing physically and everything procedurally.

Okoro frowned at it. "You broke contact."

"I removed a contact-contaminated sensor."

"You invalidated the series."

"It was invalid before I touched it."

She stripped the probe head from its housing and turned it under her headlamp. The contact plate looked clean. Too clean. The thin bead of conductive gel she'd applied six minutes earlier had drawn in from the edges — not evaporated, not frozen — pulled itself inward into a single dome at the center, as if the surface tension holding it there had changed underneath her.

She tilted the lamp. The dome trembled.

Her hand was steady. She checked that first, out of habit, before she let herself think about what else could move a bead of gel that didn't want to be moved.

"Look at this," she said.

Okoro didn't move toward her.

"Engineer."

He came down beside her at last, knees popping through the insulated gear, and she held the probe out between them so the light caught it properly.

"Gel contraction," he offered.

"No."

"Dehydration. Field exposure."

"No."

"Then what."

She touched the swab tip to the bead and watched it climb the cotton fibers, then stop, arrested halfway, forming a hard meniscus across the surface like a skin. The cotton past that line stayed dry. Nothing crossed it.

Her forearms had gone tight under her sleeves before she noticed she was tensing them.

"The wetting behavior changed," she said.

"Locally."

Okoro was staring at the swab now instead of at her. "That's not thermal."

"No. That's an interface effect. Contact angle. Adhesion. Could be permittivity, if whatever's happening is moving through the gel itself." She didn't finish the sentence the way she wanted to, because the next word she had for it wasn't a word she was ready to put into a survey log with her name attached.

The lieutenant's voice sharpened behind her. "Define worse."

She didn't answer that yet. She reached into the open equipment case for the backup contact probe and found the foam cutout empty, a clean rectangular absence where a labeled instrument should have been sitting.

Her head came up. "Where's B-three?"

Okoro blinked at her. "What?"

"My secondary contact probe. It was in this case when we came down the shaft."

The lieutenant's gaze drifted toward one of the security men posted at the light stand, and Sanderson felt something in her go cold in a way that had nothing to do with the cavern temperature and everything to do with recognizing a reflex she'd seen before, in other rooms, from other people.

"No," she said. "Don't look at him. Pull the chain-of-custody log."

Okoro thumbed through his tablet, slower than she liked. "B-three was removed at surface intake."

"It was cleared for descent. I signed for it."

"It was reclassified as redundant."

"It isn't redundant if the primary instrument is returning interface behavior that shouldn't exist."

"You can continue with available equipment," the lieutenant said.

She laughed once, and it came out badly through the closed comm loop, more breath than sound.

"Available equipment isn't a scientific standard."

"It's the standard you have."

The hum deepened under them, a shift she felt in her sternum before she registered it as sound.

Everyone in the group went still. The work lights softened at their edges without visibly dimming, and the shadows under the gantry rail stretched a beat too far before snapping back to where they should have been.

She looked at her tablet.

The basalt probe had begun to fall.

There was no physical reason for it to fall. The rock beneath it held enough residual heat to keep the contact bracket above freezing regardless of what the air was doing. It should have held steady, or climbed. Instead the line bent downward in a slope too smooth for environmental noise, as though something were pulling heat out of the rock along a path her instruments weren't built to see.

Okoro's voice came through low. "That's a sink."

"Don't name it yet."

"It's pulling heat."

"It's appearing to pull heat."

"From the rock."

"From something coupled to the rock. I don't know what."

"Into the structure?"

She watched the matrix instead of answering right away. One blue-white seam lit near its base, held, faded. A second seam lit three meters higher a beat later — not the way current chased along a conductor, but closer to the rhythm of a process registering a load and responding to it on its own schedule.

"Or through it," she said finally, and left it there.

Her tablet buzzed against her glove.

A command overlay dropped across the top edge of her live feed, black text on a translucent field, the kind of automated banner that meant someone's system had decided her raw data needed a label before she'd finished collecting it.

PRELIMINARY FIELD BEHAVIOR: ENERGY-DAMPENING RESPONSE

She went still, reading it twice to be sure she wasn't misreading the first pass.

Okoro saw it a beat later. Whatever crossed his face, he put it away fast, but not fast enough.

She turned the tablet toward him. "Did you enter that?"

"No."

"Your team?"

"No."

"Then who's classifying my live data while I'm still standing here collecting it."

The lieutenant reached for the tablet. She rotated her shoulder, keeping the screen out of his reach.

"Don't touch my instrument."

"That instrument is a controlled site asset."

"That instrument is the only reason you know your site is lying to you."

"Doctor."

"No. This is exactly how people end up dead in clean rooms with polite labels on the wall. Somebody sees a number they want, names it before anyone understands it, and then builds a procedure around the name instead of the behavior."

His jaw worked once under the clear shield.

"You will lower your voice."

The Core Throttling

The cavern answered before he could add anything to that.

A thin cracking sound traveled through the mat under her knees. Not ice — she knew ice, she'd spent enough winters learning its particular register — this was tighter, more brittle, closer to strained metal.

She looked down. Frost had built along the ridges of the rubberized mat, but only in straight lines, and the lines ran parallel to the seams visible on the matrix wall thirty meters distant. As she watched, the pattern sharpened, white tracing itself across black rubber, turning a disposable safety surface into a rough, unwilling copy of the geometry sitting in the dark beyond the lights.

Okoro stepped back too fast and nearly went down on the frosted ridges.

The lieutenant said something into his radio. Static broke the words apart before they cleared the mouthpiece.

Sanderson dropped back to one knee, tore a sample sleeve free of its packet, and shoved it toward Okoro without looking up. "Hold the light on the mat edge."

Nobody moved.

"Hold the light."

He grabbed the lamp with both hands and aimed it, and the beam shook enough that she had to track the frost line by memory between passes. She worked a sterile swab under the raised edge and scraped, and the mat resisted for a moment before giving up a curled shaving of rubber rimmed white along one side. She sealed it into the sleeve before the frost had a chance to sublime, or reverse itself, or do whatever it was about to do next.

Her tablet chirped again.

The basalt probe's fall stopped. The reading held low for a four-count, then climbed back toward its earlier value along a curve too clean, too symmetrical, to be anything but a process reaching a defined endpoint.

Whatever it was, it had started. It had run. It had finished on its own terms, not because anyone had intervened.

Okoro's breathing filled the open comm channel, ragged and audible. "We need to report that."

"We need to preserve the raw series first."

"It's already on site telemetry."

She looked at the overlay still sitting across her screen, unmoved, waiting.

"No," she said. "It's already being rewritten."

The lieutenant stepped between her and the matrix, close enough that she had to tilt her head back to find his face behind the shield. "Pack your equipment. This survey is concluded."

"On whose authority."

"Colonel Underwood."

"Then tell Colonel Underwood the baseline failed."

His expression didn't move. "That's not the wording I was given."

"Use mine."

He keyed his radio again. Static ate the first syllable of his call sign, the way it had eaten the first syllable of everything said down here since the borehole broke through.

Sanderson sealed the sample sleeve, pressed it against the housing of her hidden recorder long enough to capture a timestamp on it, and slid it into the inner pocket of her suit where her own body heat would keep the seal pliable instead of brittle. Her hand stayed against the pocket half a second longer than the task required.

Not long enough to look like she was hiding something. Long enough that the sample would still be hers, in fact if not yet in any log, until somebody made her surrender it.

Behind the lieutenant's shoulder the matrix went dark along one seam and lit along another, patient, indifferent to the argument happening in front of it.

On her tablet, the classification changed again.

ENERGY-DAMPENING RESPONSE became
FIELD UTILITY CANDIDATE.

She read it twice, the way she'd read the first version twice, because some corrections in a career like hers were worth confirming before you let yourself react to them.

Then she looked past the lieutenant's shoulder, past the frost lines already starting to soften and blur along the mat, at the black geometry buried under several hundred meters of Antarctic ice and rock, at basalt giving up its heat to something that refused to warm in return.

It wasn't dormant. She had proof of that now, sealed in a sleeve in her inner pocket and scrawled across a data series somebody upstairs had already tried to rename twice in the last four minutes.

Someone above her, in a room she hadn't been shown, had just decided that the safest word for a failed baseline was *utility*.

The Core Throttling

She didn't have a probe left to test what that word was going to cost.

Chapter 9: Diagnostic Spike

The rail was wet.

Sanderson had wiped the frost from it herself, twenty minutes earlier, dragging her sleeve along the steel to clear a mounting line for the clamp. It should have stayed dry after that. The cavern air ran eleven degrees below freezing, cold enough that her exhale crusted white on the inside of her hood, cold enough that the diagnostic case had picked up a rime of frost on its latches within the first hour. The metal under her glove was colder than the air around it, the way exposed steel always was in a place like this, drawing heat out of anything that touched it.

Water did not survive that math. It should have gone solid inside a minute, glassed over, gone the milky white of all the other ice sheeting the gantry supports.

It hadn't. A thin clear line ran along the top edge of the rail, and it trembled.

She watched it for longer than she should have, then made herself stop watching it and finish the clamp.

Below the platform, past the rail and the drop and forty meters of black air, the matrix took up most of

The Core Throttling

the cavern floor in overlapping planes that didn't quite agree with each other about where their edges were. The floodlamps mounted along the gantry threw hard white light out across the space and lost it fifty meters short, the beams thinning into a blue nothing that wasn't shadow and wasn't distance, just absence. Sanderson had stopped trying to describe it to herself in the debrief reports. Words for that kind of dark always came out sounding like a campfire story, and campfire stories got you pulled off assignment.

The hum came up through the platform decking and into her knees. It had been there since the lift doors opened. She had stopped noticing it around hour six the way you stop noticing tinnitus, and then it had started again, and she suspected that meant something about intensity rather than her attention.

"Baseline package is not locked," she said.

Kline didn't look up. His tablet sat in a heated sleeve clipped to the diagnostic rack, and his thumbs moved across the glass in corrections too small and too fast, the movements of someone compensating for a lag that wasn't in the software. His badge said KLINE, no first name, the way all the site badges were printed, and the skin across the bridge of his nose had split where his goggles sat, a thin dark line healing wrong in the cold.

"I have lock on command telemetry," he said.

"That's not the same thing."

He glanced at her. Long enough for her to register the pink rim around both eyes, the kind that came from three, four days without real sleep. "Doctor. The test profile is a low-amplitude ping."

"Against what response curve?"

"The approved one."

"There is no approved response curve. Nobody has run this test before. That's the entire reason we're standing here." She bent back over her own rig and walked the isolation channels again, one at a time, the way her advisor had taught her twenty years ago before any of this, before Iceland, before the word matrix meant anything but a grid of numbers. Channel one, stable. Channel two, stable. Channel three had begun climbing by fractions of a degree with no change in probe contact, a slow drift with no cause she could locate. Channel four sat at a flat negative value and had not moved from it in four minutes.

That wasn't a reading. That was an instrument telling her it had run out of categories for what it was seeing.

"The site's run surface diagnostics for nine days," Kline said. "This is the same class of test, just moved to direct contact."

"Surface diagnostics on a structure you hadn't physically touched."

"We've touched it."

"You've contaminated it."

Behind her, Sergeant Pell shifted his weight. Six feet of insulated armor, the kind of patience that came from a training pipeline built around not intervening until a scientist actually became the problem. He'd been assigned to her since the lift down. He hadn't said six words to her in that time. She could feel him deciding, right now, whether this qualified.

She checked the anchor line on the rig. The locking pin hadn't seated.

Of course it hadn't.

The rail had gathered its skin of condensation again, and her glove slid on the pin when she tried to force it home. She stripped the glove back with her teeth, felt the air bite the exposed skin, and pressed her bare thumb into the pin until the cold of it went past uncomfortable into something closer to burning.

It clicked. Halfway.

Not enough.

"Hold execution," she said.

Nobody answered.

She looked up, past the rack, to the observation glass thirty feet along the gantry. Colonel Underwood stood behind it with two site engineers and a communications officer, a loose crescent of people arranged around her the way subordinates arranged themselves around someone who had already made the decision that mattered. Underwood showed no visible sign of impatience. That was worse than impatience would have been. Impatience was a thing you could argue with.

Sanderson keyed her throat mic. "Command, this is Sanderson. Baseline package isn't secured, and channel four is invalid. Hold execution."

Static, once.

Underwood's voice came back fast enough that Sanderson understood she'd already been waiting for the call. "State whether the invalid channel indicates a personnel hazard."

"It indicates I don't know what we're measuring."

"That is not my question."

"It should be."

A pause. Short. Controlled, the way everything about Underwood was controlled.

"Diagnostic will proceed at minimum amplitude."

Sanderson turned to Kline. "Do not fire that pulse."

He still wouldn't meet her eyes. "Sequence is command-authorized."

"Then un-authorize it locally."

"I can't."

"You can remove the bridge relay. Physically. Right now."

His left hand drifted a few inches toward the rack, an instinct starting to form, and then it stopped. Not because he didn't understand what she was asking. Because Pell had taken one step closer behind her, and because Underwood stood behind the glass with her hands clasped and her face doing nothing, and because the entire station had been engineered, corridor by corridor, badge scan by badge scan, to make refusal feel like a mechanical impossibility rather than a choice.

"Kline." Lower now. "Listen to it."

The hum had changed.

Not louder. Narrower. It had drawn down into a tight band of pressure that sat behind her jaw and under her back teeth, the kind of frequency you felt in the bone before you registered it as sound. She looked at the live trace on her tablet. The baseline line had stopped being a line. It had begun notching at even intervals, each drop cut off square at the bottom instead of curving, the shape of a signal being sampled by something that expected time to move faster than it was moving.

The rack chirped.

Kline's face lost color.

"Sequence start," he said.

"No."

The first pulse went into the matrix.

There was no flash. No surge of light climbing the black planes below. The cavern simply subtracted a small, specific amount of ordinary certainty from the space around the platform, and everything in that space adjusted to the loss at once.

Sanderson felt her own weight change before any alarm caught up to it. Her boots lost their grip against the decking, not a slip, a lightening, the sole

contact thinning out from under her like the platform had exhaled. The ache went out of her knees. The diagnostic case strap lifted off her shoulder — not far, an inch, maybe less — and hung there wrong, held by something other than her own posture. Frost dust rose off the grating in a pale sheet and stayed up, grains hanging and trembling instead of falling back through the mesh.

Channel one spiked.

Channel two inverted, the trace flipping below its own baseline and holding there.

The seismic line folded on itself, a fold too clean and too symmetrical to be instrument failure. Instrument failure looked like noise. This looked like a decision.

"Abort," Sanderson said.

Nobody moved.

She got her bare hand around the rail. The wet line along the steel broke apart into individual beads, and the beads lifted off the metal, each one holding its shape in the air like something suspended in oil.

"Abort the test."

Kline hit the tablet twice, hard. "It's not taking input."

"Pull the bridge."

"The relay's sealed."

She turned for the rack and nearly went down doing it. Her body hadn't gone weightless — that might have been simpler, a single clean state she could compensate for. Instead every motion arrived with the wrong return. She pushed off with her foot and the platform gave back too little resistance. Her torso overcorrected into the empty space that should have had weight behind it. The anchor harness caught her a beat late, the strap biting into her collarbone after her balance had already gone.

Behind her, Pell swore, low and steady, the way career soldiers swore when the swearing was informational rather than emotional.

A wrench came up off the deck between them and hung, turning slowly end over end.

It should have fallen already. It should have hit the grating, bounced once, and slid toward the drain channel the way every dropped tool on this platform had slid for nine days. Instead it climbed another six inches, hesitated at the top of that arc like it was deciding something, and began to drift toward the open edge of the platform.

Sanderson lunged for it on reflex, before her mind had finished processing that reaching for it made no sense.

The harness caught her mid-lunge.

The rail clamp she'd fought with a minute ago slipped free entirely.

Pain went up through her thumb where the half-seated pin tore the skin along the nail bed. She got two fingers around the wrench anyway and slammed it down against the deck, hard, as if the force of the gesture could shame gravity back into behaving.

"Readout," Underwood said, over the open channel.

Sanderson stared through the glass. "Are you serious right now?"

"Readout."

Kline answered because she didn't. His voice had gone thin. "Local inertial variance detected. Point three percent. Point three five. Point four."

The numbers changed the room on the other side of the glass. She saw it even distorted through reflection and the angle of the light — the two engineers leaning in toward their own screens, the communications officer's head turning toward Underwood instead of toward the platform. Nobody over there was looking at the floating frost. Nobody was looking at Pell

fighting to keep his rifle sling from lifting off his shoulder, or at the wrench she still had crushed against the deck under her palm.

They were looking at the number.

"Point four percent?" Underwood asked.

"Localized to the cavern volume," Kline said.

"Maybe platform-adjacent. I don't have external confirmation."

Sanderson forced herself toward the rack, one argued step at a time. Her knee came up too fast on one step and too slow on the next. She caught the rack's edge with her shoulder instead of her hand and drove her bleeding thumb into the manual breaker.

Nothing.

The breaker was already showing local lockout, a status she hadn't set.

"Command has hard control," Kline said.

"Then command can hear this." She keyed the mic with the side of her wrist, thumb useless now, blood starting to thread down into the seam of her glove. "You are not looking at a deployable field effect. You are looking at a local constant drift induced by forcing an

unknown thermal-processing system past its tolerance. Shut it down."

Underwood didn't answer.

The second pulse hit before the first one's aftershock had finished moving through the platform.

The hum dropped below the range of hearing. Sanderson felt it instead, in her sternum, a sudden absence of pressure so total that the vibration following it seemed to come from somewhere behind her ribs rather than through the air. The platform flexed under her boots, a slow structural give she felt travel the length of the gantry. The floodlamps along the ceiling supports dimmed in sequence — not all at once, a rolling failure, each pool of white light fading and then climbing back before the next one in line went dark.

Kline's tablet tore loose from its tether cable.

He grabbed for it and missed. It rose, rotated once in the air with the slow tumble of something filmed underwater, and struck the underside of the upper handrail exactly as the pulse released its hold on the room. The screen took the impact at full weight. The glass went into a spiderweb of white fracture lines from corner to corner.

Then everything came down.

Not far. The platform was still the platform, four feet of clearance at most between where things had lifted and where they belonged. But it came down hard, all at once, gravity reasserting itself in a single instant rather than a return.

The wrench slammed into the grating an inch from Sanderson's boot. Frost dust hit the deck like thrown sand. Her knees buckled under weight that had gone from absent to full in a fraction of a second, and she went down hard on one hip. Pell's rifle cracked against the rail as its sling snapped taut. The diagnostic case dropped against her thigh and hit with enough force to numb the muscle from hip to knee.

Somewhere to her left, Kline cried out, short, and went down.

His wrist had taken the return wrong. He'd had a hand braced against the rack frame when the fall came, and his body's weight had gone through that arm at an angle wrists weren't built to carry. He lay on his side on the decking, one arm folded under him at an angle that had nothing to do with how a body chose to rest.

The Core Throttling

His tablet, cracked screen still lit, skidded across the grating and stopped against the toe of Sanderson's boot.

The pulse ended.

The hum came back in layers, low first, then higher, uneven, the sound of a machine climbing back out of a load it had not wanted to carry.

For three seconds nobody spoke.

Then the alarms started, all of them at once, the platform's warning system finally catching up to what had already happened and finding no useful way to describe it.

Sanderson crawled to Kline first. Her thumb had opened further in the fall and was bleeding freely now into the seam of her glove, warm for a moment before the cold started working on it, before the blood began to thicken at the edges of the tear. Kline was breathing through his teeth in short pulls, eyes wide, doing everything he could not to look down at his own arm.

"Don't move it," she said.

"I didn't—" He swallowed hard. "I didn't think it would—"

"I know."

It came out too gentle. Too late to be useful to him and too small to cover what had actually happened on this platform in the last ninety seconds. She heard the wrongness of it as soon

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as it left her mouth and didn't have anything better to put in its place.

Pell crouched on Kline's other side, his face gone pale under the edge of his balaclava in a way his training clearly hadn't prepared him to hide. "Medical's inbound."

Sanderson looked past him, toward the observation glass.

Underwood hadn't moved. Her hands were still clasped behind her back. Her eyes weren't on Kline, weren't on the blood working its way down Sanderson's wrist, weren't on the last of the frost dust settling back onto the grating in a thin gray film.

She was reading the gravity number.

It sat frozen on Kline's cracked tablet, six inches from Sanderson's boot, boxed in emergency red.

LOCAL GRAVITATIONAL VARIANCE:
0.401%

Below the platform, somewhere down in the dark the floodlamps couldn't reach, a line of blue-white light moved once through the matrix's upper planes and went out.

Sanderson understood it then, with a clarity that felt colder than the rail had been under her bare hand.

The Core Throttling

The test had broken a man's wrist. It had destroyed a piece of equipment she'd need weeks to replace out here. It had proven, in the space of two pulses, that the structure below them could move one of the rules people mistook for the fixed floor of the world.

And on the other side of that glass, none of that made it more dangerous to Underwood.

It made it worth more.

Underwood's voice came over the open channel, flat and unhurried, the same tone she might have used to confirm a supply manifest.

"Diagnostic complete. Preserve all data."

Chapter 10: Ancillary

The platform still smelled like ozone and machine oil under the cold, the way it had since the pulse, and Sanderson had stopped trying to decide whether that was real or something her body had invented to explain what had just happened to it.

Kline said he was sorry before the medic had the shears fully through the sleeve, and he kept saying it, smaller each time, like something bleeding out on its own schedule.

"I should have pulled it."

"Don't move your arm." Sanderson kept her voice level because his wasn't.

"I saw it coming. I saw the numbers."

"So did I. So did everyone standing at that rack." She found the pulse below his glove cuff with two fingers because the medic needed both hands for the splint. Fast, but strong. Good enough for now.

"Numbers aren't the same as knowing when."

The medic didn't look up from the fracture. He had the flat, procedural calm of someone who had learned to work inside classified injuries without asking

what caused them. His kit was lashed to the platform decking with two bungee cords, doubled, as though even a roll of gauze couldn't be trusted to obey gravity down here anymore. He worked with short, economical motions, cutting past the wrist, exposing a swelling already gone the color of a bad bruise two hours old instead of two minutes.

"Radius, probably. Maybe the ulna too." He said it to Sanderson, not to Kline, the way medics sometimes reported to whoever in the room could hear it without falling apart. "Splint first. X-ray isn't happening down here."

"Understood."

Sanderson crouched lower to keep her face level with his. His color was bad — not the gray of cold alone, but the thinner gray underneath it, the one she'd seen on people going into shock before their vitals caught up to the fact. She kept two fingers on his pulse point anyway, a small continuous confirmation that the numbers on the monitor weren't lying about him the way the rack's display had started lying about the room.

Kline's eyes tracked between them, glassy, still hunting for someone to absolve him.

"I had my hand on the relay," he said. "I had my hand right there."

"Then the platform moved before you did," Sanderson said. "That's not the same as failing to act."

He didn't seem to hear the distinction. Shock rarely left room for one.

"Doctor. Your hand."

She looked down without meaning to.

Blood had dried the torn glove to her thumb in a stiff dark seam. The split ran from the nail bed into the webbing where the clamp pin had opened it, and it had gone from sharp to a low, insistent throb now that she'd stopped moving the hand to work. She had no attention left to spend on it.

"Later."

"Now," the medic said, not arguing, just noting it, the way he'd note a vital sign.

"No."

Pell crouched on Kline's other side with his rifle slung tight to his chest, jaw set. He'd gone quiet after the fall in a way that wasn't sympathy exactly. Something in his face had shifted when the wrench came off the deck and hung there, and the platform slammed back under all of them at once. He could file a report under any

heading command wanted. His body had already logged what happened to it.

Sanderson had felt the same thing, a half-second of her own weight going wrong in a way no inner ear was built to interpret, and then the return, harder than the departure, like the room had remembered where things belonged and enforced it all at once. She still didn't have a clean way to describe the return phase. Words like *snap* and *correction* both implied intention she had no evidence for yet.

Around them the cavern kept doing what it always did, indifferent to the four humans clustered on the platform. The hum underneath the ice was still there, the same low frequency she'd logged since the first descent, though something about it had changed register since the pulse — not louder, not different in pitch she could name, just narrower, as if it were being listened to through a smaller aperture than before. She filed the impression without trusting it. Impressions weren't measurements, and she had promised herself a long time ago that she would stop treating them as interchangeable, especially now, especially here.

Above the platform, the observation glass changed color.

Clear to smoky gray, in about two seconds,
swallowing whatever was on the other side of it.

"That's new," Sanderson said.

Pell followed her eyes up. "Automatic privacy
screen."

"During a medical response."

He didn't answer that.

The medic locked the splint around Kline's wrist
and Kline hissed and turned his face toward the decking
grate. Frost dust had settled into the sweat along his
temple, gray-white, not melting.

"Can he walk?" Pell asked.

"No. Litter."

Sanderson stood too fast. Her knees misjudged
the platform, some leftover confusion in the muscle
memory from the weight change during the pulse, and
she caught herself against the equipment rack with the
injured hand. White crowded the edge of her vision. She
swallowed the sound that wanted to come out and put
her attention somewhere it would do more good.

The diagnostic rack was still running.

That mattered more than the hand did.

Channel one had settled into a thermal curve, noisy but plausible, the kind of noise she could account for with contact resistance and cold-soak drift. Channel two was reporting values eight-tenths of a second behind the station clock, consistently, tenth after tenth, like it was reading a different second than everyone else in the cavern. She'd checked that offset twice already against her own watch and against the wall clock by the lift, and both agreed with each other and not with channel two. Channel three still held the gravity trace from the event, unedited, a spike and a slow asymptotic return that hadn't been in any model she'd built before arriving at Cold Horizon. Channel four — the negative line that shouldn't exist, the one she'd flagged during first calibration and been told to ignore — had come back with a flat message instead of a number.

SENSOR DOMAIN ERROR.

Not a failed reading. A reading the instrument had no domain to express. She had seen sensors fail. She had seen them saturate, drift, drop bits, die outright. She had never seen one refuse a category.

She reached for the storage key in the rig's side panel.

A red light came on above the port before her fingers touched it.

She stopped and looked at it.

That drive was hers. Local removable storage, part of the package she'd built into the rig herself, because every field station she'd ever worked had a way of turning ugly data into acceptable categories before anyone with judgment got to see it raw. It wrote frames before the smoothing pass. Before the station software decided what counted.

It should not have had a permission gate on it at all.

ACCESS CONTROL: COMMAND MIRROR
ACTIVE.

"Pell."

He was helping the medic unfold the lift litter.

"What."

"Who put a command mirror on my rig."

He glanced at the rack, then up at the gray glass.

"Not my lane."

"It became your lane the day you stood behind me during calibration."

"I don't write the permissions."

"No. You enforce the ones somebody else wrote."

Kline made a sound that wasn't an apology this time. The medic set the litter down beside him without comment.

Sanderson wiped the blood off her thumb with the edge of her sleeve and hit the storage release again. The light held red.

The rack's main display refreshed on its own.

For half a second the raw gravity numbers were still up: 0.401 percent variance, radius uncertain, first pulse 4.7 seconds, second pulse 6.2, return acceleration outside anything in the model. Then the screen collapsed the data without asking her and opened a form instead.

INCIDENT CLASSIFICATION: PLATFORM SLIP /
EQUIPMENT DISPLACEMENT.

She went still, reading it twice to be sure it said what it said.

"Platform slip," she said out loud.

Pell's eyes moved to the screen. The medic's did too, and then away, faster.

She tapped the form open with her good hand. It wanted witness identifiers, an injury classification, equipment damage, environmental conditions. There was a field for the gravity event. It wasn't missing. It had been folded under a collapsed menu labeled ANCILLARY FIELD VARIANCE,

one tap deep, the kind of interface choice that only mattered if someone was trying not to be noticed.

She'd built forms like this herself, years ago, back when she still believed a good template could keep an incident report honest by forcing every category into view. She knew exactly how much design went into deciding what stayed expanded on first load and what got tucked one tap below it. Nobody collapsed a menu by accident. Somebody had decided, ahead of the pulse, ahead of Kline's wrist, ahead of any of it, which categories would be visible to a person filling this out in a hurry and which would require the presence of mind to go looking.

She had not had the presence of mind. She'd had to go looking anyway.

Ancillary.

Kline had a fractured wrist because a diagnostic pulse had moved gravity across the cavern floor, and the system had already decided which word to file that under.

The lift arrived with a hydraulic groan that shook a little frost loose from the shaft rails. Two more security personnel stepped off it with a second medic behind

them. One of them carried a hard case stamped with the same hazard chevron Sanderson had seen on the capacitor banks during unloading, back when this had still looked like ordinary heavy equipment.

Not a medical case. The proportions were wrong for it, too flat, too wide, and it moved in the man's grip like something with a defined interior structure rather than loose supplies shifting inside foam.

"What is that," she said.

The man carrying it didn't answer. He didn't even look at her directly, which told her more than an answer would have. People who could refuse to acknowledge a question were usually operating under instructions specific enough to name the question in advance.

"Doctor, step clear," Pell said.

"No."

"Step clear."

"Not until I secure the raw data."

"Command said preserve all data."

"That" — she pointed at the screen without looking away from him — "is not preservation."

His mouth tightened but he didn't say anything else.

The platform speaker clicked on.

"Dr. Sanderson."

Underwood. Not raised. Not rushed.

Sanderson looked at the opaque glass because there was nowhere else to look.

"Your objection has been recorded,"

Underwood said.

"Then you have an inaccurate recording."

"The event is under technical review."

"The event injured a member of your engineering staff and produced measurable gravitational variance in an occupied space."

"It produced a localized field response under controlled amplitude."

Sanderson laughed, once, and it came out wrong in the cavern, too sharp, too human against the hum underneath everything. "Controlled. Your engineer is being carried out on a litter."

"Which is why medical response has priority right now."

"You don't get to use his injury as cover for ignoring the warning that caused it."

"Your warning is in the record, Doctor."

"My warning is buried three menus deep under a word your engineers picked because it wouldn't trigger a review."

"Ancillary Field Variance is a standard classification."

"It's a standard classification for equipment jitter. Not for a room where the floor stopped agreeing with the ceiling for six seconds."

There was a pause on the line, longer than the ones before it. Not a concession. A recalculation.

"We will discuss classification after Kline is stable," Underwood said. "Your priority is the same as mine right now."

"My priority is a truthful record. I don't know what yours is anymore."

"My priority is that nobody else gets carried out of that cavern before we understand what put him on a litter."

Underwood's tone didn't change register at all across the sentence, which was somehow worse than if it had. "You may not like the vocabulary. The medical response does not depend on the vocabulary."

"The next diagnostic pulse might. If your engineers write this down as a slip, they'll schedule the next one the same way they scheduled this one."

Underwood didn't answer that directly. "Lower medical is waiting on you, Doctor."

Pell shifted his weight. Beside the litter, the other security personnel had stopped pretending not to listen. Then the medics lifted Kline, and he cried out despite trying not to, a short broken sound that cut through anything else being said in the cavern. Even the hum under the floor seemed to drop for a second, or she imagined it did, which was its own kind of data problem.

She turned back to the rack instead of answering Underwood again.

The compliance form had auto-saved.

She killed the screen. Not the rack underneath it. The screen.

The acquisition lights stayed lit in a small green row behind the casing, still writing raw frames, still a few seconds ahead of whatever layer formatted them for command. A few seconds. That was the whole window she had.

She got down on one knee at the base of the rig and felt for the calibration weight housing under the lower module.

Her thumb screamed when she pressed the release catch, so she used the heel of her other hand instead. The catch didn't move. Cold contraction, or

The Core Throttling

stress from the pulse, or the universe deciding this was the object it would make difficult today. She hooked two fingers under the bracket lip and pulled.

Nothing.

The medics were already moving Kline toward the lift, one at each end of the litter, moving carefully over the deck seams.

Pell saw what she was doing.

"Doctor."

She set a boot against the rack frame for leverage and pulled again. The housing gave with a dry metallic crack, sudden enough that she nearly went over backward.

"Doctor, stop."

"Not yet."

The calibration weight came free into her palm. A dull cylinder about the length of two fingers, stamped with a serial number and a mass certification, indistinguishable from forty others in her kit. Behind a threaded insert no field technician would think to check unless they'd built the rig themselves, it held an independent storage core, enough capacity for the raw thermal frames, the electromagnetic capture, and the timing variance from the last twelve hours.

She had built it into the rig on her own time, off any procurement list, because she had learned early in her career that any dataset worth defending eventually got asked for by someone who had no intention of returning it. She had never expected to need the contingency here, at a site with this much armed personnel per square meter of cavern, and she registered, distantly, that the part of her that had built it anyway had understood something about Cold Horizon before the rest of her caught up.

She closed her fist around it.

Pell took one step toward her.

She looked up at him. "Are you going to seize a calibration standard from an injured scientist in the middle of a medical evacuation?"

"Depends whether it's a calibration standard."

"It's labeled as one."

"You just pulled it out of a hidden housing."

"I designed the rig. The housing isn't hidden from me."

"That's not an answer."

"It's a fact."

The lift doors opened behind Kline's litter, warm station light spilling across the platform decking, turning the frost dust briefly gold before the cavern's dark took it back.

Underwood's voice came over the channel again. "Escort Dr. Sanderson to lower medical. Her equipment remains in place for command review."

Sanderson slid the calibration weight into the outer pocket of her parka.

Pell watched her hand do it.

He was close enough to stop her. He had the authority. He had a rifle and a rank and every institutional justification available to him in that cavern.

"Lower medical," he said instead. "Now."

Not permission. Not help, exactly.

A decision to look somewhere else with his hands.

She picked her ruined glove up off the platform deck and followed the litter, leaving small dark marks along the yellow safety rail where her thumb touched it.

Inside the lift, Kline stared at the ceiling with his pupils too wide and his lips gone the color of the ice outside.

"I'm sorry," he said again, to no one in particular.

She leaned close enough for the medics to hear but not so close that the ceiling speaker would separate her words from the noise of the cable drum.

"Remember what moved first."

He blinked at her, unfocused. "What?"

"The frost. The wrench. Your tablet. Not the platform under your boots. Not your footing."

His breath caught.

"Remember the order it happened in. Somebody's going to ask you, and they're going to want the wrong answer."

The lift began to climb.

Through the narrowing gap between the platform and the shaft wall, Sanderson watched the diagnostic rack recede below them, lit by floodlight, abandoned in the middle of the cavern floor. Two command engineers were already crossing toward it, one of them carrying the hazard-marked case.

The older one moved without the nervous energy the others had. His badge caught the lift light for a second before the gap closed: RANE. He looked at the rack. Not at Kline, not at the retreating lift. That told her where the priority sat, if she'd had any doubt left.

The Core Throttling

Rane opened the case.

Not repair tools inside it.

Interface locks. The kind built to cut a system off from whoever had been operating it.

The lift doors sealed. For a few seconds there was only cable noise, the medics' breathing, and Kline's pain held between his teeth in small, involuntary sounds.

Then the station intercom came on overhead.

"Attention lower operations. Diagnostic Event One is classified successful. All personnel preserve instrumentation for Field Utility Review."

Sanderson closed her hand around the calibration weight in her pocket until the metal warmed with fresh blood from the split in her thumb.

Successful.

That was the word they'd landed on.

The lift kept climbing, the shaft walls sliding past streaked with rime, carrying them up out of whatever the cavern had just decided to call itself.

The doors opened onto the lower medical alcove, a narrow prefab bay walled in white composite panel that never quite lost the smell of the crate it had shipped in. Two nurses were already waiting with a rolling gurney, and the transition

from litter to bed happened in the practiced silence of people who had done it before, in worse light, with less warning.

"Move him through," the lead nurse said, and the medics did, and Kline disappeared behind a privacy curtain that was ordinary blue cloth, not smoked glass, not automated. Sanderson noticed the difference and didn't know yet what to do with noticing it.

Pell stopped at the threshold instead of following her in.

"I'll be outside," he said.

"Guarding the door or guarding what's behind it?"

He didn't answer that either. He was getting good at not answering.

A second nurse took Sanderson by the wrist, gently, professionally, and turned her hand palm up under the alcove's overhead light. The glove had to be cut away in strips, the fabric fused to the wound in three places, and each strip came away with its own small percentage of pain that Sanderson catalogued and set aside rather than reacted to. The split ran deeper than she'd let herself register on the platform, down past the

first knuckle crease, the edges gone pale and waterlogged from an hour and a half of cold and pressure and use.

"This needs closing, not just wrapping," the nurse said.

"Butterfly won't hold that."

"How long."

"Fifteen minutes if you sit still."

"I don't have fifteen minutes."

"You don't currently have a choice," the nurse said, not unkindly, and reached for a suture kit.

Sanderson sat. She kept the injured hand extended and the other one closed around the calibration weight in her pocket, and she watched the curtain that hid Kline rather than the needle that was about to go into her thumb. The first pass of local anesthetic burned worse than the original cut, a bright chemical sting that made her eyes water for a reason unrelated to anything Underwood had said, and she was grateful for it in a distant way, because it was a pain with an obvious cause and a known duration, unlike almost everything else that had happened to her since the diagnostic sequence began.

Somewhere behind the curtain a monitor started chirping in the ordinary rhythm of a stable pulse, and she let herself take that much as fact, because it was one of the few things in the last hour she was allowed to.

Through the alcove's outer window, thick and double-paned against the cold beyond the rock, she could see a sliver of the corridor leading back toward the lift. Pell stood in it with his back half-turned, rifle still slung, watching the shaft doors rather than her. Waiting for someone. She didn't know yet whether it was command, or Rane, or simply the next order in a sequence she hadn't been shown the rest of.

The nurse tied off the last stitch and began wrapping the hand in gauze that would need changing by morning.

"Word of advice," the nurse said, quiet enough that it wouldn't carry past the curtain rail. "Whatever you're carrying in that pocket. Don't let them see you check it again."

Sanderson looked up.

The nurse didn't elaborate, and didn't wait for a response. She taped off the dressing, packed the kit, and moved on to the next task on a list Sanderson wasn't part of, leaving her alone with the chirp of Kline's monitor, the low hum that hadn't stopped since they'd come down here, and a fist closed around evidence that no longer had anywhere safe to go.

Sanderson flexed the bandaged hand once, testing what the stitches would allow. Not much. Enough to hold a stylus, maybe, if she kept the grip loose and didn't ask the thumb to bear weight. Not enough to run the rig again without help, and the rig wasn't hers to run anymore anyway, not with a command mirror sitting on it and two engineers standing over it with a case full of interface locks.

She thought about Rane's badge catching the light, and the way he'd looked at the rack instead of at Kline. She thought about the eight-tenths-of-a-second offset on channel two, which nobody had explained and which she had not yet had a chance to write down anywhere that wasn't already compromised. She thought about the word *successful*, repeating over the intercom in a voice that hadn't asked anyone in that cavern whether they agreed with the assessment.

Down the corridor, past Pell's half-turned shoulder, a door opened and closed with the particular weight of someone who outranked the people already in the room. She didn't look up right away. She finished flexing the hand, catalogued the pain at a four out of ten, filed it, and only then raised her eyes to see who command had sent to collect the version of events they preferred.

Chapter 11: Curated Logs

The incident file reached Cavanaugh's workstation twelve minutes after the alarms stopped.

He clocked it against the wall display before he opened anything, out of habit more than suspicion — a discipline left over from a decade of watching agencies decide what to release and when. Twelve minutes was fast enough to be automated and slow enough to look considered. Both possibilities interested him. Neither was provable from a clock alone.

The viewer opened in the format Cold Horizon used for everything it didn't want copied: a locked frame, a gray classification banner across the top, no export function, no print function anyone admitted to, no way to reposition the window on the screen. Title first.

DIAGNOSTIC EVENT ONE: PLATFORM
STABILITY DEVIATION

He read it twice and said nothing the first time.

Behind him, the soldier — HAYES, no first name offered, none asked for twice — stood at the angle Cavanaugh had learned to expect from the good ones: close enough to read the screen, far enough that he

could claim later he hadn't been looking. The data bay had cooled since the spike, not dramatically, but enough that the server-rack windows had grown small frozen crescents along their seams. The air smelled of burnt dust laid over old coffee, and somewhere under the raised floor a maintenance fan clicked, skipped a rotation, clicked again, on a rhythm nobody had bothered to fix in the weeks Cavanaugh had been here.

"Platform stability," he said, finally.

"I don't generate the labels, sir."

"No. I imagine somebody upstairs does that for you. Efficient of them."

Hayes didn't answer. He rarely did when Cavanaugh gave him an opening that wasn't a direct question, which was its own kind of information — the man had been briefed on what not to confirm, which meant somebody thought there was something worth not confirming.

Cavanaugh's hands were cold enough that the first keystrokes felt distant, like typing through a glove he wasn't wearing. He'd stopped noticing the temperature of the bay around hour three of his shift and started noticing it again now, the way attention returned to a body once the adrenaline of the alarms had drained out of it. His wrist ached faintly where he'd braced himself against a console during the spike — nothing like

Sanderson's break, nothing worth mentioning to anyone with a clipboard, but present enough to remind him his body had been in the room too, not just his eyes.

He put his hands on the keyboard and didn't move them for three full seconds. He was aware of the workstation logging that stillness the same way it logged everything else: which panes he opened, how long each stayed live, whether his search terms would read later as curiosity or as an attempted intrusion. So he started where a compliant analyst would start, and let the machine watch him behave like one.

Incident summary.

Personnel injury: one wrist fracture, one minor hand laceration. Equipment displacement: localized. Environmental contribution: platform condensation, vibration, low-traction surface. Field anomaly: under technical review. Operational conclusion: diagnostic successful within safety envelope.

He clicked into the technical appendix. Hayes shifted his weight behind him — not stepping closer, just resettling, the way a man does when he's been standing in one position too long and doesn't want to admit it.

The gravity variance hadn't been removed. It had been demoted. Four fields deep, tagged as an ancillary note attached to equipment movement, worded so that a reader skimming for liability would see nothing more than a cart that had rolled somewhere it shouldn't have.

ANCILLARY FIELD VARIANCE: 0.401%

LOCALIZED

No hazard flag. No event correlation. No primary tag standing over it to tell a reviewer this mattered.

He opened the timing chain.

The pane lagged before it rendered — not the stutter of network traffic, which he knew the shape of, but something slower and more deliberate, a request leaving his terminal and going somewhere he had no visibility into and coming back with a yes shaped like a warning.

*Diagnostic authorized: 09:42:17 Pulse sequence armed: 09:42:21
Sanderson warning logged: 09:42:28 Pulse sequence executed: 09:42:29
Local variance onset: 09:42:29 Local variance logged: 09:42:41 Medical
response initiated: 09:43:02*

He stopped on the shared second between onset and execution. That could happen. Automated capture was fast enough to fold two events into the same tick without anyone lying about it. But the variance hadn't been logged until twelve

seconds after onset, and the figure attached to it — 0.401 — carried more decimal precision than raw sensor output usually managed. Somebody had already reconciled it against an external baseline before it ever reached the file in front of him. That reconciliation had to live somewhere.

He opened the sensor reconciliation pane.

ACCESS DENIED.

Hayes said nothing, which by now told Cavanaugh less than it might have from someone else.

"That was quick," Cavanaugh said anyway.

"Problem?"

"No. Just finding the edge of the furniture."

He backed out and pulled the visible telemetry package instead — thermal, acoustic, electromagnetic, cable motion, platform vibration, personnel tracking, laid out in the kind of exhaustive detail that was meant to look like transparency. It was the sort of package built by people who understood that volume could stand in for candor if nobody had time to read all of it. He scanned it for anything under mass, anything under acceleration. Nothing, except lift-safety boilerplate that

used the word for reasons that had nothing to do with a cavern floor moving.

He paused on the acoustic file longer than the others. The band had narrowed during the pulse window — not silence, just a tightening of the range the sensors had picked up, the kind of thing an engineer might dismiss as equipment saturation if they'd never seen it happen twice. He didn't know enough about acoustics to argue the point with anyone who did. He knew enough to notice that nobody in the file had flagged it either way.

He searched the permitted file for "0.401." One result — the same one he'd already seen.

He searched "grav." No results.

He searched "variance." Seventeen hits, every one of them equipment.

He typed "CONST" and hit enter.

The viewer froze.

Not a system crash — the fans kept turning under the floor, the cursor kept its steady blink, the room stayed exactly as cold as it had been a second before. Just a held stillness, a status wheel that rotated once in the corner and vanished.

No results found.

He looked at the timestamp in the lower right of the screen. Then, briefly, at the dead half of the monitor glass, where the reflection showed Hayes watching his hands now instead of the screen.

Cavanaugh switched to keyboard navigation — tab, tab, shift-tab, enter, arrow down — the kind of movement the interface hadn't been built to expect, because whoever designed it had assumed people would click exactly what they were handed and nothing else. He worked down past the visible content of the appendix into the metadata layer attached to the ancillary field note. Not the note itself. The tags underneath it.

Most were ordinary: PLATFORM,
DIAGNOSTIC, ENVIRONMENTAL,
EQUIPMENT_DISPLACEMENT,
PERSONNEL_INJURY. One field in the array sat blank — not empty in the way an unused field was empty, but occupying space the way a filled slot does after its contents have been pulled and the shape left behind.

"Can I get coffee?" he asked.

Hayes blinked. "What?"

"Coffee. The substance in the pot that's pretending to be coffee."

"You're not authorized to leave the bay."

"I'm not touring the reactor. The pot's right there." He pointed without turning his head, deliberately wide of the actual counter by a few degrees, and watched in the glass as Hayes's eyes flicked toward it to check the distance for himself.

He used the half-second.

Alt-left. Local cache preview. Shift-F6. Metadata focus. Enter.

For less time than it took to finish reading it, the hidden tag rendered.

CONST_DRIFT_LOCAL

He kept his face where it was. Not enough time to memorize the surrounding structure. Enough to know the name existed and that it had a place reserved for it in a system that otherwise pretended today's event was the first of its kind.

The interface blanked the field and returned him to the appendix with a message dressed up as an apology.

FIELD NOT AVAILABLE UNDER CURRENT
ACCESS LEVEL

"Never mind," Cavanaugh said. "Probably not worth the paperwork."

"What isn't?"

"The coffee."

Hayes's eyes narrowed by a degree Cavanaugh doubted the man knew he'd shown. Cavanaugh let his hands rest in his lap and thought about what he now had, which was not the file, but the fact of the field. Cold Horizon hadn't invented a category for the gravity anomaly today, on the spot, out of institutional panic. It had a slot for it already, sitting in the hidden layer of a database template, sized and shaped for repeat use.

You didn't build a category like that after your first event. You built it after enough events that somebody needed a place to file them without frightening whoever read the summary above it.

He let that settle without chasing it further. Hayes had drifted half a step toward the counter, close enough to the coffee pot now that Cavanaugh wondered whether the request had planted something, a small itch the man would scratch later once he decided it cost him nothing. Cavanaugh didn't push it. A second use of the same trick would look like a trick.

He opened the telemetry routing diagram instead, because the workstation allowed it, and because

the workstation allowing it told him something too. Command let people look at architecture diagrams because architecture diagrams made a room feel informed while hiding the one pipe that actually mattered. The visible chain ran cavern instruments into Lower Ops, then Command Review, then Archive — clean, hierarchical, the kind of diagram that belonged on a briefing slide with a laser pointer trained on it.

He expanded the archive node. Three downstream branches. Two of them labeled: TECHNICAL REVIEW, MEDICAL INCIDENT PACKAGE. The third had no name at all, just a blank rectangle sitting where a description should have been. The connecting line into it was drawn the same weight as the other two — not thinner, not dashed, nothing to mark it as lower priority in whatever software had rendered the chart. Somebody had wanted the diagram to look complete without making it readable.

He hovered the cursor over it. Nothing. He clicked.
ACCESS DENIED.

"You seem to be getting a lot of those," Hayes said.

"It's a classified site. I'd worry more if I weren't."

"You trying to access something outside your lane?"

"I'm trying to determine whether a broken wrist is the least interesting thing that happened in a cavern four miles under the ice."

Hayes's expression didn't change, but his weight did — left foot to right, small enough to be nothing on its own, the kind of tell Cavanaugh had learned to read only because he'd spent years watching people who thought they didn't have tells.

"That sounded like a yes," Hayes said.

"No. That sounded like an analyst."

Hayes let it sit for a moment, weighing whether to escalate the exchange into something that would need to be reported, and apparently decided against it. Cavanaugh had watched him make that calculation twice already today. It was the most useful thing about him — not the silence itself, but the visible cost of deciding to keep it.

Cavanaugh opened the workstation's system clock next and set it beside the timestamp column from the incident file. The terminal clock ran 0.8 seconds behind the event log — not a general drift he could blame on hardware, but one isolated specifically to the record of the pulse. The visible file had been corrected

to station time somewhere after acquisition. The terminal itself hadn't been. That gap meant the formatting pass — whatever smoothed the raw capture into the clean narrative on his screen — had happened afterward, as a separate step, by someone deciding what the record should say once the numbers were already in.

He pulled up a maintenance shell.

Hayes took one step closer this time. "Sir."

"Relax. Read-only."

"I'll decide what read-only looks like."

"Not technically. But go ahead."

Hayes leaned in anyway, close enough that Cavanaugh could hear him breathing through his nose, and Cavanaugh let him, because the maintenance shell only exposed what nobody had thought worth restricting — keyboard response, display refresh, storage latency, local clock, session ID. Beneath the clock sat a sync history. The last sync had failed at 09:42:29. The next had succeeded twelve seconds later, the same twelve seconds the official file logged as the gap between onset and the variance appearing in the record.

He opened the sync error.

The maintenance shell was not built for reading comfort — white text on black, no formatting, no line wrap worth

trusting — and he had to scroll twice to find the entry that mattered among a column of routine failures that meant nothing: three dropped pings, a storage warning that had cleared itself, a firmware check that had run twenty minutes late. Buried among them, one line, truncated.

TIMEBASE DISAGREEMENT WITH—

The rest had been redacted, and for a moment Cavanaugh nearly smiled at whoever had built the redaction, because they'd designed it to behave like deletion when it only behaved like an outline. Enough black shapes stacked in enough rooms, and eventually you could reconstruct the furniture underneath them without ever seeing the furniture itself.

He copied nothing. He wrote nothing down — no note, no photograph, nothing that could be found on him later and used to prove what he'd seen rather than merely what he'd guessed.

Instead he shifted his chair a few degrees with pressure from his knee, angling the dead half of the monitor until it caught the workstation behind him: an unattended engineering seat, its screen unlocked,

The Core Throttling

someone having left a diagnostic dashboard open during the alarms and never come back to close it out.

Data scrolled there in columns too small to read directly. Not too small, though, to recognize as structure rather than noise.

Event ID. Time. Local field. Magnitude. Containment. Disposition.

A list. Not a single incident logged once and forgotten. A ledger, with the shape of something that had been maintained.

The unattended screen refreshed. For less than a second, one row lit red before the display dropped straight into a password prompt. Cavanaugh caught three fragments before it vanished.

CH-09 / MATTER COHESION / PRIOR

Prior.

The bay door opened before he could hold the thought in place long enough to turn it over. Two command engineers came in with a sealed equipment case carried between them, condensation beading along its seams, a new hazard tag slapped crooked over an older one that hadn't been fully peeled away. Neither of them looked at Cavanaugh. That, more than anything either of them said in the next thirty seconds, told him where their attention actually was.

Cavanaugh recognized one of them by the way he carried his shoulders — Delacroix, from the platform crew, the man who'd been closest to Sanderson's rig when the alarms started. He hadn't seen him since. The other engineer he didn't know by name, only by the fact that he kept his eyes on the case handle and nowhere else, the particular discipline of someone who had been told exactly what not to look at and had decided the safest way to obey was to look at nothing at all.

"You need this bay cleared?" Hayes asked them.

"No," one of the engineers said. "We need Terminal Three."

Terminal Three. The unattended workstation.

Cavanaugh looked back down at his own screen a half-second before either engineer could catch him looking at theirs.

The nearer one badged into Terminal Three, palm code following the badge, and the dashboard flashed back to life before he killed the open view with a single command. His first attempt at the keystroke missed the field entirely — his hand shaking just enough that the input landed one row off. Cavanaugh knew hands. Not medically. Professionally, the way you learn

The Core Throttling

anything by watching enough people do it under pressure: ordinary stress hid itself in the face first, and only showed in the hands when something underneath the ordinary had gone wrong. This wasn't ordinary stress.

The engineer plugged the sealed case into the terminal. The room lights dipped — one long flicker, sustained just past the point where Cavanaugh would have called it a normal brownout — and then steadied. Somewhere behind the wall, a second fan joined the first, both of them clicking on staggered rhythms now, and the frost crescents along the rack windows seemed, for a moment, to sharpen at the edges, though he couldn't have said whether that was the cold or his own attention doing the sharpening.

Delacroix didn't look up from the case. His partner glanced once at Cavanaugh's workstation, found nothing worth reporting, and went back to watching the handle.

On his own screen, the controlled viewer threw a notification.

SESSION SCOPE REDUCED

A second followed it.

SEARCH TERMS LIMITED BY COMMAND

REVIEW

Then a third.

DIAGNOSTIC EVENT ONE ARCHIVE
UPDATED

He opened the incident summary again, already knowing before he finished loading it that something in the text would have moved. The field-anomaly line had changed.

Field anomaly: expected within diagnostic envelope.

He sat with that for a moment — the future rewriting itself backward into a document that had said something different thirty seconds earlier, while he was still looking at it — and felt the particular narrowing that came before he did anything he couldn't undo. It wasn't panic. It was closer to inventory: Hayes on his right, close enough to react. Two engineers at Terminal Three, both facing away from him now, absorbed in their own case. The badge reader on the door behind him, requiring a code he didn't have. The camera mounted over the coffee counter, angled to cover most of the room but not the exact seat he was in. The floor vent under the third server rack, ticking every time the fan under it missed a rotation, loud enough to cover a keystroke if he needed one covered.

He couldn't take the file. Couldn't export it, couldn't photograph it with a camera watching the room and a soldier watching his hands.

He could still make the system show him where the lie was sitting, even if it wouldn't let him carry the proof out with him.

He tried the revision history.

ACCESS DENIED.

Expected, and useful only in that it confirmed the history existed at all. He tried the print stylesheet next — a function nobody used anymore, since nothing at Cold Horizon printed out of a controlled viewer, but the option had never been removed because some compliance package years ago had required a template and nobody since had bothered stripping it out.

The stylesheet loaded with its field mappings intact. Most of them were generic: `incident.title`, `incident.summary`, `injury.classification`, `equipment.damage`, `environmental.notes`. And then, unrendered on the page but still referenced in the template because whoever had suppressed the visible display hadn't thought to clean the code underneath it —

```
const_drift_local.value  
const_drift_local.duration const_drift_local.radius  
const_drift_local.trigger const_drift_local.disposition
```

Disposition.

The drift had a disposition category. A menu with more than one outcome on it — contain, suppress, exploit, repeat, he didn't know the actual options, only that a category like that didn't exist unless somebody upstream had needed to choose between them more than once.

The viewer closed itself.

Not a crash. A decision, made somewhere outside the room, executed cleanly and without apology.

His session dropped back to the landing page, stripped down to three tiles: APPROVED BRIEFING MATERIALS, PUBLIC GEOLOGICAL SUMMARY, MEDICAL INCIDENT ACKNOWLEDGMENT. He sat looking at them for a second longer than he needed to, the way a man looks at a door that has just been shut in a specific, considered way rather than slammed.

Cavanaugh's hands had gone stiff in his lap, more from stillness than cold now, and he flexed them once under the desk where the ceiling camera wouldn't

catch the movement as anything but idle fidgeting. He was aware, distantly, of being hungry — the kind of hunger that arrived hours after a meal he couldn't remember eating, displaced by everything that had happened between then and now. He filed it under things that would matter later and not at all right now.

Hayes set one hand on the back of Cavanaugh's chair.

"Command says your current task is complete."

"Command's optimistic."

"You're to remain seated until escort transfer."

"Where."

Hayes didn't answer that. He rarely answered questions shaped like that one, which by now Cavanaugh understood as an answer of its own — not because Hayes was hiding something personally, but because nobody had told him enough to be worth extracting anything from.

Across the bay, one of the engineers locked the sealed case while the other wiped Terminal Three with a command key, leaving the screen black and glossy enough to hold a reflection. Cavanaugh caught his own face in it for a moment — tired, deliberately flat, in the early stages of understanding just how large the thing on the other side of today's paperwork actually was.

He looked back at the three approved tiles. No incident file. No telemetry. No appendix. Nothing left on the screen to prove, to anyone who hadn't been sitting in this exact chair for the last forty minutes, that the workstation had ever shown him 0.401 percent localized gravitational variance, a metadata tag with no public name, or three characters — CH-09 — attached to something called matter cohesion and marked *prior*.

He didn't need the file anymore to know it had existed. That was the trouble with trying to hide a pattern from someone who'd spent a career reading for a living. Absence had edges of its own, and a patient enough reader could trace the shape of a document by cataloguing everything it had been careful not to say.

He ran the fragments once, in order, the way he would have laid out a case file if he'd been permitted paper. Twelve minutes to a prewritten title. A variance number too clean to be raw. A metadata field with a name and no public content. A routing branch with no label at all. A clock correcting itself by less than a second, in exactly the gap where the story needed room to be edited. A red row on a stranger's screen reading *prior*. None of it, on its own, would survive a hearing.

The Core Throttling

Together it had a shape he didn't like, and he'd learned a long time ago not to distrust a shape just because it was inconvenient.

The bay intercom clicked once before the announcement started.

"All personnel, prepare for Field Utility Review at 1400. Diagnostic Event One data package has been accepted."

Accepted. Not analyzed. Not reviewed. Accepted, the way a shipment gets accepted at a loading dock by someone whose job is to sign for boxes, not open them.

Cavanaugh kept his hands still in his lap. Under the raised floor, the maintenance fan clicked, missed a rotation, clicked again, indifferent to everything that had just happened six feet above it. On his own dead screen the workstation clock corrected itself by 0.8 seconds without asking him first, quietly closing the last visible gap he'd been using to measure the distance between what happened and what the record said had happened.

Across the room, before Terminal Three's screen finished going dark, one line surfaced in the glass and held there just long enough to be read once.

CONST_DRIFT_LOCAL: RECLASSIFIED FOR
OVERRIDE MODELING

Kalamazoo

Then the reflection dropped to black, and Hayes was still watching him, and nobody in the bay said anything else for a long time.

Chapter 12: Heat Sink Model

The medical alcove had two cots, one monitor bolted crooked to the wall, and a sink that coughed twice before it remembered it was supposed to produce warm water. Someone had built the room for injuries that ended, not injuries that kept happening. Sanderson sat on the near cot with her thumb out and let Ariza clean it because he had put himself between her and the door and did not look like he intended to move.

"Flex," he said.

She flexed. The pain came up bright enough to make the fluorescent tubes look sharper than they were.

"Again."

"I need my hand."

"You have one."

"I need both."

"You should have thought of that before you tried to open a rack with your thumb."

She looked at him. He didn't look away. ARIZA, the tape on his chest said. He had the face of a man who had treated frostbite, concussion, and command decisions in roughly equal volume and stopped being surprised by any of them years ago.

"That was a joke," he said.

"It wasn't funny."

"No."

He wrapped the thumb tight enough to stop fine motion and loose enough that it still throbbed. The dressing held white for three seconds before a crescent of blood came through the gauze from underneath.

"Stitches?"

"If it opens again." He lowered his voice without changing his expression. "You should let command review the equipment properly."

"I did. They reviewed it into someone else's custody."

"They review everything into someone else's custody."

Ariza taped the last edge of the dressing down and checked his watch, an old analog piece with a scratched crystal, the kind of thing a man carried because command couldn't confiscate a habit. "You've got maybe forty minutes before the swelling tells you how bad it actually is. Ice, if you can find any that isn't load-bearing."

"Funny."

"I try."

He capped the antiseptic and set it back in its drawer with more care than the motion required, lining the label square to the edge as if the drawer itself might be inspected later. Sanderson watched him do it and understood, a beat late, that the tidiness was its own kind of defense — something to control when nothing else in the building answered to him.

From the far cot, Kline made a sound that wasn't quite a word. Sanderson was up before Ariza could stop her. The room tipped a fraction — not the floor this time, just her own blood pressure, no food since the aircraft, the cold sitting in her joints, adrenaline with nowhere left to go.

She crossed to him anyway.

Kline's eyes tracked the ceiling instead of her face. His wrist sat in a foam brace, the color under his cheekbones gone flat and waxy. "I saw the timestamps."

"Don't talk if they medicated you."

"They didn't. Not yet."

"Then talk carefully."

He swallowed, and his breath came shallow, fast, working around something. "The pulse didn't start the drift. The first one."

Sanderson went still.

Ariza glanced at the door.

"The frost lifted before command execution," Kline said. "Half a second. Maybe less. I thought I imagined it."

"You didn't."

"I thought maybe my eyes—"

"Your eyes were better than the rack."

That settled him more than comfort would have. He turned his head toward her, waiting.

"It means the structure responded before the imposed pulse registered on our system," she said.

"To us?"

"No." She pressed the bandaged thumb into her palm until the dressing pulled taut. "To the load."

Kline's eyes finally came off the ceiling and found hers. "Are you going to tell them?"

"I already did. Twice."

"And?"

"And they wrote it down."

He almost laughed, a short exhale through his nose that cost him something, judging by the way his face tightened around it. "That's it? Written down?"

"Written down means it exists somewhere. That's not nothing." She heard herself say it and wasn't sure she believed it.

Ariza appeared at her shoulder with a blanket, the thin foil kind that made noise every time it moved. "He needs to stop talking now."

"I'm stopping," Kline said, and closed his eyes before anyone could argue with him.

The door opened. Pell, windburn still raw on his cheeks from the descent though none of them had been outside since. "Dr. Sanderson. Command wants you in the instrument quarantine room."

"They can wait."

"Now."

Ariza said, "She's under treatment."

"She can walk ten meters."

"She can reopen the hand."

Sanderson was already moving. "I need to see what they did to my rig."

Pell stepped aside for her. That was new. Not much of a shift. Not enough to call help. But he no longer occupied the doorway like a piece of furniture waiting for an order.

The instrument quarantine room had been storage once — she could tell by the shelf brackets still bolted along one wall, painted over and abandoned mid-job. Somebody had added a grounded floor mat, a strip of hazard tape, and three cameras bracketed into the corners like an afterthought that had become policy. The floor was colder here than the corridor, concrete instead of composite decking, and it came up through the soles of her boots in a way that made her think, briefly and uselessly, about how far below them the drill shaft still ran.

Her diagnostic rack sat under floodlights in the center, access panels open, cabling pulled loose and re-routed through a portable interface she didn't recognize. A man stood beside it with a tablet, badge finally readable this time: RANE. A second technician was seating lockout tabs into the analog ports one at a time, methodical, unhurried, the kind of unhurried that came from being told exactly how much time he had and using all of it.

Something in her chest went from cold to hot in under a second.

"Take those out."

Rane didn't turn. "This equipment is under preservation order."

"You're modifying it."

"We're securing it."

"You're changing the interface state after an anomalous event."

He looked at her then. Older than Kline by fifteen years, too alert for a man who should have been asleep six hours ago, with the particular economy of someone who had learned which questions not to ask in front of the wrong people. "The data's been mirrored."

"That's the problem."

Pell stayed by the door. Two more security personnel stood outside the glass, close enough to be visible without anyone having to say why.

Sanderson crossed to the rack and held out her uninjured hand. "Give me the raw thermal package."

Rane checked his tablet without really needing to. "Command has it."

"Command has the formatted telemetry. I want the raw thermal frames."

"Not available."

"Not available to you, or not available because someone erased it."

His jaw set. She let the silence sit there and do its work.

She stepped closer. He moved to block the rack. She stopped an inch short of him and looked past his shoulder at the panel behind, not his face.

Panel C. Closed. It should have been open if anyone had located the acquisition buffer — it wasn't hidden, just unlabeled, tucked behind a redundant board nobody who hadn't built the rig would think to check. The calibration weight housing sat empty, because she'd made it empty on the way down here. The visible storage key was locked under command seal. The analog trace buffer, though — the deprecated field-verification channel wired in for exactly this kind of failure — that was still sitting inside the temperature bridge where it had always been, doing nothing, waiting.

"I need five minutes," she said.

"No."

"Two."

"No."

"Then stand there and listen while I tell you what your formatted package erased."

Rane said nothing. Good.

Behind him, the second technician had stopped working and was watching now instead, one lockout tab still pinched between two fingers, forgotten. He hadn't been told to stop. He'd stopped anyway.

She pulled the calibration weight from her jacket pocket. Pell saw it. Rane saw it. Something in the room's attention reorganized itself around the small dull cylinder in her palm.

"What is that," Rane said. Not a question, exactly.

"A mass standard."

"That's not all it is."

"No."

He held out his hand. "Doctor."

"If I hand this over, your report becomes the only version that exists."

"Command can compel transfer."

"Then compel it." She looked at the nearest camera. "On record."

Nobody moved.

The hum came through the walls, quieter here than it had been in the cavern but still present, a low pressure that

gathered in fillings and screw heads and the backs of her teeth. She set the calibration weight down on the grounded mat and worked the end cap loose with her bandaged thumb braced against her palm. It hurt enough to prickle sweat under her collar. She kept going. Pain was just data arriving at an inconvenient time.

The storage core slid free into her palm.

Rane inhaled once, short.

"You should think about this," Pell said, very quiet.

"I am thinking."

"You plug unauthorized storage into a command system, that's extraction. They'll call it that."

"I'm not plugging it into their system."

She unzipped the side pocket of her field case and pulled out the battered tablet — the one with the crack running through one corner from the gravity return, from when she'd caught the rack instead of herself. It powered on slow, flickered twice, then settled into her own local interface. Ugly. Simple. No network icon. No compliance skin over the top of it.

Rane reached for it.

The Core Throttling

Pell stepped half a pace forward, close enough that the movement counted as something.

Sanderson looked between them. "This tablet has no transmitter. You tagged it at intake — check your own manifest. If I falsified that, charge me. If I'm reading my own instrument's own output, stand back."

Rane hesitated because the sentence had too many witnesses attached to it now, and none of them were his.

She inserted the core.

Twelve hours of raw acquisition blocks loaded across the cracked screen, most of them corrupted, incomplete, or misaligned in ways that made her exhale instead of tense. Clean data would have meant someone had already been inside it. This was ugly in the way that told her nobody had touched it yet.

She called up the thermal map from first contact.

Rock wall: elevated. Matrix contact: below ambient. Matrix interior, inferred: no emission. Adjacent strata: heat depletion gradient.

Then the sequence from the diagnostic spike.

The numbers sat there looking wrong in a way she hadn't fully processed the first time, in the cavern, with everyone shouting.

"Look," she said.

Rane didn't move.

"Look, or stop pretending this is a technical review."

He looked.

She enlarged the thermal gradient. "Before the pulse, the matrix surface was already drawing heat out of the adjacent rock. Not much at the contact point — we're only seeing an exposed interface — but enough to build an asymmetry."

"Heat sink behavior's already in the preliminary report."

"No. Your report says energy dampening."

"Functionally—"

"Don't use that word with me." It came out sharper than she meant. Through the open door she saw Kline turn his face toward the sound. She brought her voice down. "A dampening field is what command wants, because it sounds deployable. This isn't dampening. It's transfer."

"To where?"

"I don't know."

"That's not useful."

"That's the most useful thing I've said since we landed."

His mouth opened, then closed on whatever he'd been about to say.

She dragged two traces together on the screen — thermal depletion and hum amplitude. They didn't align cleanly. Nothing in the raw data ever did. But the relationship held anyway: heat absorption climbing before the hum narrowed, dropping through the gravity event, spiking afterward as the structure recovered.

"Before the diagnostic pulse, absorption increased," she said. "It saw the load coming before our own system registered execution. Or it was already under load from something your test added onto. Either way, the pulse didn't make it stronger. It made it compensate."

Rane's frown was involuntary. "Compensate for what?"

"Thermal stress. Processing stress. Possibly both. I need the deep-crust exchange data and the pre-breach seismic package."

"You're not cleared for pre-breach."

"Then you're not cleared to tell me I'm wrong."

"That's not how clearance works."

"No. It's how command wants clearance to work. There's a difference, and you know it, because you just told me the packet doesn't cover something you didn't expect to see."

Rane's eyes went to the second technician, who had gone back to pretending to work, tab pinched between two fingers and not moving. "You want the pre-breach seismic data so you can build a model of what the site looked like before anyone touched it."

"Yes."

"That model won't match the current structural survey."

"I know."

"You already think it won't match."

"I think whoever wrote the current survey started from a conclusion and built backward. I want to see what the ground actually did before there was a conclusion to protect."

He looked toward the camera bracket like it might intervene for him. "Doctor, this is drifting outside the scope of review."

"No. The system is drifting outside the scope of your review."

She pulled up the gravity variance trace. The tablet lagged, painted the line in segmented jumps, and for one full second the graph rendered upside down before correcting itself. That didn't reassure her. Nothing about the correction looked like a fixed bug. It looked like the software catching up to something that had already happened.

"The first physical response wasn't projection," she said. "It was a stability correction. Local gravity dropped point four percent for under seven seconds. That sounds small until you put people, platforms, and load-bearing structure inside the radius."

"If it can reduce effective gravity—"

"There." She pointed at him. "That's the mistake."

"What mistake."

"If it can reduce gravity, you think that means you can make it reduce gravity on command. The data says the reduction was coupled to stress recovery. You didn't steer anything. You injured it, and it protected itself badly."

Pell's eyes dropped from the screen to the floor, the way a man remembers his own boots losing weight without wanting to.

Rane leaned closer despite himself now, pulled in against his own resistance. "You're assuming intent."

"No. I'm assuming a limitation."

She switched back to the heat map. "It's drawing thermal energy from below and around it and emitting essentially none. After the pulse, absorption spikes, and then the output channels collapse instead of opening. That's throttling behavior."

"That's machine language."

"That's physical behavior. Throttling means a system limiting throughput to avoid failure. Heat exchange, computation, field stabilization — it doesn't matter which. I'm not naming the mechanism. I'm telling you it's operating at a limit."

The lights flickered once, all three tubes at the same time, then steadied.

Everyone looked up.

The hum held its pitch through it.

Sanderson didn't wait for anyone else to move first. She opened the raw audio overlay, then the thermal response, then the field variance, three traces that were each bad in a different way, contaminated by the same event from three separate angles. She aligned them by the pre-pulse frost lift instead of the command execution timestamp.

The pattern sharpened. Not clean. Enough.

Heat absorption rose 0.6 seconds before the official pulse execution. The hum narrowed 0.4 seconds before that. The first field variance arrived close to simultaneous with the command timestamp — but only because the command timestamp had been normalized after a sync failure that nobody had flagged in the packet she'd been handed.

Her stomach dropped a half-inch.

"There," she said.

Rane stared at the screen like it might resolve into something more forgivable. "That can't precede execution."

"Official execution."

"No."

"Yes."

"Then the clocks are wrong."

"Yes."

"Which clocks?"

She looked at him until he understood what he'd just asked out loud, in a room with three cameras in it.

The room felt smaller after that, though nothing in it had moved.

A bad clock was an equipment problem, filed and forgotten. A system in which separate physical processes

Kalamazoo

disagreed about the order events had happened in was something that didn't have a form to file it on.

"That's not in the packet," Rane said, quieter now.

"No. It was classified as platform instability."

He stepped back from the tablet like the graph itself had become something he didn't want touching him.

Pell's radio clicked against his collar. He tilted his head, listened, and his face closed by degrees, one muscle at a time, until it had the same flatness Ariza wore in the alcove.

"Doctor," he said. "End the review."

"I need to copy this."

"No."

"I'm not finished."

"Command's transferring the rig to Field Utility Review."

Rane's head came up sharp. "Already?"

Pell didn't answer him.

Sanderson shut the tablet lid over the storage core before either man could reach for it. "They can't model an override off contaminated interpretation."

"They can," Pell said.

That landed worse than an argument would have.

The quarantine room door opened again. Two security personnel came in with a sealed transport frame, one of them carrying a chain-of-custody clipboard, the other a lockbox with a combination dial already spinning shut.

Sanderson picked the calibration core up off the mat.

Pell held out his hand.

Neither of them moved for a second.

He didn't want to take it. That much was visible in the set of his jaw, the way his hand stayed half-open instead of grasping. Not mercy. Not an alliance either, not yet. Just recognition of something he hadn't asked to carry.

Then the wall speaker crackled on.

"Sergeant Pell. Secure all unauthorized storage media."

Underwood's voice, smoothed flat by the bad speaker, arriving at the exact point where an institutional choice needed to become somebody else's hands doing the work.

Pell's expression emptied out.

"Doctor."

Sanderson closed her fingers around the core. "This is the only raw record that shows a pre-execution response."

"Then put it in your report."

"They'll bury the report."

"Probably."

The word sat between them a moment before anyone reacted to it. Pell had said it. Rane had heard it. She had too. Nobody repeated it.

The transport team started rolling the diagnostic rack toward the door. One wheel caught on a seam in the mat and stalled, and the man pushing it leaned into the frame without slowing his pace, the way people move when they've been told a deadline exists but not what happens if they miss it. One of the lockout tabs caught on the edge of the floor mat and tore free with a small plastic snap that nobody stopped to pick up. The wheels squealed across the threshold, leaving two black skid marks on the grounded mat that would still be there tomorrow, evidence of nothing anyone would ask about.

The clipboard man logged the transfer without asking her name, already knowing it, and held the lockbox up for the camera the way a courier holds up a signature. Custody was a word people used when they wanted a transfer to sound clean. What she'd watched was closer to confiscation with paperwork attached, and she didn't correct the distinction out loud, because there

was no one left in the room who would have written it down differently.

Sanderson set the core into Pell's open palm. Not because she trusted him with it. Because refusing here ended with the core taken anyway, the tablet seized on top of it, and her hand hurt badly enough afterward to put her out of every room that still mattered.

Pell's fingers closed around it.

"Evidence transfer witnessed," he said, not looking at the camera bracket.

She watched his hand instead of his face. "Remember where it came from."

He gave no sign that he'd heard her, which wasn't the same as not hearing.

The tablet still sat open on the counter where she'd left it, screen showing the last trace she'd aligned before shutting the lid too late to stop them seeing it. Three lines. Heat absorption. Hum narrowing. Local variance. All three beginning before the point where the official record said the event had started.

The screen flickered once. For a moment the graph redrew itself in blocky right-angled segments that had nothing to

do with any software she'd written, then smoothed back into its ordinary render.

Then it went dark on its own, battery indicator sitting at forty percent, no reason for the shutdown that she could see.

Sanderson stood in the cold quarantine room with her thumb throbbing under the dressing and the taste of metal at the back of her mouth, the kind that came from cold air and adrenaline rather than anything she'd touched.

She had no proof left in the room with her. The rack was gone. The core was in Pell's pocket now, wherever that ended up. What she had was a model, built out of misaligned traces and a half-second of frost lifting before anyone had told it to.

The matrix wasn't projecting force outward the way a weapon would. It was absorbing. Compensating. Narrowing its own bandwidth under load.

Trying not to fail.

And somewhere above her, in a room she wasn't cleared to enter, command was already drafting an override out of the same symptom she'd just watched them refuse to read correctly.

Ariza's voice reached her from the corridor, telling someone Kline needed to lie flat and stop talking. She flexed her thumb once, testing the dressing, and the pain arrived exactly where it had before — sharp, clarifying, entirely useless to anyone but her.

Rane hadn't left. He stood near the empty rack outline still marked on the floor by four indentations in the mat, tablet held loose at his side, not typing anything into it. He looked at the dark screen on the counter the way a man looks at a door he's already decided not to open.

"Doctor."

She didn't answer right away.

"The clocks," he said. "If someone asks me directly. What do I tell them?"

"Tell them the truth. Tell them you don't know which one is wrong."

"That's not an answer they'll accept."

"It's the only one I have."

He didn't argue with that. He gathered his tablet under one arm, checked the empty ports where the analog buffer had been before the transport team took it, and walked out without looking back at her. The second technician followed, lockout tab still in his hand, unused.

Sanderson stood alone with the grounded mat and the two black skid marks and the counter where her tablet sat dark. She picked it up, turned it over once to check the crack hadn't spread, and put it back in the side pocket of the case she no longer had a rack to attach it to.

The hum was still there. It had never stopped being there, not since the first descent, not through the pulse, not through any of the last two hours — a pressure under the floor that the building had simply learned to live inside, the way a body learns to live around an old injury that never quite heals.

She left the quarantine room and did not look at the cameras on her way out.

Chapter 13: Maintenance Outage

The clocks agreed at nineteen hundred hours exactly, and that was the first thing wrong.

Cavanaugh had spent eleven days inside Cold Horizon watching every timepiece in the facility disagree with every other by some small, deniable margin — a workstation clock four seconds ahead of the wall panel, the wall panel eleven seconds behind the security terminal, the command message ticker running its own private arithmetic in the corner of the screen. He had stopped mentioning it after the second day because Hayes stopped pretending to care. Then, without warning, without a countdown, every display in the data bay swung to the same second and held there. Workstations. Wall clocks. Security panels. Environmental monitors. The ticker.

19:00:00.

He watched the numbers settle and felt the skin along his forearms tighten.

Hayes caught the look before Cavanaugh could file it away. "Something wrong?"

"Usually."

"With the clocks."

"They're very confident."

"That's not an answer."

"No," Cavanaugh said. "It's an observation."

Hayes had been assigned to him long enough now that the distinction cost him something. His jaw worked once and he didn't answer.

A maintenance crew came through the bay doors with gray cases and the particular posture of men who had been told not to look at anyone under escort. Two of them went straight to Terminal Three — the station that had, for eleven minutes two days ago, shown Cavanaugh a live reclassification of a physical constant before someone upstream noticed and pulled the feed. They didn't open the housing to diagnose it. They started disconnecting cables.

Not repairing. Removing.

One case had a tag zip-tied to the handle, printed in the flat sans-serif the facility used for anything it wanted to look procedural.

TIMEBASE RECOVERY / LOWER OPS
SCOPE: CLOCK DOMAIN RECONCILIATION
PRIORITY: COMMAND

Clock domain reconciliation. Someone in an office above ground level had chosen those words for a

form, and the form had made it down here inside a tool case, and now two men in gray coveralls were carrying out whatever the words actually meant. Cavanaugh turned the phrase over twice and put it in the part of his memory that didn't need permission to keep things.

Hayes noticed him watching. "You're staring."

"They gave it a nice name."

"What did."

"The repair."

Hayes followed his eyes to the crew and found nothing that looked like a story. "Scheduled maintenance."

"After a gravity event."

"After a platform stability deviation."

Cavanaugh turned enough in the chair to look at him directly. "Did that feel better to say?"

Hayes's face didn't move. His jaw did, once.

The overhead lights flickered — not a dim, not a browning-out, but a rolling stutter that ran the length of the bay out of sync with itself, panel by panel, as though each fixture had briefly forgotten the others existed. Server-rack LEDs blinked down the row in a delayed cascade. Somewhere past the far wall a relay opened with the flat metallic weight of a door bolt thrown by something much larger than a door.

Kalamazoo

Every screen in the bay froze.

Every clock held at 19:03:14.

Hayes lifted his radio. "Control, Data Bay Two experiencing—"

Static answered him, thick and even, the kind that meant a channel had been muted rather than lost.

The maintenance crew looked up at the same moment, all four of them, which told Cavanaugh more than anything on the tag had. Men surprised together were more honest than logs. Logs could be built to lie in advance. A flinch couldn't.

Above the bay door, a status panel changed from green to amber.

MAINTENANCE WINDOW ACTIVE
LOCAL ACCESS RESTRICTED REMOTE
MIRRORING PAUSED

He read it twice to be sure of the third line before it disappeared.

Remote mirroring paused.

Less than three seconds. Then the panel corrected itself, quietly, as if it had misspoken.

MAINTENANCE WINDOW ACTIVE
LOCAL ACCESS RESTRICTED

Better. Cleaner. A sentence with the interesting half removed.

Absence had edges, if you were paying attention when it happened, and he had been.

Hayes started toward the door. "Stay seated."

"I was going to enjoy the clock failure from right here."

"Stay seated."

The maintenance lead swore under his breath and jammed a cable back into Terminal Three's housing. The screen stayed black. A younger tech tapped a handheld unit against his palm, checked it, tapped it again. For half a second its display showed a network map — lower operations broken into small isolated shapes, each with a thin line running up toward a command node, most of the lines drawn in red — and then the screen went to gray.

Hayes tried the radio again. "Control, respond."

Static.

Then a voice, torn into pieces by whatever was doing this to the signal: "—all lower sectors hold position. Clock reconciliation in progress. Do not initiate manual—"

Nothing after that. The channel dropped back into static and stayed there.

Manual what.

Cavanaugh sat with the unfinished sentence and let the room work out, on its own, what rules still applied.

The bay door lock clicked open. No alarm. No acknowledgment tone. Just the small mechanical surrender of a latch that had lost whatever was telling it to stay shut.

Hayes heard it. So did Cavanaugh.

The soldier's eyes went to the door and back.

"Do not."

"I haven't moved."

"Keep not moving."

The door wasn't the problem. Doors were visible; doors made people defensive, made them count heads and watch exits. The floor was the problem, and nobody was watching the floor.

Under Terminal One, where a cooling fan had been ticking badly since noon, one of the raised-floor grates sat half a centimeter proud of its neighbors. Not enough to write up. Enough that whatever service harness ran beneath it had shifted during the gravity drop two days ago and never fully reseated. He'd watched the maintenance crew step around that exact

The Core Throttling

panel twice already this evening without either of them saying why.

He put a hand flat on the desk, as if bracing himself against nothing.

Hayes's attention flicked to the door again — another lock somewhere down the corridor had just answered the first one.

Cavanaugh slid his shoe under the edge of the grate and lifted.

It didn't move.

He pressed down on the far edge first, unseated it, then lifted again. The panel rose with a dry scrape that the maintenance crew's noise mostly covered.

Hayes turned back toward him.

Cavanaugh let the panel drop.

It was an ugly sound. Louder than he'd meant.

Hayes stared at him.

"Sorry," Cavanaugh said. "Foot slipped."

"You're seated."

"That's what makes it embarrassing."

The maintenance lead snapped at no one in particular, "Can somebody get us a hardline reset off Junction C?"

The younger tech said, "Junction C's locked."

"Everything's locked. That's the point of saying hardline."

Hayes drifted a step toward them, the way soldiers moved toward anyone who sounded like they understood the actual problem. "Do you need clearance for that?"

The lead looked at him like he'd asked whether the fire needed a permit. "I need the local bus up before command decides we broke it."

Hayes hesitated. One beat. Long enough.

Cavanaugh lifted the grate again, more carefully this time, and looked into the gap. Cable bundles, a film of gray dust, a drop too narrow for a body but wide enough for a hand and a forearm. One bundle, tagged in yellow heat-shrink: AUX-TERM / C-JUNCTION, running toward the wall behind the coffee counter.

Auxiliary terminal. Junction C. Not a rumor. A physical run of cable with a destination printed on it.

He lowered the panel before anyone turned around.

The lights stuttered again, and this time the screens across the bay didn't freeze — they rebooted, one after another, into flat gray shells with no command

banner, no compliance header, just bare prompts. On his own workstation, the change happened last.

LOCAL SESSION RESTORED REMOTE
MIRRORING UNAVAILABLE ENTER MAINTENANCE
CREDENTIAL

The field blinked, waiting.

He didn't touch it.

"Hands where I can see them," Hayes said, closer now, sharper.

Cavanaugh raised both a few inches off the desk. "You make it sound intimate."

"Hands."

He set them beside the keyboard, palms down, and left the prompt alone.

Maintenance credential. Not analyst. Not visitor. The outage had stripped the system down to something smaller and older than institutional rank — for the length of this window, it wanted proof of a specific job, not proof of who you were allowed to be.

Across the bay, the younger tech fumbled his credential pouch open with two shaking hands and a stack of temporary cards slid out onto the floor. Three of them skated under the

wheeled cart holding Terminal Three's ruined housing. One slid free of the others and stopped near the leg of Cavanaugh's chair.

Hayes saw it land.

Cavanaugh saw Hayes see it.

Neither of them moved.

The tech cursed and dropped to his knees to gather the cards. Hayes bent automatically toward the nearest one, then caught himself half down, remembering — a half second too late to matter — that he wasn't supposed to touch access media that wasn't his.

For that half second the card lay face up.

MAINT-C / POWER THERMAL SHIFT 19-
03 VALID LOCAL ONLY

Cavanaugh took it in the way he took in a filing error: not memorized, filed. Expiration format. A four-digit suffix. A color band along the top edge. A scratch through the second digit of the printed code. A manufacturer's mark small enough to miss along the bottom edge. Not enough to forge the card. Enough to know the shape of what Junction C would accept.

The tech snatched it back up. "Thanks."

Hayes hadn't touched it. "Secure your credentials."

The tech's ears went red. "Yes, Sergeant."

Cavanaugh looked at his own blank prompt and didn't type a code he didn't have. That was how people got caught — guessing at a class they'd only glimpsed. The card had told him a category, not a password. The part worth keeping was three words: VALID LOCAL ONLY.

Some systems, under outage conditions, dropped back to class-based permission before they asked who specifically was standing in front of them. Not because the engineers who built them were careless. Because a power plant, or a drilling platform, or whatever Cold Horizon actually was underneath its cover story, needed bodies who could isolate a breaker or close a valve when the link to command went dark, and it needed them fast enough that individual authentication became a luxury.

Cold Horizon was a prison until the systems that ran it hurt. Then, for a window measured in minutes, it needed hands more than it needed to know whose hands they were.

The intercom clicked on, flat and evenly spaced. "Lower operations, maintenance window extended. Personnel remain at assigned stations. Command mirror restoration in eight minutes."

Eight minutes.

Hayes's shoulders eased a fraction — someone had finally given him a number to hold onto.

Cavanaugh felt the opposite settle into his chest.

Eight minutes meant the gap upstairs was real, not decorative. Eight minutes meant that whoever needed a local system open would be using it now, while it was open. Eight minutes meant the auxiliary paths in this building were live, watched badly by people distracted with their own emergencies, and would be sealed harder than before the moment the window closed.

He let his eyes drift to the coffee counter without turning his head.

Behind it, a wall vent breathed warm air in slow pulses. On the third pulse the vent cover shuddered against its frame with the same rhythm as the bad fan under Terminal One. Below the counter, a service plate had been reset at some point with the wrong hardware — bright stainless screws standing out against a wall of matte black fasteners, the kind of small mismatch that meant someone had been in there recently and hadn't cared, or hadn't had the right screws, or hadn't expected anyone to look twice.

Junction C was behind that counter. Of course it was.

"Stop looking around," Hayes said.

"Then stop putting interesting things in rooms with me."

"Sir." The word had gone from courtesy to warning somewhere in the last two days, and it finished the turn now.

"Fine." Cavanaugh let his shoulders drop, let his hands stay open and visible, let his face settle into the specific look of a man deciding, visibly, not to be a problem. Annoyance was allowed. Interest wasn't.

Hayes studied him for five full seconds before the bay door opened again and Colonel Underwood walked in, and the room changed faster than the lights had.

The maintenance crew straightened without being told. Hayes shifted half a step, guard reorganizing itself into subordinate. Underwood wore the same cold-weather uniform she always wore, the same economy of motion, the same expression Cavanaugh had started disliking on principle rather than evidence. Her eyes moved once across the dead terminal, the open tool cases, the amber panel over the door, the position of every body in the room, and then came to rest on him last, which he understood to be deliberate.

"Mr. Cavanaugh." Her voice carried the same flatness it always did. "You're being relocated for the maintenance interval."

"To where?"

"A holding office."

"Does it have a better view of the lies?"

Hayes went still in a way that had nothing to do with posture.

Underwood didn't. "You've been given access appropriate to your role."

"My role keeps changing."

"Your role is to analyze approved data."

"That's not analysis. That's obedience with charts."

One of the maintenance techs found something urgent to do with a cable.

Underwood stepped closer, not fast, not threatening in any way a report could describe. "Do not confuse restriction with deception."

"Do not confuse control with comprehension."

He heard it leave his mouth and knew, half a syllable too late, that it had cost him more than it was

The Core Throttling

worth — not because it was untrue, but because it told her exactly which direction he'd been looking.

Underwood heard the same thing. For the first time since he'd met her, something crossed her face that wasn't quite interest and wasn't quite anything else he had a name for.

"Escort him," she said.

Hayes took his left side. A second soldier came in from the corridor and took the right. Neither touched him yet. That told him they were expecting compliance and had a separate procedure ready in case they didn't get it.

Cavanaugh stood. His knee caught the edge of the chair on the way up.

The misaligned floor grate under Terminal One shifted, a fraction of a millimeter, audible only to whoever was already listening for it.

No one else noticed.

Underwood had already turned toward Terminal Three. "How long until command mirror restoration?"

"Six minutes," the maintenance lead said, "if Junction C holds."

"If?"

"Local bus is unstable."

"Stabilize it."

"Yes, ma'am."

Hayes started him toward the door with a hand that didn't quite touch his shoulder. As they passed the coffee counter, the wall vent pulsed warm air across his ankle and the mismatched stainless screws caught the amber emergency light. One of them wasn't seated flush. Behind the plate, for less than a second, a green diagnostic LED blinked in a pattern that didn't repeat.

He didn't slow down.

The corridor was colder than the bay, and darker in the particular way that meant half the fixtures had gone to amber and the other half hadn't gone anywhere. Badge readers along the wall showed local mode or nothing at all, no pattern he could find in which was which. Two engineers hurried past with a coil of fiber tether slung between them and didn't look at him once, which told him whatever had broken wasn't contained to the room he'd just left.

At the first junction, the route to the residential modules ran right.

Hayes turned left.

Wrong direction. Cavanaugh let it sit without asking why, the way he let most wrong things sit until they explained themselves or got explained for him.

"Where's the holding office?" he asked anyway.

Hayes didn't answer.

The second soldier said, "Temporary."

"Everything here is temporary."

They passed a glass panel overlooking the lower operations atrium, three stories of it visible in a single sweep — security teams moving fast between stairwells, engineers clustered at breaker cabinets, command staff with tablets held low against their bodies like something they didn't want dropped. On a wall display at the center of it all, up for less time than it took him to read it before someone below cut the feed, a countdown:

MIRROR RESTORE: 05:12

And under it, one more line, gone almost as fast:

AUXILIARY ACCESS PATHS ACTIVE

He looked away before Hayes could follow his eyes there.

The escort stopped outside a small windowless office, an old mechanical lock retrofitted years ago with a keypad that had

never quite matched the door around it. Hayes badged the reader. The panel blinked amber.

LOCAL MODE — ENTER CODE

Hayes entered one. Rejected. He entered it again, more carefully. Rejected again.

The second soldier lifted his radio. "Control, escort transfer delayed at Office L-12. Local keypad not accepting."

Static answered him too.

Hayes swore under his breath and dug the old mechanical key off his ring. It went into the lock and stuck halfway.

Cavanaugh almost smiled. He kept it off his face.

The facility had gone back to what it actually was underneath the routine — not an institution with total authority, but a set of aging systems under cold stress and instability it hadn't chosen and couldn't fully explain, every locked door only as locked as the next system's willingness to agree that it should be. For six minutes, agreement was optional.

Down the corridor, a maintenance tech appeared moving fast, a coil of hardline cable slung over one shoulder.

"Junction C?" Hayes called after him.

The tech didn't stop. "Other way."

Hayes looked down the empty corridor where the man had come from, then at the stuck key, then at the radio still hissing static in the second soldier's hand. For the first time since Cavanaugh had been assigned to him, his attention split three directions at once and held in none of them.

Cavanaugh looked at the keypad. The rejected code was still visible, if you knew to look — grease from a hundred thumbs worn into a handful of keys in a pattern that meant something to somebody, just not to him. Not yet useful.

"Go get someone with command override," Hayes told the second soldier.

The soldier left at a jog.

One guard now. A locked door that wasn't going to open. A failing local system. And somewhere behind a coffee counter three corridors back, an auxiliary junction sitting open with no remote mirror watching it.

Cavanaugh tilted his head slightly, as if listening to the ventilation.

Under the floor, under the walls, the hum that had been running through the station's structure since long before Diagnostic Event One rose a half-step, low and patient, present

the way a held breath was present. The amber lights along the ceiling blinked once, together. The corridor clock above the office door corrected itself from 19:03:14 to 19:09:02, held there for two seconds, dropped back to 19:03:14, then went dark.

Hayes noticed. So did Cavanaugh.

The soldier's hand started toward his radio again.

"Your clock just blinked," Cavanaugh said.

Hayes looked up at it.

Cavanaugh moved.

Not far, not clean. He drove his shoulder into Hayes's side low and fast, because Hayes had been braced for a hand and got weight instead. The soldier hit the wall hard enough to knock the air out of him, more startled than hurt. Cavanaugh caught the key ring before it could fall, wrenched the stuck mechanical key free of the lock, and ran back the way they'd come.

Hayes shouted something behind him that didn't resolve into words before the corridor lights died for a full second.

He used the dark. Cut right at the junction, shoulder clipping the cold wall hard enough to numb his arm to the elbow. A radio barked somewhere back the

way he'd come. Boots hit metal decking, closing. He didn't try the data bay door — too obvious, too watched. He went to his knees at the service plate under the coffee counter, jammed the bent-but-usable key between one stainless screw and the frame, and twisted.

The screw held.

He twisted harder. The key bent further, metal complaining under his palm.

"Come on," he said, to no one.

The plate popped free with a crack far louder than he wanted, and behind it Junction C blinked green in the dark, a palm-sized auxiliary terminal running on whatever local power the outage hadn't touched.

MAINTENANCE WINDOW ACTIVE REMOTE
MIRRORING PAUSED LOCAL QUERY MODE
AVAILABLE

Boots hit the corridor behind him, closer.

He put his fingers against the screen. Cold. The terminal asked, in the same flat sans-serif as the tag on the tool case, for a credential class.

He typed MAINT-C.

The screen held a beat too long.

Then it accepted.

Kalamazoo

Not because anyone had cleared him for it. Because the facility was hurt in a way it hadn't planned for, and for the length of this window it needed a body more than it needed to know which one.

A menu opened.

POWER THERMAL TIMEBASE ARCHIVE
ROUTING FIELD EVENT CACHE

He had seconds. Maybe fewer than that.

He selected FIELD EVENT CACHE.

The screen filled with folders, plain block text against the gray shell interface, no icons, no decoration, nothing built for a visitor.

DIAGNOSTIC_ONE
CONST_DRIFT_LOCAL
MATTER_COHESION_PRIOR
REALITY_BLINK_ARCHIVE

He stopped breathing without deciding to.

Hayes came around the corner behind him, radio gone, both hands free now.

"Step away from the panel!"

Cavanaugh braced one hand flat against the cold wall to steady himself and opened the last folder.

Chapter 14: Reality Blink Files

Hayes held the sidearm low-ready, both hands, muzzle tipped at the grating between them instead of at Cavanaugh's head. That restraint was the worst part. A man pointing a weapon at your face was reacting. A man holding it at the floor was following a procedure, and procedures did not stop because someone found a folder they weren't supposed to see.

The maintenance plate hung open on two remaining screws. Behind it, the auxiliary terminal sat wired into the service wall on its own dead-end power loop, no larger than a tablet, its housing scuffed gray from a decade of gloved hands that were never supposed to touch it. No command wrapper. No compliance skin. No polite redaction bar scrolling apologies across the bottom of the screen. Whoever had built this terminal had built it for someone doing actual work, not someone being managed.

At the top edge, small green numbers counted down.

REMOTE MIRROR RESTORE: 02:41

Cavanaugh did not know what a mirror restore did to local cache. He knew what a countdown attached to a live

terminal usually meant, which was that whatever he could see now, he would not be able to see later.

The corridor ran narrow and low-ceilinged, service conduit bolted along one wall, structural bracing along the other, everything under the same institutional amber that made faces look tired even when they weren't. Maintenance window. The one stretch of the night cycle when foot traffic thinned to almost nothing and half the automated systems ran their own internal checks instead of watching the people who worked around them. He had planned to use exactly that gap. He had not planned on Hayes using it first.

No one else was in the corridor. That fact, more than the sidearm, told him how this would be handled if it went wrong. Quietly. Procedurally. Written up afterward by someone who hadn't been standing here.

He opened the folder labeled
REALITY_BLINK_ARCHIVE.

"Close it," Hayes said.

"Can't."

"Close the panel."

"Technically also can't." His fingers were already moving, and stopping them now would have taken longer than finishing.

"I will put you on the floor."

"You should. Give it two minutes."

"Now."

The folder rendered in blocky gray text, the kind of interface built for function rather than for anyone's comfort. No titles. No summaries. Event identifiers stripped down by local caching to strings that meant nothing and everything at once.

RB-CH-03 / STORAGE LEVEL D / MATTER
COHESION DISPUTE RB-NRO-17 / PACIFIC RELAY /
STRUCTURAL PASS-THROUGH RB-ICE-01 / HEKLA /
FINANCIAL-THERMAL COUPLING RB-CH-09 /
CAVERN SUPPORT / PRIOR RB-GRID-22 / SOUTHERN
TRANSMISSION / MICRO-COLLAPSE

He read ICE-01 twice before his mind caught up to what his eyes had already registered. Iceland. Hekla. The word sat there doing nothing dramatic, and the room narrowed around it anyway.

"Step back," Hayes said, closer now.

"There are prior events."

"I don't care what there are."

"You will."

Hayes crossed the last two feet of grating. Cavanaugh had one selection to make before contact, and he made it on instinct rather than certainty: RB-CH-09, because he'd seen the fragment reflected off the glass at Terminal Three days ago and had been unable to confirm it existed at all. Prior mattered more than complete. If he tried to grab everything, he would end up with nothing he could hold onto when they took the screen away.

The file stuttered before it opened. A line of warning text crossed the top.

LOCAL CACHE INCOMPLETE DISPLAY
DEGRADED

"Good enough," he said, mostly to himself.

The first frame loaded, and it was not a report. It was video, low resolution, timestamped, the kind of footage nobody watches until something goes wrong.

A lower cavern service lane, before the gantry work had finished. Temporary lighting on stands. Raw scaffold still bearing manufacturer stickers. A tool cart parked beside a poured concrete curb, the kind meant to keep equipment from rolling into a drainage channel.

The Core Throttling

Two men moved through frame in cold-weather harnesses with the loose, unhurried motion of people who had stopped believing the camera watched anything worth seeing.

A timestamp ran in the lower corner. It skipped.

09:17:44 became 09:17:42, then 09:17:45.

One of the men pushed the cart toward the curb. The front wheel reached it.

The wheel did not strike the curb, and it did not ride up over it. It went into the concrete, perhaps half an inch, as though the curb and the wheel had briefly failed to agree on which of them occupied that space. The cart kept moving. The worker's hands closed harder on the handle before his face changed, his body registering wrongness before his mind found a word for it.

Then the curb remembered what it was.

The wheel came back out with enough force to throw the cart sideways. Tools scattered across the deck plate. One struck the wall hard enough to leave a dent Cavanaugh could see even through the compression artifacts. The worker went down holding his wrist against his chest. Someone off-camera shouted something the audio track had already lost.

The clip froze on its own.

He became aware, distantly, that Hayes had stopped closing the last of the distance between them. Whatever was on the screen had bought a few extra seconds nobody was fighting to reclaim yet.

Cavanaugh tapped forward.

A still image replaced it. A close shot of the curb afterward: concrete shattered outward in a rough half circle, and inside the shatter pattern, a wheel groove pressed into the surface with a depth that no impact from above could have produced. A measurement overlay sat beneath the image in small type. Three-axis displacement. Material cohesion variance. Reconciliation lag.

Reconciliation.

Not a word from a weapons program. Not a word an engineer used to describe bad flooring. It was the kind of word you used when something had disagreed with itself and then corrected the disagreement after the fact, the way a database resolved two conflicting entries by picking one and discarding the other.

Hayes reached him.

Cavanaugh dropped his shoulder — not into Hayes, into the loose maintenance plate — and it swung shut over Hayes's leading hand long enough to buy a half second. Hayes grabbed a fistful of parka at his collar instead and hauled.

Cavanaugh kept his thumb on the terminal glass.

The screen backed out of the video to the file tree. His hand slipped on the next tap and opened RB-GRID-22 instead of what he'd meant to select.

A spreadsheet loaded. No video this time. Rows of transmission stations, voltage dips recorded in milliseconds, a column of regional grid events with dates running back further than he expected. As the window resized, a hidden column flashed into view along the right edge.

PROCESSING DROP CORRELATION

It vanished before he could read the values under it.

Hayes slammed him into the wall beside the service cavity. Pain traveled through his shoulder and down into his ribs on the same side, a familiar ache with a new edge to it. The terminal blurred in his vision. Hayes got a forearm under his jaw and started peeling him sideways, away from the wall.

"You're done," Hayes said.

"Not remotely."

Kalamazoo

He hooked one finger into the terminal's frame and pulled himself toward the wall instead of letting Hayes pull him away from it. Plastic cracked along the edge of the housing. The screen stayed lit.

REMOTE MIRROR RESTORE: 01:58

He opened the Iceland file.

It took longer than the others to load, the progress bar crawling as if the request had to travel somewhere far away and come back with an answer. Hayes shifted his grip from the collar to Cavanaugh's wrist, and the pressure landed in the small bones there with a specificity that told him Hayes could break something without meaning to, simply by following training that assumed compliance would come before damage.

Text arrived first.

RB-ICE-01 / HEKLA FACILITY INCIDENT
CROSS-LINK: SUB-ZERO SIGNAL
CORRELATION: THERMAL EXTRACTION /
FINANCIAL RECORD REWRITE / NODE
COUPLING DISPOSITION: CONTAINED
PUBLICLY / UNRESOLVED INTERNALLY

Below the text, a graph rendered in stages: processing load along one axis, a local temperature collapse plotted against it, then a third trace for data divergence, a fourth for signal amplitude. The amplitude trace had the same squared-off notches he'd watched Sanderson circle in red pen on a printout two days ago, the ones she hadn't been able to explain and hadn't stopped looking at since.

He'd sat across from Sanderson in the equipment bay two nights ago while she traced that same amplitude pattern with a gloved finger over a printout, saying nothing for a long time before admitting she didn't have an instrument built to explain what she was looking at. He hadn't had a name for it then either. He had one now, even if the name only raised further questions: node coupling.

He scanned for a line short enough to carry out of the room in memory, because memory was the only storage device Hayes couldn't confiscate.

NODE COUPLING SOURCE: CH-PRIMARY /
ANTARCTIC MATRIX FAILURE MODE: EXTERNAL
DEMAND EXCEEDS LOCAL THROUGHPUT SYSTEM
RESPONSE: THROTTLE / PRIORITIZE / RECONCILE

Hayes twisted the wrist he was holding. White pain climbed Cavanaugh's arm and briefly erased the screen from his attention.

"Step away," Hayes said again, quieter this time, which was somehow worse than the shouting.

Cavanaugh laughed once, a short exhale with no real humor in it, because he didn't have the breath to spare on anything longer. "You still think I'm reading against command."

"You are."

"No." He held the next three lines in his eyes as they flickered, repeating them under his breath the way he used to memorize plate numbers before he trusted a phone to do it for him. "I'm reading against extinction."

Hayes shoved him harder. His shoulder caught the edge of the wall cavity, and the terminal's display cut to black for a full second.

It came back with the file tree scrolled further down than he had taken it. A folder sat centered on the screen, as if something on the other end of the mirror restore had wanted him to see it before the connection closed.

OVERRIDE FIELD UTILITY / DEPLOYABLE
SCENARIOS

Subfolders unspooled beneath it.

GRAVITY SUPPRESSION TEST PROFILES
ELECTROMAGNETIC HARDENING APPLICATIONS
MATTER COHESION DISRUPTION MODEL
REGIONAL FIELD CONTROL ESCALATED COMMAND
OVERRIDE

The language was flat and administrative, and that was what made it land the way it did. Nobody writing those folder names had been trying to hide what they contained. They had simply renamed the danger until it filed under utility instead of hazard, the way a body count became a casualty estimate, the way an accident became an incident. He had spent enough years reading documents built this way to recognize the pattern from three words in: language chosen not to conceal a fact but to change which drawer the fact got filed in, so that anyone reviewing it later would reach for the wrong set of assumptions before they ever reached the content.

Hayes saw it. Maybe not all five lines — Cavanaugh couldn't be sure how much of the angle reached him — but enough. His grip loosened for half a second, an instinctive flinch rather than a decision.

Cavanaugh used the half second to open the folder's metadata instead of its contents, because metadata loaded faster than anything else on the screen and he might not get a second chance to ask the terminal a question.

Created before Diagnostic Event One. Modified twelve minutes after Diagnostic Event One.

Linked evidence:

CONST_DRIFT_LOCAL RB-CH-09 RB-
GRID-22 THERMAL_ABSORB_PROFILE
SUBZERO_SIGNAL_MATCH

He looked at Hayes rather than the screen now.
"They had the scenarios before the test."

Hayes's face had gone the color of the corridor lighting, flat and without warmth.

Boots sounded from the far end of the corridor. A second soldier came around the corner at a jog, and behind him, Underwood's voice arrived over an open channel, calm in the particular way that made calm sound like the most dangerous setting available to her.

"Secure Cavanaugh. Do not permit additional terminal access."

REMOTE MIRROR RESTORE: 00:49

Cavanaugh did the only thing left that was still his to do. He let Hayes's own grip drag him a foot off the wall, absorbed the motion instead of fighting it, and then drove backward into the loose service plate with what strength his shoulder had left. The plate cracked against the terminal housing. The display spidered across one corner but held. The impact also knocked something else loose — a small maintenance storage cartridge, no larger than a matchbook, popped halfway out of a slot beneath the screen.

Not a drive. A cache wafer. Cheap, disposable, probably encrypted, possibly already corrupted by the same cache degradation that had chewed through half the archive. It might hold nothing. It might hold three seconds of video. Evidence didn't need to be complete to matter — it only needed to exist somewhere Underwood couldn't reach.

He lunged for it.

Hayes drove a knee into his thigh and took him down before his fingers closed. Cavanaugh hit the grated floor shoulder-first, and the impact forced the air out of him in one hard burst that left him gasping against the metal. The second soldier arrived at a run, boots loud on the grating, hands already extending.

His fingers found the wafer anyway, curled around it in the half second before either soldier reached his arm.

It was hot. Not warm from a housing that had been running all night — hot enough that it bit into his palm the way a stove element did in the first second before a person's hand understood what it was touching and pulled back. He didn't pull back. He folded his fist closed around it and kept it there.

The second soldier caught his right arm. Hayes caught the left. Between them they rolled him onto his stomach with an efficiency that spoke of drills rather than anger, and pinned him against the floor. His cheek pressed into the grating. Dust and metal grit stuck to his lower lip. Somewhere below the floor plate, he thought he could feel a low vibration that might have been machinery and might have been nothing at all, his nerves manufacturing a rhythm out of adrenaline.

The terminal, still lit inside the wall cavity, gave a single flat beep.

REMOTE MIRROR RESTORED

The display refreshed itself in one motion. Every open folder closed at once. The archive tree disappeared

entirely, replaced by a clean command screen with three lines centered on a black field.

ACCESS VIOLATION RECORDED LOCAL
CACHE PURGED AUTHORIZED PERSONNEL ONLY

Of course. The system hadn't been defeated. It had simply forgotten, on schedule, the way it was built to. Somewhere in whatever process governed the mirror restore, someone — or something — had decided three minutes was an acceptable exposure window, and had built the recovery to close it without needing a human to notice in time.

Hayes knelt beside him, breathing hard enough that Cavanaugh could hear it over his own pulse. "What did you take?"

Cavanaugh kept his fist closed against his chest, tucked beneath his own weight where neither man could see it without moving him.

"Nothing."

Hayes reached for the hand anyway.

The corridor lights dropped by half, not out, just dim enough to register as a change rather than a failure. Nobody moved for a second. It wasn't that the dimming itself meant anything specific. It was that everyone in the corridor had learned, over the last several weeks, that darkness here was rarely

only darkness. It usually meant something below the floor plates had shifted its draw, or its output, or its patience.

Cavanaugh used the pause. He opened his fist enough to let the wafer slide down into the fold of his sleeve, where the parka's insulation had bunched near the elbow during the fall, and closed his hand again around nothing.

Hayes forced the hand open a second later.

Empty.

The second soldier swore under his breath, more at the situation than at Cavanaugh specifically.

Underwood arrived without hurrying, which was, as far as Cavanaugh had ever been able to tell, the entire method behind her authority. Other people created the urgency. She appeared afterward and inherited whatever conclusion the urgency had already reached. She took in the sprung panel, the purged terminal glowing its clean gray warning inside the wall, Hayes's strained face, Cavanaugh flat on the grating.

"Report," she said.

She didn't look at Cavanaugh while she said it. She looked at the terminal, then at the sprung panel

screws scattered across the grating, cataloguing the physical damage before she asked about the intangible kind.

Hayes got to his feet. "Subject accessed an auxiliary maintenance terminal during the mirror outage."

"Duration?"

"Unknown. Under three minutes."

"Files?"

Hayes hesitated a half beat too long.

"Files," Underwood said again, no louder than before.

Cavanaugh turned his head against the grating far enough to see her boots. "Platform stability deviations, mostly."

Underwood looked down at him without any particular expression. "Search him."

The second soldier hauled him upright hard enough that his vision pulsed white at the edges before it settled. Hayes went through his pockets first, methodical, working top to bottom the way he probably had a hundred times in training against a mannequin that never bled and never lied to him. Parka pockets, empty except for a folded glove liner. Interior zip pocket, empty. He moved to the cuffs of the parka sleeves next, running two fingers around each one, checking for the kind of flat object a person might tape against skin.

Cuffs, collar, waistband, boot tops. He came up with only the bent maintenance key Cavanaugh had lifted two nights earlier, and pocketed it with the kind of controlled anger that looked, from the outside, almost like care.

The wafer burned a thin line against the skin of his forearm where it had settled into the sleeve seam near his elbow.

Hayes checked the sleeve. His hand moved fast, closer to a formality than a search, and it registered fabric instead of the wafer tucked against the inner fold where the parka had bunched. He moved on without pausing.

Underwood stood a few feet back with her hands loosely clasped in front of her, watching Cavanaugh's face during the search rather than Hayes's hands, which told him more about her method than anything she'd said aloud. She wasn't interested in whether he had something. She was interested in whether he believed he still did, and whatever showed on his face while Hayes worked was, to her, more reliable data than the search itself. He kept his eyes on a point past her shoulder and thought about checksum

gaps, file trees, event order, Sanderson saying heat before hum before gravity variance in that clipped voice she used when she was trying not to sound as frightened as she was. He thought about a wheel entering concrete and the concrete snapping back into a shape that wanted everyone to believe it had only slipped.

He did not think about his sleeve, and that was, he understood later, the only reason it worked.

Hayes stepped back. "No media located."

Underwood said nothing for two beats longer than the silence needed.

Then the wall intercom clicked on with the flat tone that meant a scheduled announcement rather than an emergency one. "Command to lower operations. Field Utility Review moved up to 2130. Override scenario package locked for senior clearance."

Cavanaugh felt the wafer's heat fade slightly against his arm as it cooled toward the same temperature as his skin.

Override scenario package. Locked now — but not before he had seen five subfolder names and one line of metadata that placed their creation before a test the official record insisted no one could have anticipated.

She let a long moment pass before she spoke again, long enough that the second soldier shifted his weight, uncertain

whether he was still supposed to be restraining anyone. Then she turned toward the ruined panel. "Seal Junction C. Remove the auxiliary terminal."

A technician near the far wall spoke up without quite raising his hand. "Ma'am, that will interrupt local recovery."

"Remove it."

The technician glanced once at the wall cavity, then away, the look of a man deciding the argument wasn't his to finish. "Yes, ma'am."

Underwood faced Cavanaugh again, close enough now that he had to tip his head back to hold her eyes. "You have mistaken access for understanding."

"No." His voice came out rougher than he intended, scraped by the floor impact and the arm still pinned behind him. "You have mistaken hazard for capability."

He hadn't planned to say it. It came out the way the wafer had ended up in his sleeve, as something his body decided before his judgment weighed in on whether it was wise.

Her expression didn't move. It occurred to him, watching her not move, that Sanderson had said

something close to that same sentence to her days ago, in a different room, and that Underwood had filed it away the same way she was filing this — as noise from someone she'd already decided not to hear.

"Escort him to residential confinement," Underwood said. "No terminals. No shared spaces. No contact with Dr. Sanderson."

Hayes and the second soldier turned him down the corridor by the arms, not gently, not brutally either, just with the same procedural weight that had characterized everything since the sidearm first came out. His shoulder throbbed in time with his pulse, and the thigh Hayes had driven a knee into during the takedown had already stiffened enough that he had to lead with the other leg to keep an even pace. Nobody offered to slow down for it. Cavanaugh let them walk him. There was nothing left worth resisting for in this stretch of hallway. He had less than a file. Less than proof. He had fragments — reality blinks stitched together out of a curb that didn't stay solid and a cart wheel that remembered gravity a half second late, matter cohesion failures logged under language built for spreadsheets, a grid in the southern hemisphere losing milliseconds to something called a processing drop, an Iceland facility whose disposition read contained publicly and unresolved internally,

and override folders built with scenario names before Diagnostic Event One had finished producing its injuries.

Fragments were how anything true survived inside a system built to purge itself on schedule.

The corridor ahead of them ran the same amber length it always did, doors spaced at even intervals, each one marked with a stenciled designation instead of a name. He'd walked it a dozen times without paying attention to how far confinement quarters sat from anything useful — no terminal access along this stretch, no shared spaces, a design decision made long before anyone here had a reason to need it. Someone had planned for this hallway to hold a person who knew too much. That planning had happened before Cavanaugh existed to them as a problem, and the thought sat uncomfortably alongside everything else he was carrying out of Junction C.

Behind him, the loosened maintenance plate gave up its last screw and struck the floor plating with a thin, unimpressive clatter, the kind of sound that would mean nothing to anyone reviewing an incident report six months from now.

Inside his sleeve, against the crook of his elbow, the cache wafer had gone from a burn to a dull, retreating warmth. It left a narrow reddened line on his skin where the heat had sat too long, the kind of mark that would still be visible tomorrow if anyone thought to look for it, and that no one currently walking him down this corridor had any reason to.

Chapter 15: Sanderson's Independent Storage Core

Sanderson knew about the storage core before anyone told her, because the room around her had gone quiet in the wrong way.

Ariza taped off the new dressing on her thumb without the usual instruction not to move it. Sergeant Pell had positioned himself outside the medical alcove instead of inside it, arms crossed, watching a point on the far wall that did not require watching. When the second detail arrived — two soldiers, evidence bags, a handheld scanner — they walked past her tablet, sitting open on the rolling tray, and went straight for her coat.

They were checking her pockets, not her equipment. That told her something had already been taken, and that whoever had taken it believed the important part was still on her.

"Open your hands," the tech said.

She held them up. One bare. One wrapped in gauze that had gone from white to pink somewhere in

the last hour, the stain spreading from the knuckle outward like a slow tide.

He looked at the bandage and, for a half second, looked like he regretted his job.

"Palms."

She turned them over.

"Outer pockets."

Ariza said, "She's under medical care."

"She's under security review."

"For a laceration?"

"For unauthorized media handling."

Sanderson found Pell through the doorway. He hadn't moved. His right hand, the one that had been curled into a fist since the alcove door closed behind him twenty minutes ago, hung open at his side now. Nothing in it. The calibration-weight core he'd been carrying wasn't there anymore, and she had not seen where it went.

"Where did you take it?" she asked him.

His eyes came to her and stayed there.

The tech said, "That's not part of this review."

"Wasn't talking to you."

"Evidence intake," Pell said.

Two words. She turned them over the way she'd turn over a reading that didn't match the calibration curve — not because the words were wrong, but because of which words he hadn't used. Not command review. Not the phrase that had been stamped across every classified folder in this station since the diagnostic spike. Evidence intake had a procedure attached to it. A chain. Command review didn't.

She held his gaze one beat longer than the exchange required.

He'd heard the distinction land. She could see that much on him, the small flattening around his mouth that wasn't quite a wince.

The tech crouched and ran the wand down her boots. It chirped at the zipper pull on her parka, at a rivet near the left pocket, at the metal clip on the wrist wrap. Each time it did, he stopped and checked, as though a hidden datachip might announce itself as politely as a knife would.

"Arms out," he said.

"No."

Pell's weight shifted in the doorway.

The tech looked up. "Doctor."

"Right hand's injured."

"Left, then."

She extended it. The wand passed shoulder to wrist and back with no sound. He checked the sleeve seam with two gloved fingers, found nothing, and moved to the right arm. He hesitated over the dressing the way a man hesitates over a door he already knows is locked.

Ariza stepped into the gap between the tech and her hand. "You open that bandage, you own the bleeding that comes after."

The tech looked past him, to Pell.

"Skip the dressing," Pell said.

The tech didn't like it. He finished the pass anyway and bagged the wand tip when he was done, like her clothing might have contaminated the instrument.

"No media located," he said, to no one in particular, and left.

Pell stayed.

For a few seconds the only sound in the alcove was the medical monitor over Kline's cot, tracking a heartbeat that had finally slowed into something like sleep, and underneath it, low enough that she'd stopped consciously registering it hours ago, the hum that lived in the walls of this station and never fully left

them. Kline's splinted wrist lay across the blanket at an angle that made her think, unhelpfully, of a broken instrument arm left in its cradle.

"Evidence intake means chain of custody," she said, quiet enough that it wouldn't carry past the doorway.

"It means storage."

"It means signatures."

"It means I did what I was ordered to do."

She looked past him at the corridor where the tech had gone. "You did one millimeter less than what you were ordered to do."

"You shouldn't measure it that way."

"It's what I do."

He almost said something back. His radio interrupted first, a burst of static resolving into a voice pitched for broadcast rather than conversation. *All lower operations personnel, Field Utility Review begins in fifteen minutes. Nonessential civilian movement suspended.*

Pell touched the earpiece, listened past the end of the message, and came back blank.

"Dr. Sanderson. You're to remain in medical until transferred."

"Transferred where?"

"Residential module."

"Confined."

"Transferred."

"They already have Cavanaugh."

He didn't answer that, which was its own answer.

She stood.

"Sit down," Ariza said.

"I need the instrument annex."

"No."

"I'm not asking."

Pell came a step into the room. "Doctor."

"Your intake core proves the raw data existed. It doesn't prove what was in it."

"That's the point of evidence."

"No. Evidence proves what survives handling. Nothing more than that." She reached with her uninjured hand for the field tablet on the tray. Ariza made a sound like he was going to object, but she already had it against her chest, both arms around it the way you'd carry something you didn't trust the floor to catch. "The acquisition buffer wrote to three locations during the event. The storage core was the cleanest copy. It wasn't the only one."

His eyes dropped to the tablet and stayed there half a second too long.

She let him look. "If command finds the second trace before I do, they'll call it corrupted and overwrite it during preservation. That's standard procedure for anything that doesn't fit the report they've already written."

"How many copies did you make?"

"I didn't make copies. I built redundancy into the acquisition path before the event even happened, because I don't trust single points of failure in a room that behaves like this one does."

"That's a distinction Mr. Cavanaugh would enjoy."

"Then he's rubbing off on me."

Pell glanced toward Kline's cot, then back.

"Three minutes."

"Sergeant," Ariza said.

"Three minutes before I'm required to remember I heard any of this."

She didn't thank him. Gratitude would have made it a favor between two people instead of a gap in a

procedure, and gaps in procedure were the only kind of room this station still had left for her.

Night cycle didn't bring darkness to Cold Horizon. It brought amber — low strip lighting along the floor track, red status plates over the sealed doors, and behind glass in the rooms she passed, the flat blue glow of instruments still running their assigned tasks long after the humans who'd started them had gone past the point of being useful. The corridor air carried more heat than the cavern above it and still managed to feel like something she didn't want against her skin.

Pell walked behind her, close enough to stop her if it came to that, far enough that the arrangement could still be called an escort instead of what it actually was.

The instrument quarantine annex sat two doors down. One camera bracketed above the frame. One badge reader. A mechanical override lock beneath it, the kind that predated every system in this building by at least two design generations. A printed evidence seal ran diagonally across the jamb, the kind of paper strip that broke the instant anyone tested it and could not be reapplied to look intact.

The seal was unbroken.

That, at least, was something.

Kalamazoo

Pell badged the reader.

The panel came back amber. ACCESS
SUSPENDED.

He tried it again. Same result.

"Command suspension," he said.

She looked past him at the mechanical override.

"You have a key for that."

"Not for this door."

"Then you have a problem."

"You have a problem."

"I have a bad hand and a short window. Not the
same thing."

Something moved through the wall — not a
sound exactly, more a pressure change she felt in her
back teeth before she registered it as anything else. The
corridor lights dropped and came back up half a beat
later than they should have. The badge reader's display
cycled from amber to a flat gray.

LOCAL MODE.

Pell stared at it.

"Cold Horizon is developing a sense of timing,"
she said.

"It isn't a person."

"I know that."

He didn't argue further. He pulled a second key from the ring at his belt — older than the one he'd used on the main door, brass under a worn black coating, the edges rounded from years of a thumb finding it in the dark. He fit it to the mechanical lock.

It didn't turn.

"Of course," she said.

"Quiet."

The key caught halfway and stopped. She could hear the pins inside grinding against something that wasn't quite rust and wasn't quite ice — some combination the station's humidity cycle had built without anyone's permission. Pell reset his grip and forced it. The evidence seal stretched taut across the widening gap, held for a moment longer than she expected, and then tore with a small dry sound, paper giving up all at once.

He flinched at the noise.

"Bothers you," she said.

"Yes."

"Should."

The annex smelled like cold circuitry and open storage cases, a smell she associated with rooms that had recently had

something taken out of them. Her diagnostic rack was gone. The main sensor crates were gone. What remained was what someone had judged worthless enough to leave: torn cable wrap, foam packaging with case-shaped depressions pressed into it, a cracked stylus, a snapped lockout tab from the rig housing, two gray bins stenciled PENDING DISPOSITION.

She stood in the doorway and, for one long second, thought she'd guessed wrong.

Then she saw it — upside down on the bottom shelf of a rolling cart, half hidden behind an empty crate, its casing scuffed and its status tape torn halfway off. The analog bridge module.

Not flagged. Not tagged. Not important, if you'd never had a reason to build the thing yourself.

She crossed the room before Pell had the door shut behind them.

"Two minutes," he said.

"I thought I had three."

"You spent one on the lock."

She pulled the bridge free with her good hand. The casing had warped somewhere during the diagnostic event — heat, most likely, though she hadn't logged an

The Core Throttling

ambient reading high enough to explain the degree of it — and one seam had split wide enough to show the green edge of an internal board underneath. The module was obsolete by design, and that was the entire reason it had value now. No wireless handshake. No auto-sync. No command mirror sitting between the sensor and the record, deciding on its own authority what counted as signal and what counted as noise. It caught raw voltage the way instruments used to, before anyone built systems clever enough to be untrustworthy.

She set the tablet on the cart and unfolded its cracked stand.

The first cable had a crushed connector.

The second was missing a pin.

The third had gone stiff with cold and refused to uncoil past a half-loop.

"Come on," she said, to the cable, to the room, to nothing in particular.

Pell had positioned himself at the door, listening to the corridor instead of watching her.

She stripped the outer insulation off the stiff cable with her teeth. The plastic didn't peel the way it should have; it cracked, and a fleck of it stuck to her lip before she spat it out. She used the broken edge of the lockout tab as a blade to finish

the job. Her thumb throbbed under its dressing with every motion that required gripping, and when she looked down she saw the gauze had gone from pink to a deeper, wetter red at the seam.

The connector wouldn't seat cleanly at the module's port. She braced the bridge against her hip, pushed, and the tablet slid across the cart's uneven surface toward the edge.

Pell caught it before it went over.

They both stood still for a second, tablet suspended between his hand and the drop.

He held it out to her without saying anything.

"Useful," she said.

"Fast."

She seated the connector at an angle that wasn't right but was close enough, and powered the tablet on.

The screen flickered. The battery reading came up at nine percent, then dropped to seven, then climbed back to eleven for no reason the hardware would ever explain to her. She didn't trust the number. She trusted it less with every second it kept changing.

The local acquisition interface came up and reported no compatible module. She switched the input

manually. The first attempt returned static, an unbroken gray field with no structure in it at all. The second returned a column of values that meant nothing — not corrupted exactly, just disconnected from any physical quantity she recognized. The third produced a timestamp.

It was wrong by eight-tenths of a second.

Her breath changed without her deciding it should.

"There you are," she said, to the timestamp, which did not answer, because nothing in this room ever did on the first attempt.

Pell leaned in despite himself, close enough that she could hear him breathing through his nose.

She pulled the bridge's cache. Not a clean dataset — a residue trail, analog capture off the thermal probes sampled badly, drift baked into every reading, gaps where the hardware had saturated and simply stopped recording rather than report a value it couldn't trust. It was ugly in exactly the way that made her believe it.

The first graph opened.

Thermal absorption, rising before the command execution timestamp. Not after. Before.

The second.

Acoustic response, narrowing across the band in a way that matched nothing in the equipment's calibration history.

The third.

A spike in field noise that led the gravity variance by a margin too clean to be coincidence and too small to be anything she'd call proof.

Then a fourth file appeared under the others, its name stripped down to a channel number with no description attached.

CH4_DOMAIN_ERR_RAW.

The impossible line. The one that had thrown values with no unit, no scale, no category the instrument's firmware had ever been built to hold.

She opened it.

At first there was nothing on the screen but saturated black — not a picture of black, a total absence of any value the tablet knew how to render. Then it applied a basic voltage map to the raw signal, and the black resolved into a series of square notches spaced at intervals that lined up, when she checked them against her own pulse, with the hum she could still feel through the floor.

"What am I looking at?" Pell asked.

"I don't know yet."

"Reassuring."

"It isn't." She leaned closer to the screen. "That's the problem."

The channel hadn't failed the way instruments usually failed — a burned sensor, a severed lead, a firmware fault throwing garbage values at a fixed rate. This had responded to something outside the thermal domain entirely, arriving through the probe's physical housing rather than its intended input. Not heat. Not cold. Something closer to a boundary condition passing through the metal the way a pressure wave passes through water — as if the instrument had made contact, however briefly, with a surface that wasn't behaving like a surface should.

She overlaid it against the gravity event.

The notches started before the pulse did.

She overlaid the corrected heat-absorption curve.

The notches deepened as the absorption climbed, in step, like two recordings of the same thing taken with different instruments.

She overlaid the acoustic narrowing.

The pattern locked into place — not perfectly, not with the kind of clean correlation she'd have needed to put in a report, but enough that her hands had gone still on the tablet's edge without her telling them to.

"Channel four wasn't broken," she said.

Pell's radio clicked. He raised one finger, asking for silence without looking at her.

A voice came through, thin and chopped by static. *Pell, report location.*

He didn't answer right away.

Pell, report.

He keyed the mic. "Medical corridor. Dr. Sanderson secure."

Confirm visual.

He looked at her.

She held still.

"Visual confirmed," he said.

A pause, then: *Prepare transfer to residential confinement.*

"Copy."

He released the mic and looked at the door instead of at her.

"Thirty seconds," he said.

She snapped a cable tie off with her teeth and used it to lash the bad connector in place, buying herself a margin she didn't have time to calculate properly. "I need this copied."

"To what?"

She looked around the room — no drives, no cards, nothing built to hold data that wasn't already accounted for on someone's manifest. Foam packaging. A broken lockout tab. The cracked stylus, its cap hanging by a thread of plastic.

She picked up the stylus and turned it over. Inside the cracked shell sat a small manufacturer's calibration chip, meant for nothing more than pressure sensitivity on the tip, useless to any ordinary interface and, if you were desperate and knew how to talk to hostile hardware, writable.

She split the cap the rest of the way with the lockout tab.

"That's storage?" Pell said.

"Barely."

"How much does it hold?"

"Enough for one good lie detector, if I'm lucky."

The tablet dropped to five percent.

She selected four files — pre-pulse thermal rise, acoustic narrowing, the raw channel-four trace, the timing-offset map — and the transfer estimate came back too large. She stripped

metadata. Still too large. She cut resolution on the graphs she could afford to lose detail on, and kept only the aligned interval spanning eight seconds before execution to six seconds after the gravity event returned to baseline.

Copy estimate: forty-one seconds.

"No," Pell said.

She started the transfer anyway.

The progress bar advanced in small, sick jumps that didn't correspond to any consistent rate.

Nine percent.

Twelve.

Twelve.

Twelve.

Pell crossed to the door and eased it open a few inches. Amber light cut a bar across the floor. Voices somewhere down the corridor, indistinct. Boots, more than one pair.

"Stop it," he said, without turning around.

"Can't."

"Stop it, or I'll have to."

"Tell them the lock jammed."

"It didn't."

"Tell them I reopened the wound reaching for something."

"You did."

"There. Accurate report."

The bar reached forty-one percent and held there long enough that she thought the transfer had already failed silently. Then it jumped to fifty-six.

The tablet's screen blinked once, a full-frame flicker like a dropped frame in old film.

Battery: three percent.

A message box opened over the graphs.

INPUT SIGNAL UNSTABLE — PRESERVE RAW
CACHE?

"Yes," she said, barely above a whisper, and pressed confirm before she'd finished deciding whether that was the right call.

The tablet froze.

Pell turned back from the door. "Doctor."

"Don't touch it."

"Doctor."

Boots, closer now. Two sets, maybe three, moving with the flat unhurried rhythm of people who already knew where they were going.

The progress bar jumped to seventy-eight percent, held, climbed to eighty-one—
and stopped.

TRANSFER FAILED PARTIAL WRITE
RETAINED

She felt the sound she made before she was aware of making it, something between a breath and a word that didn't finish forming.

Partial wasn't nothing. Cavanaugh built entire theories out of less than partial.

She pulled the stylus chip free before the interface could finish reporting its own failure. It was small enough to disappear under a thumbnail, cold, almost weightless in her palm — a fragment of plastic and silicon that might hold eight seconds of a phenomenon nobody in this building had a name for.

She looked for somewhere to put it.

Not a pocket. They'd scan the pockets again, and this time they might not stop at a chirp.

Not the tablet. Too obvious, and dying besides.

Her injured hand.

The gauze had already loosened where blood had soaked through the inner layer, the wrap gone soft

and slightly separated at the seam near her palm. She worked the chip beneath the outer layer, against the heel of her hand, and pulled the wrap tight again over it. The pain that came with the pressure was sharp enough to make her vision narrow for a second, bright and specific, nothing like the dull ache that had been sitting under the dressing for the last hour.

Pell saw her do it. He closed his eyes once, briefly, the way a man does when he's decided not to have seen something and knows the decision won't hold.

The annex door opened behind him.

A corporal stood in the gap, another soldier a half step behind. "Sergeant. Command wants them transferred, now."

Pell stepped clear of the doorway. "She reopened the wound."

The corporal's eyes went to the bandage, to the fresh stain spreading past the old one. "Medical can handle that upstairs."

"Medical will handle it where I say medical handles it." Ariza's voice arrived from the corridor before he did, and he pushed past the corporal to take Sanderson's wrist — carefully, more carefully than the moment strictly required, and turned the bandaged hand palm-down so nothing under the wrap had reason to shift. His eyes moved from the blood to her face to

Pell and back to the dressing. He wasn't a stupid man. Cold Horizon, it seemed, had failed to staff itself with quite enough of those.

"You tore it," Ariza said.

"Yes."

"On what?"

"Bad procedure."

"That your official statement?"

"It will be, if you write it down that way."

"We have transfer orders," the corporal said.

"And I have a patient with active bleeding."

Ariza kept her wrist steady, angled away from the corporal's line of sight. "You want your name on the line that says command pulled her before bleeding control?"

The corporal did not want that. Nobody in this building wanted their name on the wrong line of a record system that had already shown a willingness to rewrite every other line around it.

"Five minutes," Pell said. "For the dressing."

The corporal held there a second, then nodded once. "Five."

The detail withdrew into the corridor, and their voices dropped to the low, procedural register of people filling out time rather than arguing it.

Sanderson kept the tablet pressed against her side like something too heavy to carry any other way. Its screen had gone dark. Behind her, the analog bridge sat open on the cart, warped and useless-looking again, indistinguishable from the rest of the debris in the room. The stylus cap lay cracked among the packaging foam. Nothing left in the annex looked like proof of anything.

That, she understood, was the only reason it had survived this long.

Ariza guided her back toward medical with a hand at her elbow that was more support than escort. Pell fell into step a pace behind, close enough to hear, careful not to close the last of the distance.

She didn't look at him when she spoke.

"Evidence intake will show the clean core."

"Maybe."

"The tablet will show a failed transfer."

"Probably."

"The partial write won't authenticate on its own. No unbroken chain, no clean hash, nothing a review board would accept without a fight."

"No."

She flexed the fingers of her injured hand, just enough to feel the chip's hard edge press into torn skin under the gauze.

"But it will show the order things happened in," she said.

Pell didn't answer that one at all.

She kept her eyes on the corridor ahead — the amber floor lights, the red status plates over doors that had no reason to be sealed except that someone with more authority than her had decided they should be, the station narrowing around all three of them the way it had been narrowing since the diagnostic spike, turning her injury into a line item and her warning into something with a schedule attached.

For the first time since the gravity event had thrown her against the rig housing, she felt something colder than fear and considerably more useful.

She still had raw sequence.

The Core Throttling

Not enough to win an argument with anyone in this building.

Enough to make their version of events harder to say out loud without contradiction sitting right behind it.

Chapter 16: Underwood's Utility Framing

They left him his boots.

That was the detail Cavanaugh kept returning to as the second search moved down his arms, across his ribs, into the seams of a parka they had already taken away. Hayes's man was thorough with the parka — turned it inside out, ran a gloved thumb along the zipper track, checked the collar for a wire that wasn't there. He was less thorough with the boots. He gripped the ankle, pressed once at the arch, and moved on.

The corridor before the review room had been the residential level, low ceilings, condensation beaded along a seam where two pipe runs crossed without insulation. His own breath had shown white for eleven steps. He had counted them without deciding to. It was a habit left over from a job that used to reward counting things.

Cavanaugh sat now at the end of a table too narrow for the number of people it was meant to seat, a security man on either side, his wrists unbound and

useless in every way that mattered. Someone had turned the room's heat up since he'd last been through it, or he had simply lost the ability to tell warmth from adrenaline. The cache wafer had been against his skin for nine minutes before he found somewhere to put it. Nine minutes was long enough for the material to lose its warmth. Short enough for the mark to stay in the leather. When Hayes had walked him through the residential corridor he had stopped to retie a boot that had not come undone, and slid the wafer beneath the inner lip of the sole. A bad hiding place. Obvious if they pulled the boot. Fragile if he stepped wrong. Too warm at first against the sole of his foot, then too cold, then simply present, a weight he had to keep forgetting on purpose. He had thought about it exactly once since sitting down, and made himself stop.

Evidence was not comfort. It only felt that way when you had nothing else.

He flexed his hands under the table, out of the nearer guard's eyeline, and found them steadier than he expected. That was worth noting too. Fear had a way of announcing itself through the extremities before it reached anywhere useful.

Underwood stood near the wall display at the far end of the room, arms loose at her sides, watching three diagrams resolve on the screen behind her. The diagrams were clean.

Kalamazoo

Clean meant someone had already decided what the numbers were for.

The header at the top read:

FIELD UTILITY REVIEW DIAGNOSTIC
EVENT ONE / APPLICATION PATHWAYS

Not failure review. Not hazard review.

He counted the room without turning his head more than the search allowed. Two site engineers. A medical liaison who hadn't looked up since the door closed. Three officers — logistics insignia, operations insignia, and a collar device Cavanaugh didn't recognize, which told him more than the ones he did. A remote command window sat in the display's upper corner, audio only, initials in place of names, clearance tags in place of faces. No one photographed people before asking them to authorize something permanent.

There was a chair at the near end of the table with a folded blanket on the seat, the kind issued from medical stores, still creased from a shelf. Nobody had removed it. Nobody was going to sit in it either.

Sanderson's chair was empty.

He noted that before anything else in the room, because it was the fact most likely to still matter in an hour.

The room smelled faintly of solder flux and the particular staleness of recycled air run through filters past their service interval. A single monitor at the back wall displayed a status column he couldn't read from the angle he'd been given, rows of green resolving into amber further down the list. Nobody at the table was looking at it. He filed the column's existence and let the numbers go.

Underwood said, "Summarize revised interpretation," and the lead engineer cleared his throat like the sound might buy him time.

He'd printed nothing. Nobody printed here who didn't want paper to become a record somebody else could find. His tablet sat tethered to the table by a blue compliance cable, locked to the room's isolated network.

"Diagnostic Event One demonstrated a localized field variance within the cavern volume," he said. "Magnitude peaked at approximately point four zero one percent effective gravitational deviation. Duration under seven seconds."

"Cause?" A remote voice, flattened by the feed.

The engineer paused before he answered, and Cavanaugh watched him choose which version of the truth was safe to say out loud.

"Causal pathway remains under investigation. However, correlation with the imposed pulse is strong."

Strong. Not complete. Not sufficient. A bridge word — the kind of word people reached for when the actual finding wasn't ready to carry the weight being put on it.

"Operational significance?" Underwood said.

The engineer touched his tablet. The display changed to a simplified sphere ringing the exposed matrix, gradients shaded in careful, reassuring color.

"The event suggests the structure can induce or mediate localized gravitational and electromagnetic change under external excitation. Current estimate indicates a field envelope that may be expandable with increased input amplitude."

Cavanaugh shifted his weight without meaning to. The boot sole pressed the wafer into the arch of his foot, a small, specific pain he filed and set aside.

Hayes, behind him: "Sit still."

He did.

The Core Throttling

Underwood turned her head partway toward him, not all the way. "Mr. Cavanaugh. You have an objection."

"I have several. Would you like them categorized by severity or admissibility?"

The logistics officer frowned at that. The operations officer didn't react at all. Underwood did neither.

"You are present because your prior work identified signal correlations," she said. "You are not present to obstruct review."

"If this is a review, where is Dr. Sanderson?"

"Under medical observation."

"She can walk."

"She can also bleed."

The medical liaison's eyes dropped further, to a tablet he had stopped reading some time ago.

Cavanaugh let the silence run a beat past comfortable. "She's the only person in this facility with a defensible thermal model."

"The model is being evaluated."

"By people renaming heat stress as field potency."

"Colonel," the remote voice said, "remove the consultant if he cannot remain procedural."

Underwood didn't look at the screen. "He will remain procedural."

"That was optimistic," Cavanaugh said.

"Continue," Underwood told the room, and the engineer brought up the next slide.

This one drew the matrix as a node, arrows entering from below and from the station side. The lower arrows read GEOTHERMAL / DEEP-CRUST EXCHANGE. The station-side arrows read EXTERNAL EXCITATION. The space around the whole thing was labeled RESPONSE POTENTIAL.

Nothing on the slide was, strictly, a lie.

That was usually how the durable ones survived.

"Where's the throttling layer," Cavanaugh said.

The room went quiet in a way that had shape to it, a silence with edges, the kind that happened when a question landed somewhere it hadn't been budgeted for. The engineer didn't answer fast enough. His eyes went to the tablet, then to Underwood, then back to the tablet, a small circuit that told Cavanaugh exactly who in the room he thought could authorize a real answer.

Underwood did. "Define your term."

"I didn't invent it. Your own archive did — throttle, prioritize, reconcile. Those were system labels tied to Iceland and to the local field events here."

Hayes's hand found the back of Cavanaugh's chair. A warning, not yet a grip.

"Those records are outside your current clearance," the operations officer said.

"That doesn't make them less relevant."

"It makes your possession of them a violation."

"I don't possess them."

Not false. Not enough, either.

Underwood studied him for a second longer than the exchange required. "Then we will not base policy on them."

"You're already basing policy on the same events. You've just cut the parts where the logs say the system is compensating, not obeying."

Something crossed the lead engineer's face — small, involuntary, a flick of the eyes toward Underwood and away. Cavanaugh caught it. So, he was fairly sure, did she.

"Dr. Rane," Underwood said.

The engineer straightened. "Ma'am."

"Do the unformatted records establish autonomous compensation?"

Rane's fingers found the edge of the tablet and stayed there without moving it. He didn't want the room to hear his answer. That, Cavanaugh thought, was the first honest thing the man had brought into this meeting.

A cooling fan somewhere behind the wall display cycled up, ran for four seconds, and dropped back to idle. Nobody reacted to it. Cavanaugh did, because idle systems didn't usually need to cool.

"They indicate pre-execution response."

The logistics officer looked up. "Before the pulse?"

"The timestamp chain is compromised."

"Is that a yes or a no?"

Rane's throat moved before the words did. "It's evidence of a response not fully explained by our command signal."

Static bled through the remote channel. Underwood lifted one hand slightly and the noise settled, as if the gesture itself had authority over the audio feed.

Cavanaugh watched her in the pause that followed, because Underwood under pressure told him more than Underwood giving orders. She didn't bristle.

Didn't deny it. Didn't rush to fill the silence with volume. She took the bad fact, held it a moment, and turned it the way a machinist turns a part to find the angle that fits the jig.

"That supports the central concern," she said.

Sanderson would have hated that sentence on its own terms — not for what it claimed, but for how easily it could sit beside a burn chart and not look out of place.

Rane blinked. "Ma'am?"

"If the structure responds before our logged execution, it is monitoring load conditions within or adjacent to our operating environment. If it monitors load, it can be induced to reveal response thresholds."

"Or overloaded," Cavanaugh said.

"Threshold testing is how operational limits are established."

"With bridges. Engines. Batteries. Materials. You know what those are before you test them to failure."

"And if the object under test can alter gravity, electrical behavior, and material cohesion without a conventional delivery system, refusing to characterize it is also a strategic decision."

There it was. Not greed. Not madness. A framework, built the way command frameworks were always built, out of the

worst-case assumption and the fear of being the one who hadn't acted on it.

"You don't know it's a power," Cavanaugh said.

"I know what happened to the wrench. The platform. Dr. Kline's wrist."

"Kline."

Underwood paused.

"His name is Kline. Not doctor. Not platform casualty. Kline. He was on the gantry crew for eleven months before this. He has a wife stationed at McMurdo who doesn't yet know the cast goes to the elbow."

Something tightened faintly at the corner of Underwood's mouth, the first expression Cavanaugh had gotten out of her since she'd walked into the room. "I know his name."

"Then use it before you turn his injury into a test parameter."

The medical liaison's pen stopped moving against the edge of his tablet. He didn't look up, but his hand had gone still, and stillness in that particular hand was its own kind of statement.

For the first time the calm thinned enough to show what was underneath it. "Do not mistake command responsibility for indifference."

"Then do not mistake injury for opportunity."

The remote voice came back, older, male, filtered flat, out of patience. "Colonel. Move to override feasibility."

Underwood held his gaze one more second. Then she turned back to the room. "Override feasibility."

Rane brought up a new slide, and Cavanaugh felt the register of the meeting shift under him — everything before this had been clearing a throat for the thing already decided somewhere above the remote window, before any of them had sat down.

ESCALATED COMMAND OVERRIDE FIELD
ENVELOPE STABILIZATION / RESPONSE CAPTURE

Beneath the header: capacitor banks, a power-routing diagram, a pulse ladder, a cavern delivery architecture rendered in the same reassuring gradients as everything else.

He recognized the capacitor banks. Sanderson had seen them on the aircraft manifest, filed under a line that claimed survey support. She'd read the weight figures back to him from memory in the mess hall two nights ago, annoyed at the math, certain the numbers were wrong for anything as simple as

backup power. They hadn't been wrong. They hadn't flown in redundancy. They'd flown in the hardware to force the thing, packed under a label chosen by someone who understood exactly how far a euphemism could travel before somebody like Sanderson checked the arithmetic.

The diagram itself was drawn with the same restraint as the others — no warning color, no hazard iconography, just clean vector lines connecting boxes labeled with model numbers instead of purposes. A capacitor bank was, on paper, indistinguishable from any other capacitor bank. It was only the destination of the lines running out of it that made the thing dangerous, and destinations were the easiest part of a diagram to leave unlabeled.

"Override package remains within installed power capacity," Rane said, his voice drier now, sanded down to procedure. "Primary capacitor bank can deliver a stepped excitation profile across matrix contact points Alpha through Delta. Expected output is a stronger field response, potentially mapable in real time."

"Expected by what model?" Cavanaugh said.

Rane didn't answer.

Underwood did. "The model currently available."

"That's not a model. That's a wish with voltage."

Hayes moved half a step closer, close enough that Cavanaugh felt it more than saw it.

"Mr. Cavanaugh."

"The matrix drew heat before the pulse. Your logs classify throughput throttling. You have matter-cohesion events during processing drops. Grid micro-collapses in the archive. Clocks disagreeing about event order. None of that says field-control asset. It says stressed system."

The operations officer leaned in. "A stressed system can still reveal capabilities under pressure."

"A dying pilot can still move a control stick. That doesn't make turbulence a weapons platform."

Rane looked at his tablet and didn't touch it. The room had gone very still, the kind of stillness that happens when people are counting who will have to answer for what's about to be said.

Underwood came around the end of the table. Not fast. Not for effect. She stopped across from him, the table between them, no illusion that they were equals in the room.

"You believe we should suspend all active testing."

"I believe you should stop forcing a system whose failure modes include matter becoming negotiable."

"Noted."

"That's it?"

"Your objection is noted."

"You say that like it stores consequences somewhere."

"It does."

"In a file you can reclassify."

Her voice didn't rise. "Mr. Cavanaugh, I have personnel under the ice. A structure of unknown origin beneath a military installation. Proof of local physical-law deviation. Unexplained events tied to global infrastructure. And a strategic environment in which any other state discovering a comparable capability would not wait for our philosophical comfort. My responsibility is not awe. It is containment."

"Containment isn't the same as control."

"No," she said. "But uncontrolled knowledge isn't safety either."

He didn't answer right away, because for a moment he understood her, and that was the part that

frightened him more than the slide behind her. Underwood was wrong. She was not, however, a fool, and he could not hate a framework the way he could hate a person. She had taken the same set of facts he had and run them through a different machine, one built for a different kind of survival. If she'd been stupid he could have stopped listening. Instead he had to keep listening, and that cost time he didn't have.

He thought of the way NRO briefings used to close, back when he still trusted the shape of them — a summary slide, a recommendation, a sentence that assigned the risk of doing nothing a number nobody could verify. Underwood's sentence did the same work in different words. It made the absence of action into its own kind of action, weighed and logged and defensible in front of whatever board eventually asked. He'd written versions of that sentence himself, years ago, for programs he no longer had clearance to name. He didn't offer that out loud. It wouldn't have changed the room, and it would have cost him something he wasn't ready to spend yet.

"Colonel," the remote voice said.

Underwood turned from him. "Proceed with readiness classification."

Rane's hand hovered over the tablet. Cavanaugh watched the small tremor in it before Rane brought it under control.

"Readiness classification. Conditional."

The operations officer frowned. "Conditional?"

"Thermal model unresolved. Timebase discrepancies unresolved. Field response not repeatable under controlled baseline."

Underwood watched him say it and didn't move.

"Recommend limited hold," Rane said, "pending integrated review with Dr. Sanderson's raw acquisition data."

A small rebellion. Technical. Cautious. Almost invisible if you weren't looking for it. Cavanaugh was looking for it. The room seemed to wait to see whether it would survive contact with the next question.

"Can the override package proceed without integrated review?" Underwood asked.

Rane closed his eyes once.

"Yes."

The rebellion ended there, in one syllable.

"Then classify readiness as provisional operational," the remote voice said.

Rane entered the field. The display updated.

OVERRIDE READINESS: PROVISIONAL
OPERATIONAL

Cavanaugh felt the wafer in his boot the way you feel a splinter you've stopped trying to work free — present, small, and completely unable to help him yet.

"Schedule," Underwood said.

The operations officer answered. "Capacitor conditioning can begin within six hours. Full stepped excitation window available after final cavern access lockout."

"Move it up," the remote voice said.

Underwood looked toward the audio window, the first time she'd addressed it directly. "Rationale."

"Security breach risk. Consultant interference. Unknown external monitoring. The longer we wait, the less containment we retain."

Containment as acceleration. Fear translated directly into a schedule, with nothing between them. Cavanaugh had seen the same conversion happen in briefings before, always dressed differently, always arriving at the same place — a threat too vague to define precisely enough to argue against, and a deadline precise enough that nobody had time to try.

"How far," Underwood said.

"Readiness review in ninety minutes. Execution package prepared for command authorization thereafter."

Rane looked, for a moment, genuinely sick. The medical liaison opened his mouth and then didn't say whatever had gotten as far as his throat.

Cavanaugh didn't look at Hayes. He looked instead at the room's two exits, the camera housings bracketing the display, the network jack visible beneath the table's edge, the way Underwood's attention kept splitting between the ceiling-mounted audio and the people actually breathing the same air she was. Too many guards for the space. Not enough of them watching the same direction at the same time. No Sanderson. No clean way out of the room that didn't end with someone deciding what to do about him on the spot.

He wasn't going to stop this here.

He needed her data. She needed what he'd carried out in his boot for nine cooling, aching minutes.

Underwood closed the meeting in three sentences.

"Rane — revise the package for stepped excitation. Operations — move capacitor conditioning to priority. Security — place Cavanaugh and Sanderson under controlled movement pending final authorization."

Controlled movement.

Not arrest. Not yet. The words staged the consequence before anyone's body had to carry it out. Somewhere in whatever policy manual governed this place, controlled movement probably came with its own careful definition, its own permitted radius, its own list of doors that would no longer open to his badge. He didn't know the definition. He'd know it soon enough by which doors failed first.

Hayes put a hand on his shoulder, and this time didn't pretend it was a suggestion.

As Cavanaugh stood, the wall display behind Underwood stuttered. For half a second the field envelope diagram broke into rectangular blue-white segments, each refreshing a beat after the one before it — the same panel pattern from the borehole feed, the same geometry he'd watched come apart in the cavern footage two nights ago. Most of the room didn't register it happening at all. The logistics officer was already reaching for his own tablet. The medical liaison was still looking at the floor.

Underwood did.

So did he.

The display corrected itself.

OVERRIDE READINESS: PROVISIONAL
OPERATIONAL

She looked at the screen for a moment the way a person looks at a machine that has just said something in a language she has decided not to understand.

"Continue," she said.

Hayes walked him toward the door, two fingers at his shoulder blade, not gripping, just present, the way a hand rests on a door you expect to have to close in a hurry. The second guard fell in a half-step behind. Cavanaugh kept his weight even between both feet, careful now, walking like a man who had simply been sitting too long, not like a man carrying nine minutes of heat and cold pressed flat under his arch.

At the threshold Cavanaugh looked back once. Rane was still seated, staring at a tablet he wasn't touching, both hands flat on the table on either side of it as if the surface needed holding down. Underwood stood in front of the screen with her hands clasped behind her back, spine straight, face giving away nothing

at all. Behind her the remote command window's audio hissed once, some far-off room settling into its own silence.

The remote command window closed.

For one second before the display went dark, a status line surfaced beneath the override diagram, small enough that only someone already watching for it would catch the words before they vanished.

FIELD RESPONSE CLASSIFICATION UPDATED:
CAPABILITY

Then the Cold Horizon seal filled the screen.

The door took its time sealing behind him, the pneumatic hiss dragging half a second longer than the doors upstairs, as if the mechanism down here had its own opinion about how fast anything should be allowed to close. Hayes turned him left, away from the residential corridor, toward a stretch of hallway Cavanaugh hadn't been walked through before. New corridor. New camera housings, unmarked, spaced closer together than the ones near his quarters.

"Where," Cavanaugh said. Not a full question. He'd learned that much about Hayes — half sentences got him further than complete ones.

"Holding," Hayes said. "Yours and hers. Separate."

"Separate."

"Ninety minutes, Cavanaugh. Colonel's orders don't leave a lot of room for creative interpretation."

Ninety minutes matched the number from the room. Hayes had been listening after all, or had already been told, which was its own kind of answer.

The corridor bent, and the temperature dropped again, air pulled from somewhere closer to the cavern than the surface levels ever were. Cavanaugh walked it at an even pace, weight distributed, breath even, cataloguing exits he wasn't going to be able to use for at least an hour, maybe less. His foot ached faintly with every other step, a small mechanical throb keeping time against the sole of his boot, a fact he had no plan for yet beyond enduring it until enduring it stopped being enough.

Somewhere behind him, several floors up and several rooms away, a capacitor bank he had never seen was already beginning, very quietly, to charge.

Chapter 17: Sanderson's Warning

They had walked her down two levels to reach the room, past a checkpoint that hadn't existed on her last transit of that corridor, manned by someone who checked her badge against a list rather than a reader. She'd asked what the list was. He hadn't answered. She filed that the way she filed most things now, without expecting it to resolve into anything before it needed to.

The briefing room had no independent display ports. Sanderson noticed that before she noticed anything else, because it meant the wall screen answered to whoever sat at the front table and to no one else.

The second thing she noticed was the guard, standing inside the door rather than outside it. Outside meant traffic control. Inside meant something else.

The third was her own data, already on the wall. Someone had taken her diagnostic rack output, stripped the raw timing traces out of it, and enlarged what remained around a single field-effect diagram carrying a label she hadn't written.

LOCAL FIELD RESPONSE POTENTIAL
DIAGNOSTIC EVENT ONE

She stopped two paces past the threshold and read it twice, the way she'd learned to read numbers that had been rounded before she got to them.

Her hand throbbed under Ariza's dressing. The partial-write chip sat beneath the outer wrap, smaller than a fingernail clipping, pressed flat against swollen tissue by the compression bandage. Each heartbeat pushed the edge of it a fraction deeper. She kept her arm at her side and did not look at it.

Colonel Underwood stood at the front table. Dr. Rane sat to her left with his tablet closed in front of him, and two operations officers sat further down, one of them already reviewing something on a laptop angled away from the room. A third chair, empty, sat pushed slightly out from the table on Underwood's other side, as if someone had meant to occupy it and then hadn't. A remote command window occupied the upper right corner of the wall display, black background, white text. Initials only — J.M., no rank given. The audio channel carried a faint hiss, the particular texture of an open line with nothing coming through it yet.

Cavanaugh was not in the room.

She catalogued that too, filed it next to the missing ports and the early guard, and didn't let it show on her face. He'd been pulled somewhere else, or kept somewhere else, and either way the effect was the same: whatever she said in the next several minutes, she'd be saying it without him in the room to hear it firsthand, which meant it would reach him later, if it reached him at all, filtered through whatever version of this meeting made it into a log.

"Dr. Sanderson," Underwood said. "You requested review time."

"I requested a halt."

"You requested review time. In language that required a command response."

"I can simplify it. Stop the override."

One of the operations officers shifted in his chair, a small motion, chair leg against floor. Underwood didn't move at all.

"You have five minutes," Underwood said.

"No," Sanderson said. "You have five minutes before capacitor conditioning commits the system to an input profile you can't characterize."

Rane looked down at the table.

She set her field tablet on the surface in front of her. The screen was cracked corner to corner, a white fracture line running through the glass like a fault trace. She'd tried pairing it to the room twice already in the corridor outside, out of habit more than hope, and both times the handshake had timed out before completing. Now, with the table's own display active behind her, the system refused it on principle instead of on signal. A small window opened on the wall, polite and gray.

UNAUTHORIZED DEVICE DETECTED
PRESENTATION MODE RESTRICTED

She looked at Underwood. "Convenient."

"Security protocol."

"Evidence suppression protocol."

"You may present verbally."

"Science doesn't improve when you strip it down to testimony."

"Then be precise."

She turned the tablet so the table could see its own screen instead, small and dim, running at six percent after Ariza had let her use a medical charger for ninety seconds under the pretense of checking the dressing for electrical burns. The display showed the

aligned partial interval — thermal absorption climbing, acoustic narrowing beneath it, the channel-four domain error, and beneath both, the execution timestamp offset after sync correction. The graph looked wrong in the way graphs look wrong when the instrument recording them has started disagreeing with itself. It had survived only because it looked too damaged to matter.

"Diagnostic Event One was not a successful field test," she said. "It was a stress response."

Underwood folded her hands on the table. "Define stress."

"The structure increased thermal absorption before your system logged pulse execution. Acoustic narrowing followed within the same interval. Channel four recorded a domain error in the thermal probe substrate before the local gravitational variance appeared. The field response came after load compensation began. Not before it. Not from it."

The nearer operations officer spoke without looking up. "You're working from a partial copy."

"I'm working from raw acquisition fragments. Your formatted package reordered the event before I saw it."

Rane said, quietly, to the table, "There were timebase discrepancies."

She looked at him until he had to look back.

"Say it plainly."

His eyes went to Underwood first. Underwood said nothing, which was its own kind of permission and its own kind of warning, and Rane exhaled through his nose.

"Station time and local acquisition time diverged during the event."

"How much?"

"Variable."

"How much, Rane."

"Eight-tenths of a second, on at least one lower-ops system. Possibly more in the cavern feed. We haven't finished reconciling it."

"Eight-tenths of a second," Sanderson said, "is enough to make an effect look simultaneous with a command that actually followed the system's own pre-response."

The remote channel clicked, a small mechanical sound ahead of the voice.

"Dr. Sanderson. Your conclusion assumes the acquisition fragment is valid."

"My conclusion assumes I should trust raw instrument behavior over a formatted incident report that reclassified a gravity event as platform instability."

Underwood's eyes sharpened, a fraction, at that. Sanderson let the silence hold instead of filling it.

The room had been too warm when she walked in, the particular overcorrected heat of command spaces built for people who'd just come up off the ice. Sweat had gathered under the tape at her wrist since then. Her thumb ached at the second knuckle where nothing was actually broken. She pressed her hand flat against the table edge and felt the chip bite harder.

She changed the tablet display to the thermal map, and the interface lagged for a second and a half doing it — the cracked digitizer registering her thumb somewhere left of where she'd actually touched, a small daily indignity she'd stopped reacting to days ago. The map resolved. Blue-green gradient across a cross-section she'd built herself, mesh by mesh, from three separate probe strings, two of which had failed partway through acquisition and had to be patched with interpolation she'd flagged in the metadata rather than hidden.

"Look at the heat path," she said. "Not the field envelope. Not the diagram Field Utility likes to project on walls. Heat, first. The rock adjacent to the matrix runs warmer than

the matrix's own contact surface. The exposed matrix face emits almost nothing. After the diagnostic pulse, absorption spikes hard. If this were an externally controllable dampening field, you'd expect output proportional to input. That's not what the instruments recorded."

"We observed proportional field response," the operations officer said.

"You observed a wound swelling after you pressed on it."

Nobody spoke for a second that felt longer than a second.

"Analogies aren't data," Underwood said.

"No. They're what I use when command keeps misusing the data."

"Then return to the data."

Sanderson advanced the graph one frame.

"The matrix is functioning as a heat sink, or a thermal exchange interface tied to processing load. It's already operating near a limit. Throttling appears to reduce throughput when thermal or computational stress rises. That throttling correlates with local physical-constant drift."

"Appears," the remote voice said.

"Yes. Appears. Because I'm not going to pretend the uncertainty went away because someone needs permission to fire capacitor banks into a buried structure."

Rane was staring at the tabletop now, at nothing.

Underwood's posture hadn't shifted. "What is your failure projection?"

"Unknown."

"Define bounded risk."

"There is no bounded risk."

"Doctor." A single word, and a warning folded into it.

Sanderson heard the warning and went past it anyway. "You are forcing energy into a system that's actively managing thermal imbalance and local physical-state coherence. If the matrix is damping its own throughput to preserve stability, an override could do one of three things. It could force a larger compensatory response. It could collapse the local exchange gradient. Or it could trigger a protective mode built for system integrity, not for the survival of anyone standing near it."

The nearer operations officer set his pen down.

"Precedent for the third option?"

"Iceland," she said. "Partial. We didn't have instrumentation deep enough there to confirm it, only afterward damage consistent with it."

"Consistent isn't confirmed."

"No. It's what I have. It's more than you had before you asked."

He didn't answer that. He picked the pen back up.

"Scope."

"Cavern, at minimum."

"At maximum?"

"You don't have the evidence to ask that question safely."

"Ask it anyway," Underwood said.

Sanderson looked once at the remote window — the hiss on the line hadn't changed — and back to Underwood. "Crustal equilibrium disturbance across the deep-interface zone. Regional electrical instability. Expanded constant drift. Matter-cohesion failures, if the processing drop propagates far enough. A southern-hemisphere thermal event, if the exchange path collapses badly enough."

The logistics officer, three seats down, had gone the color of the wall behind him.

The operations officer hadn't. He was frowning the way a man frowns when he's sorting a catastrophe into categories he already has folders for.

"Probability?" he asked.

"I can't assign one."

"Then we can't operationalize it."

"You can operationalize not doing the thing."

The remote voice again, flat: "Colonel, the witness is advocating paralysis."

She turned toward the speaker mounted above the display. "The witness is advocating not stabbing the only system in this room that may be keeping physics aligned."

The room changed. Not because anyone had understood her. Because the sentence had crossed some threshold from technical warning into a category of consequence large enough to threaten careers, and careers were a language this room spoke fluently. The logistics officer's pen stopped moving. The nearer operations officer glanced, briefly, at the remote window, as if checking whether the initials up there had heard the same sentence he had. They gave nothing back.

Underwood stepped away from the table and crossed to the wall display. She touched one control. Sanderson's cropped field diagram came back, bright and clean and wrong.

"Diagnostic Event One produced a measurable field response," Underwood said. "The response was temporary, localized, and survivable."

"Kline's wrist may dispute survivable."

"Dr. Kline is receiving treatment."

"His name is Kline."

Underwood paused, a half-second longer than the sentence required, and something crossed her face that Sanderson couldn't name and filed anyway. Someone else had said almost the same words in another room. Cavanaugh. She was almost sure of it, and the certainty settled into her like ballast — not comfort, exactly. Something adjacent to it.

Underwood continued as though the pause hadn't happened. "The response produced no sustained structural damage to the station."

"Because it was small."

"And therefore a stepped profile may establish tolerance."

"No. A stepped profile may push the system from compensation into failure."

"Or define the envelope."

"You don't define the envelope of a system by provoking its emergency responses while you're standing inside it."

Underwood turned from the display. "Doctor. Your data is not policy."

Sanderson felt that sentence land somewhere below her sternum, more than she heard it. It was clean. It was honest. It was the truest thing anyone in the room had said all night, and it was fatal.

Her whole professional life ran on the assumption that measurement, if gathered carefully enough and defended long enough, eventually forced reality into a room with power and made power adjust. Not quickly. Not cleanly. But eventually, if you were patient and thorough and did not flinch from the numbers. Underwood had just told her, without raising her voice, that the weight of evidence didn't matter unless command chose to lift it.

Sanderson looked down at the tablet. Battery four percent.

She thought, briefly and without wanting to, of every prior instance where this had worked — the calibration drift she'd caught on the Reykjanes array before anyone would admit the array needed recalibrating, the contamination event at White Sands that hadn't been believed until she'd run the same sample through three independent instruments and put all three results in front of the same panel. Patience had been the method. Repetition had been the method. Command had adjusted, eventually, because the numbers had nowhere else to go and neither did the people holding them.

There was nowhere else for these numbers to go. That was the difference, and she understood it standing in the heat of that room with her hand throbbing under the dressing: not that the evidence was weaker this time, but that this room had no mechanism through which evidence became consequence. It only had a mechanism through which evidence became a line item, formally entered, safely contained, forwarded to a review board built to take its time.

She could keep arguing. She could sharpen the failure projection further, ask Rane to repeat the

timebase discrepancy for the remote channel's benefit, reach for thermal runaway, domain instability, unmodeled field propagation, nonlinear coupling, and watch each phrase get translated, in real time, into a risk category that left the override schedule intact. She had watched people do that with her data before. She could watch it happen again.

Proof had reached the room. The room had not become a different room because of it.

She closed the tablet.

"Doctor—" Rane started.

"No," she said.

He stopped.

She put the tablet under her injured hand, weight resting on the chip through the dressing, and leaned forward over the table. Rane still hadn't opened the override readiness file in front of him. His hand sat flat on the closed cover, fingers spread, not moving.

"You are not characterizing the matrix," she said. "You are forcing it to reveal its damage thresholds. Your own records show prior matter-cohesion failures during processing drops. Your own archive ties the Iceland signal to this node. Your own diagnostic injured personnel and moved local gravity. You have evidence that the structure pre-responds to load and that it is

already throttling. If you proceed and something happens, the event that follows is not an accident."

"Is that an accusation?" Underwood asked.

"It's a warning. With witnesses." Sanderson looked once down the table, at the logistics officer, at the operations officer, at Rane. "Three of them, now, besides the one who isn't recording anything."

The remote window's audio hissed on, unchanged.

The remote channel clicked once more.

"Colonel."

Underwood didn't answer the voice right away. Sanderson watched the decision form — not in Underwood's face, which gave up nothing, but in the room around her. The guard at the door shifted his weight from one boot to the other. Rane looked away from the table entirely. The nearer operations officer closed one file on his laptop and opened another, unhurried, a man clearing his desk before the end of a shift. The remote window stayed open, faceless, patient, the hiss on the line unchanged.

Underwood had not been waiting to be convinced. Sanderson understood that now, a beat too

late to matter. Underwood had been waiting to document that Sanderson had been heard.

"Dr. Sanderson," Underwood said, "your warning is formally entered."

"Stop the override."

"No."

Rane's jaw tightened. He didn't look up.

"Your raw acquisition material will be reviewed by Field Utility personnel," Underwood said. "Until that review is complete, you are relieved of independent instrument access."

Sanderson laughed once, low, no humor in it at all — failure, venting through the wrong opening. "You can't review what you're already misusing."

"Your continued involvement is becoming a safety concern."

"To the site?"

"To the operation."

"At least you can still tell the difference."

Underwood looked past her, to the guard. "Escort Dr. Sanderson to medical. Then residential confinement, pending final authorization."

The word arrived softly, the way the worst ones usually did.

Confinement.

The guard came off the wall.

Sanderson picked up the tablet.

"Unauthorized device remains here," the operations officer said, not looking up from his laptop this time either.

She looked at him until he did look up. "Come take it."

The guard stopped halfway around the table, waiting on something — a nod, a word, some cue Sanderson couldn't see. Underwood gave it to him with her eyes alone.

"Doctor," Underwood said.

Sanderson held the tablet against her chest with her good hand, four percent bleeding down toward three while she stood there. The device itself was worthless as evidence now; the chip that mattered was under gauze against her palm, and the tablet mattered only because they wanted the compliance of taking it from her, the small procedural theater of it.

Her pulse hammered against the cut beneath the dressing.

The guard reached for the tablet. Up close he looked younger than his posture suggested, and uncertain about the specific mechanics of taking a piece of equipment out of an injured woman's hands without making it look like what it was.

She let him take the edge of it.

Then she didn't let go.

For one absurd second the two of them stood like that, fighting quietly over a cracked field tablet, while somewhere below the floor a buried structure moved heat through ice and stone at a rate no one in the room had bothered to log correctly, and command prepared to push voltage into whatever answered it.

The screen flickered between their hands, pressure against the fracture line distorting the last frame of the display.

PRE-EXECUTION RESPONSE DETECTED

The guard saw it. The nearer operations officer saw it, his laptop forgotten for exactly as long as the line stayed lit. Underwood saw it too — Sanderson watched her see it, watched nothing at all happen on her face because of it.

Then the screen went dark. Battery gone.

Sanderson let go.

The guard nearly overbalanced from the sudden absence of resistance, a small stumble, human and unimportant and, in its way, satisfying.

"Medical," Underwood said, even as ice under load.

Sanderson turned toward the door.

Passing Rane, she stopped. He didn't look up. His hand rested on his tablet, beside the override readiness file he hadn't opened yet, knuckles pale against the case.

"Don't give them clean numbers," she said, quiet enough that it wouldn't carry to the front of the room.

Rane didn't move.

"Clean numbers make bad decisions look precise."

The guard's hand found her elbow. She pulled free of it before he could steer her and kept walking on her own line toward the door.

In the corridor the air was colder, and it steadied her more than anything said inside the room had. Amber lights ran along the floor in slow, timed pulses toward medical, and past medical, toward the residential module

that would become a cell as soon as someone found paperwork gentle enough to justify the lock.

Behind her, before the door sealed, she heard the remote voice one last time.

"Proceed to final readiness."

And then Underwood, unhurried, unshaken. "Prepare the override package."

The door closed on the rest of it.

Sanderson kept walking, the guard half a step behind her now, not touching her again. Her legs felt the walk more than she expected — the adrenaline that had carried her through the last ten minutes draining out through her knees, leaving something closer to the fatigue she'd been holding off since before the briefing started. She hadn't eaten since before the diagnostic. She didn't let herself think about that either.

The dressing on her hand had gone damp somewhere near the end, sweat or blood or both, and the chip beneath it dug in with each step, a small hard proof of something that wasn't going to be enough by itself. Proof had reached the room and the room had produced a lock instead of a change of course. Whatever the chip carried would need somewhere else to go, someone else to carry it, some route that didn't pass

through Underwood's table or the remote channel with its faceless patience.

Ariza was waiting outside medical when they reached it, arms crossed, reading something on the duty log that made her look up too fast when Sanderson came through the door with an armed escort behind her. She didn't ask what had happened. She'd clearly already been told, or told enough.

"Sit," Ariza said, nodding at the exam table, and started peeling the outer wrap of the dressing back with a care that had nothing to do with the guard standing three feet away pretending not to watch.

Sanderson sat. The guard took up a position by the door, replacing the earlier one, part of whatever rotation kept a body between her and every exit from here forward. She held still while Ariza worked, and did not look down at her own hand, and counted the amber lights she could see through the open doorway instead — four, five, six — until Ariza's fingers found the edge of the inner wrap and stopped, waiting for permission before going further.

"It's fine," Sanderson said, before Ariza could ask. "Leave it."

Ariza looked at her for a second longer than the instruction required, then re-taped the outer wrap without lifting it, and didn't ask again.

"Residential," the guard said, from the doorway, when Ariza finished. Not unkind. Just the next word in a sequence he'd been given.

The walk from medical to the residential module took four minutes she counted without meaning to, boots on decking, then boots on the colder composite flooring of the connecting corridor, then boots on carpet thin enough to still feel the composite underneath it. The module door was standard, keyed, no different from any of the others she'd passed a hundred times without registering them as anything but doors. It registered now. The guard didn't follow her in. He stood in the frame instead, waiting while she crossed the small room and set down nothing, because she was carrying nothing anymore that they hadn't already taken or already known about.

"Someone will bring dinner," he said, which was either a kindness or a script, and she didn't have the energy left to tell which.

The door closed. She heard it seat into the frame, and then, a moment later, a second sound underneath the first — mechanical, brief, unmistakably a lock rather than a latch. She

stood in the middle of the room and listened to it happen, and didn't test the handle, because testing it wouldn't change what she already knew.

She sat down on the edge of the bunk, finally, and let her injured hand rest palm-up on her knee, and looked at the dressing instead of at the door. Under the tape and the gauze, against her palm, the chip was still there. Small. Intact. Waiting on a use she hadn't found yet.

She flexed her fingers once, slowly, and felt it shift, and did not open the dressing to check on it, because there was nothing in this room that could see her do it, and for the first time since the briefing started, that fact worked in her favor instead of against her.

Chapter 18: Proof Streams

The badge reader rejected Hayes twice before it took the second soldier's card, and by then nobody standing outside Residential Module B believed that was an accident of hardware.

Hayes swore at the panel. The second soldier reported it into his collar mic. The corridor speaker gave back three seconds of static and half a sentence from some other level of the facility, a voice discussing a supply manifest that had nothing to do with any of them.

Then the door across the hall opened on its own schedule.

Sanderson stood in the gap between her guard and Ariza. Her field tablet was gone. Her parka hung open. One hand was wrapped in gauze already gone pink at the knuckle.

Cavanaugh registered the blood before the face. Then the face. Then the way she held the hand against her ribs, not the angle of someone favoring an injury but someone protecting it.

Hayes stepped into his sightline. "Eyes forward."

"That's anatomically difficult."

"Forward."

Sanderson looked past her guard's shoulder without changing expression, and that stillness told him more than a reaction would have. She'd already priced in the guard's presence, Hayes's shoulders, the fact that anything said here traveled through at least one uniform before it reached anyone who mattered.

Ariza said, "She needs the adjacent room. I'm not walking her farther than necessary if that dressing opens."

"Orders say separate quarters," Hayes said.

"Orders say medical observation until transfer. I'm medical."

"You're not security."

"No, Sergeant. I went to school."

Sanderson's guard coughed once into his glove and said nothing.

"They are not to communicate," Hayes said.

"They won't," Ariza said. "B-12 for her, B-10 for him. If they manage a conversation through two locked doors and an armed hallway, arrest the architecture."

Cavanaugh nearly liked him for that.

Hayes glanced at the ceiling camera. It panned left, then right, then stopped on the blank wall and

stayed there. Four people watched it do this. Nobody mentioned it.

Under the floor, the hum climbed and thinned into a band that found Cavanaugh's molars. The lights went from amber to white and back, once, like a held breath released too fast.

Hayes made the call people make when every option is bad and only one requires an explanation before morning.

"B-10," he said to Cavanaugh. "Inside."

Sanderson's guard opened B-12.

They passed close enough to smell each other — antiseptic on her hand, burnt dust worked into his sweater, cold metal on both of them, a smell the facility never quite let go of.

"Bad presentation?" Cavanaugh asked.

She didn't look at him. "Excellent presentation. Failed audience."

Hayes put a hand on his shoulder and moved him through the doorway. "No talking."

The door closed.

Six meters by three. Too warm. Too bright. Bolted cot, bolted desk, a medical camera in the upper corner, a vent near

the ceiling, a sink with no mirror. No terminal. Nothing loose enough to matter on a first pass.

He gave it a second pass.

The vent cover had four screws. Three matched. One didn't. The desk had a seam at the back where a service cable had once run and been removed. The camera's status light blinked in a pattern — two green, one red, four green, one red — that wasn't a stable state and wasn't a dead one either.

He sat on the cot.

The wafer was still under his insole, against his heel, and it had to come out before the next search reached that far.

He waited until Hayes's shadow slid off the frosted glass in the door, then bent to his laces as if adjusting them. The knot was ugly, double-tied hours ago in worse conditions, and his fingers didn't want to cooperate. Cold. Impact bruising from the cart. Fatigue sitting in his hands like sediment.

The wafer had worked itself under the inner sole near the heel. He eased it out with two fingers — a chip of black ceramic no bigger than a clipped fingernail, one edge cracked, a smear of boot adhesive on the flat side.

The Core Throttling

It might be dead. It might be encrypted past anything he could reach. It might hold nothing but cached thumbnails and a violation log nobody would ever read. It might be enough.

The camera clicked.

He closed his fist around it and looked up.

The status light went red. Then green. Then red.

The door speaker crackled. "Hands visible."

He raised both hands. The wafer sat between two fingers, angled behind his palm. Not hidden. Delayed.

"Both hands flat on the cot."

He put his right hand down.

The left moved slower.

Something knocked through the wall — not the door, the shared wall with B-12. Three taps. A pause. One tap. A pause. Two taps.

Sanderson.

"What was that," Hayes said through the speaker.

He set his left hand flat and let the wafer slide under the thin government pillow in the same motion.

"What was what."

"Noise."

"The architecture confessed."

Silence on the line.

"Do not test me," Hayes said.

Too late for that. He waited.

Three taps. One. Two.

Not a code they'd agreed on. Not words. A question shaped like an opening.

He got up and went to the sink, and the camera tracked him a fraction of a second late. Lag. Useful. He turned the faucet on. The pipes hammered somewhere behind the wall, and under the hammering the knock came again — three, one, two.

He answered with the heel of his hand against the wall below the sink. Two. Two. One.

A pause on her end. Then one tap.

Yes. No. Repeat. He had no way to know which.

They weren't prisoners with a code between them. They were a physicist and an analyst improvising against plumbing, which made it clumsy enough to be honest. He needed language, not rhythm.

The pipe entered the wall through a collar that had gone loose with age — not loose enough to detach, loose enough to carry sound. He crouched and put his mouth close to it, voice barely above the water noise.

"Can you hear me?"

Nothing but hammering.

"Sanderson."

A long pause. Then, thinned by metal and distance:

"Barely."

His knees didn't feel entirely reliable for a second. He nudged the faucet up to cover more of the exchange.

"Camera?"

"Facing the door. Audio uncertain."

"Assume it's live."

"Then stop breathing like you ran into something."

"I ran into several things."

"Useful things?"

Something close to a smile got as far as his mouth.

"Some."

The door speaker clicked. "Step away from the sink."

He shut the water off. The silence that followed came too fast, too complete.

He went back to the cot and sat. The speaker stayed dead. Through the wall, Sanderson knocked once, twice, then stopped.

He needed another cover, and the room only had one candidate: the heater under the cot, which cycled every forty seconds with a rattle in the housing. He waited through the dead

interval, counted the first metallic cough of the next cycle, and went back to the pipe.

"I found the reality blink archive."

Something touched her wall softly on the other side.

"Say that again," she whispered.

"Reality blinks. Matter cohesion failures. A tool cart passed through concrete and snapped back."

A beat before she answered. "Processing drops?"

"Yes. Matrix throttling. Iceland's linked to this node. The failure mode's logged as external demand exceeding local throughput."

"External demand."

"That's the language in the file."

"Not command input?"

"Not human command. System language.
Throttle. Prioritize. Reconcile."

The heater died mid-word. He stopped talking. Footsteps passed in the corridor — two sets, unhurried. One stopped outside his door. The latch clicked and didn't open. He sat and looked at nothing in particular until the steps moved on.

The heater cycled again.

"Your turn," he whispered.

"My raw trace shows pre-execution response. Thermal absorption rises before the official pulse. The hum narrows. Channel-four domain error. Then gravity variance."

"Command saw it?"

"Enough to ignore it."

"Underwood?"

"Entered the warning. Proceeded anyway."

"Of course she did."

"No." The word came through the pipe sharper than the rest. "Not of course. That assumption gets us killed. She understood the warning and converted it into threshold language. They're not dismissing the data, Jason. They're weaponizing the response."

He closed his eyes.

It fit. Underwood didn't need to deny the heat-sink model. She only needed the part of it that looked like it could be steered, and she could let the rest sit unexamined in a folder somewhere with a classification stamp on it.

"I saw the override folder," he said. "Deployable scenarios. Gravity suppression. Electromagnetic hardening. Matter cohesion disruption. Regional field control."

The pipe carried nothing for a moment.

"Matter cohesion disruption isn't a capability," Sanderson said. "It's a failure mode."

"They updated the folder after Diagnostic Event One."

"They had it before the event?"

"Yes."

"How long before?"

"Metadata predates the test. Content was modified after."

The heater rattled under the cot, and he leaned in closer to it.

"That means Diagnostic Event One wasn't discovery," she said. "It was confirmation."

"Yes."

"And Kline was inside their confirmation."

"Yes."

The name sat in the wall between them for a while.

Her breathing changed on the other side of the pipe. Not tears — she wouldn't spend air that way somewhere a microphone might turn it into evidence against her composure. Anger, maybe. Or grief kept too disciplined to show itself where it could be logged.

"Do you have the files," she asked.

"Fragment. Maybe. A cache wafer."

"Where."

"Not saying that through plumbing."

"Good."

"Your raw trace?"

"Partial. In the dressing."

"In the —"

"Don't make that sound."

"I was going to say efficient."

"No, you weren't."

"No."

The heater cut off. He went back to the cot, picked up the pillow as if trying to get comfortable, and worked one thread loose at the seam with his thumbnail. The wafer slid into the gap where fabric met stitching — not secure, only better than his fist. He set the pillow back down with the seam toward the wall.

The camera's status light blinked red. He looked at it. It blinked green.

The speaker clicked. "Cavanaugh."

"Yes."

"Step back from the wall."

"I'm on the cot."

"Stay there."

"I can tell you've done this before."

"Not with anyone enjoying it this little."

The speaker went dead. It wasn't Hayes's best line, and Hayes knew it wasn't. Something in him was fraying at the edges where the badge readers and the panning camera and the static-filled speaker kept reminding everyone that the facility no longer fully agreed with itself.

Useful, if it lasted.

The heater started again before he reached the pipe. Sanderson spoke first.

"They're moving me to confinement once medical clears the hand," she said.

"Same."

"We need to combine the proof before that happens."

"We're combining it."

"Operationally, I mean. Your fragment shows what the matrix is doing at the system level. Mine shows the event order physically. Apart, they call yours unauthorized and mine corrupted. Together they show

pre-response, throttling, processing drops, and matter failure in one line."

"Still not enough to stop Underwood."

"It might be enough to stop Rane."

He turned that over. "He hesitated in the Field Utility Review."

"Good."

"Then classified readiness as provisional operational."

"Bad."

"Under remote command pressure."

"That matters."

"Only if we can reach him."

"We may not need to. He needs one defensible reason to delay."

"Underwood overrides him."

"Then he needs one reason to contaminate the override package."

He almost laughed despite the room. "Doctor Sanderson."

"What."

"That sounded illegal."

"It sounded calibrated."

The heater stopped. The latch clicked.

He moved away from the wall and sat upright on the cot a half second before the door opened.

Hayes stood in the frame with two security personnel behind him and a handheld scanner at his side.

"Up," Hayes said.

Cavanaugh stood.

"Hands against the wall."

"You just told me to step away from it."

"Now I'm telling you to touch it."

"Consistency's dead."

Hayes pushed him into position and ran the scanner from collar to boots. It chirped at his belt buckle, at the pale line on his wrist where a watch used to sit, at the metal eyelets on his boots. Hayes held the wand over the left boot a second longer than the right.

Not long enough.

He moved to the cot. Cavanaugh kept his face still and thought about nothing at all, which took more effort than thinking about something would have.

The scanner passed over the mattress. No chirp. The blanket. No chirp. The pillow.

It chirped.

Hayes stopped.

Something in Cavanaugh's chest changed shape and didn't change back.

The scanner chirped a second time. Hayes lifted the pillow. Nothing lay underneath it, because the wafer wasn't under it — it was inside the seam, buried in fabric where the metal detector could catch the faint edge of a conductive fleck but where an eye would only find a manufacturing pucker.

Hayes squeezed the pillow in both hands. Cavanaugh did not look at his own hand.

Hayes turned the pillow over. The seam held.

The technician behind him said, "Could be the fire-retardant tag."

Hayes found it — a stiff foil-lined inventory label half-sewn into the far corner — and ran the wand across it. It chirped again. The tech shrugged.

Hayes did not look satisfied. Dissatisfaction wasn't evidence. He dropped the pillow back on the cot.

"Sit."

Cavanaugh sat.

Hayes came close enough that his next sentence didn't need volume. "If you have something, I will find it."

"I believe in you."

"I don't need you to believe in me."

"No," Cavanaugh said. "You need me to have nothing. That's a different problem."

Hayes left without an answer for that one. The door closed. Cavanaugh listened for the lock to engage before he let his shoulders drop half an inch.

Through the pipe, once the heater cycled back on: "That was loud."

"Almost."

"Do you still have it?"

"Yes."

"Then listen. The heat model alone says the matrix is stressed. Your archive says processing drops correlate with matter instability. The Iceland line says external demand exceeds local throughput. Put those together — what do you get?"

He looked up at the ceiling. Too bright, too blank, built by people who didn't want anyone in this room to have anything to imagine with.

"Not a weapon," he said.

"More."

"An overloaded stabilizing system."

"More."

The Core Throttling

He followed it the rest of the way. Thermal absorption rising before command input. The hum narrowing before the gravity drift showed up on anyone's readout. Matter cohesion failing during the processing drops, not independent of them. Iceland as a coupling point instead of a source. External demand exceeding throughput. Throttle. Prioritize. Reconcile.

"The matrix isn't causing the failures as an attack," he said.

"No."

"It's preventing a worse one."

The pipe gave him nothing for a long moment. Then:

"Yes."

The word didn't make the room feel any bigger. If anything it made the walls come in closer, because if the matrix was holding something back, then Underwood's override wasn't a controlled experiment running inside a black site. It was a wrench forced into the one mechanism still keeping a failing system from finishing its failure. Diagnostic Event One hadn't uncovered a capability. It had made the stabilizer flinch, and somebody upstairs had written the flinch down as usable.

He pressed his knuckles against the cold pipe until it hurt enough to be a fact instead of a thought.

"Can your data show that," he asked.

"It can show compensation before excitation. It can show throttling behavior. It can show command's timeline is false or incomplete."

"Mine can show prior failures and override intent predating the test."

"Together, maybe enough to force a delay."

"Through whom?"

"Rane."

"Pell, maybe."

A pause. "Pell?"

"He didn't seize the core cleanly, did he."

"No."

"Hayes has seen enough at the terminal to know command's lying to somebody, but he'll ride orders until they fail under his own boots."

"That leaves Rane."

"And Ariza."

"Ariza?"

"He cares about names."

"Good standard."

The corridor intercom cut in, flat and even. "All residential transfers prepare for controlled movement. Final readiness window begins in forty minutes."

Forty.

He put one hand flat against the wall. On the other side, Sanderson tapped once.

Not a code this time. Presence, plain and unadorned.

He answered once.

The camera clicked. He didn't look at it.

The old model — gather enough, route it through the right channel, let institutional process do what institutions were supposedly for — had died somewhere between the platform stability lie and the utility review that classified a stabilizer's distress as provisional operational readiness. Sanderson had warned them. He had put the concealment on the record. Underwood had proceeded anyway, on schedule, as if the warning had been a formality to clear before the real decision.

Evidence hadn't moved command. It might still move the people command was built out of.

He leaned toward the pipe one more time. "Can you walk?"

"Yes."

"Run?"

"No."

"Climb?"

A longer pause than the others. "Badly."

Kalamazoo

"That may be enough."

"Jason."

It was the first time in the facility she'd used his given name, and it stopped him more than the intercom had.

"What."

"If this becomes physical, we're not trying to seize the system. We're trying to stop them from forcing it."

"I know."

"Do you?"

He looked at the pillow seam hiding a cracked wafer, at the camera with its light doing its two-green-one-red arithmetic, at the vent screws that didn't match, at the sink pipe carrying a woman's voice through a wall in a building engineered specifically to prevent exactly that.

"No," he said. "But I'm catching up."

The intercom clicked on again, closer this time, a different speaker in the chain.

"Security teams to Residential Module B. Prepare subjects for separation."

He pulled the pillow into his lap and tore the last thread loose with his thumbnail. The wafer dropped into his palm.

On the other side of the wall, Sanderson knocked twice.

He answered once.

Then the locks disengaged, both doors at once, and the hallway filled with boots.

Chapter 19: House Arrest

The doors unlocked together.

Cavanaugh registered that before he registered anything else — before Hayes's shape filled the frame, before the scanner light swept the room. Transfer moved one subject at a time, two guards per corridor, a medical escort trailing behind if command still wanted the fiction of care to hold. Two locks disengaging on the same half-second meant someone above Hayes's pay grade had traded procedure for speed.

He folded the cache wafer into his palm before the door finished opening.

Hayes came in first. No sidearm this time — baton on the belt, scanner in the left hand, shoulders set the way they set on a man executing an order he had not been permitted to improve.

"Hands visible."

Cavanaugh showed both.

The wafer sat flat under his index and middle fingers, hidden only because Hayes was hunting for defiance and not for stillness. Stillness had been the first thing anyone had ever taught him, long before Cold

Horizon, before Iceland, before the record itself had started developing holes. People searched hardest where they expected motion.

The second guard swept the sink, the cot, the vent, the pillow. He found the torn thread at the pillow seam and held it up as though it were a confession.

"Government textiles," Cavanaugh said, "are fragile."

"Quiet."

Across the corridor, Sanderson's door opened. He heard it rather than watched it — latch, hinge, a guard talking too fast, her answer pitched too low to parse. Then Ariza, closer than command would have liked.

"She is not cleared for aggressive handling."

"She's cleared for transfer."

"She has an active wound."

"Then walk with us."

"I intended to."

Hayes glanced toward the sound. Cavanaugh used the half-second to work the wafer up his left sleeve, into the gap between the inner seam and the compression cuff. Less secure than the pillow. More mobile.

The scanner chirped at his chest zipper. His belt. His boot eyelets. His left cuff.

Hayes stopped.

Cavanaugh lifted his eyebrows, the closest thing to an alibi he had left.

The scanner chirped again.

The second guard turned from the vent.

Hayes took his wrist. Not hard. Precise, which was worse. He folded back the outer sleeve — nothing — pressed the seam — nothing visible. The wafer had ridden half an inch further up the cuff than he'd planned, caught in a pucker where the parka lining still hadn't recovered from the fight two days earlier.

Hayes's fingers moved closer.

The corridor lights dipped. Not out. Down, the way they'd been dipping since the diagnostic spike, a change everyone on the station now felt in the same wordless register. Cavanaugh saw it land in Hayes's grip — the pause, the recalculation no manual covered. A narrow band of hum came up through the floor and the scanner answered with one thin false tone.

Hayes looked at the scanner instead of the wrist.

Cavanaugh pulled his hand back.

"Static?"

Hayes said nothing.

"Cold, dry air. Old building. There's a matrix under the floor. Could be anything."

Hayes's hand closed around the scanner until the knuckles went pale, and for a moment Cavanaugh thought he would check the cuff again anyway, out of spite or thoroughness. He didn't.

"Move."

They brought him into the corridor.

Sanderson stood outside B-12 with Ariza at her shoulder and two security personnel close enough to keep her from Cavanaugh but not close enough to block the sightline. Her bandaged hand was tucked against her ribs. Her face had the particular gray of someone running on pain management and will alone. The guard behind her carried her cracked field tablet in an evidence sleeve, holding it the way people hold things they don't understand and don't want to be blamed for later.

Good. Let them have the dead thing.

Her eyes went to his cuff. Too fast for anyone else in the corridor to register. Not too fast for him.

She still had the partial chip in her dressing. He still had the wafer. Two pieces of evidence, two separate vulnerabilities, no way to combine them into anything safer than what they already were.

"Both subjects to Residential Lockdown Suite,"
Hayes said.

"Suite," Sanderson said. "Generous."

The guard nearest her said, "Doctor."

"Was that unclear?"

They moved as a group toward the far end of the corridor — not toward lower operations, not toward medical, not toward the lift. Residential Lockdown Suite was simply a better name than *cells* for rooms with locks on the outside. Cavanaugh mapped as they walked out of habit more than hope: a camera bracketed above the emergency shower, a panel seam low on the wall, two overhead pipes stenciled *potable* and *graywater*, a speaker mounted lower than it should have been, one access hatch under a tamper seal held down by screws that predated the station's current security contract.

Old screws mattered. He filed it and kept walking.

The intercom cut in overhead. "All departments, final readiness window opens in thirty-seven minutes. Capacitor conditioning continues under restricted command authority."

Sanderson stopped walking.

The guard behind her bumped her shoulder. She didn't move.

"Careful," Ariza said.

Sanderson looked past him, at Hayes. "Capacitor conditioning has already begun?"

Hayes didn't answer.

"Field Utility Review moved it up," Cavanaugh said.

She turned to him fully. "How far?"

"Readiness review in ninety. That was before this."

Underwood's voice arrived through the intercom before Hayes could shut the exchange down. "Security, maintain separation."

Hayes took Cavanaugh's arm. "Move."

"If the capacitor bank is conditioning," Sanderson said, louder now, not to Hayes but past him, "thermal preload has already changed."

The wall speaker clicked. Underwood: "Dr. Sanderson, your warning has been recorded."

"My warning assumed you hadn't started."

"Move them."

Hayes pushed him forward, and the corridor tilted for one breath — not visually, physically, a small roll through the floor like the station had become a ship under ice. Cavanaugh's

shoulder hit the wall. Sanderson caught herself against Ariza. One guard reached for the overhead pipe and closed his hand on air, because the pipe had been there a fraction of a second earlier and now wasn't quite in the place his hand expected.

Gravity came back without announcing itself.

A ceiling light popped. Glass ticked down somewhere at the far end of the hall.

No one moved for a second. No one needed Sanderson to explain what had happened. She explained it anyway.

"That's with conditioning only."

Hayes's jaw tightened. The guard with the evidence tablet looked down at the dead screen in its sleeve as if it might offer him a reason to stop.

The speaker held silent for three seconds.

Then Underwood: "Proceed."

That single word did more damage than the gravity shift had.

The guards moved again. Residential Lockdown Suite B turned out to be a pair of rooms facing each other across a short vestibule behind a second security door — the kind of design that came from someone

who had stopped trusting a single lock. Outer gate. Inner vestibule. Individual rooms. A camera in the vestibule. A speaker. No shared sink pipe running between the two rooms this time. Command had learned faster from architecture than it ever had from Sanderson's readings.

They put her in the left room. Him in the right.

Before the door closed, she said, "Jason."

Hayes stepped between them.

Cavanaugh held still.

"Remember the order," she said, level, like she was reading it off a chart.

"Inside," Hayes said.

"Heat," Cavanaugh answered. "Hum. Drift."

Something in her face changed, just enough for him to catch it and no further.

The door closed.

The room on the right was smaller than B-10 and newer. No sink. No vent cover within reach from the floor. The cot was welded instead of bolted. The camera sat behind a protective dome. The light had no visible switch. The walls were composite panels over a metal frame, seamless enough that there was nothing for a fingernail to start on.

Someone had built this room after somebody else had gotten out of a different one.

He sat on the cot. The wafer pressed against his cuff, a small hard fact he kept returning to the way a tongue returns to a chipped tooth.

No terminal. No pipe running to Sanderson's side. No obviously loose screw. Only himself, the camera, and a clock recessed behind clear plastic above the door.

Thirty-six minutes, if the intercom hadn't lied. He could wait and hope somebody outside the room decided in his favor.

That option had died somewhere back in the briefing rooms, three, four days ago now, he'd lost the exact count.

He stood and inspected the door. Composite, magnetic seal, electronic lock, a mechanical override plate accessible only from the outside. Concealed hinges. The gap under the door let in vestibule light and nothing thicker than a credit card. A ceiling vent ran along one wall — a perforated strip, not a grille, too high to reach without standing on the cot.

He stood on the cot.

The Core Throttling

The camera clicked, adjusting.

The vent strip was cold under his fingers, held by recessed star screws. No tool for those. He checked the cot next. Welded legs, a thin mattress zipped into a vinyl sleeve, and under the sleeve a metal support grid that flexed slightly when he pressed it. Not much give. Enough to squeal if he forced it.

The speaker above the door came on. Hayes.

"Step down."

He stayed where he was. "Checking the climate control."

"Step down. Now."

He stepped down.

The trouble with having no tools was that the room had been built on a wrong assumption about what counted as one. Mattress seam. Cot grid. Clothing zipper. Boot shank. A cracked ceramic wafer. Human bone, in a situation desperate enough. He wasn't there yet.

Across the vestibule, muffled but clear enough through the wall, Sanderson's door speaker clicked.

"Do not touch the dressing," a guard said.

"Then stop asking me to unwrap it."

Good. They were still after the chip.

Bad. They might get it.

Cavanaugh drew the wafer from his cuff and looked at it properly for the first time since the drilling site. Black ceramic, one corner cracked, a contact edge along the long side, no casing, nothing stamped on it except a maintenance mark too small to have meant anything to whoever had discarded it: *C-JUNC TEMP CACHE*. Bend it and it would snap. Swallow it and it might survive him and nothing else would. Keep it on his person and the next search would find it — Hayes had already come within a knuckle's width once.

Evidence needed custody. Custody needed a plan. He had neither.

The station shook again — short, lateral, over almost before his weight had shifted to compensate. The overhead light flared blue-white for a fraction of a second, a color shift and not a flicker. The clock behind its plastic cover jumped from 19:54 to 19:51, held there, then corrected itself to 19:55.

Clock disagreement, during conditioning.

He studied the housing. Clear plastic cover, four screws mounted outside the enclosure but accessible from inside the room — nobody had planned for a

The Core Throttling

detainee who cared about the clock. Too high to reach standing on the floor. Reachable, maybe, from the cot.

The cot was welded down.

The mattress wasn't.

He folded it lengthwise and wedged it against the wall below the clock. It slumped immediately. He tried again, forcing the lower edge between the cot frame and the wall, and it held, badly, the way a bad idea holds together for exactly as long as you need it to and no longer.

The speaker clicked. "Cavanaugh."

He didn't answer.

He stepped onto the folded mattress. It slid under him. He caught the wall with one hand, swore under his breath, and bit down on the inside of his cheek hard enough to draw blood. The bruise from the earlier corridor slam complained through his shoulder.

"Step away from the clock."

He reached the plastic cover.

No tool.

He took the wafer between thumb and forefinger and wedged its cracked corner into the head of the first screw.

Bad idea. He turned it anyway, gently.

The wafer slipped.

He turned harder.

The screw gave a quarter-turn. So did the crack in the ceramic, a little further along.

"Security to Lockdown Right."

No time to reconsider it. He backed the first screw out halfway and moved to the second. The wafer's edge chipped; black ceramic dust marked his fingers. The second screw turned once and stuck. He pried at it with the metal zipper tab from his inner fleece, bending the tab nearly flat in the process.

Footsteps, out in the vestibule.

He didn't need the whole cover off. Just a gap.

He yanked.

The plastic cracked down one side and the housing popped open far enough to show a maintenance cavity behind it — wiring, a status board, a backup battery the size of his thumb, and a narrow service void running laterally through the wall in the direction of Sanderson's room.

Not a passage. Nowhere near wide enough for a hand.

Wide enough for a wafer.

The Core Throttling

He slid it in behind the clock wiring and pushed. It traveled three inches and stopped.

The door lock clacked.

He pushed again with the bent zipper tab.

The wafer moved another inch.

The door opened.

Hayes came in fast. Cavanaugh dropped the tab into his sleeve and turned as if he'd only been steadying the broken cover.

Hayes looked at the clock. At the folded mattress. At him.

"What did you do?"

"Time was bothering me."

Hayes grabbed his collar and hauled him off the mattress. He landed badly, one knee taking his weight first, pain flashing up through the joint.

Hayes pinned him to the wall. "What did you put in there?"

"A complaint."

The second guard checked the cavity with a flashlight. Cavanaugh watched the beam cross wires, battery, cracked plastic, dark space beyond. The wafer had gone too far in to see.

Maybe far enough that a hand couldn't reach it either. Maybe not, if the guard had a hook or the patience to find one.

The guard reached in with two fingers.

Cavanaugh's stomach tightened around nothing.

Across the vestibule, Sanderson shouted, "Do not remove that dressing!"

Everyone in the room stopped.

The timing wasn't an accident. She'd heard his door open, heard the scuffle that followed, and spent her own leverage to split the room's attention in half.

Hayes swore. "Secure her!"

The guard pulled his fingers out of the clock cavity, empty.

Hayes shoved Cavanaugh toward the cot. "Sit down."

He sat.

The second guard left the clock cover hanging open and stood on the folded mattress, angling the flashlight deeper into the cavity a second time.

Sanderson's door banged open across the vestibule. Ariza's voice carried through both walls at once. "You tear that bandage and she loses the hand!"

The Core Throttling

"Hold her still," someone said.

Cavanaugh rose without deciding to.

Hayes hit him across the chest with a forearm and drove him back onto the cot.

"Sit."

He sat, because there were two guards in this room and at least two more in hers, and nothing he did here would help her there.

He listened instead. A struggle, not large. Sanderson: "It's medical packing." Ariza: "I need clean gauze." A guard: "Command wants the wrap checked." Then Pell, cutting through from farther down the vestibule: "Then check it under medical protocol. Not like a cargo pocket."

Pell.

Good. Maybe.

The speaker in both rooms came alive with Underwood.

"Sergeant Pell, report."

A pause, longer than it needed to be.

"Medical dressing disrupted during transfer. Dr. Ariza is re-securing."

"Was unauthorized media recovered?"

Another pause.

"No media recovered."

Cavanaugh exhaled through his nose, quiet, or quiet enough that he thought it was.

Hayes heard it anyway. He turned slowly.

"You two coordinate again, and I will have her moved to the upper module."

Cavanaugh looked at him. "You felt the floor tilt."

Hayes's eyes hardened.

"You felt it," Cavanaugh said. "That was conditioning. Not the override. Ask yourself why they want the two of us kept apart before the override even starts."

Hayes stepped closer. "I'm asking myself why you keep making everyone's job harder."

"Because your job is about to become impossible."

Hayes held the stare a second longer than the exchange needed.

Then the intercom sounded. "Final readiness window begins in thirty minutes. Override capacitor system at thirty percent charge."

The room light flickered again. This time it left a rectangular afterimage burned faintly on the wall —

The Core Throttling

blue-white, sharp-edged, as though the light had refreshed itself in panels rather than simply failing.

Hayes looked at it.

Cavanaugh said nothing. He didn't need to.

The second guard climbed down off the mattress.

"Clock cavity clear."

Hayes kept looking at the wall.

"Sergeant," the guard said again.

Hayes blinked. "Seal the room."

They left the mattress flat on the floor and the cracked cover hanging open above it. The door locked behind them.

A moment later, across the vestibule, Sanderson's door locked too.

Silence gathered in the space between the two rooms, the kind that had weight to it, that made him aware of his own breathing.

Then one knock, from her side.

Alive.

He answered with one knock.

The wafer sat somewhere in the wall void, three or four inches shy of wherever it needed to reach. The partial chip was still in her dressing, if Pell and Ariza had kept the search sloppy enough to fail on purpose or by accident, he couldn't tell which

and didn't need to yet. They were separated, watched, stripped of anything that looked like a tool, and roughly thirty minutes from an override Underwood meant to push through on a system that was already flinching under far less.

House arrest had done what two days of argument hadn't.

It had made the next step obvious.

Cavanaugh stood under the broken clock and studied the dark gap behind it. The wafer hadn't traveled far enough to reach her side. Not yet.

But the void ran the width of the wall, straight through, and it didn't stop at the seam between the two rooms — it kept going, the way a lot of things in this station kept going past where the blueprints said they should.

He pulled the bent zipper tab out of his sleeve, tested the edge of it against his thumb, and listened to the hum under the floor narrow into something that had started, in the last few minutes, to sound almost like it was trying to say something.

Then he went back to widening the crack.

Chapter 20: Lock Bypass

Cavanaugh learned the wall was not solid by listening to it fail.

Not loudly. The station didn't extend him the courtesy of an obvious weakness. It clicked. It hummed. It flexed a half degree and then lied about having flexed at all, the way institutional systems behaved in the hour before someone above them decided a thing was an emergency.

The clock cavity above his door backed onto a service void — a horizontal channel no wider than his wrist, threaded with low-voltage lines and a timing bus somebody should have replaced before deciding to bury command authority under four miles of ice. The cache wafer had stopped somewhere inside that channel two hours ago. It hadn't reached Sanderson. It hadn't fallen back to him. It sat there, evidence of nothing to anyone who didn't already know what it was, while below them the override capacitor system kept charging.

He stood on the folded mattress and worked the bent zipper tab into the seam around the clock housing.

The tab slipped.

His knuckles hit plastic.

Kalamazoo

He swallowed the sound before it became a word, because the camera dome in the corner couldn't register pain, but Hayes might be listening on something that could.

The room speaker clicked. Static. No voice behind it.

He held still.

Across the vestibule, through wall and sealed door, something moved in Sanderson's room. Too soft for a guard. A scrape, a pause, another scrape. Then one knock.

Alive. Working.

He answered with one knock against the clock frame.

Below the station, the hum narrowed — not louder, just closer, as if a wide sound had been made to stand in a smaller room. The light over his head shifted toward blue-white and stayed there a half second past where it should have snapped back. The clock display jumped.

FINAL READINESS WINDOW: 00:28:14

Then corrected itself.

AUTHORIZED PERSONNEL REMAIN CLEAR OF
LOWER ACCESS

He looked at that a moment. The clock no longer seemed sure whether it was a clock or a warning board, and neither option struck him as reassuring.

He jammed the zipper tab under the housing's lower edge and pried. The plastic cracked along its existing fracture line. A triangular piece broke free and dropped onto the mattress with a sound too sharp for the size of the room.

The camera clicked.

He looked at it. "Scheduled maintenance."

The speaker stayed silent.

Good or bad, silence bought time, and time was the only currency left in the room.

He reached into the cavity with two fingers. Cold wire. A film of dust. A burr of unfinished metal that opened a line across the side of his index finger before he registered it as sharp. No wafer yet. He pushed deeper, shoulder against the wall, and the bruise from the corridor slam three hours ago lit up his neck in protest.

There.

A flat edge. Ceramic.

The wafer had wedged against a cable tie.

He pinched it and pulled. It gave a quarter inch and then stopped, harder than before.

"Of course," he said, to no one.

Sanderson knocked twice through the wall.

He had no way to answer in words. No shared pipe. No vent between them. Only the structure itself, carrying knuckle against metal, and even that was a language with maybe four words in it.

He knocked three times. Wait.

Then he hooked the tab into the cable tie and sawed.

The tab bent. The tie didn't.

Footsteps entered the vestibule.

He let go of the wafer, dropped from the mattress, and sat on the cot with the particular stillness of a man who has just been moving and knows it looks wrong. Too late for that, probably. He did it anyway.

Hayes opened the door.

He looked first at Cavanaugh, then at the broken clock housing.

"You were told to leave it."

"I ignored that."

The Core Throttling

Hayes came in with one man behind him. The second guard kept his rifle slung, hand resting near the grip in the way of someone who has been told this is not yet the moment for it.

"Stand up."

He stood.

"Back wall."

He went to the back wall.

Hayes climbed onto the mattress and put his flashlight into the cavity. The beam crossed wire, loose plastic, the cut cable tie, the smear of blood Cavanaugh had left on the metal burr. Hayes angled the light deeper.

Cavanaugh watched his face instead of the light.

Hayes saw something. Whether it was the wafer itself, or only the cut insulation, or the general idea that a room can hold evidence of a man trying to leave it — Cavanaugh couldn't tell from the outside, and Hayes gave him nothing to work with.

The corridor lights flickered.

The second guard turned his head toward the vestibule — the reflex of a body that has heard something behind it its whole life and never once been wrong to check. Cavanaugh moved on that reflex, not on the light.

He shifted his weight onto the loose edge of the mattress.

It slid under Hayes.

Hayes came down off the wall — not a fall, just a landing bad enough that his left knee buckled under him. Cavanaugh didn't run first. He drove his cut hand into the cavity, hooked two fingers around the wafer, and tore it past the cable tie. The ceramic edge opened a second line across his palm on the way out. He folded his fingers over it and kept them folded as Hayes hit the floor.

The second guard turned back a half second too late. Cavanaugh put both hands into his chest and shoved him into the doorframe. Nothing clean about it. The rifle sling caught on the frame's edge and yanked the man sideways — one second bought, no more.

Hayes grabbed his sleeve.

Cavanaugh twisted out of the parka, kept the wafer crushed against his palm as the sleeve dragged clear of it, and caught the wafer in the cuff of his fleece just as the coat came away in Hayes's fist. Cold air found his arms through the thin fabric. He put his shoulder into the door and stumbled into the vestibule.

Sanderson's door stood open.

She was inside it, Ariza's medical scissors in her good hand, blood soaked through the outer wrap of the bad one. Pell filled the doorway, one arm braced across the jamb, holding back a security corporal who hadn't yet decided whether the physician, the patient, or the sergeant was the actual problem in front of him.

For half a second everyone in the vestibule saw everyone else.

Then it broke.

"Down corridor," Cavanaugh said.

Sanderson didn't ask which way. She ducked under Pell's arm, hit the vestibule wall with her shoulder, recovered badly, and ran with a limp he didn't like the look of. Ariza shoved the corporal backward with more force than her medical training accounted for.

Hayes shouted, "Lockdown breach!"

The vestibule alarm tried to answer him. It produced one stretched, low tone and then collapsed into static, and the matrix hum underneath the floor swallowed what was left of it.

Cavanaugh caught Sanderson's uninjured arm to pull her toward the junction. She shook him off before he'd finished the motion.

"I can walk."

"Good. Do it faster."

"Running and walking are different categories."

"Collapse them."

Behind them Hayes came out of the room with the parka still fisted in one hand and something worse than anger in the other. Pell stepped into his path — not a block, not defiance exactly, a hand raised like procedure still had a shape that could stop a body in motion.

"Subject injury," Pell said. "Medical risk."

Hayes nearly went through him. "Move."

"Medical risk."

"Move."

Pell moved. Too slowly to matter, but slowly enough that it would read, later, on whatever record survived this, as an attempt. That was the kind of help that lasted about ten seconds before paperwork swallowed it whole.

They reached the corridor junction. Left led to lower medical and the main lift — blocked by now, certainly. Right led into residential support: laundry, waste processing, service ladders, every system command hadn't designed with visitors in mind.

He went right. Sanderson followed.

"Where?" she asked.

"Service crawlspace."

"You have a map?"

"I have a memory of three partial diagrams and a strong institutional suspicion that nobody maintains laundry exhaust."

"That is not a map."

"It's a philosophy."

The corridor narrowed and dropped by a few degrees. Pipe runs crowded the ceiling, wrapped in foil insulation split in places to expose yellow fiber underneath. Amber lights pulsed at knee height along the floor. Every third panel carried a camera; half of them had gone dark in the last flicker, and the ones still working moved with a delay he could count.

One second.

Two.

He pulled Sanderson into a maintenance alcove just ahead of a camera's pan.

She hit the wall beside him and hissed through her teeth.

"Hand?"

"Everything."

"Can you climb?"

"I answered that already."

"I was hoping the answer had improved under gunfire."

"No one is shooting."

"Yet."

A security team crossed the corridor ahead at a run, boots hammering the deck plate, moving toward residential confinement rather than away from it — toward the room they'd already left. Hayes's report had gone somewhere useful, at least for now. Temporary, and he knew it.

He crouched at the access panel in the back of the alcove. Tamper seal. Two screws. One keyed latch.

"No tool," Sanderson said.

He held out his hand. "Scissors."

"They are medical."

"They are metal."

She passed them handle-first. The blades were too blunt and too short for the screw recess. He tried the first screw anyway. It stripped on contact.

Sanderson closed her eyes briefly. "Do not make the screw worse."

"It started worse."

"Give them back."

He did. She knelt with the bad hand held tight against her ribs and worked the scissor tip not against the screw but against the tamper seal itself, slicing the adhesive edge, peeling it back until the latch pin showed, then wedging the closed blades in behind the pin.

"Push the panel inward," she said.

"It opens outward."

"Not if the latch is pressure-bound. Push."

He pushed. The panel gave a fraction of an inch inward. She levered the pin. The latch popped, louder than either of them wanted.

Both of them went still.

No footsteps answered it.

The panel swung outward.

"Your philosophy has an access hatch," she said.

"I'm broad-minded."

Inside wasn't a passage. It was a duct chase — a cramped rectangular gap between wall skins, cable trays along the floor, pipe supports crossing the top. Warm air moved through it in stale pulses that carried lint, coolant, and something underneath both of those that had been scorched at some point and never quite finished airing out.

Sanderson looked at the opening.

"No."

"Yes."

"I said badly, not vertically compacted."

"Then be bad at it quickly."

A voice reached them from the corridor.

"Service alcove!"

He shoved the panel wider and went in first.

The chase took his shoulders badly. Metal edges pressed into ribs already sore from the corridor slam. A cable bracket caught his belt before he'd moved two feet, and he was pinned face down in warm lint for one humiliating second while Sanderson climbed in behind him.

"Move," she said.

"I am exploring constraints."

"Move."

He reached back, tore the belt free, lost a strip of skin at his hip doing it, and kept crawling. The duct magnified every breath. Every scrape of fabric on metal sounded like a tool dragged across a microphone. Behind Sanderson, the alcove panel banged against the wall.

"Stop!" A guard's voice, closer now.

Sanderson pulled the panel most of the way shut behind her — not fully closed, just enough to ruin the angle for anyone trying to see through it.

The first shot didn't come. Security still wanted them alive, for now, which meant command hadn't yet decided they were worth killing inside its own infrastructure.

He crawled. The chase turned left after six meters — he knew that much from the partial diagrams and from the heat. Warm exhaust meant laundry or environmental recovery, and every station designer who'd ever lived hated wasting a pipe run, which meant recovery connected somewhere useful below.

Behind him Sanderson's breathing had gone rough.

"Hand?" he asked.

"Ribs. Keep moving."

The chase narrowed further. A cable bundle hung lower than the last stretch. He flattened and pushed under it; it caught his shoulder, dragged along his back, lit up the bruised ribs on the way past. He heard Sanderson swear when the same bundle snagged her bandaged hand.

"Stop," she said.

He stopped.

"Caught?"

"Yes."

He couldn't turn to see it.

"Can you free it?"

"No."

Footsteps reached the alcove behind them.

Voices. A panel scraping against its frame.

He backed up two inches and caught her shoulder with his boot.

"Sorry."

"Later."

He reached behind him, blind, and found cable, then sleeve, then gauze. She made one sharp sound when his fingers found the bad hand under it.

"Sorry."

"Now is later."

He found the cable loop hooked under the bandage edge, not tight but wrong-angled. He pulled it up. It resisted — something in the wall hummed, and the cable stiffened under his fingers as if the cold, or the current, had changed inside its sheath while he held it. He pulled harder. It released.

She shoved forward.

A flashlight beam cut into the duct behind them.

"Visual!" someone shouted.

The Core Throttling

He crawled faster. The chase dropped without warning. His left hand found air where floor should have been. He nearly went down a vertical service step face first and caught himself on a pipe that turned out to be hot enough through the glove liner to register as pain before it registered as heat. He bit down on the sound and swung one leg over the edge. Ladder rungs. Bare metal. No cage around them. Three meters, at least, maybe more.

"Drop," he whispered.

"Where?"

"Ladder."

"That is not a where."

He went down two rungs, then three. Condensation slicked the one under his right boot. His foot went. He caught himself with the burned hand and saw white for a second.

Above him Sanderson reached the opening. A flashlight beam struck the top of the shaft.

"Hands visible!" a guard shouted down into it.

She looked down at him. He looked up at her.

Then the station lights went out.

Complete dark. No amber. No blue. No warning strip glow anywhere in the shaft — only the hum, suddenly wide and low under everything, and the sound of men above them

discovering that rifles, flashlights, radios, and eyes were all built for a world that was behaving less reliably by the minute.

"Now," he said.

She came down the ladder by feel. Too fast. Her boot caught his shoulder; he nearly lost the rung; she grabbed his collar with her good hand and hauled herself against the ladder rail, muttering something that started as a prayer and finished as something closer to profanity.

The lights returned at half power. The flashlight above flickered and did nothing useful with what it had.

He dropped the last meter and hit the grated landing hard enough to send pain up through both knees. She landed beside him worse, stumbling into the wall. He caught her arm before she hit the railing. For one second they stood chest to shoulder in the narrow service shaft, both of them breathing too loud for the space.

Then the speaker system came alive, distorted, a half second behind whatever had triggered it.

"Security breach in Residential Support. All lower-level access points seal immediately. Override capacitor system at forty-six percent charge."

Sanderson's head came up. "Forty-six."

"Bad?"

"At fifty, induced field contamination begins if the delivery bus is already coupled."

"English."

"They're close to the point where shutting off the hardware stops being enough."

"Then we don't go to hardware."

"No." She pushed off the wall. "We go to the matrix."

He didn't like that answer, mostly because it was the only one left standing.

Below them the shaft opened into a darker corridor lined with insulated pipe and emergency signage. One sign pointed toward LOWER OPERATIONS ENVIRONMENTAL. Another toward CAVERN ACCESS SUPPORT.

He chose the second.

Behind them, metal rang as someone entered the duct chase above.

They ran. Not well. The floor sloped; condensation had frozen in patches along the walls. Sanderson slipped once and caught herself on a steam pipe, yanking the bad hand back

before it made contact. His lungs burned on the dry recycled air, and fear had made every breath shallower than it needed to be.

A security door ahead began closing. Slowly. A heavy sliding slab with a yellow hazard stripe, descending from the ceiling as if the facility had decided it wanted to chew them in half rather than simply stop them.

He sprinted. Sanderson stayed three steps behind. He reached the door and put his shoulder under the descending edge — stupid, and necessary in equal measure. It hit his back and drove him down onto one knee. The motor ground on somewhere above. A pressure alarm shrieked. He got both hands under the slab and pushed.

"Go," he said.

She dropped and rolled under the gap, the bad hand clamped to her chest, the dressing dragging a red mark across the yellow line on the floor.

"Now you," she said, from the other side.

"I noticed."

The door pressed lower. His back screamed under it. He shifted sideways, tried to slide beneath it, and caught his belt on the threshold.

The Core Throttling

Not again.

She grabbed his wrist and pulled. The belt held. The alarm's pitch changed.

He fumbled for the buckle. His fingers slipped on it. She leaned back through the narrowing gap and drove Ariza's scissors into the strap. The blades skidded off the first pass.

"Cut it."

"I'm trying."

"Try sharper."

"They're medical scissors, not a saw."

The strap split. He slid through as the door slammed the last few inches into the floor behind him, catching the heel of his boot hard enough to tear the rubber.

For three seconds he lay on his back, staring up at the sealed door, breathing like someone he didn't fully recognize.

Sanderson sat beside him, pale, the scissors still in her hand.

"Belt loss," she said.

"Acceptable cost."

"Your boot?"

He looked down. The heel hung half torn from the sole.

"Philosophically intact."

She almost smiled. Almost.

Then the wall lights flickered blue-white again, this time in rectangular segments marching down the corridor toward them in sequence, one panel lighting and dimming before the next caught. Both of them went still.

The segments passed over their bodies like a scanner with no visible source. The hair lifted on his arms. Sanderson raised her injured hand slightly and studied the bandage, as if it might report something the walls wouldn't.

"What is that?" he asked.

"Not command hardware."

The panels went dark in sequence. The corridor returned to plain emergency amber.

Ahead, through a small reinforced window set into the next hatch, he could see a lower access lift cage and, past it, the black drop of the shaft toward the cavern.

Behind the sealed door, boots finally arrived — too late to matter, and Hayes's voice came through the metal anyway.

"Cavanaugh!"

He got to his feet with effort. Sanderson braced against the wall and did the same.

He looked at her hand. "Can you hold a cable?"

"No."

"Can you tell me where to hold one?"

"Yes."

"That may be enough."

A new alarm started somewhere under the floor, deeper than the others. The intercom stuttered and then carried Underwood's voice, thin and flattened by distance and static both.

"Lower security teams, prevent cavern access. Nonlethal authorization granted."

Sanderson looked at him.

"Nonlethal," he said.

"For now."

They crossed to the lift. On the wall beside it, the access panel flashed between two states, holding neither one long enough to mean anything:

LOCKED BY COMMAND

LOCAL FIELD ERROR

LOCKED BY COMMAND

LOCAL FIELD ERROR

He put one hand on it.

"Don't," Sanderson said.

"Why?"

"It may arc."

"Helpful timing."

He pressed the release anyway. The panel sparked, burned the pad of his fingertip, and the lift gate juddered open halfway. He shook his hand once and didn't look at it. Sanderson pushed the gate the rest of the way open with her shoulder.

Cold air breathed up out of the shaft below, carrying the hum from the matrix in a fuller register now — not a sound under the station anymore, but a pressure rising to meet whatever came down to it.

He stepped into the cage. She followed.

The gate stuck on the torn heel of his boot. He kicked it clear.

The cage began its descent before either of them had touched a control.

Above them, behind sealed doors and orders that had already stopped making sense to the systems meant to carry them, the facility went on shouting into itself.

The Core Throttling

Below, the cavern was already waiting.

Chapter 21: Descent Under Countdown

The lift cage dropped without accepting a command.

Sanderson knew because Cavanaugh tried twice. The first time the panel simply ignored his thumb. The second time it sparked, and a black crescent bloomed under his nail. He pulled his hand back, shook it once, said nothing. The cage kept falling.

"Stop touching things," she said.

"You say that like it's listening to someone else."

"It may be."

He looked at her. She hated that the answer wasn't rhetorical, that she'd meant it, that some part of her brain had already filed the lift's behavior under *consistent with prior compensation events* before she'd decided whether that category meant anything.

The shaft ran past concrete and frost-limned conduit, past structural ribs and small dark windows onto levels that were supposed to sit below human comfort and above the cavern's reach. Amber

emergency strips lined the walls. Every few seconds one lagged behind the others by a fraction of a second, leaving a rectangular afterimage hanging in the air until the eye let it go.

She held the rail with her good hand. Her bad one stayed pressed to her chest, wrapped in blood-stiffened gauze, the partial-write chip a hard seam against her palm. Every shudder of the cage drove it deeper. She had stopped being grateful for the pain somewhere around the second landing. Pain wasn't clarity. It was noise wearing a body.

Cavanaugh stood near the gate with one shoulder lower than the other. His left boot heel had come half loose and flapped when the cage bucked. He kept trying to shift his weight off it and kept failing. His eyes moved over the shaft: access labels, inspection plates, cable runs, the small printed data he'd spent two days learning to read faster than anyone wanted him to.

Still mapping. Still pretending the body attached to him wasn't in the room.

The intercom above the gate broke into static.

"—all unauthorized personnel—lower access—seal immediately—"

Then Underwood, clean the way command voices always managed to sound clean after everything else had degraded.

"Cavanaugh. Sanderson. Stop the lift at Support Level Two and await escort."

Cavanaugh glanced at the dead panel. "It's not taking requests."

"Do not proceed to cavern access."

Sanderson leaned toward the speaker grille.

"Then halt capacitor conditioning."

A pause, longer than the last one.

"Conditioning remains under command authority."

"That is not a safety response."

"No," Underwood said. "It's a command response."

The line went dead.

"She's becoming concise," Cavanaugh said.

"She was always concise. Now she's stopped pretending we're part of the process."

The cage passed a marked landing — SUPPORT LEVEL TWO — without slowing. Through the mesh she saw two security personnel with restraints and a

portable field scanner, one of them reaching for the gate as it went by. His glove brushed the wire, slid, and he stepped back fast, startled by something in the motion that wasn't his.

Cavanaugh watched the landing recede above them.

"That wasn't us."

"No."

"Facility?"

"Maybe."

"Matrix?"

"Maybe."

"You hate that word."

"I hate the accuracy."

The cage slowed at the next level. Not one she'd been cleared for. Black stencil over yellow paint: CAPACITOR SERVICE / CAVERN BUS ACCESS — AUTHORIZED FIELD PERSONNEL ONLY. The air changed before the gate finished opening — warmer, electrically dry, a faint copper taste against her back teeth, the kind of air that sat around live high-voltage equipment whether or not anything was actively discharging.

The gate stopped halfway and jammed.

Cavanaugh grabbed the mesh with both hands.

"Wait," she said.

He waited. That counted, tonight, as a form of progress.

She crouched, awkward with the bad hand, and found the lower track. Ice had grown along the inner groove — not frost, a clear hard bead line, while the surrounding metal stayed warm enough to sweat. She brought the back of two fingers near it, not onto it. Cold came off the beads in a way that didn't match the air around them.

"Thermal inversion on the track."

"Can we get through?"

"Yes. Don't touch the clear ice."

"Usually solid advice."

He put his shoulder to the gate above the frozen rail and shoved. It gave six inches, screamed against the track, stuck. She went through the gap first, bandaged hand catching the mesh, pain breaking her vision into white grain for a second. Cavanaugh caught her elbow, then let go when she looked at him.

"Sorry."

"Later."

He squeezed through after her. The torn heel snagged the bottom rail; he kicked free and left a strip of rubber behind on the floor plate.

The service level wasn't a room. It was a long gallery running parallel to the cavern delivery bus, insulated conduit bolted to the walls in bundles wide as tree trunks. Warning strobes fired at intervals overhead. They were supposed to be red. Half of them came out blue-white now, not from the bulbs but from something the light did on the way down.

At the far end, a blast door stood open onto the cavern gantry and the dark below it.

Between them and the door: six capacitor banks in two rows behind waist-high barriers, hazard symbols and classification bands stenciled across each cabinet. Charge indicators climbed in yellow increments.

49.

One screen jumped to 51, held, corrected back to 49.

Sanderson stopped walking.

"Bad?" Cavanaugh asked.

"The bus is already coupling."

"You said fifty."

"I said *if* it was already coupled."

"Is it?"

She didn't answer. The floor answered for her, a hum she felt through her boots before she heard it.

A technician in an arc-flash suit stepped from behind the second bank and froze. Fiber tether in one hand, insulated torque tool in the other, helmet shield up, eyes red-rimmed and too wide.

"You're not supposed to be here."

"Neither is the override."

His radio crackled. "Service team, report intruder location."

He looked at the radio.

Cavanaugh moved before the man chose. He crossed the floor fast enough to make the technician step back, not fast enough to look trained. The torque tool came up between them — accident, not defense — and Cavanaugh caught his wrist and drove the tool down. The technician resisted for exactly as long as it took Sanderson to reach them.

"Stop."

Both men looked at her.

"Stop making this harder," she said to the technician. "How far into conditioning?"

"I can't—"

The Core Throttling

"How far?"

His mouth worked. His eyes went to the ceiling camera.

She stepped into its field and lifted the bandaged hand.

"That camera can't tell you whether the bus is safe. I can. How far?"

The display ticked to 50. A low tone sounded through the gallery.

He swallowed. "Capacitors at fifty. Bus pre-coupled to Alpha and Beta. Gamma isolated. Delta's intermittent."

"Why is Delta intermittent?"

"Field noise."

"From the matrix?"

He nodded once.

Cavanaugh let go of his wrist. The technician didn't run. That made him either frightened or useful, and Sanderson had stopped expecting to know which until later.

"Where's the local disconnect?"

"Command locked it."

"Mechanical?"

"Behind Bank Four."

"Can it interrupt conditioning?"

"Not cleanly. Not after bus pre-couple."

"What can it do?"

"Drop Gamma. Maybe Delta, if it's not arcing."

"What happens to Alpha and Beta?"

"Command fires them, or they bleed through the dump stack."

"Dump stack capacity?"

He looked away.

She closed her eyes for half a second. "Not enough."

"No."

The wall speaker clicked on. "Service personnel, clear the gallery. Security teams inbound. Override package remains active."

The technician flinched.

Cavanaugh looked toward the open blast door.

"We need the cavern."

"We need to break the delivery path."

"Can we?"

"Not fully."

"Partially?"

She looked at the banks, the rising numbers, the Delta warning flickering out of rhythm with the hum underneath everything. Partial had become the operating

principle of the night. Partial proof. Partial access. Partial trust.

"Maybe enough to reduce the first hit."

"That sounded like a yes."

"It wasn't."

He turned to the technician. "Bank Four."

The man's eyes went to the camera again.

"They're going to fire this into the only system holding local physics together," Cavanaugh said. "Pick a side that still has a floor."

The technician looked at him, then at her. The display hit 52.

"Behind Bank Four," he said, and ran.

She followed. Cavanaugh followed her.

The floor vibrated harder near the fourth bank. Heat rolled off the cabinet in waves; the barrier rail beside it was cold enough to frost at the bolt heads. Hot metal, cold rail, warm air, cold breath — energy moving in directions no code had ever priced for.

Behind Bank Four, a narrow service space held a gray disconnect lever under a clear cover with two seals and a command lockout bar.

The technician cracked the first seal with shaking hands. "This is logged."

"Everything is logged," Cavanaugh said.

"Not like this."

Sanderson looked at the lockout bar. "Can you remove it?"

"Not without command code."

Cavanaugh pulled the bent zipper tab from his sleeve. "Define without."

"No," she said.

"It worked on the clock."

"This is high-voltage hardware."

"So, less forgiving."

The technician said, "We can't pry it. It's ceramic-interlocked. Break it wrong and the disconnect faults closed."

She looked at his torque tool. "The cover hinges."

"What?"

"Not the lock. The hinges."

He blinked, then understood, and jammed the tool into the hinge pin. It slipped. He tried again; his glove skidded. Cavanaugh braced the cover with both hands and felt vibration coming through the plastic, fast and fine, like something with a pulse.

"Don't touch the metal bracket," she said.

"I wasn't planning to."

"You were about to."

"I was exploring."

The pin moved.

A voice from the gallery entrance. "Step away from the equipment!"

Security.

The technician froze. She grabbed his wrist with her good hand and forced the tool back into the hinge. "Do not stop now."

He turned toward the shout. The pin dropped.

The cover swung open on one side, still locked on the other, enough to expose the lever. Yellow insulated handle.

MANUAL ISOLATION / GAMMA-DELTA BUS.

She reached for it.

Cavanaugh caught her sleeve. "Hand."

"Left."

"Your left is bad."

"My right is useful."

The technician said, "It takes force."

"Then me," Cavanaugh said.

"You pull when I tell you. Not before. If Delta is arcing when it drops, it can flash through the cabinet."

"Tell me something comforting."

"No."

She crouched beside the status window and used its reflection to see the arc indicator inside the cabinet. The hum narrowed, widened, narrowed. The Delta light strobed out of sync with the charge display.

54.

Security started down the gallery. Pell wasn't with them. Hayes was, and his face changed when he saw the open cover.

"Cavanaugh!"

She kept her eyes on the indicator. "Wait."

Cavanaugh put both hands on the lever. The handle vibrated under his palms.

"Wait."

Hayes ran.

The technician backed off, then didn't, staying beside the hinge with the tool still in his hand, looking like he might be sick.

The Delta light went steady.

"Now."

The Core Throttling

Cavanaugh pulled. The lever didn't move. He pulled harder, his hurt shoulder shaking, the muscle in his neck standing out in a cord. The handle shifted an inch and stopped. Hayes closed to ten meters.

Sanderson stepped in beside him and put her good shoulder under his arm, adding what weight she had. Pain went up her bandaged hand where it pressed her sternum.

"Again."

They pulled together. The lever came down with a sound like something breaking inside a wall.

Light first, not sound. A thin blue arc snapped inside Bank Four, crawled across the inner insulator, died against the dump stack with a bang that landed in her chest like a fist. She went backward into Cavanaugh. He caught her badly and they both hit the barrier rail.

The displays dropped.

GAMMA: ISOLATED. DELTA: FAULT /
BLEEDING. ALPHA: CHARGING. BETA: CHARGING.

Overall charge fell from fifty-four to forty-three.

Then it started climbing again.

The technician stared at the numbers. "You bought minutes."

"How many?" Cavanaugh asked.

"Maybe six. Maybe less. Command can reroute Alpha-Beta for initial strike."

Hayes reached them with two security personnel behind him.

"Hands up!"

The technician raised both. Cavanaugh didn't.

She looked down the gallery toward the open blast door and the dark past it. The partial disconnect hadn't stopped the override. It had cut two limbs off the first strike and left the rest of it angrier.

The matrix hum changed under the floor. Wider, if she let herself use the bad word. Stressed, if she used the better one.

Hayes's weapon came up, low and controlled.

"Step away from the bus."

"You saw the charge drop," Cavanaugh said.

"I saw sabotage."

"You saw us reduce the hit."

"I saw you break into restricted infrastructure during an active command operation."

She pushed upright against the rail. "Sergeant. If Alpha-Beta fires at full charge into a pre-stressed matrix,

the compensatory response may propagate beyond the cavern."

Hayes's eyes cut to the displays. A flicker of something. Not enough.

The speaker again, Underwood this time. "Security, detain them immediately. Engineering, restore Gamma-Delta path if possible."

The technician's whisper: "Not possible."

Hayes looked at him.

"Not before initial window."

"Then secure the remaining bus," Underwood said. "Override proceeds on Alpha-Beta."

Sanderson felt the sentence land somewhere below her ribs.

"She knows it's worse," Cavanaugh said, "and she's proceeding."

Hayes aimed at the floor between them. "Move."

The station shook. Not like the corridor tilt earlier. Bigger. The whole gallery flexed; the capacitor banks rattled in their mounts. One strobe tore loose and swung on its cable, throwing red and blue-white light across Hayes's face in a rhythm that didn't match anything electrical.

Behind him, the blast door began to close. Slowly. Command sealing the lower path from somewhere she couldn't see.

She looked at Cavanaugh. He looked at the door. Hayes caught both looks.

"Don't."

Cavanaugh moved. Hayes lunged for him, and she put herself in the gap with the only asset she had left — her own body, already broken in enough places that adding one more felt almost administrative. He checked himself before hitting her full force, some reflex of decency or training she'd remember later and not now, and the half-stop let Cavanaugh past.

She followed, because staying behind turned an interruption into a capture, and a capture turned the evidence into an autopsy.

A hand closed on the back of her shirt. She twisted. Pain tore through the bandaged hand as the hidden chip cut skin again, fresh and immediate. She swung Ariza's medical scissors backward without looking. The blunt blades hit a glove, not flesh. The grip loosened.

She ran. Badly.

The blast door was half down.

Cavanaugh reached it first and dropped to one hip, sliding under the gap. His boot heel caught and tore free, another strip of rubber left behind on the deck. She hit the floor after him with no grace at all, the impact slamming her shoulder, her bad hand flaring white behind her eyes. She shoved the scissors ahead of her and dragged herself under as the door kept closing.

Hayes shouted something she didn't catch.

Cavanaugh caught her under both arms and pulled. The door edge passed over her boot with less than an inch to spare.

It sealed behind them with a boom that cut off the gallery alarms mid-note.

For a moment there was only dark.

Then the cavern cold rose around them, different from the gallery's dry heat — a cold with weight to it, settling rather than blowing.

She rolled onto her back and stared at the ceiling of the access throat, frost growing in rectangular patches along the ribs overhead, each patch a slightly different shape, none of them explained by any airflow she could name.

The intercom inside the sealed section sputtered, failed, then delivered one last message through the door, Underwood's voice flattened by the metal between them.

"Override execution window adjusted. Alpha-Beta strike in seventeen minutes."

Cavanaugh was breathing hard beside her.

"Seventeen," he said.

She flexed the injured hand and felt the chip under the dressing, sharper now, wet with new blood.

"Then we have to reach the console in less than that."

Ahead of them the access throat opened onto the cavern gantries, the dark falling away in a way that made the ceiling lights above the throat look like the last ordinary thing in the station.

The matrix hum filled the space, no longer under them.

Around them. Through them. Waiting under every step they hadn't taken yet.

Chapter 22: Cavern Crossing

From the observation deck, the cavern had seemed enormous and contained, a hole the size of a cathedral tamed by cable runs and floodlight brackets and the general insistence of command glass. From the lower access throat, with the blast door sealed behind them and the emergency lamps guttering in their housings, none of that held. The space did not get bigger. It got rougher — ice, then rock, then scaffolding bolted into a material that should not have accepted bolts. Cable trays ran toward the exposed matrix like something with circulation. A gantry no wider than a hospital corridor crossed dark air above a floor Sanderson could not locate by sight or by sound.

The matrix filled the lower half of the world, planes of titanium-silicon interleaved at angles that made her eye keep sliding off the wrong edge. It did not glow. It absorbed the floodlight and gave back thin blue-white lines only where something in it was under load.

The hum sat in her sternum before she consciously registered it as sound.

Her boots had stopped transmitting temperature some while back. She noticed the absence secondhand, the way she noticed most things now, filed after the fact rather than in it.

Her breath came out in short columns that dissolved before they reached shoulder height, which was ordinary, and which she made herself confirm as ordinary, because down here ordinary was a thing that needed checking. The floodlights the drilling crews had rigged along the upper scaffold threw long doubled shadows across the ice, one shadow from each fixture, and where the two overlapped the ground looked briefly solid in a way she knew it wasn't.

Cavanaugh stood at the gantry gate, weight off his ruined boot, face the color of the emergency lighting.

"Seventeen minutes," he said.

"Less, now."

"Thank you."

"You wanted precision."

"I wanted hope with decimals."

She checked the panel bolted beside the gate. The readout cycled through itself.

CAVERN ACCESS: RESTRICTED FIELD
INSTABILITY: LOCAL ALPHA-BETA STRIKE:
00:15:42

The countdown dropped out mid-digit.

FIELD INSTABILITY: LOCAL

"Command killed the clock," Cavanaugh said.

"No. The panel lost authorization to show it."

"Is that a useful distinction."

"Not to us."

She pulled the scissors from her sleeve and started on the outer gauze wrapping her hand. Cavanaugh turned at the sound of the fabric.

"What are you doing."

"Securing the chip."

"It's currently inside an injury."

"It's currently inside a wet one."

Blood had softened the adhesive enough that the chip had begun to migrate under the wrap, and she could feel the edge of it now with every flex of her fingers. If it slid free on the gantry it was gone into two hundred meters of dark. If command took her before she found somewhere better, it would be found exactly where she'd put it. Neither was acceptable. She peeled the gauze back anyway.

The pain arrived clean, without the buffer the outer layer had been providing.

Cavanaugh looked away, then made himself look back.

"That's bad."

"It's small."

"Those aren't the same sentence."

She worked the chip loose from the wet inner layer — a dark flake, one bright contact edge, small enough that losing it seemed almost more likely than keeping it — and folded it into two clean strips torn from what gauze remained dry. The bundle went into the torn seam at her collar, where the parka lining had split during the crawl through the vent.

"Better?" Cavanaugh asked.

"Less terrible."

"I'll take upward trends where I find them."

Something struck metal on the far side of the sealed door behind them, twice, followed by a voice cut off before it became words. Security had reached the barrier. They would go around it or through it, and the facility had a great many locks and a great many people paid to defeat them.

Sanderson took the gantry rail with her good hand. It was warm on top. Frost rimed the underside.

She let go.

"What," Cavanaugh said.

"Don't trust the rail."

"That was the whole plan."

"New plan."

They started across.

The gantry moved under their weight in a way it had not been built to move — not sway, not the ordinary give of steel under load, but a delay, a half-beat where the metal seemed to check with something else before agreeing to bend. Cavanaugh compensated badly, favoring the torn boot. Sanderson compensated badly in the other direction, one hand useless at her side, the cavern floor somewhere her inner ear refused to place.

The far path split. Left toward the diagnostic platform and the two command-installed contact points, Alpha and Beta. Right, along the older excavation line, where bare rock met the matrix without any clean boundary between them.

The console was down that way. The site plans didn't call it a console. They called it a nonhuman interface structure, four words chosen by someone who understood that naming a thing honestly gave it a kind of standing.

Cavanaugh pointed left. "Alpha-Beta?"

"Contact points. Not the console."

"Right, then."

"Right."

A line of blue-white light crossed the matrix below them, and the gantry dropped.

Not far. Two centimeters, maybe. Enough that both of them grabbed for the rail neither of them was supposed to trust. Sanderson caught herself before her hand closed on it. Cavanaugh didn't. His palm hit warm metal and slid on a film of condensation that had no business forming in air this cold, and he went down hard on one knee.

The grating under him bowed.

"Don't move," she said.

"Strongly considering it."

The bow deepened by degrees and then stopped.

She crouched and looked through the grating at the support bracket underneath. It hadn't broken. It had thinned — not eroded, not physically reduced, but wrong to look at, its edge trading places with shadow in a rhythm that matched the hum almost exactly.

The cart wheel in the prior file had done something like this. So had the concrete under it, according to the report she hadn't been allowed to keep a copy of.

"Shift left," she said.

"Toward the rail?"

"Toward the main beam. Slow."

He moved. The grating held until his weight crossed onto the beam, and then the bowed section snapped flat with a crack that carried into the dark on either side of them.

Someone shouted, muffled, past the sealed door.

Cavanaugh got up. "One of your failure modes."

"Yes."

"Contained?"

"For now."

"You need new words for that."

They kept going. She logged the bracket's location against the rest of the gantry — third support from the gate, on the side nearer the older path — without writing it anywhere, because there was nothing left to write on that command hadn't already confiscated or that the crawl through the vent hadn't already ruined. Memory would have to serve as the ledger. It was a bad ledger. It was the one she had.

Cavanaugh limped a half-step behind her the rest of the way to the stairs, favoring the boot in a rhythm she'd started to predict without meaning to.

The older path descended by a switchback of steps cut into a scaffold frame anchored partly to human-drilled rock and partly, without any engineer having signed off on it, to protrusions of the matrix itself. The clamps sat against the nonhuman material without a scratch mark anywhere on either surface. The steel around them had grown frost in patches too regular to be weather — small rectangles, evenly spaced, as if something had measured out where cold was permitted to form and where it wasn't.

She kept one hand on the scaffold rail where the scaffold rail was actually scaffold, and let go of it wherever the rail crossed onto matrix. Cavanaugh copied her without asking why, which she registered and did not comment on. His breathing had changed since the gantry — shallower, timed to his good leg rather than to any need for air.

She counted steps, because looking down too long made her balance unreliable in ways that had nothing to do with fatigue.

Seven. A landing. Nine. A landing. Eleven.

At the second landing the lights failed.

Not all at once, and not gradually — the human floodlights first, then the emergency strips, then every small indicator light either of them had on their bodies: the zipper pull on Cavanaugh's jacket, the dead ghost of her tablet's standby glow, the service labels stenciled along the scaffold rail. All of it went dark in the same instant.

The matrix did not.

It wasn't illuminated. It was defined. Blue-white stress lines traced the edges of planes ahead of them that no light source had been showing a moment before — cable runs overhead, dust suspended mid-fall, and Cavanaugh's breath, which had stopped in front of his face and held its shape.

"Don't move," she whispered.

He froze on the step below her.

The cloud of his breath hung there, refusing to disperse or fall, its outer edge resolving into a faint grid.

Then the air seemed to forget what shape it had been holding.

The vapor collapsed into ordinary mist. The lights returned.

Cavanaugh's voice came out very quiet. "Was that a blink."

"Local reconciliation lag."

"Better?"

"No."

She looked down the remaining flight. Three landings below, a wrench lay half-swallowed by the metal grating, its handle standing clear, its jaw buried through the floor as if the two objects had negotiated a settlement and then frozen mid-signature. No one had dropped a tool here since the diagnostic spike, as far as either of them knew.

Cavanaugh saw it at the same time she did.

"From the prior event?"

"Or ours."

"Can we use it?"

The question was absurd enough that her first instinct was to say no by reflex. She looked again instead.

The wrench was real. Steel, ordinary, the jaw threaded through the grating near a maintenance hatch handle. If they could free it without losing fingers, it was the first tool either of them had held since the scissors.

"Maybe."

They descended to the landing. The handle was cold enough to fog faintly in the warmer air of the lower

cavern. She crouched beside it and studied the grating where it met the jaw. No bend. No visible deformation anywhere in the metal. It had simply overlapped the tool and then reconsidered, the way the bracket upstairs had reconsidered its own shape.

Cavanaugh reached for it.

"Stop."

He stopped an inch short.

"If it reconciles while your hand's on it, you lose skin at best."

"At worst?"

"You don't want the technical answer."

"Correct."

She tapped the handle with the closed scissors. It rang — an ordinary sound, ordinary recoil, the handle shifting a fraction under the blow. She hit it again, harder, and the jaw moved inside the grating and the whole landing creaked under them.

"Not good," Cavanaugh said.

"It may release under a load change."

"Here?"

"Carefully."

"Your favorite word."

She moved to the far edge of the landing, weight over the support beam. Cavanaugh took the opposite side. The landing tilted less than a degree. The wrench handle quivered against the grating.

"Now," she said.

He kicked the handle sideways with the torn heel of his boot. The wrench snapped free and the grating closed behind it with a sound like teeth meeting.

Cavanaugh stumbled into the rail. It held.

The wrench skidded to a stop against her boot. She lifted it with the scissor tips first, waited through a full second in which nothing happened, and then wrapped it in a strip of bandage before taking it in her hand.

"Tool acquired," Cavanaugh said.

"Don't call it progress."

"It is progress."

"It's also evidence of matter-cohesion failure under active load."

"Can it be both."

"Yes."

"Then I'll take both."

They kept descending. The wrench hung from a bandage loop at her belt now, knocking against her hip with a small dead weight that felt, absurdly, like reassurance. She had carried worse tools into worse readings. She had never carried one that had needed to be argued out of the floor.

The next flight was shorter, six steps to a landing barely wide enough for the two of them standing shoulder to shoulder. Cavanaugh's flashlight — confiscated, then somehow not confiscated after the vent crawl — threw a weak cone across standing frost that had grown in ridges along the grating, each ridge the width of a finger, spaced with the same unnatural regularity as the patches on the clamps above. She crouched and touched one with a gloved knuckle. It didn't melt. It didn't feel cold enough to justify existing.

"Anything?" Cavanaugh asked.

"Frost that isn't behaving like frost."

"Is that new?"

"No. It's just the first time we've had a light steady enough to see it."

He didn't answer that. She hadn't expected him to.

Below them the matrix filled enough of the cavern that distance stopped meaning anything reliable. Some of its surfaces

mirrored the gantry back at them as a distorted black line. Others carried ridges too fine to focus on, the kind that made her eyes water within seconds of trying. The hum changed by location — widening near bare rock, narrowing near the command cable runs, thinning into something closer to pain along the approach to Alpha and Beta.

"Listen," she said.

Cavanaugh stopped.

"What."

"The contact path is stressing it already."

"I hear hum."

"You hear one thing. I hear load moving."

"That sounded arrogant."

"It was accurate."

He let that stand, which told her something about how far they'd come from the observation deck.

She stopped at a joint where two of the matrix's planes met without a seam, close enough to touch, and didn't. The hum here had a texture she hadn't logged anywhere in six weeks of readings — not a pitch so much as a pressure that changed depending on where she stood relative to the join, stronger a half-step to the

left, weaker a half-step to the right, as though the load moving through the structure had a shape and she was walking through the edge of it. She had no instrument left to confirm that. She had her own body, which was not a calibrated instrument, and which she distrusted for exactly that reason even as she used it.

"We should keep moving," Cavanaugh said.

"I know."

She didn't move for another three seconds, and he let her have them.

A status monitor mounted ahead of them flickered on.

ALPHA-BETA CHARGE: 61% STRIKE WINDOW:
00:10:38 GAMMA-DELTA: ISOLATED / FAULT BLEED

Their interruption upstairs had bought them under seven minutes.

"Can the console stop Alpha-Beta?" Cavanaugh asked.

"I don't know."

"Can it tell us what Alpha-Beta will do?"

"Maybe."

"Worst maybe yet."

She moved faster.

The path narrowed where the human scaffold ended and the older structure began. What came next was not a bridge in any sense an engineer would recognize — a broad rib of matrix

material emerging from the cavern wall and descending into the larger mass below, a maintenance rope bolted along one side by people who had wanted a handhold without wanting to call the surface underneath it a walkway.

No rail. No grating. Nothing to say the rib cared one way or the other whether they crossed it.

Cavanaugh stopped at the edge. "This is the way."

"Yes."

"To the console."

"Yes."

"Was there another route?"

"Probably, before command routed everything through the contact gantries."

"Comforting."

"You asked for the map."

"I regret asking."

She stepped onto the rib.

The surface resisted her boot with a delay she hadn't felt on the gantry — fainter, more intimate, as though pressure traveled somewhere before it decided how to answer. She kept her weight centered and did not

look down. The rope beside her trembled without any wind to move it.

Cavanaugh followed, one hand hovering near the rope without touching it, his torn boot flapping softly with every step. In the scale of the cavern the sound was obscene.

She kept her eyes on a fixed point ahead — a seam in the rib's surface, dark against dark, that didn't move when she blinked and so served as something to hold her balance against. Below the rope line, where the rib widened into the larger mass, the matrix carried faint parallel ridges like the ones on the frost above, evenly spaced, as though whatever process laid down the frost patterns and whatever process shaped this surface answered to the same instruction. She had no way to test that. She filed it as a question instead of an answer, the way she'd been trying to file everything since the diagnostic spike, and kept walking.

A radio crackled somewhere behind the sealed door, too distorted to carry words, cut off before it resolved into anything they could use. Underwood's people, or Underwood herself. It didn't matter which. What mattered was that the sound meant a door, somewhere, had opened.

Halfway across, the matrix beneath them lit blue-white.

She felt it before she saw it: a vertical drop in temperature, then heat coming back up from somewhere below, then pressure behind her eyes like a fast change in altitude. She stopped. Cavanaugh stopped a half-step too late and caught her shoulder.

The rib ahead resolved into panels — not physically, but visually, each one refreshing a fraction of a second after the one before it, a grid of blue-white edges marching toward their feet.

"Back?" he asked.

"No."

"Forward?"

"No."

"Excellent."

The grid reached a loose maintenance bolt lying on the rib three meters out. The bolt blurred, briefly doubled, offset by the width of a finger, and then one of the two versions of it dropped away. The remaining bolt fell flat with a sharp tick against the surface.

"Reconciliation front," she said.

"Can we cross it?"

"After it passes."

"How do you know that?"

"I don't."

The grid reached her boots. Cold traveled through the soles and up through bone into her hips. The wound under the bandage burned as if the chip in her collar had gone hot on its own. Her vision broke into dark rectangles at the edges, each one refreshing a beat behind the one beside it.

For one instant Cavanaugh was not beside her.

Then he was.

His face hadn't changed expression. It had changed position — half a step farther back than it had been, his eyes wide with the look of a man who had felt something happen to him without being able to name it.

"Did you move?" she asked.

"No."

The grid passed. Sound came back a beat late — the hum, then breath, then the distant alarm behind the sealed door, then, last of all, her own pulse.

She looked at the thin white line that had appeared along his jaw. Not a cut. Not frostbite. A mark like the kind pressure left against glass.

"You blinked," she said.

"I noticed."

"Pain?"

"Define pain."

"Later."

They crossed the rest of the span.

The console stood in a concave recess larger than a house, where the rib entered the greater mass of the matrix and the human-installed barriers and warning tape looked almost offensive against the material behind them. It was not a console in any sense the word implied. No keyboard, no screen — a waist-high plane of dark material ridged in nested arcs, and above it a shallow field of suspended dust that would not settle no matter how long the air around it stayed still.

Portable terminals sat near the barrier, all dead. Fiber tethers ran from them back toward the sealed door. One had been cut, or burned. Another simply vanished into the matrix surface without a connector at the end of it.

"This is how we talk to it?" Cavanaugh asked.

"No," she said. "This is how someone already tried."

A cracked helmet lay near the tape. Old — dust settled inside the fracture, a smear of insulating foam

gone brittle along the rim, a grid pattern etched into the visor from the inside out. Evidence of prior contact, unfiled or buried in one of the archives Cavanaugh wasn't supposed to have opened.

Sanderson crouched beside it without touching it yet. No tag. No evidence marker, no chain-of-custody sticker of the kind command applied to everything it wanted accounted for later. Whoever had worn this had either come before the paperwork existed, or after someone had decided the paperwork would be a liability.

"Nobody logged this," Cavanaugh said, reading the same absence she was.

"No."

"Which means command doesn't know it's here, or—"

"Or does, and left it."

He looked at her. "Which."

"I don't know yet."

She lifted the helmet by the chin strap, turned it once under Cavanaugh's flashlight, and set it back exactly where it had lain. Not evidence she could carry. Evidence she could only witness, for now, and remember the position of.

She stepped past the tape.

The console reacted before she touched it. The suspended dust arranged itself into squared, offset layers, a three-dimensional error pattern hanging above the dark plane. The chip against her collar warmed. The wound in her hand pulsed in time with it.

Cavanaugh saw her flinch. "Hand?"

"Data."

"What?"

"The chip. It's reacting."

"To the console?"

"To the field, maybe. Proximity, maybe. The raw sequence, maybe. I don't know."

Behind them the monitor updated through a burst of static.

ALPHA-BETA CHARGE: 68% STRIKE
WINDOW: 00:08:02

Cavanaugh pulled the cache wafer from his cuff — cracked, black, barely bigger than the chip against her collar, marked along one edge with his own blood from when the ceramic had cut him during the escape.

"Mine too," he said.

A thin blue line moved along the broken edge of it.

She looked at the console, the arranged dust, the two damaged fragments of proof that neither the facility nor whatever built this place had wanted them to keep. The matrix wasn't asking anything of them. It had no language for asking. But the fragments in their hands had come out of its own failures — its stress responses, its attempts to reconcile what the station had done to it with whatever reality it was trying to maintain underneath. They weren't keys. That would have been too easy.

They were matching scars.

She thought, briefly and without deciding to, of the report she'd filed six weeks ago on thermal equilibrium and the eleven-page silence that had followed it. Nobody had needed her to be wrong. They had only needed her to stop saying it out loud. Down here, with a piece of herself wrapped against a piece of Cavanaugh's blood, saying it out loud didn't feel like the point anymore.

Cavanaugh watched her weigh the wafer in her palm. "If this does something we can't undo—"

"Then it does something we can't undo whether we try or not. Alpha-Beta doesn't wait for us to finish deciding."

"That wasn't a yes."

"It wasn't a no."

"Give it to me," she said, and held out her good hand.

Cavanaugh hesitated only long enough to check her face, then set the wafer in her palm.

She pressed it against the bandaged chip at her collar and stepped toward the console.

The hum climbed until it stopped being sound and became pressure instead.

Somewhere above them, past the sealed access path, the first capacitor discharge warning began.

Chapter 23: Direct Interface

The first warning did not sound like an alarm. It sounded like something in the walls remembering how to be afraid.

Sanderson felt it before she heard it, a tone too low for the cavern's speakers and too even to be the hum they had all stopped noticing days ago. The portable terminals ringing the interface recess woke in sequence, screens flaring pale blue, then died again, one after another, as if the floor beneath them had withdrawn permission. Across the cavern the Alpha and Beta charge lamps burned yellow through a haze of frost and distance.

She stood at the edge of the dark plane with Cavanaugh's cache wafer closed in one fist. The partial-write chip sat against her collarbone, wedged under torn fabric where she'd shoved it an hour ago without thinking about whether that was a good place to keep a piece of evidence.

Both fragments were warm.

Not dangerously warm. Not yet. The heat sat in them the way held breath sits in a chest.

Cavanaugh's eyes were on the charge lamps. "How long?"

"Less than eight minutes."

"That's not an answer."

"It's the only honest one I have."

The console occupied the center of the recess, if console was even the right word for architecture that had grown rather than been built: a waist-high shelf of the same titanium-silicon body as the cavern walls, its surface interrupted by ridges that resolved into nothing recognizable as controls and too much sense as boundaries. Dust hung above it in thin, ordered layers. Each layer shifted a fraction when she exhaled.

She kept her hands at her sides.

The last time anyone had forced this thing to answer, Kline's wrist had come apart at an angle wrists did not go, and gravity in the recess had turned negotiable for four full seconds. She knew more now. Knowledge didn't make it safer. It removed the excuse of not knowing.

Somewhere above them, buried in the pack she'd dropped at the mouth of the recess, the diagnostic rig sat dead, its housing cracked along one seam from the fall on the gantry stair. She hadn't stopped to check it. There had been no version of the last hour where stopping

made sense. What she had instead was two damaged fragments of memory and whatever the console decided to do with them, and that inventory felt thin measured against eight minutes.

Cold came up through the recess floor in a way that had nothing to do with any reading a sensor would give her. She had stopped trusting her own sense of cold three days ago, after the first diagnostic spike made her skin report frost where the instruments recorded nothing. She trusted it now only as information about her own body. Not about the room.

"What does contact do," Cavanaugh said.

"I don't know."

"What do you expect."

"Expectation is how you design a bad experiment."

"What do you fear."

She looked at the dark plane a moment longer than she needed to. "That it reads the fragments as stress records and tries to reconcile them through us."

"That was specific."

"It's a better class of ignorance than the alternative."

Another tone rolled through the recess, and the suspended dust flattened into a square that ignored the curve of the ceiling above it. For half a second the air between her and

the console looked gridded — subdivided, transparent, wrong. Then the panes folded into each other and were gone.

Cavanaugh's jaw tightened. "That wasn't command hardware."

"No."

"Then it knows the strike is coming."

"Or the strike path is already loading into it."

She unwrapped the chip from her collar. Blood had wicked into the gauze overnight, dried at the edges, tacky in the middle. She wiped the contact edge once against her sleeve, leaving a rust-colored smear along one corner. Cavanaugh took the used gauze from her without asking and wound it around two fingers, pressing until the cold set it before the wet could spread.

"You don't have to do that," she said.

"Yes, I do."

She looked at him. Bruised along one cheekbone, one boot split at the sole, still turning his head toward the dead terminals as though there were still routes to read in a place where routes had stopped meaning anything. He wasn't calm. He didn't look like a man performing calm for her benefit either.

"Immediate goal," she said.

"What?"

"Immediate goal. We're not commanding this thing and we're not explaining ourselves to it. One action: interrupt or damp the Alpha-Beta coupling before the strike lands."

"Good. Obstacle?"

"It may not be built for human contact. Command may fire before we understand what we're touching. And it may protect its own stability over ours, if those turn out to be different things."

"Evidence state?"

"Partial raw timing sequence. Partial reality-blink cache. Both corrupted. Both tied to prior stress events."

He let out a short breath that wasn't quite a laugh. "Are you writing a report?"

"I'm keeping us from pretending this is a plan."

"Fair."

She set the wafer and the chip side by side on the dark plane.

Nothing happened.

The warning tone kept going, patient in a way that felt worse than urgency would have.

"Good nothing?" Cavanaugh asked.

"Don't assign it a moral value."

The two fragments sat there, absurd against the scale of everything around them — human storage media, a cracked corner, a bloodied contact groove, objects that by every rule she understood should have needed human hardware to mean anything at all.

The surface changed under them.

Not light. Texture. The ridges rose and sank by microns and then by millimeters, the give of something like black liquid being held in a shape it didn't want. The wafer tipped. The chip buzzed faintly against the plane. Overhead, the dust dropped in a single sheet and then stopped an inch above the console, suspended.

She made herself not step back.

The fragments sank. Only halfway — the interface didn't take them so much as close around their lower edges, holding both upright in slots that hadn't existed a breath earlier.

"That's either very good or very bad,"
Cavanaugh said.

"Yes."

A pattern started in the dust above them. At first it read as noise — layered squares, gaps, blocks missing

The Core Throttling

where they shouldn't be. Then it rotated without any of it moving, the way a shape rotates when the fault is in your own eyes and not in the object. Lines connected the chip to the wafer, the wafer to the ridges of the console, the ridges down into the body of the matrix through pulses that traveled inward rather than across.

The air pressure dropped. Her ears popped once, hard.

The pain in her hand sharpened into something with edges.

She caught the console's rim with her uninjured hand before her knees gave. The surface was colder than any metal she'd touched and warmer than her own skin, both at once, the contradiction arriving as sensation a full second before her mind caught up to name it. Her fingers stuck to the ridge — not frozen. Recognized.

Cavanaugh's hand closed on her shoulder. "Sanderson."

"Don't pull me."

"You're shaking."

"Don't pull me."

The dust snapped into three dimensions.

It wasn't a hologram — there was no light source for one — just particles arranged into a volumetric map of the cavern that kept expanding past the walls of the cavern itself.

Rock. Ice. Station levels stacked overhead. The gantries. The capacitor banks. Beneath all of it, a thermal path running down into crustal gradients she had no instrument capable of confirming. Around the matrix's outer boundary, a thin shell marked by small interruptions, like static caught mid-collapse.

The map wasn't built for her. It was a system using whatever local matter it could recruit to carry shape, and she happened to be standing in the way of it.

She saw the heat move first.

That probably saved her from standing there in awe.

Deep rock fed thermal exchange into the matrix — not the way a power plant converted heat into work, but as a balancing channel, heat going in and something she didn't have a word for coming back out, if out even applied to a body that didn't emit. Load rose whenever the local environment diverged from whatever baseline the matrix maintained. It spent time and precision and energy holding the physical behavior around it inside some tolerance she couldn't see the edges of.

She was looking at throttling. Not the label. The shape of it. Processing channels narrowing. Error

regions triaged, the small discrepancies allowed to sit for seconds or minutes or, for all she knew, years, while the larger divergences got handled first. The reality blinks weren't attacks. They were late corrections, arriving after the fact because the fact had outrun the correction.

She tried to say some of that out loud.

Nothing came out.

Cavanaugh's grip tightened. "Talk to me."

The map lurched. The Alpha and Beta contact points brightened — not with light, with density, like two crude needles reaching from command's capacitor line toward the matrix's outer shell. The representation tightened around them, routing capacity toward the expected strike. Everywhere else in the map, small error nodes dimmed.

Prioritize. Throttle. Reconcile.

She understood enough of it to be afraid in a specific, unhelpful way.

"It's reallocating," she said.

"Because of the strike?"

"Before the strike. To survive it."

"Can it?"

She looked at the deep thermal path. It was narrowing — not because the rock had gone cold, but because the

exchange itself couldn't scale fast enough to meet what was being asked of it. Computation was finite. Heat was finite. Correction was finite. Local precision was the thing that got sacrificed first, before anything collapsed outright.

"No."

The word left her quietly, almost apologetically, as if she owed the machine something for the answer.

The interface responded to it.

The dust map collapsed into a single vertical column — not symbols exactly, though her mind kept trying to read them that way, because language was what a human brain reached for when consequences got too large to hold as pure behavior. She made herself refuse the translation and watch the behavior instead.

One sequence showed the Alpha-Beta strike entering at full profile.

The map shed coherence outward from its edges. Lights failed across the station. Gravity variance spread through the cavern. Support structures, gantries, cable anchors, and — rendered only as heat shapes — human bodies, all showed the same failure mode: matter losing its cohesion in patches that grew. Then the

The Core Throttling

representation jumped outward past the ice, past the ocean, through markers she recognized as regional grid infrastructure. Not an explosion. Not a blast radius. A cascading loss of local correction while the matrix tried, and failed, to hold a boundary larger than itself.

A second sequence showed Gamma-Delta isolated and Alpha-Beta reduced. Smaller collapse. Still catastrophic in the cavern. Still, by any reasonable estimate, fatal to everyone in it. But the deep thermal path stayed attached. The matrix held. Regional propagation dropped to something the map treated as survivable.

A third sequence showed no strike at all. The matrix kept throttling. The error nodes persisted, unresolved, the countdown lines sliding outward rather than closing. Delayed failure. Not solved. Deferred.

None of the three sequences resolved into anything she could call good. She catalogued them anyway, the way she'd catalogue three failed calibration runs — not because one of them was usable, but because ruling out two of three was still information, and information was the only currency she had left that hadn't already been spent.

"What is it showing?" Cavanaugh asked.

"Outcomes."

"Can we choose one?"

"I don't know."

The warning tone changed pitch, climbing into a register that made her back teeth ache. Across the cavern the charge lamps flared brighter. Somewhere behind the sealed access path, a wall speaker cut through static in fragments:

"— Alpha-Beta alignment — final — hold personnel clear —"

Cavanaugh's head turned toward the gantries.

"They're aligning."

The dust map answered by routing more density toward Alpha and Beta.

She pressed her uninjured hand harder against the console's edge. Pain shot up through her palm where the microscopic ridges bit into skin she hadn't realized was already broken.

"No," she said. Not to command. Not to the thing under her hand. To the situation itself, in general, the way you'd say it to weather.

The map shivered. The partial chip's embedded edge brightened. The wafer followed a half-second later. Their two corrupted histories surfaced inside the larger

map as small, distinct threads: pre-execution response, thermal rise, hum narrowing, local drift on one side; prior reality blinks, matter-cohesion failure, the Iceland coupling, override scenarios on the other.

The interface had read the scars. Or it was using them as coordinates. She wasn't sure there was a meaningful difference.

Cavanaugh's eyes moved across the map faster than hers. "There."

"What."

He pointed without touching. "Routing. The strike package isn't hardwired straight to Alpha-Beta."

"That's command software. You can't read command software off a dust cloud."

"No — look under it. The physical route. The bus path's waiting on a local timing handshake. It needs a response from the matrix to phase-lock before it fires."

She followed his finger. He was right in the way analysts were sometimes right — not because he understood the physics underneath it, but because he could see one sequence waiting on another with a piece of institutional stupidity wedged between them. Command's strike needed a phase lock so the discharge would land inside what they believed was a response window. Without the handshake, the strike would dump badly — locally,

through the stack, less efficiently into the matrix, she genuinely didn't know which.

"Can we disrupt the handshake?" he asked.

"Maybe."

"How?"

The map didn't answer that directly. It took her another second to notice what she'd been avoiding noticing, because she hadn't wanted to turn it into language too soon: the interface was displaying the phase-lock signal as a narrow timing band tied to the station's corrected clocks. The same clocks that had failed during the diagnostic spike. The same eight-tenths of a second Cavanaugh had found and command had quietly normalized out of the record.

"They need agreement," she said.

"Who does?"

"Command hardware and the matrix's response model. The strike's scheduled against a predicted window. If the timing disagreement resurfaces —"

"The strike misses the window."

"Or dumps somewhere safer."

"Safer how?"

"Less directed."

"Less directed into what?"

"The dump stack, if we're fortunate."

"That's a heroic amount of *if we're fortunate*."

"Do you have a better option?"

"No."

The ridges under her hand shifted again. A new structure formed in the dust: a timing chain. Station clock. Command mirror. Local cavern sensors. Matrix response. Capacitor gate. The weak link wasn't the capacitor at all — it was the reconciliation layer command had bolted on after the diagnostic spike. Clock domain reconciliation. The maintenance outage. Junction C. All the small fictions that let the station agree with itself.

Cavanaugh got there before she said it. "We break time again."

"Not time. Their agreement about time."

"I'll take the correction if we live long enough to argue about it."

The map traced a pathway back through the cache wafer. Field event cache. Timebase error. Reality-blink archive. A record of the station once failing to agree with itself. The wafer wasn't a key. It was proof that disagreement had already happened, once, and left a mark. If they forced that mark into

the current timing chain, the matrix might refuse a clean phase lock rather than grant one.

Her hand had started to go numb.

The surface had climbed over her fingers without her noticing when.

She looked down. It wasn't swallowing her hand — not exactly. Dark ridges had formed around her fingertips and the side of her palm, holding rather than pulling. Fine blue-white lines moved beneath her skin, not across it, tracing something that might have been blood vessels or might have been nerves or might have been neither.

"Pull back," Cavanaugh said.

"Not yet."

"Your hand."

"I know."

"Sanderson."

"If I break contact now, we lose the map."

"If you don't, we might lose you."

She looked at him. "This is the first accurate model we've had of any of this."

"And the worst possible reason to die."

That nearly broke her concentration, because he wasn't wrong, and she didn't have an answer ready.

The interface pulsed. The map narrowed back to the timing chain and lit two actions inside it.

First: hold the wafer's cache path against the console long enough to insert the old timebase disagreement into the active phase-lock query.

Second: route the raw thermal sequence from her chip in as proof that command's execution timestamp was false.

Translated badly into anything human: make command's system distrust its own clock at the exact moment it needed to be certain.

She could do the second part. Cavanaugh, maybe, could do the first.

She reached for the wafer to lift it clear.

It didn't move.

The console held it embedded, upright, unyielding.

"Of course," she said.

"What."

"It needs both fragments. Not one and then the other. Both, together."

"Can it copy them instead?"

"It might already be doing something worse than copying."

She tried anyway. She got two fingers under the wafer's cracked corner and pulled, gently at first and then with her full weight behind it, until the effort made the wound in her hand flare bright enough to blur her vision at the edges. The wafer didn't shift a fraction of a millimeter. It was, she understood distantly, like trying to pull a tooth that had already decided to stay.

The material around the fragments rose. The wafer cracked, audibly, a sound like a knuckle popping in the cold. Cavanaugh flinched at it. The crack filled with blue-white light. The chip's edge did the same a beat later. Dust compressed over both of them into a thin, dense bar.

Then the console itself changed.

A ridge below the fragments opened, and what lay underneath wasn't wiring or circuitry but something layered, thinner than foil, too deep to resolve by eye. Perfectly black. Cold radiated out of it with the flat authority of a vacuum chamber door.

A contact point. Built for nothing with hands.

She understood what it wanted before her thoughts had finished forming the sentence, because her skin understood it first: continuity. The fragments, the raw sequence, the current state of the system, and a human body willing to hold the interpretation open long enough to choose which contradiction mattered.

She hated it. Hated the shape of what was being asked of her, the way it dressed itself up as something almost mythic — a machine turning a person into a working part of itself.

Then she looked again at the alternative sitting in the dust map. Full strike. Local collapse. Regional propagation. The matrix's survival uncertain. Human survival not even a variable in the equation.

She put her injured hand over the open contact point.

Cavanaugh caught her wrist. "No."

"Yes."

"It's already hurt."

"Then I've got fewer decisions left to make."

"That's terrible logic."

"It's field logic."

"Joanne."

Her name in his mouth made the cavern briefly, sharply less abstract than it had been a second before.

She held his eyes. "Hold me upright."

"That's the plan?"

"No. That's your part of it."

The warning tone jumped again. Behind them, a half-dead command terminal blinked awake long enough to show a single line — **STRIKE WINDOW: 00:03:12** — before dying again.

Cavanaugh let go of her wrist. He moved behind her instead, both hands braced around her upper arms, not restraining, not pulling, just there, a support she could reject if she needed to.

She lowered her hand.

The contact point took the bandage first. She felt the blood in it freeze on contact, a sensation like a held breath turning solid. Then the chip warmed hard against her collar, the wafer split the rest of the way through, and the pain stopped being pain and became something closer to geometry.

She didn't scream right away. Her body seemed to have forgotten, for a moment, how.

Cold ran through the wound first, then heat, then pressure with no direction she could name. Blue-white lines climbed under the skin of her hand. The

chip's raw sequence unfolded through her — not as memory, as order: heat, hum, drift, not pulse, heat, hum, drift, false timestamp, correction layer, command lie. The wafer answered with what Cavanaugh had carried this far: reality blink, a cart wheel spinning through solid concrete, a grid's micro-collapse, the Iceland coupling, an override folder, deployable scenarios, metadata timestamped before the test it claimed to record.

The matrix didn't understand their anger. She was fairly sure of that much. It understood contradiction.

She pushed the contradiction into the timing chain.

Command clock: execute.

Local acquisition: not yet.

Station reconciliation: agree.

Raw sequence: false.

Reality-blink cache: this disagreement existed before.

The interface amplified all of it at once.

The cavern lights failed. Cavanaugh's hands tightened on her arms hard enough to leave marks she'd find later. The entire station seemed to drop eight-tenths of a second out from under itself — not literally, there was no clean way to think that sentence, but sound arrived before the vibration that should have caused it, her breath returned to her chest before she'd finished exhaling it, and the warning light across the cavern

Kalamazoo

flashed a beat after its reflection had already appeared on the console's surface.

The dead terminal behind them screamed back to life.

**TIMEBASE FAULT PHASE LOCK
FAILED ALPHA-BETA GATE HOLD**

She tried to pull her hand back.

The interface didn't let go.

"It worked," Cavanaugh said.

"No."

The contact point opened deeper. The dust map expanded past the cavern, past Antarctica, past anything she could still hold as ordinary geography. For one second that felt much longer than a second, she saw not a planet but a resource boundary — a managed volume, a local sector, correction routes, allocation channels, maintenance queues, human history rendered as a set of constraints running inside something with no apparent interest in names or nations or grief.

Then a new signal entered the map. Not from command. Not from the station.

External instruction.

The Core Throttling

The matrix reacted the way a machine without fear still manages to convey terror: reallocation, purge, defragment, reduce sector load.

Her knees gave.

Cavanaugh held her up.

The interface released her hand a half-second later, almost as an afterthought. She fell back against him, dragging blood and frost out of the contact point with her. The chip was gone. The wafer was gone. Her bandage had burned away in a neat rectangle over the wound, the skin beneath it pale and dry and, worst of all, without pain.

Across the cavern, the Alpha and Beta charge lamps dropped from yellow to red.

Not discharged.

Held.

One of the dead command terminals rebooted on its own. A single line appeared on the screen.

**EXTERNAL MAINTENANCE INSTRUCTION
RECEIVED**

Cavanaugh read it out loud. His voice had gone thin.

She could barely stand on her own. Her hand had stopped hurting entirely, and that frightened her considerably more than the pain had.

Kalamazoo

The terminal updated again.

**LOCAL NODE DEFERRAL REQUEST:
DENIED**

Then:

**DEFAGMENTATION CYCLE:
INITIALIZING**

Beneath them, the hum changed. It had been a warning a minute ago. Now it sounded, unmistakably, like work.

Sanderson stared at the terminal until the letters stopped rearranging themselves in her vision. Her hand, when she finally looked at it directly, showed a rectangle of skin gone the color of old paper, dry at the edges where the bandage had burned away, and she made herself note that fact the way she would have noted it on someone else's chart. Not hers yet. Just a fact.

The charge lamps stayed red. Held, not fired, not stood down — a state the station's own signage had no color for.

Cavanaugh's hands were still on her arms. Neither of them moved to change that.

"We didn't stop it," she said.

"No."

"We changed what it's doing instead."

He didn't answer that. Above the recess, somewhere in the dark stack of levels between them and the surface, something in the matrix's body had already begun to work through whatever instruction had just arrived from outside the station, outside the mountain, outside anything either of them had clearance to imagine.

Chapter 24: Unredacted Terminal

The interface recess still smelled like the phase-lock discharge — ozone and scorched insulation, thinner now but not gone, sitting under the recycled air the way a fire left its trace in a building long after the smoke cleared. The portable rail lighting had failed on one side of the cavern sometime in the last several minutes, and nobody had been in a position to notice exactly when.

The words on the terminal did not resolve into meaning for three full seconds.
DEFRAGMENTATION CYCLE: INITIALIZING sat on the glass the way foreign alphabets sat on glass — recognizable as symbols, useless as language.
Cavanaugh's mind kept sliding off it.

Then Sanderson's weight dropped against his side, and the sentence stopped mattering.

He got an arm under hers before her knees finished buckling. "Joanne."

"I'm conscious."

"That was not the most reassuring version of that sentence."

"Hand."

He looked down. The bandage was gone — not torn free, not burned away in any pattern fire or friction produced. Cut cleanly out, a neat rectangle of gauze simply absent, the skin beneath left to account for its own exposure. The gash across her thumb had stopped bleeding. Under the surface of the skin around it ran a thin blue-white branching, faint as frost forming inside glass rather than on it.

"Don't look at it like that," she said.

"How should I look at it."

"Later."

"Everything is later."

He didn't press it further. There was nowhere to put a bandage back even if he'd had one — the kit was two levels up, behind a door that no longer answered to either of their credentials. He noted the condition of her hand the way he noted anything he couldn't act on yet, location and state without a conclusion attached, and kept his arm where it was.

The cavern hum changed register. Not louder. Busier — the difference a server room made when load shifted from idle to something with priorities. Around them, black material threaded with blue-white lines moved in nested sequences, and Cavanaugh felt the movement land in his molars and the small joints of his hands. Sort. Route. Discard. Preserve. He didn't

Kalamazoo

have proof for those verbs. He had the sensation of a process working under a deadline, and the sensation was enough to write itself into his body without his permission.

Across the cavern the Alpha and Beta contact lights still burned red.

Held. Not the same as safe. The phase lock they'd broken upstairs had bought the two indicators a kind of stalemate, nothing more — no green, no confirmed disengagement, just red holding steady in a way that hadn't been true twenty minutes ago. He'd take it. He didn't trust it.

The nearest command terminal cycled through a second reboot. The Cold Horizon seal assembled, fractured into square blocks mid-render, and dropped into a diagnostic shell Cavanaugh had never held clearance to see, not at any tier, not on any system he'd touched in eleven years of reading other people's classification decisions for a living.

No compliance skin. No command wrapper. No redaction bars. He'd spent his career learning to read the space around information as carefully as the information itself — what a form omitted, which fields a requester

wasn't permitted to see, the particular blankness of a line item that had been scrubbed rather than simply never filled in. This screen had none of that architecture. Nothing here had been built to be looked at by someone like him, which was, he understood distantly, the reason it was readable at all.

A file tree opened on its own.

LOCAL NODE STATUS EXTERNAL
ALLOCATION SECTOR MAINTENANCE DEFERRAL
HISTORY HUMAN-READABLE BRIDGE

He read the last line twice before saying it. "Human-readable bridge."

Sanderson's head came up. Her face had gone the gray of something left too long outside its rated temperature range. "Read it."

"You should sit down first."

"Read it."

He lowered her against the base of the interface plane instead of arguing further. She caught the edge with her good hand and pulled herself upright against it, breathing through her teeth. The injured hand stayed in her lap, fingers curled slightly inward, the branching under the skin brightening and dimming on no schedule he could find.

He turned back to the screen.

A prompt blinked above the tree.

DISPLAY TRANSLATION LAYER:

DEGRADED SOURCE MODEL: LOCAL NODE /

LIMITED REPRESENTATION WARNING:

SEMANTIC LOSS HIGH

Then the branches opened.

LOCAL NODE STATUS came first.

NODE DESIGNATION: CH-PRIMARY

FUNCTION: LOCAL PHYSICAL-STATE

RECONCILIATION SCOPE: ANTARCTIC SECTOR

/ SOUTHERN HEMISPHERIC BOUNDARY

SUPPORT / TEMPORAL HISTORY CONTINUITY

ROLE: ERROR CORRECTION / PARAMETER

STABILIZATION / RESOURCE MEDIATION

CENTRAL AUTHORITY: NOT LOCAL

He read it once for the words and again for the shape of what wasn't there. No mention of weapons systems. No mention of power generation. No claim to anything resembling a brain for the whole of the operation. Every briefing he'd sat through since Iceland had built toward one of those three categories or another, stacked assumptions on top of assumptions because the alternative had been to admit no one in the

building knew what they were looking at. None of the assumptions were on this screen.

"It's a local node," he said, because the cavern needed a human voice on the record even if no one but Sanderson could hear it.

She had her eyes shut. "Not central."

"No."

"Read the scope line again."

He did.

"Temporal history continuity," she said.

"That sounds like—"

"Do not translate too far."

"I wasn't going to."

"You were already halfway there."

He had been. His mind wanted a story where the data wanted to stay flat: timelines, simulations, history managed the way a duty officer managed a shift log. The thought was too large to hold cleanly and too dangerous to make into a sentence he could say out loud twice. He put it down and looked back at the screen instead of at the shape of it.

EXTERNAL ALLOCATION opened next.

RESOURCE CLASS: PROCESSING / THERMAL
EXCHANGE / PRECISION BUDGET CURRENT

STATUS: REDUCED ALLOCATION REDUCTION

SOURCE: EXTERNAL LOCAL REQUESTS:

THROTTLE EXCEPTIONS / DENIED PRIORITY

UPDATE: SECTOR LOAD MINIMIZATION

MAINTENANCE MODE: FORCED

A line beneath it flickered, corrected itself,
flickered again.

REASON: GLOBAL RESOURCE

REALLOCATION / LOCAL SECTOR

EXPENDABILITY ELEVATED

Something moved through Cavanaugh that the
cavern's temperature had nothing to do with.

"Expendability elevated," he said.

Sanderson's mouth went thin. "That is not a
scientific category."

"It's a translation."

"It is still obscene."

He read the block again, slower. Reduced
allocation. Throttle exceptions denied. Every anomaly
Sanderson had logged since they'd arrived — the heat
sink readings that didn't add up, the acoustic dampening
under load, the clocks in the lower cavern disagreeing
with the clocks two decks up by margins too small to

explain and too consistent to ignore — sat under this one line like footnotes finally attached to a source. Not malfunction. Not damage. A budget, cut from somewhere else, and a system trying and failing to get an exception approved.

DEFERRAL HISTORY opened.

LOCAL NODE DEFERRAL REQUESTS: 9,448
OLDEST ACTIVE REQUEST: 11,706 YEARS MOST
RECENT REQUEST: 00:00:41 AGO PRIMARY BASIS:
SENTIENT CIVILIZATION DEVELOPMENT /
OBSERVER CONTINUITY / HIGH-COMPLEXITY
CULTURAL STATE EXTERNAL RESPONSE: DEFERRAL
TEMPORARY / RESOURCE NONCOMPLIANT

He stopped breathing for a count he didn't track.

Sentient civilization development. Observer continuity.
High-complexity cultural state.

Not affection. Not mercy, not in any shape a person would recognize walking into it. But not indifference either — something had been asking, on some schedule that made eleven thousand years look like a rounding error, for more time. The thing under the ice had not been resting. It had been in a negotiation it was losing by inches, for longer than any human institution had existed to notice.

Eleven thousand years put the oldest request somewhere before the earliest cities he'd studied in a survey course two decades ago, back when he'd still believed the past was a fixed quantity rather than a line item under review. He didn't know what to do with that comparison and set it down before it could turn into something he'd have to defend later.

Forty-one seconds. The most recent request had gone out forty-one seconds before the screen rendered it. He looked at the terminal clock, then at Sanderson, then back at the number, and didn't say it aloud.

Sanderson said it before he could. "It's been buying time."

He looked at her. The branching under her skin brightened once, held, faded. He turned back to the terminal because looking at her any longer would let the scale of the thing get into him in a way he didn't think he could use yet.

SECTOR MAINTENANCE opened.

MAINTENANCE ACTION:

DEFRAGMENTATION / LOAD REDUCTION /
ERROR MAP CONSOLIDATION LOCAL
CONSEQUENCE: LOSS OF LOW-PRIORITY

DETAIL / CONTINUITY COLLAPSE / PHYSICAL-
STATE REWRITE HUMAN IMPACT: NONSURVIVABLE
IN AFFECTED REGIONS AFFECTED REGIONS:
UNSTABLE / EXPANDING INITIAL WINDOW: 29 DAYS
17 HOURS 12 MINUTES ACCELERATION FACTOR:
HUMAN OVERRIDE ATTEMPT / FORCED PHASE
DISRUPTION / ALPHA-BETA COUPLING

His hands went cold against the terminal casing.

Twenty-nine days. Not collapse tonight. A window, closing on a schedule that had apparently existed before either of them walked into this cavern, and their partial success upstairs had not kept it from opening. It had changed the shape of the phase lock. It had held the strike. It had also, according to the line in front of him, fed the acceleration factor.

He read INITIAL WINDOW again, then ACCELERATION FACTOR, then back, trying to find a reading where the two lines didn't sit next to each other the way they sat next to each other. He couldn't. The number had been twenty-nine days before they touched anything tonight, and it was still twenty-nine days now, and somewhere in the difference between those two facts was a version of events he didn't want to be the one who'd written it.

"We caused part of it," he said, before he'd decided to say anything.

"No." Sanderson's answer came out hard, immediate.

"Human override attempt. Forced phase disruption. Alpha-Beta coupling."

"Underwood caused the coupling."

"We broke the phase lock."

"To stop the strike."

"It still accelerated."

"If a man fires into a pressure vessel, and you knock the barrel aside before it punctures the core, you did not cause the damage because the ricochet tripped an alarm."

He looked at her. She was gray, unsteady, angry enough to stay conscious through it.

"Don't take command's arithmetic into your blood," she said.

"That's not what this is."

"It's exactly what this is. Underwood will tell you the same math in a different order tomorrow, and you'll believe her less, and that will not make you correct now."

He didn't have an answer that survived contact with that. He let it sit for one breath before the cavern moved again.

Not a quake. A sorting motion — far below, the blue-white lines shifted in bulk, and patches of the matrix surface went dark in sequence, as if something had withdrawn its attention or its current from those sections in a specific order. The air pressure in the recess changed enough to register in his ears. Dust lifted from a ledge, hung, and settled.

A terminal at the far end of the cluster woke.

Then another.

Then every dead command screen in the recess lit with the same tree, no redactions on any of them.

Sanderson saw it happen. "It's using their terminals."

"Or their terminals are inside its fault field."

"Does the difference help us."

"No."

He crossed to the second terminal. HUMAN-READABLE BRIDGE sat at the top with a scrolling error log underneath it.

TRANSLATION GENERATED FROM LOCAL
HUMAN SYSTEMS MODEL LIMITATIONS SEVERE
ONTOLOGICAL MAP INCOMPLETE CAUSAL

LANGUAGE APPROXIMATE AGENCY

LANGUAGE APPROXIMATE TIME LANGUAGE

APPROXIMATE

"That's a remarkable disclaimer," he said.

Five lines admitting the translation in front of him might not mean what it appeared to mean, stacked one under the other, in a system that had shown no other interest in qualifying itself. He didn't know what to do with a machine that hedged. Every human process he'd ever audited hedged in the opposite direction — certainty inflated, caveats buried in footnotes nobody read. This read like the opposite habit, and he had no shelf to put that on either.

Sanderson braced her good hand against the interface plane and tried to stand. He crossed back and took her weight without asking. She let him take it. That was worse than if she'd fought him.

"Copy what you can," she said.

"With what."

She looked at the row of terminals. Every one of them ran on command infrastructure. Every one kept logs somewhere command could reach and purge. No drives sat loose on the consoles. No tablet, no chip, no

wafer. Whatever proof existed had gone into the interface itself, and the interface answered to nobody in this room.

He looked down at the cluster instead of at the screens. Old habit surfaced, the one that had kept him employed for over a decade: panic accomplished nothing, but architecture always left a shape somewhere, even when the content was gone. Every terminal wrote crash logs to a partition. Every display update passed through a buffer. Every sealed command system kept a maintenance spool whether anyone remembered to check it or not. Not a copy. A scar. A scar would be enough, if they lived long enough to point at it.

It was the same instinct that had gotten him this far in the first place — not evidence, not proof, just the residue proof left behind when someone tried to take it away. A form filed under the wrong retention schedule. A timestamp that didn't match the narrative built around it. He'd spent a career reading absences. This cavern was the first time an absence had been actively manufactured in front of him while he watched.

He opened the local system shell.

It let him in.

Maybe command authorization had collapsed with the rest of the network. Maybe the node underneath had no concept

Kalamazoo

of clearance levels to enforce. Maybe the terminal itself was already dying and no longer cared who touched it.

The shell listed storage partitions.

COLDHORIZON_OS

COMMAND_MIRROR

FIELD_TERMINAL_CACHE PRINTER_SPOOL

Printer spool.

He opened it. Empty. He almost laughed before he saw the device list under it: three disabled thermal label printers, one plotter wired to a survey package, and one incident-report printer physically installed in the command room above them, offline since the outage that had started this.

He couldn't print anything. He didn't need paper. He needed the spool to register an attempt.

He routed the open file tree to the plotter.

The terminal hesitated.

PRINT DEVICE UNAVAILABLE QUEUE

JOB? Y/N

He typed yes.

JOB QUEUED SPOOL ID: CH-PLOT-7791

Sanderson watched him from the floor. "That doesn't save the file."

The Core Throttling

"It saves the attempt."

"Is that enough?"

"No idea."

He queued LOCAL NODE STATUS. CH-PLOT-7792.
EXTERNAL ALLOCATION. CH-PLOT-7793. DEFERRAL
HISTORY. CH-PLOT-7794. SECTOR MAINTENANCE.

The terminal froze on the fifth job for two full seconds
before accepting it.

CH-PLOT-7795.

A line appeared under the queue.

REMOTE MIRROR RESTORE ATTEMPT
DETECTED COMMAND PURGE PENDING

"Time," Sanderson said.

"How much?"

"Not the kind I can give you."

The screens around them began to flicker in a shared
rhythm. Command authority was reasserting itself through the
dead systems — not a body walking into the room, but software
climbing back along the edges of whatever fault had let it out in
the first place. Two screens threw up red banners.

UNAUTHORIZED DISPLAY STATE PURGE IN
PROGRESS

A third screen joined them. Then a fourth, faster than the first two, as if the process had located its own rhythm and no longer needed to relearn it terminal by terminal. Cavanaugh didn't have a count of how many seconds separated the first purge warning from the last. He didn't have time to make one.

Cavanaugh opened the spool directory and started renaming the queued jobs. Not with anything that pointed at itself. Not node status, not maintenance, not deferral. He typed incident language instead, the kind that could sit in a system for years without drawing a second look.

PLATFORM_STABILITY_DEVIATION_AP
PENDIX_A
MEDICAL_ACKNOWLEDGMENT_KLINE
ENVIRONMENTAL_CONDENSATION_REPORT
FIELD_UTILITY_REVIEW_VISUALS

He copied each spool ID into the next file's metadata notes, one to the next, a chain instead of five loose links. If someone purged the content, maybe the names would survive somewhere in a log of what had once existed. If the names survived, maybe someone with his particular habits could find the absence later and

follow it back. Absence had edges if you left enough of them lying around.

He typed fast, aware of how thin the logic was even as he built it. A retention audit eighteen months from now, run by someone bored enough to notice five job IDs clustered in the same nine-second window under four unrelated department headers. A clerk cross-referencing a medical acknowledgment against a name — Kline — that didn't correspond to anyone on the current roster. It wasn't proof. It wasn't even a lead, not yet. It was a set of small inconsistencies scattered where an inconsistency-hunter might eventually trip over them, and he was the only kind of person he could think of who might actually go looking.

The first screen went black.

The second followed it.

The third started to go.

Sanderson's hand closed on his shoulder. "Enough."

"One more."

"No."

"One more."

He opened the human-readable bridge disclaimer and queued it under a title that would mean nothing to anyone auditing the spool later.

COFFEE_INVENTORY_ROTATION

Sanderson looked at him.

"Nobody reads those," he said.

The terminal accepted it. CH-PLOT-7796.

Then it purged every visible file at once.

The command seal snapped back onto all six screens in the same instant. For half a second the cavern looked ordinary again — locked terminals, classification banners, warning tape strung along the rail, command infrastructure arranged to suggest it had never lost the room.

Then the screens failed completely, one after another, in a line down the cluster.

Behind them, the sealed blast door groaned.

Security was cutting through, or command had found a bypass from the other side. Either way, time had stopped being an abstraction and turned back into something with a door attached to it.

He turned toward Sanderson. "Can you move?"

"Yes."

"You're lying."

"Yes."

He put her good arm over his shoulder and took the rest of her weight. She didn't fight the position. That, too, was a bad sign, filed next to the others he didn't have time to sort yet — the missing bandage, the branching under her skin, the color that hadn't come back into her face since the interface first took hold of her. He couldn't do anything about any of it down here. He filed it anyway, because filing it was the only thing he knew how to do with a fact he couldn't act on.

The nearest terminal rebooted a final time — not into the command shell, not into the Cold Horizon seal, but into a stripped maintenance diagnostic with one line still lit.

DEFRAGMENTATION CYCLE: 29D 17H 09M

The minutes were already running.

Sanderson saw it too. "It isn't ending here." She blinked, as if the sentence had come from somewhere other than her own mouth.

"What?"

"Nothing. The countdown."

He didn't push it. The words had landed with a texture that didn't belong to her voice, and he had no spare capacity to examine that right now.

The blast door groaned again. A line of white light widened along its lower edge.

Underwood's voice came through a distorted speaker, thin but level. "Cavanaugh. Sanderson. Step away from the interface. Medical and security teams are entering."

Medical first. Security second. Somewhere in the last several minutes, Underwood had learned which word to put in front, and Cavanaugh registered the ordering the way he registered any change in how an institution chose to phrase itself.

He looked at the dead screens. No file tree left visible on any of them. No fragments. Only five queued ghosts sitting under false names in a dying command system, and whatever the matrix had left running under Sanderson's skin.

He called toward the widening gap in the door. "You need to halt all override activity."

A pause, then Underwood's answer. "The override is suspended."

Sanderson lifted her head. "Suspended."

"Pending review."

Cavanaugh had read enough forms like the one that phrase implied to know its shape before anyone wrote a word on it. Suspended, in the language

The Core Throttling

Underwood used for a living, meant a switch left within reach.
Not thrown.

He almost closed his eyes at that.

The door opened another inch.

Sanderson said, quiet enough that it barely reached him,
"She still thinks it's her decision to make."

The cavern hum shifted again, deeper in the rock than before. Around them, the blue-white lines withdrew further into the matrix's lower layers, away from the human contact points, away from Alpha and Beta, away from the interface recess where the two of them stood. The node was reallocating itself under instruction from somewhere neither of them could see, pulling back from every surface a person could touch.

Cavanaugh watched the last visible thread sink below the surface of the wall nearest him and not come back. He counted three full seconds after it disappeared before the wall went dark the way ordinary rock went dark, and he had no framework for what to do with three seconds like that except to notice them.

Leaving. Or getting ready.

He tightened his arm around Sanderson's waist.

Beside them, the terminal clock — dead a moment before — flickered back on and corrected itself to a number neither of their bodies wanted to trust.

Kalamazoo

29D 17H 08M

The display went black.

Boots on the far side of the door now, more than one set, moving at the pace of people who'd been briefed on a hazard they didn't fully believe in yet. A gurney wheel caught on the door track and someone swore at it, low and practiced. Cavanaugh heard Sanderson's name inside the noise before he heard his own.

The blast door opened, and the light came in white and full, and neither of them moved toward it first.

Chapter 25: The Grid in the Sky

The blast door opened wide enough for guns first.

Then floodlight. Then people.

Security came in low, spreading across the access throat in the overcontrolled movements of men who had been told nonlethal and heard maybe anyway. Medical followed two steps behind them, Ariza at the front with a trauma bag in one hand and something harder in the other. Colonel Underwood came in last, without a weapon raised, without slowing for the men who had theirs up.

Cavanaugh kept his arm around Sanderson.

Her weight had changed since the interface. Not physically. She leaned into him with the discipline of someone who had decided that collapse could be rescheduled. Her injured hand hung uncovered from the knuckles down, the blue-white branching under the skin faint now, but there.

Every muzzle in the room found that hand.

Ariza saw it and swore under his breath.

Underwood saw it and said nothing.

"Step away from the interface."

Cavanaugh looked back at the dark plane behind them. The fragments were gone. The dust had settled. The suspended

map, the node status, the unredacted terminal tree, the external allocation lines — all of it had folded back into black material and dead screens. The interface looked inert enough to write a report about.

"That's your problem," he said.

Hayes moved in from Underwood's left, weapon low, face pale with exhaustion and something unresolved. Pell wasn't with the entry team. If Underwood noticed that the evidence-intake sergeant had dropped out of the visible chain of custody, she didn't spend a word on it here.

"Hands visible."

Cavanaugh lifted the one that was free.

Sanderson lifted her injured hand an inch and nearly went down.

Ariza pushed past Hayes. "Move."

Hayes didn't.

Ariza looked at him. "Block me again, Sergeant, and I will write the cleanest medical complaint this continent has ever seen."

Hayes stepped aside.

Ariza reached Sanderson and stopped short of the injured hand, two fingers hovering near her palm

The Core Throttling

without contact. For the first time since Cavanaugh had met him, the medic looked uncertain.

"Doctor. Can you feel this?"

"Pressure. Cold. No pain."

"That's not good."

"I know."

He opened the trauma bag.

Underwood's gaze stayed on the dead terminal cluster.

"What did it show you?"

Cavanaugh laughed once, badly. "That's your first question?"

"It's the one that matters now."

"No. The first one that matters is whether every capacitor path is dead."

Underwood didn't answer that.

Sanderson's head came up. Cavanaugh felt the movement through his shoulder.

"Colonel."

"The Alpha-Beta strike was suspended after phase-lock failure."

"Suspended."

"Yes."

"Not cancelled."

"Systems are being safed."

"By whom."

"Engineering."

Sanderson said, "If they're safing through command lockout, they're preserving the option to fire."

Rane came up behind the security line, helmet off, hair flattened with sweat. He looked smaller without the presentation screen behind him. Smaller and more useful.

"Gamma-Delta's still isolated," he said. "Alpha-Beta gate's held open but charged. Dump stack's unstable. If we kill it wrong we could get an uncontrolled bleed through the contact bus."

Sanderson closed her eyes. "How much charge."

"Seventy-one percent peak. Decaying. Sixty-eight now."

"Bleed profile."

"Ugly."

"That's not a profile."

"No," Rane said. "It's an honest preliminary word."

Underwood looked at him. "Can you secure it?"

His eyes went from Underwood to Sanderson's hand, then to the dead interface. "Not alone."

The answer changed the room, and not because Rane had defied her. Because he'd finally said out loud what had been true for hours: command was no longer the organizing intelligence inside Cold Horizon. The facility had power, locks, ranks, weapons, procedure. It didn't have enough understanding to spend any of it correctly.

Sanderson tried to step forward. Her knee folded.

Cavanaugh caught her.

"No," Ariza said.

Sanderson pushed at Cavanaugh's arm with her good hand. "If Alpha-Beta bleeds into the bus after the node's started defragmentation routing, it may accelerate the maintenance cycle further."

Underwood's mouth tightened. "Further than what."

Nobody answered that either.

Cavanaugh looked at the terminal cluster. Black screens, all of them. He'd queued the unredacted lines to dead spool jobs under false names. Maybe they still existed somewhere. Maybe a command purge had already eaten them. Maybe the local node had left enough scarring in the system that deletion would leave an edge to find later. Maybe none of it would matter if

Underwood restored the override architecture before anyone above her understood what had already been done in this room.

He said, "Twenty-nine days."

Underwood turned to him.

"The maintenance cycle begins in twenty-nine days and change," he said. "External instruction. Local deferral denied. The override attempt accelerated it."

The cavern seemed to hold still around that sentence.

Underwood looked at Sanderson, not at him.

"Confirm."

Sanderson's answer came through clenched teeth. "The terminal translated it as defragmentation. Load reduction. Error-map consolidation. Human impact nonsurvivable in affected regions. Affected regions unstable. Expanding."

One of the younger security men made a sound he probably hadn't meant to make.

Underwood absorbed it. Cavanaugh watched her do it — not denial, not belief. Integration. The dangerous kind. She would add the countdown to her map of responsibility and keep looking for a lever.

"Evidence?"

Cavanaugh almost admired the reflex.

"Gone from the visible terminals. Possibly preserved in spool attempts. Possibly not."

"Then we have testimony."

"We have Sanderson's hand."

Ariza looked up sharply. "You don't use patients as evidence."

"I'm not," Cavanaugh said. "It's still true."

Sanderson said, "Rane needs to dump Alpha-Beta through the stack without re-coupling to the matrix."

Rane nodded once. "And without trusting station time."

"Clock domain reconciliation is compromised," Cavanaugh said. "Use local analog delay. No phase window."

Rane stared at him.

"What," Cavanaugh said. "I listen."

Rane looked at Underwood. "I can rig a dirty bleed through physical isolation if I get authority over command gating."

"Risk."

"Dump stack failure. Fire. Local arc. Damage to whatever bus is left."

"Matrix coupling?"

"Less likely than doing it clean."

Underwood took one second too long.

Sanderson heard it. "Colonel."

"I'm deciding which failure I can still put my name under."

"You can't control this by preserving the thing that caused it."

Rane said, "Ma'am. Clean control is the dangerous option here."

That landed harder than Sanderson's warning had, because it came from inside the machinery of command. It also gave Underwood something belief hadn't: a defensible local basis to disobey a remote order without calling it disobedience.

Underwood gave one nod. "Do it."

Rane moved before anyone could reinterpret the order.

Hayes looked at Underwood. "Subjects?"

Underwood's eyes stayed on Rane as he crossed toward the terminal cluster. "Medical custody. Armed escort. They stay until the bus is safe."

"Command said separate confinement."

"I'm command on site."

The words carried weight. The ceiling speaker proved they were contested a half second later.

A remote voice cut through, distorted by whatever the station's damage had done to it. "Colonel Underwood, suspend local deviations and secure the consultants pending senior review."

Underwood didn't look up. "Senior review is late."

"Repeat order. Secure the consultants. Preserve override assets."

Rane stopped halfway to the service panel.

Cavanaugh watched the old chain of command reach for the room and fail to close a hand around it.

Underwood keyed her throat mic. "Override assets are the hazard. Local command proceeding with emergency safing."

"Colonel, you are not authorized to —"

The speaker snapped into static.

Not off.

Overwhelmed.

The hum moved through the cavern in a broad pressure wave that made every rifle strap tremble against its owner's shoulder.

Underwood looked at the dead speaker. Then at Rane.

"Proceed."

Rane ran.

The next seven minutes were not a victory. They were work.

Rane and two engineers stripped the command gate out of the Alpha-Beta loop with insulated tools and shouted fragments of instruction that didn't sound like instruction so much as triage. Sanderson sat on a portable equipment case because Ariza and Cavanaugh had put her there and refused to let her stand, and gave direction in a voice that thinned with every sentence. Analog delay. No station clock. Don't wait for phase alignment. Bleed before the next hum narrowing. Watch the rail frost, not the digital indicator.

Cavanaugh did what he was best at and hated most. He translated between systems that didn't trust each other. He told Rane when Sanderson meant no because she'd gone too quiet to say it with any force. He told Underwood which phrases from the remote channel were traps built to preserve control rather than solve anything. He told Hayes to move his men off the contact path before their boots became conductive evidence of a mistake nobody would survive long enough to write up.

The Core Throttling

Hayes obeyed the last one.

That mattered. Not enough to absolve him. Enough to put men where they would live.

When Rane finally opened the dirty bleed, the capacitor banks screamed.

The sound came through the sealed blast door as a tearing howl that started electric and turned mechanical partway through, like something with a voice deciding it needed a different one. Light flashed through the access throat — not blue-white matrix light. Human light. Arc flash. A fire suppression system triggered somewhere above them, choked on a pressure fluctuation, restarted in bursts.

The matrix responded.

Blue-white lines rose across its planes and withdrew — not toward Alpha and Beta, but deeper, away from every point of human contact. The hum widened once, painfully, then dropped into a register Sanderson felt through Cavanaugh's arm before he heard it himself.

Rane's voice came over a handheld radio, half swallowed by static. "Alpha-Beta charge falling. Fifty-two. Forty-eight. Dump stack holding. Fire in Bank Two containment. No contact surge."

Sanderson closed her eyes.

Cavanaugh didn't let himself feel anything yet.

"Defragmentation?" Underwood asked.

No terminal answered.

Then the dead screen nearest the interface flickered.

DEFRAGMENTATION CYCLE: 29D 16H

51M

Still moving.

Sanderson opened her eyes.

Underwood stared at the display the way a person stares at a number on a form they've already signed.

"The cycle continues," she said.

"Yes."

"Can it be stopped?"

"I don't know."

"Can it be delayed?"

"I don't know."

Underwood hated the answer. Cavanaugh could see that much — not because it challenged her authority, but because honest uncertainty was unusable in the systems that had built her. She had to act without knowing. Everyone in the cavern did. The difference

was that the price of pretending otherwise was now printed on every dead screen in the room.

The cavern shook again, weaker but wider.

Dust sifted down from the rock overhead.

Not a collapse. Not yet.

Ariza looked up. "We need to move her."

Underwood nodded. "Evacuate lower cavern."

Sanderson said, "No."

Cavanaugh bent toward her. "This is one of the times no means yes."

"The interface —"

"The interface is done with us."

She looked at it anyway. The dark plane gave nothing back. Not dormant. Withdrawing, the way a light withdraws when the current behind it is needed somewhere else.

Cavanaugh helped Ariza lift her. She bit down on a sound when her injured hand brushed the front of her parka. Hayes moved in on her other side, hands not making contact until Ariza snapped, "Support under the elbow, not the wrist." Hayes did exactly that and nothing more.

They crossed back over the matrix rib in a line that would have looked absurd from above: medic, injured physicist, disgraced analyst, armed guard, commander, engineers, security

team, all of them stepping with unnatural care over the stretch of floor where human certainty had thinned to something you could see through if the light hit it right. No one used the rope. No one trusted the rail. Twice the floor panels refreshed their light in neat squares, and everyone stopped both times until Sanderson said, "Move," and they did.

At the access throat, Cavanaugh looked back once.

The interface recess was already hard to make out. Floodlights failed around it in staggered rectangles, not all at once, not in any order he could read as a fault pattern. The console receded into dark not because it had moved, but because the local node no longer had reason to spend whatever it took to make itself legible to the people standing near it.

The station carried them upward under emergency evacuation lighting that had its own failures baked in — a strip out here, a strip out there, the corridor going from white to amber to dark in patches that didn't correspond to anything Cavanaugh could tie to load or damage.

Not the main lift. That was locked and drifting between time states, according to an engineer who sounded close to the edge of something. They took a service cage past smoke-hazed capacitor levels, past medical teams working in masks, past security checkpoints where no one seemed sure whether Cavanaugh and Sanderson were detainees, witnesses, saboteurs, or simply the only two people in the building who understood enough to be dangerous.

At lower medical, Ariza tried to stop.

Sanderson refused.

"Surface," she said.

"You need treatment."

"I need sky."

Ariza looked at Cavanaugh like this were somehow his doing.

"She means it scientifically," Cavanaugh said.

"I do not," Sanderson said.

That settled it more than any argument would have.

Underwood authorized the surface transfer. No one questioned her where Cavanaugh could hear it.

The ride up through the ice took too long and no time at all. The shaft walls slid past in bands of white, black, steel, frost, the cage rattling at a frequency that seemed to change without

the machinery changing speed. The hum faded by layers but never fully went out. Sanderson sat on the floor of the lift because standing had become a kind of performance and she'd run out of the energy performance required. Cavanaugh sat beside her. Hayes stood across from them, rifle slung, eyes on the floor plates.

After a while, Hayes said, "The cart wheel."

Cavanaugh looked up.

Hayes didn't meet his eyes. "The file you opened. It was a cart wheel, wasn't it."

"Yes."

Hayes nodded once. Not apology. Not confession. A record entering another mind, filed there without ceremony, the way most true things got filed once you stopped waiting for them to feel important.

The lift reached the surface airlock.

They came out into a storm that had paused without clearing. Antarctic night held the station in hard blue-black. Floodlights burned along the perimeter, catching blowing ice crystals that moved sideways in the wind like static across a bad signal. Vehicles sat half-

The Core Throttling

buried in drift. The comms mast blinked red at the edge of visibility. Beyond it, the horizon was flat and gone.

Cold hit Cavanaugh's face like something with intent behind it. He was glad of it anyway — glad for air that hurt, because it meant there was still air to breathe that had an opinion about him.

Sanderson stepped out with Ariza on one side and Cavanaugh on the other. Against every piece of medical advice in the vicinity, she lifted her injured hand and looked at it under the floodlights. The blue-white branching beneath the skin had faded to faint lines. Not gone.

Then the floodlights flickered.

Every light across Cold Horizon went out at once.

The station disappeared.

For one second there was only the Antarctic night, wind, breath, and the low hum rising up through the ice under their boots.

Then the sky changed.

Not aurora.

Not cloud.

A grid appeared overhead in the dark — faint blue-white lines crossing from horizon to horizon in vast rectangular panels, some of them brightening a fraction of a second after

the ones beside them, some failing to appear at all until the neighboring line refreshed. The pattern covered the whole visible sky. Too high to be projection. Too geometric to be weather. Too quiet to be anything the human eye was built to survive calmly.

No one spoke. Even Underwood didn't.

The grid flickered once. For a fraction of a second, stars showed in the gaps between panels, then vanished behind the next refresh, the way a picture vanishes behind the next frame of something running faster than the eye can hold. The sky didn't break. It updated.

Sanderson's hand closed around Cavanaugh's sleeve.

Her voice barely carried over the wind. "It's not local anymore."

The grid shifted, south to north, panel by panel, at a pace that had nothing to do with wind or rotation.

A low tone came from the station behind them as every exterior speaker activated at once. Not command. Not alarm. A synthesized voice built out of the station's own emergency system, flattened by whatever translated it.

The Core Throttling

"Maintenance cycle initiated."

The floodlights came back on.

The grid stayed.

On the wall display beside the surface access hatch, where weather status should have been, a countdown appeared in plain white letters.

29D 16H 43M

Underwood looked from the display to the sky and back.

Cavanaugh watched her understand the difference between secrecy and scale.

There would be containment attempts after this. Classifications. Communications blackouts. False weather reports. Compartmentalized briefings written by people who had never stood under this. Officers above Underwood would decide the grid was information before they'd admit it was a warning. Cold Horizon would keep trying to behave like an institution, because institutions didn't know how to become anything else, even standing under a sky that had stopped pretending to be one.

But the sky had no clearance level.

The grid brightened over Antarctica, each line straight enough to make the horizon look temporary by comparison.

Sanderson leaned heavier against him.

Kalamazoo

Cavanaugh tightened his arm around her and
stared up at the world showing its seams.

The countdown kept moving.